

BRAZILIAN PORTUGUESE RHOTIC VARIATION AND DELETION
IN SALVADOR AND SÃO PAULO

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To my parents

Acknowledgements

This dissertation is the consequence of chance circumstances in life combined with the realities of the COVID-19 pandemic. In 2021, as I was nearing the stage of beginning data collection for my qualifying paper, I was faced with the dilemma of finding a topic to research. With international travel still limited, I either needed to find a way to collect data domestically, or remotely. Since I was not particularly interested in working with English and I did not have sufficient access to speakers of another language here in the United States, besides maybe Spanish, I turned to my boyfriend, Diego, who is from Salvador. I asked him what was interesting about the way they spoke in his city and after some thought, he told me that they delete the “r” a lot at the ends of words. I started looking into this and discovered the insane complexity that is Brazilian Portuguese rhotics and decided to work with this phenomenon in Salvador.

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BRAZILIAN PORTUGUESE RHOTIC VARIATION AND DELETION IN SALVADOR AND SÃO PAULO

Brazilian Portuguese has two rhotic phonemes: the alveolar flap /r/ and the historically long version which previous publications variously call velar, uvular, or glottal fricatives, or alveolar trills or approximants. This variation occurs both within and across dialects. Deletion is also common, most notably in word-final position. This word-final deletion has been attributed to African influences on Brazilian Portuguese. Salvador is the largest Brazilian city with a population predominantly self-identified as being of African descent (Brazilian census categories *preto* “black” and *pardo* “brown” or “mixed race”). Given these factors, there is a question of how race may be tied to deletion of the rhotic in this variety.

For this dissertation, 35 participants (self-identified as 21 *preto*, 6 *pardo*, 7 white, 1 indigenous) from Salvador were recorded reading predetermined stimuli of isolated words and sentences. Additionally, 10 participants (1 *preto*, 3 *pardo*, 6 white) from São Paulo were recorded and demographically matched to a subset of participants from Salvador. There are 6,383 total instances of the rhotic phoneme. Productions were classified as exhibiting deletion or for voicing, frication, flapping, and place characteristics.

The results indicate that the surface forms of the phoneme are more variable than previously cited, with palatal fricatives common in Salvador and flap + fricative variants common in São Paulo, along with other less frequent forms. Furthermore, participants identifying as *preto* or *pardo* delete the phoneme more frequently than those identifying as white. This difference is further complicated by the fact that most Brazilians, regardless of identification, have a mix of African, Indigenous American, and European ancestry, and a lack of

consistent classification of individuals into racial categories. In Salvador, glottal fricatives predominate across the board with higher deletion rates. In São Paulo, glottal fricatives predominate in onset position, but alveolar trills and approximants and flap + fricative variants predominate in coda position. When demographically matched for race, age, gender, and socioeconomic class, deletion rates are higher in Salvador, suggesting that African influences are stronger there than São Paulo. Deletion is the most common word-finally, but it occurs in all environments.

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1. Introduction

This dissertation presents a phonetic and sociolinguistic study on the production, lenition, and deletion of the non-flap rhotic phoneme in the Salvador, Bahia, variety of Brazilian Portuguese. The city of Salvador is the capital of the State of Bahia, located in the northeast region of Brazil. It is home to about 2.4 million people, making it one of the five largest cities in Brazil. In addition, about 83% of residents of Salvador identified as *preto* “black” or *pardo* “brown” or “mixed race,” racial categories associated with African ancestry, in the 2022 census, making it the largest city in Brazil with a majority identifying as such (*Instituto Brasileiro de Geografia e Estatística* 2023a, b). Additionally, people identifying as white in Salvador may still have African ancestry and have parents or siblings who identify as *preto* or *pardo*. This variety of Brazilian Portuguese is of interest because of its extensive lenition phenomenon occurring in all environments for the phoneme.

There are three main things to note regarding the history and demography of Salvador: Bahia’s role in the Atlantic slave trade, racial mixing between Africans and Europeans in Brazil, and the effects of post-slavery European immigration to Brazil. Firstly, Bahia had a dominant role in the Brazilian system of slavery, especially in its early days. Salzano and Freire-Maia (1970) called Bahia “the main center of business and distribution of slaves,” receiving the enslaved Africans directly and then distributing them throughout Brazil. Salvador, being the main port and capital city of Bahia, was for many years the main port of entry for enslaved Africans (Dixon 2016). The prevalence of Africans in Salvador from an early point in Brazil’s history established a foundation population of Afro-Brazilians which continued to be built on over the centuries until today’s demographics. Secondly, there was extensive racial mixing in Brazil that led to a present-

day situation where many people who are considered white are known to have African ancestry. At the start of colonization and continuing throughout the colonial era, the Portuguese immigrants in Brazil were predominantly male. As a result, it was common for the Portuguese men to have children with the enslaved African women (Ribeiro 1995; Telles 2002). As each subsequent generation was born, further genetic admixture occurred so that, over time, a range of different skin colors came into existence among people from the same ancestral populations (Pena et al. 2011). Thus, since everyone had the same origins, Brazilians began to classify race by physical appearance, as opposed to ancestry (like the United States) (Nogueira 1955; Harris 1964; Kottak 1967; Ribeiro 1995; Telles 2002). Today, people who are classified as white often have African ancestry, as has been shown in genetic tests (Kehdy et al. 2015; Lima-Costa et al. 2015; Magalhães et al. 2015). In fact, people who identify as white may even have black parents or siblings, as is the case for four of the seven white participants in the current study. Lastly, after the abolition of slavery, many immigrants from Europe moved to the south and southeast of Brazil, fundamentally altering the demographics by increasing the white population in these regions (Ribeiro 1995; dos Santos and Hallewell 2002). However, Salvador received very few immigrants by comparison, maintaining its demography of predominant African descent. Overall, the combination of these three factors means that variety of Salvador has had of some the greatest African influences among varieties of Brazilian Portuguese. Further details are provided in chapter 3 of this dissertation.

This research is important to the field of linguistics because of the complex phenomena that occur with rhotics. In Brazilian Portuguese, the wide range of possible ways to pronounce this rhotic makes the language of particular interest to linguists. Additionally, this dissertation

has found substantial variation in the surface forms of the rhotic, many of which have not been documented in research on other varieties of Portuguese. Finally, although there are phonological studies such as Battisti (2021), the majority of contact-oriented studies on Brazilian Portuguese have focused on morphosyntactic phenomena. This dissertation adds a significant phonetic contribution to the literature.

The demographics of Salvador are of particular importance because linguists have frequently cited African influences as being among the most significant factors in the development of Brazilian Portuguese. Such African influences include word-final deletion of the rhotic, one of the key observations examined in this dissertation (Mendonça 1973; Reis 2015; Clements 2018). Since the phenomenon of rhotics in Salvador has not yet been extensively studied, the present study aims to contribute to the existing knowledge of Brazilian Portuguese rhotics by documenting the phoneme in this variety.

Lastly, the results from Salvador in this dissertation are compared with a second dataset collected from the city of São Paulo with the same stimuli. The city of São Paulo, as will be discussed in chapter 3, underwent significant demographic changes in the late 19th and early 20th centuries when the population went from being predominantly black to predominantly white. Consequently, today, only about 41% of the population of São Paulo identifies as *preto* or *pardo*, while 57.8% identifies as white (*Instituto Brasileiro de Geografia e Estatística* 2023a). This suggests that African influences may be weaker in São Paulo than in Salvador. If this is the case, one would expect to see lower deletion rates on the rhotic phoneme in São Paulo than in Salvador.

This dissertation is organized as follows: Chapter 2 gives an overview of rhotics and rhotic deletion in Brazilian Portuguese and refers to works on the phenomenon of rhotic deletion in other languages as well. Chapter 3 gives an overview of the history and demographics of Salvador and Brazil. Chapter 4 presents the questions and hypotheses of this dissertation. Chapter 5 describes the methodology used. Chapter 6 presents the data and statistical tests for the Salvador corpus as a whole. Chapter 7 discusses the findings of the analysis reported in chapter 6 and establishes the inventory of allophones of the rhotic in Salvador as well as the estimated deletion rates of the phoneme across prosodic environments. Chapter 8 tests the data for demographic differences among the participants from Salvador. Chapter 9 presents the data from São Paulo and statistical tests for differences between the two cities. Lastly, chapter 10 gives the conclusion and proposes future work related to this dissertation.

2. Rhotics and rhotic deletion in Brazilian Portuguese

2.1. Rhotics in Brazilian Portuguese

The Portuguese language has two rhotic phonemes: the alveolar flap /r/ and another phoneme, which varies greatly in production. While many of these productions are not segments traditionally considered rhotics, this phoneme is considered a rhotic because of its origins dating back to the Latin “r” as well as the fact that it is represented orthographically with an “r.” Historically, the distinction of two rhotic phonemes in Portuguese can be traced to an intervocalic contrast between a short and a geminate trill from Latin (Câmara Jr. 1953, 1976; Tarallo 1990; Rennie 2015). Today, in Portuguese, the existence of these two rhotic phonemes is exemplified by the following minimal pair: *caro* [karu] “dear” and *carro* [kaʔʔu] (ʔʔ = pronunciation varies) “car.” This phonemic contrast only exists in intervocalic contexts. Elsewhere, their distribution is complementary. The distribution of these two rhotic phonemes bears a resemblance to what is found in Spanish (Hualde 2004; Bradley and Willis 2012), though it is not identical. Generally, in varieties of Brazilian Portuguese, the flap occurs in onset clusters (*prato* “plate”) and intervocalically (*coração* “heart”) while the other rhotic occurs in simple onsets (*rato* “rat”), in codas (*cantar* “sing” and *arte* “art”), and intervocalically (*terra* “land”) (Barbosa 1983; Mateus & d’Andrade 2000). Additionally, the findings in this dissertation, as well as other work such as that of Rennie (2015) demonstrate that the rhotic phoneme in question may surface as a flap when it is in word-final position and followed by a vowel, effectively neutralizing in this context (e.g., *limpar as janelas* “to clean the windows”). However, this is the only circumstance where that may occur in Salvador. Given all this information regarding the rhotic phonemes in Brazilian Portuguese, the distinction between the two can be considered a

“marginal contrast” or an “intermediate relationship” between fully contrastive phonemes and allophones (Hall 2013; Hall and Hall 2016).

Both rhotic phonemes are represented orthographically as “r” except for the intervocalic non-flap rhotic, which is written “rr.” It is important to note that not all varieties of Portuguese distinguish between the rhotic phonemes, as mergers of them have been shown to exist in African varieties of Portuguese, as well as in southern varieties of Brazilian Portuguese influenced by European immigrants (Reinecke 2006; Brandão & Paula 2019; Mendes & Agostinho 2020, 2022; Agostinho 2022). Additionally, Portuguese-based creoles tend not to retain the rhotic distinction that existed in Portuguese at the time of their creolization (15th to 19th centuries) (Clements 2014c). Therefore, given the contact situation surrounding the development of the Salvador variety, it would be reasonable to think that a merger of the rhotic phonemes could have occurred. However, there is no indication in the data of this dissertation that such a merger has occurred other than the aforementioned word-final flapping before a vowel. Additionally, it is common for varieties of Spanish to maintain this marginal contrast as well (Hualde 2004; Willis and Bradley 2008; Bradley and Willis 2012; Henriksen 2015). Given the similarities between the rhotic phoneme distribution between the two languages, this provides further indication that two separate rhotic phonemes exist in Salvador.

Table 2.1: Various productions of the rhotic phoneme in question as listed in the literature.¹

Place of articulation	Possible Productions
Alveolar	[r], [r̥], [ɹ]
Velar	[x], [χ]
Uvular	[χ], [ʁ], [R]
Glottal	[h], [ɦ]

A review of the literature reveals the many ways in which the rhotic phoneme may be produced in Brazilian Portuguese, shown in Table 2.1. No variety has been documented as having all of these allophones but rather, a subset of them. Despite this, pronunciation within varieties is still variable. Additionally, there were instances in the data collected in this dissertation which are not found in Table 2.1, such as palatal fricatives. It is also important to note that speakers are not typically aware of which variant they are producing (Callou & Leite 1990; Cardoso 1999). However, based on personal interactions with native speakers, it appears that speakers do have an awareness of variants which do not exist in their variety. According to Rennie (2015), in the state of Bahia, production of the rhotic is expected to be a back fricative ([χ], [ʁ], [h], [ɦ]). The findings of this dissertation suggest that this is largely the case, but anything from palatal through glottal is possible, as well as the uvular trill, [R]. There were also a few marginal productions of other phones surfacing as the rhotic phoneme, which will be presented with the data in chapter 6. Furthermore, extensive variety outside that found in Salvador was found in the data from São Paulo, which will be presented in chapter 9.

¹ Reed and Leite (1947); Lopez (1979); Guy (1981); Barbosa (1983); Silva (1998); Mateus and d'Andrade (2000); Barbosa and Albano (2004); Osborne (2015); Rennie (2015)

2.2. Rhotic deletion in Portuguese and its connection to African influences

2.2.1. African Influences and their connection to rhotic deletion in Portuguese

Many features which developed in Brazilian Portuguese have been attributed to the languages spoken by enslaved peoples brought from Africa. These languages went on to serve as substrate influences in the varieties of Bahia (Clements 2018). Additionally, it has been suggested that African influences could have been a driver behind the divergence of European and Brazilian Portuguese (Clements 2014a). In this article, Clements (2014a) also states his belief that the high proportion of the black population led to ongoing linguistic changes in Brazilian Portuguese. As a result, since Bahia was home to a predominantly black population, African features were more strongly represented in the language there. Further supporting this notion, scholars have also generally agreed that the contact of various ethnic groups was a key component of the evolution of Brazilian Portuguese (Mello 2014). Additional evidence of African influence in Brazil comes from idiomatic expressions in the country that often have word for word equivalents in African languages, but these expressions are not present in European Portuguese (Mello, et al. 1998). Mendonça (1973) argued that these various African influences also led to phonetic alternations in Brazilian Portuguese, including rhotic deletion. Evidence of rhotic deletion among African-influenced speech can be found diachronically in Lusophone literature. Alkmim (2006) found consistent rhotic deletion in literary portrayals of slave speech ranging from the 15th through 19th centuries in both Portugal and Brazil. Some works that include such portrayals include plays by the 16th century Portuguese playwright Gil Vicente, the 1789 pamphlet *O preto e o bugio*, and the 1914 novel *Rei Negro* by Coelho Neto (Alkmim 1996, 2008; Reis 2015).

2.2.2. Relating the phonologies of languages of Africans in Brazil to rhotic deletion

While African influences have been cited as a factor in the deletion of the word-final rhotic, it is important to understand the syllable structures permitted in the languages spoken by the people taken to Brazil from Africa and determine what kind of role could have been played by the phonologies of their native languages. Clements (2018) cites Kwa and Bantu languages as well as languages from Angola as being common native languages of Africans in Brazil. Mendonça (1973) gives multiple locations of origin for Africans brought to Brazil including Guinea, Angola, Costa da Mina, Sierra Leone, Gambia, Congo, Mozambique, and Cameroon. In terms of languages, Mendonça cites Mandinka, Wolof, Yoruba, Hausa, Fula, and Kimbundu as native languages of Africans in Brazil. Similarly, Harris (1964) cites Yoruba and Ashanti (a dialect of Twi) as native languages of Africans along with Congo as a place of origin. Nishida (2003) cites Angola, the Bight of Benin, and the Bight of Biafra as places of origin along with Ewe, Hausa, and Yoruba as native languages. Lastly, Kehdy et al. (2015) give Yoruba and Mandinka peoples as likely origins of residents of Salvador given their genetic data. Keeping these various cited origins in mind, let's briefly examine the syllable structures of these languages and others found in the cited regions to see whether a rhotic consonant would be permitted in a coda in the native languages of Africans in Brazil.

To start, there are many languages from the places cited which do not allow codas at all. These languages include Yoruba, Chokwe, Kimbundu, Ewe, Ngangela, Kikongo, Kwanyama, Twi, and Umbundu. In Yoruba, the syllables may be CV, V, or N (Ogunbawale 1970, p. 192; Akalugo 1998, p. 45-47). Manyah (2002, p. 116-117) gives the same possible syllables for Twi. In Ewe, the list is similar, with addition of the possibility of a complex onset with CCV syllables (Duthie 1996,

p. 9-11). In Ngangela, syllables are limited to CV and CVV (Maniacky 2003). Some additional, more complex syllables can be found in Kikongo and Kwanyama, though codas are still not permitted. Kikongo syllables may be V, VV, N, CV, CVV, CCV, or CCVV. Additionally, all syllables with CC onsets are nasals followed by oral consonants (Tavares 1915, p. 4). Kwanyama permits the following syllable structures: V, CV, GV, CGV, and NCGV (G representing a glide). Additionally, in a small set of morphemes, syllabic nasals are permitted phonetically, though a phonemic vowel attached to the nasal is still produced in careful speech. (Ntondo 2015, p. 57-58; Halme 2004). In some languages, there are situations which may be perceived as being coda consonants, however research has established that they are not codas. For example, Martins (1990) states that all words in Chokwe end in vowels and that nasals which may appear to be in word-medial codas are actually part of a complex onset in the following syllable. Similarly, in Kimbundu, all syllables end with a vowel and word-medial nasals preceding consonants are part of a complex onset in the following syllable (Chatelain 1889, p. xxi-xxiii). Lastly, looking at Umbundu, Valente (1964, p. 26-27) states that Umbundu has no words which end in a consonant.

Given the information above, it is clear that a large number of the languages which are believed to have been native languages of Africans in Brazil do not allow coda consonants at all. Thus, it is quite plausible that second-language acquirers of Portuguese with African L1s would have had trouble producing the word-final rhotic and therefore have deleted it. With enslaved people comprising such a large proportion of the population in Brazil, particularly in Salvador, it is reasonable to believe that this could have had lasting influence on Portuguese spoken in Brazil and made rhotic deletion more common.

However, it is not the case that all of the languages did not allow coda consonants; quite a few still allowed codas. However, in some of the languages that allowed codas, there was a limited distribution of coda consonants that would have prohibited a word-final rhotic. Looking at languages which allowed a limited number of coda consonants, Susu permits V, CV, CVV, and CVC as syllable structures, where all codas are nasal consonants (Duport and R.R. 1865, p. 5; Houis 1963, p. 37). Similarly, in Mende, syllables consist of one or two consonants in an onset and a vowel in the nucleus, with occasional V only syllables as well as nasals in coda position (*Grammar of the Mende language* 1882, p. 11). Thus, in Susu and Mende, a rhotic consonant would not be permitted in coda position at all. Lastly, syllables in Mandinka may be V, CV, CVV, CCV, or CVC. In CVC syllables, the second consonant is either a nasal or the first half of geminate (Ngom 2000, p. 69-74; Balde and Drame 1982, p. 31). Consequently, the only rhotic in coda position in Mandinka would be the first half of a geminate, rather than the codas of the word-final or syllable-final rhotics in Portuguese.

While many of the potential substrate languages do not allow coda rhotics, some possible native languages of Africans in Brazil would have permitted such codas. In the Edoid languages, the most common syllable structures at the phonemic level are CV and V. However, Dëgëma and Epie also allow CCV and CVC syllables. The others still have CCV at a phonetic, though not a phonemic, level (Elugbe 1989, p. 44). In Hausa, all syllables in Hausa are consonant initial, with CV, CVV, and CVC being the possible syllable structure (Jaggar 2001, p. 23-25). Taking Fula languages into consideration, codas are permitted. The basic Pulaar syllable structure is CV(V)(C)(C) (Niang 1997, p. 35), while in Pular, the possible syllable structures are CV, CVC, and CVV (Diallo 2000, p. 48). All syllables in Wolof consist of at least one vowel and one consonant,

which may be either in the onset or the coda. The vowel may be long and there may be two consonants in a coda. Additionally, an onset may consist of a nasal consonant followed by an oral consonant (Ka 1994, p. 73-55; Diouf 2017, p. 50-52; Ngom 2003). Finally, syllables in Ewondo may be V, CV, CVC, CCV, CCVC, or N. All CC clusters are nasals followed by oral stops (but not prenasalized stop phonemes) (Redden 1980, p. 6).

Thus, there are clearly many languages spoken by Africans in Brazil which permitted codas as well. With this fact, it is clear that not all African languages would have contributed to rhotic deletion in Brazilian Portuguese. However, due to a lack of good records, many of the languages are guesses based on the general areas of Africa that people were taken from. Additionally, there is not an indication as to whether languages which would not have permitted rhotics in codas were more frequently spoken by Africans in Brazil than those which would permit codas. Nevertheless, it is still plausible that these African influences drove rhotic deletion and there is not enough evidence to reject this possibility.

Therefore, among the languages believed to be spoken by Africans brought to Brazil, there is a mix of languages which would allow rhotics in coda position and languages which would not allow such consonants in codas (or codas at all). A lack of records makes it difficult to know whether one type of language or the other tended to be more common or influential among the enslaved African population trying to acquire Portuguese. All we can be certain of is that a strong propensity for coda rhotic deletion, especially word-final deletion, exists within Brazil and Salvador, and the tendency among linguists to attribute this deletion to African substratal influences on Brazilian Portuguese cannot be rejected when examining the syllable structures of the potential languages spoken by the African people brought to Brazil.

Consequently, upon examination of syllable structures among various African languages, it is likely that the phonologies of these languages did contribute to and increase word-final rhotic deletion in Brazilian Portuguese.

It is also possible that detectability played a role in the dropping of rhotics in coda position. Clements (2014b) argued that a CV syllable structure is more detectable than others and ultimately found that the forms selected in Portuguese- and Spanish-lexified creoles tended to match his criteria for detectability (CV structure, stressed syllables, and free-standing morphemes are more detectable). Clements also emphasizes the importance of frequency in form selection as well. Given the large numbers of Africans in Brazil who would have spoken languages that did not allow rhotics in coda position, and the higher detectability of CV syllables, it is plausible that this created a tendency to delete the word-final rhotic among the enslaved population in Brazil. This, in turn, would have created a high frequency of deleted rhotics in the input for any other learners of the language brought to Brazil as well as for any children born to the Africans in Brazil who were acquiring Portuguese as their first language. Thus, it is conceivable that African influences could have played a nontrivial role in rhotic deletion in Brazil. With these possibilities and Salvador's status as the largest city in Brazil with a majority of residents of African descent, it is not surprising that Salvador developed a variety with a reputation of deleting rhotics more than other varieties of Brazil.

2.3. Other work on rhotics in Portuguese

Using interviews, Callou et al. (1996) examined rhotic variation in coda position in various Brazilian cities, including Salvador. Syllable-finally, they found a 2% deletion rate, with 55% of productions being velar, 40% being glottal, 1% being uvular, and 2% being flaps. Word-

finally, they found a 62% deletion rate, with 18% of productions being velar, 10% being glottal, and 10% being flaps. The deletion rate in word-final position was the highest in Salvador (62%) among the cities studied, with the next highest being 50% in Recife. However, the syllable-final deletion rate was comparable to the other cities which deleted at either 3 or 4%.

Regarding other work in Bahia, there are the *Programa de Estudos do Português Popular Falado de Salvador* “Program of Studies of the Popular Portuguese Spoken in Salvador” (PEPP), the *Vertentes* project, and *O Projeto da Norma Urbana Linguística Culta* “The Urban Standard Linguistic Norm Project” (NURC). All of these projects focus overwhelmingly on sociolinguistic studies dealing with morpho-syntactic variation. Among the limited phonetic work completed in these projects, the PEPP has a study on the acquisition of the trill /r/ by Bahian learners of Spanish (Gomes, 2016), and the *Vertentes* project has work on rhotacization of the lateral approximant /l/ in complex onsets and in coda position into an alveolar flap [r] (e.g., *placa* → *praca* “plaque”) (Dias 2019). A similar phenomenon, lateral rhotacization, has been noted in lusophone literature portraying the speech of Africans (Alkmim 1996, 2006, 2008). Thus, this is a potentially relevant phenomenon that could occur in Salvador due to the historical African population there. However, no such rhotacization was found in the corpus for this dissertation. Regarding NURC, which recorded speakers from five Brazilian cities, including Salvador, the current author was unable to obtain representative data from the Salvador NURC database.

With respect to work on rhotic deletion in other varieties of Portuguese, Serra and Callou (2013) compared word-final rhotic deletion in the varieties of Rio de Janeiro and Lisbon within the context of informal interviews. They found that that speakers from Rio de Janeiro deleted 76% of word-final rhotics, compared with a 32% deletion rate among Lisbon speakers.

Oushiro & Guy (2015) examined the correlation of various characteristics of speakers from the city of São Paulo in sociolinguistic interviews and found significant correlations between word-final rhotic deletion and noun phrases not marked with word-final /s/ plurality agreement, verbs not marked for 3rd person plural agreement, and verbs not marked for 1st person plural agreement. These correlating factors are also listed as African influences on Brazilian Portuguese in Clements (2018), further indicating a historic African influence in the rhotic deletion of Salvador. Cardoso (1999) interviewed four native speakers of Belém, two students and two professors, finding that morphosyntactic category and following phonological environment affected word-final rhotic deletion rates. Reis (2015) examined word-final rhotic deletion in participants from Belo Horizonte (Minas Gerais) in informal interviews, sentence reading, and isolated token reading, finding that deletion was most common in the interviews, followed by the sentence reading, and finally, isolated tokens. Vieira (2021) examined rhotic production and deletion in the Brazilian state of Espírito Santo. Lastly, Vieira and Balduino (2021) examined rhotic deletion in São Tomé and Príncipe using interviews, finding that 77.5% of word-final rhotics and 32.6% of syllable-final rhotics were deleted.

The most extensive study on Brazilian Portuguese rhotics is Rennie (2015), which examined rhotic production and deletion in a wide variety of contexts and environments for speakers from the southern portion of the state of Minas Gerais. Deletion was found to be more common before consonants than vowels or a pause and verbs were found to be more likely to delete the word-final rhotic than nouns. Additionally, deletion was uncommon in monosyllabic tokens. She also reports that lenition is occurring in all syllable-final segments of Brazilian Portuguese.

It is also important to note a parallel phenomenon of word-final /s/ deletion in Brazilian Portuguese. The rhotic phoneme and /s/ are the only coda consonants permitted in Portuguese. Guy (1981) examined the deletion of word-final /s/ in Brazilian Portuguese and found that stress and whether the following syllable began with a vowel or consonant were factors affecting deletion, with stressed syllables and syllables followed by vowels less likely to delete. The retention of word-final /s/ before vowels but not before consonants was attributed to a more general tendency to maintain a CV syllable structure. These results are similar to those of studies of word-final rhotic deletion, indicating that it may be part of a greater trend and that the factors affecting /s/ deletion affect the rhotic deletion as well.

2.4. Rhotic deletion in other languages

Where other languages are concerned, rhotic deletion is a common phenomenon. For example, English is widely known to have had a systematic deletion of the approximant rhotic /ɹ/ in coda position in several dialects (Lass 2008). Coda rhotic deletion has been found in 12th and 13th century varieties of French, where the rhotic is believed to have been a fricative. However, the pronunciation of the French rhotic in coda position was reintroduced by grammarians in the 16th and 17th centuries (Gess 1999). In Andalusian Spanish, the trill /r/ has been documented to delete word-finally and neutralize with the lateral /l/ or delete in syllable-final position (Ruiz-Sánchez 2007). Rhotic deletion in other languages can occur in onset position as well. In Samothraki Greek, the rhotic can surface as a trill, a flap, an approximant, or a fricative when produced. However, the dialect is known to delete rhotics in both simple and complex onsets and to compensate for this deletion by lengthening the following vowel. The rhotic in coda position was also once lost in Samothraki Greek as well, though it has since been

restored (Katsika & Kavitskaya 2015). Furthermore, the phenomenon of rhotic deletion is far from being limited to Indo-European languages. For example, in Hebrew, the rhotic is uvular and is typically an approximant, a fricative, or a trill, depending on the speaker and its position in the word. Deletion of the rhotic in Hebrew has been found in all positions, particularly for children acquiring the language (Cohen 2015; Cohen & Ben-David 2016).

This dissertation contributes to the existing literature on rhotics by examining an understudied feature of a widely spoken variety which can help linguists better understand the phenomenon of rhotic deletion in Brazilian Portuguese and in turn, add to the understanding of lenition and deletion within the field of linguistics. Additionally, the history and demographics behind rhotic loss in Brazilian Portuguese mean that studying the Salvador variety can offer a new perspective on the conditions leading to widespread rhotic deletion.

3. History and demographics of Salvador

Although both countries were colonized by European powers in the Americas, the colonial history of Brazil has many differences from the colonial history of the United States. Over a period of several hundred years, these differences allowed for the development of a nation whose demographics starkly contrast those of the United States and ultimately, they led to a concept of race in Brazil which is based more on physical appearance than ancestry. For this reason, differences in the population based on race in Brazil do not mean the same thing as differences by race in the United States. With this in mind, in order to fully understand the content of this dissertation, it is important to understand in the context of Brazil's history and demographics and how the sociolinguistic implications of differences within the Brazilian population are not necessarily the same as one would find in the United States.

This chapter is organized as follows: Section 1 reviews the history of Salvador and Brazil and the elements of the history that contributed to today's demographic situation. It is subdivided into history prior to abolition of slavery and history after the abolition of slavery. Section 2 presents the current demographic situation in Brazil and is divided into subsections which cover the differences in race between the US and Brazil, the variety of racial terms in Brazil, racial self-identification versus classification by others, and Afro-Brazilian movements and their effect on racial identification. Section 3 relates the demographic situation to genetic studies on the Brazilian population. Lastly, section 4 relates this information to the dissertation and what it means in the context of rhotics in Salvador.

3.1. History and how it relates to demographics

3.1.1. Colonial history and history of slavery

When the Portuguese first began to colonize Brazil in the 1500s, the area was populated by various indigenous peoples, estimated by Freire and Maia (1970, p. xv) to have been about 1 million people, the best known of which is the Tupi. At the start of colonization and continuing throughout the colonial era, the Portuguese sent predominantly men to live in Brazil (Ribeiro 1995; Telles 2002). Without Portuguese women to accompany them, the colonists looked to marry and have children with the indigenous inhabitants of Brazil. In doing so, the Portuguese took advantage of an indigenous practice in which strangers would be incorporated into the indigenous community, which the Portuguese called *cunhadismo* (literally “brother-in-law-ism”). As a part of this practice, one man was able to take many indigenous wives (known as *temericós*) and integrate himself into multiple communities. From these relationships, the Portuguese men had many children with indigenous women, solidifying their connections with these communities and establishing the presence of the Portuguese in Brazil. This practice of *cunhadismo* formed the foundation of early Brazilian society and allowed Brazil to survive as a colony (Ribeiro 1995).

Having established themselves in Brazil, the Portuguese began to enslave the indigenous people in order to exploit Brazil’s resources for the benefit of Portugal. The number of indigenous people forced into slavery steadily increased over much of the 16th century. From approximately 1540 to 1570, enslaved indigenous individuals most commonly worked on sugar plantations along the coast, particularly in Bahia (Nishida 2003, p.14; Schwartz 1985, p. 35-37). As sugar plantations continued to grow, the demand for slave labor increased as well.

Eventually, the demand for slave labor in Brazil exceeded what the Portuguese could force from the indigenous population, which was decimated by high mortality rates due to the brutality of slavery and the introduction of European diseases to Brazil. As a result, by the 1570s, importation of labor from Africa began (Nishida 2003, p. 14; Marchant 1942; Schwartz 1978; Hemming 1978; Levine 1999, p.46; Harris 1964, p. 12-13).

While colonies throughout the Americas enslaved Africans, no place had more enslaved people than Brazil. It has been estimated that one-third of Africans brought to the Americas and 60 percent of those brought in the 19th century were taken to Brazil (Levine 1999, p. 66). Many sources estimate that over the course of the slave trade, around 3.5-3.6 million Africans were taken to Brazil (dos Santos and Hallewell 2002; Goulart 1950; Mattoso 1981); however, the estimates vary, and this figure is on the lower end of the spectrum. Others have estimated slightly higher at around 4 million (Dixon 2016, p. 29; Curtin 1969) while some estimates have even been as high as 18 million. Ultimately, there is no way to know with certainty exactly how many people were brought over due to a lack of records (Salzano and Freire-Maia 1970, p.40). Nevertheless, due to the harsh conditions of slavery, there was a high mortality rate of enslaved Africans that necessitated continuous engagement with the slave trade in order to supply Brazil with the labor force it needed (Nishida 2003, p. 14; Schwartz 1985, ch. 3; Conrad 1986, p. 15-19).

Within Brazil, particularly during the early days of slavery, Bahia dominated the slavery system. It has been estimated that nearly 1.5 million of the Africans brought to Brazil were taken to the shores of Bahia (Dixon 2016; Curtin 1969). Kottak (1967) referred to the northeast as “the historical center of a plantation economy,” and said it is the region of Brazil which

received the largest number of enslaved individuals from Africa. Salzano and Freire-Maia (1970, p.41) called Bahia “the main center of business and distribution of slaves.” They state that Bahia would receive the enslaved Africans directly and then distribute them throughout Brazil.

Salvador, being the main port and capital city of Bahia, was for many years the main place where the enslaved Africans were taken to (Dixon 2016, p.29). As time went on, the importance of Brazilian sugar in the world economy waned. As a result, the center of Brazil’s economy shifted from the northeast to the south. Consequently, enslaved Africans were more commonly taken to that region than the northeast and the capital of Brazil was moved in 1763 from Salvador to Rio de Janeiro (Nishida 2003, p.15). By the time slavery was abolished in Brazil in 1888, the enslaved population was predominantly concentrated in the southeast (dos Santos and Hallewell 2002). However, despite this decrease in prominence, Salvador still continued to participate actively in slavery and the slave trade. For example, it has been estimated that, in 1823, 20 percent of all enslaved Brazilians were still living in Bahia (Dixon 2016, p. 29; Klein 1986).

As a result of centuries of the slave trade, the city of Salvador became filled with a variety of people both from Africa and of African descent and within each group, there were both enslaved and free individuals. At the same time, the Portuguese colonists continued to be predominantly male and as a result, it was common for them to have children with the enslaved African women (Telles 2002). Combining these various factors led a to socially stratified system in Salvador that laid the foundation for the eventual development of the current situation surrounding race in Brazil (Nishida 2003; Dixon 2016, p. 31-32).

Looking at the social situation surrounding the last few decades of slavery in Salvador can give us some insight into the development of these divisions. In the 1840s, slavery in Salvador began to decline as the proportion of the free population continued to grow. The proportion of the city's population that was enslaved declined from 42 percent in 1835 to 26 percent in 1855 (Nishida 2003, p. 22; Morgan 1991). Coinciding with this reduction in the enslaved population was a switch from the majority of enslaved people being African-born to being Brazilian-born by the middle of the 19th century (Nishida 2003, p. 93). This is largely due to the fact that as long as the slave trade continued (until 1851), slave owners preferred importing labor from Africa over using reproduction to build on the existing enslaved population (Nishida 2003, p. 2, 25). Ultimately, by the late 19th century, the enslaved population was predominantly Brazilian born (Nishida 2003, p. 142). As free individuals and Brazilian-born individuals began to predominate the population during the final days of slavery in Brazil, they began to achieve a higher status in society than those who were still enslaved or African born, leading to some degree of hierarchy within the African and African-descended community of Brazil. This can be further combined with the fact that many Brazilian-born individuals of African descent were mixed with European ancestry and the fact that Brazilian-born people of African descent were divided into various color categories since the early colonial days, including *negro*, *preto*, *crioulo*, *pardo*, and *cabra* (Nishida 2003, p. 25, 130-131, 147). This variety of labels eventually translated over into the post-slavery population as a means of describing race.

As the 19th century continued on, it became evident that slavery was not going to last forever in Brazil. Despite the inevitability of its demise, the elites did everything they could to prevent the end of slavery in Brazil and delay its abolition (dos Santos and Hallewell 2002). This

was largely accomplished by taking a series of small steps over the course of several decades. The first of these steps was the official ban of the slave trade in 1831, though it did not actually end until 1851. Next, in 1871, there was the Free Womb Law (or *Rio Branco* Law), which declared that children born to enslaved women shall be free, although the owners of the mothers were permitted to keep the children until age 21 in exchange for a fee. The final step toward abolition was the Saraiva-Cotegipe Law, which freed enslaved individuals aged 60 and over in 1885. Lastly, slavery was completely abolished in 1888 with the passage of the Golden Law (Nishida 2003, p. 2; Dixon 2016, p.35; Levine 1999, p. xvi, 68; dos Santos and Hallewell 2002).

At the time of the passage of the Golden Law, only about 5 percent of Brazil's population had been enslaved, down from a peak of 50 percent. The move has been seen as largely symbolic, rather than one of genuine care for the people who were enslaved (Dixon 2016, p. 35; Conrad 1972; Levine 1999, 68). Many abolitionists believed that slavery had Africanized Brazil, and they resented that it had led to a population that was predominantly black. (The term "black" will be used to refer to the combination of the Brazilian terms *preto* "black" and *pardo*, frequently referred to as *negro*, also "black," in Portuguese. The reasoning for the combination of these two categories into one is detailed later.) They wanted to abolish slavery in order to replace the existing population with a white one (dos Santos and Hallewell 2002). For some context, around this time, the first two Brazilian censuses were conducted in 1872 and 1890. The non-white population of the country in these censuses, which consisted of people of African descent and indigenous people, was 61.9 percent and 56 percent respectively. It should be noted that at this point in time, the indigenous population was severely reduced (no more

than 5 percent), so the vast majority of the non-white population was of African descent (dos Santos and Hallewell 2002; Ribeiro 1995, p. 229). Other abolitionists argued against the economic costs and the impracticality of slavery. Once slavery was abolished, they did not do anything to help protect the freedoms of the recently freed population (Levine 1999, p.70).

During much of the slavery period, many of the terms that are now used to describe race in Brazil had certain associations that were connected to place of birth and the free/enslaved status of the individual in the years prior to abolition. For example, the term *preto* “black,” which has been used in the Brazilian census since the first one in 1872, was associated with being close to slave status. This led to many people of African descent in Brazil to avoid usage of this term. However, in the last couple decades of slavery and into the post-slavery era, the associations with this terminology had begun to change. The term *crioulo* “creole” had disappeared from existence and *preto* came to mean solely the skin color of black. The term *pardo* was used for individuals of African descent with lighter skin, and it, along with *preto*, was one of two official categories in the 1872 census denoting African ancestry. Additionally, the term *negro* came to identify the general black population (*pretos + pardos*), something that continues to this day (Nishida 2003, p. 146-147; Borges 1992; Nobles 2000).

There are many similarities between the ways Brazilians described race in the late 19th century and the way Brazilians describe race today. However, as will be seen in section 2, these are far from the only terms describing people of African descent, let alone the entirety of racial terms used in Brazil.

3.1.2. Post-slavery demographic history

In the post-slavery years, two opposing factors affected the degree of racial mixing that occurred in Brazil. The first is the fact that Brazil, unlike the United States, did not have formal segregation laws. There were no laws in Brazil like Jim Crow in the United States or Apartheid in South Africa. Nevertheless, discrimination and segregation along economic lines (where the poor were overwhelmingly non-white) were prevalent in Brazilian society (Ribeiro 1995, p. 12-13, 101; Salzano and Freire-Maia 1970, p. 54). Without any such laws, Brazilian people were able to marry who they pleased and this acted as a driver for further racial mixing. However, the second factor, which reduced racial mixing, was the tendency for people to have relationships within their own color group. The tendency for relationships to be between two people of the same skin color can be traced back several centuries, with interracial marriage still being rare in the early to mid-19th century, despite this being a time where people were still being brought to Brazil for slavery and many people in Brazil were still solely of African or European descent (Nishida 2003, p. 130-131). At the same time, Brazilian elites decided to use racial mixing as a means of whitening the Brazilian population. By incentivizing European immigration and then encouraging these immigrants to mix with the black Brazilian population, they had hoped to turn Brazil into a “white country” within a few generations (dos Santos and Hallewell 2002; Seyferth 1985; Telles 2002; Skidmore 1974). This subsection will explore, from the days immediately following slavery through today, the balance between the freedom to marry any race and the tendency for relationships within color in Brazil, and the attempts to alter Brazil’s demographics through immigration and subsequent racial mixing.

With the prevalence of relationships within skin color in Brazil, this raises a question of how exactly one could have arrived at the current situation where regardless of racial identification, nearly everyone has both African and European ancestry. The way this assortative mating fits in with racial mixing is as follows: First, two people from different populations (e.g., one African and one European) have children. Next, these children find others in their generation with a similar background and then tend to have more children with those whose skin color most closely matches their own. The following generation now has a wider variety of skin colors, and they also tend to have children with those whose skin color matches their own. This repetition over the course of generations results in a range of skin colors from completely white to completely black that comes from the same ancestral populations (Pena et al. 2011).

The tendency to have relationships with those of a similar skin color has continued throughout Brazil's history and continues to this day. Kehdy et al. (2015) and Silva et al. (2015) found that after nearly five centuries of admixture, Brazilians still preferentially mate with individuals with "similar ancestry" (and its correlated phenotypes and socioeconomic characteristics). Salzano and Freire-Maia (1970 p. 55-56), using a skin color scale of 0 to 8, found that marriage within skin color was more likely than what would be expected in random pairing in the east and northeast of Brazil. Additionally, Petruccelli (2001) found that the majority of marriages in Brazil were within the same census category from the *Instituto Brasileiro de Geografia e Estatística* "Brazilian Institute of Geography and Statistics" (IBGE). However, the percentage did decrease slightly from about 81% in 1987 to about 78% in 1998, with this change being driven predominantly by more frequent interracial marriage in the younger generation. Furthermore, education level also plays a role in the likelihood to marry within one's

race in Brazil. Higher levels of education increase the chance of interracial marriage for those identifying as *preto* or *pardo*, but they decrease the chance of interracial marriage for those who are white. Looking at this on a regional level, education increases the likelihood of interracial marriage in the north and northeast of Brazil (where Salvador is located), but it decreases the likelihood of interracial marriage in the south and southeast (Petrucelli 2001).

While relationships tended to be within skin color, this was not an automatic rule which always applied. However, among the elites of Brazil, mixing races was looked down on (Ribeiro, 1995 p. 91). Of course, it is important to remember the context of how this elite was formed: by racial mixing and subsequent assortative mating which over time resulted in a group of people with white features gaining control, despite them having the same origins as the people of color in the lower socioeconomic classes.

Thus, in Brazil, there is a tradition of staying within one's race when marrying and having children, despite the racial mixing that occurred in the early days of Brazil's history. Nevertheless, a lack of legal restrictions on interracial relationships allowed for occasional further racial mixing to occur. Additionally, recent decades have seen small decreases in the tendency to marry within one's race. Ultimately, the initial racial mixing and subsequent assortative mating in Brazil allowed for the development of the complex racial situation seen in the country today.

The post-slavery era in Brazil also saw attempts by elites to whiten the country. With slavery no longer in the country, they desired to replace its labor force with European immigrants, rather than employ the existing black population in the country. This exclusion of black people from the labor market was one driver of the persisting racial wealth gap in Brazil

that still exists today. Since the last decade of the slavery era, some people had suggested the idea of bringing immigrants from places beyond Europe, such as Asia. However, the majority of elites rejected this idea on the basis that anyone non-white was inferior to white people. This notion of inferiority was embodied in the opinions of the *Congresso Agrícola* “Agricultural Congress” and speeches by elites such as Joaquim Nabuco, who was a prominent abolitionist. Nabuco also feared that Brazil would always be seen as a non-white country. Others, like Euclides da Cunha and João Batista de Lacerda, believed that European immigrants would bring “progress” to Brazil and improve its reputation among other countries (dos Santos and Hallewell 2002). Of course, adding white people to Brazil does not remove the existence of black people within it. Thus, the goal with this European immigration was to make black people disappear through racial mixing with white people. The aforementioned elites believed this would work because they thought non-white features were inferior and were unlikely to last when large numbers of white people were added to the population (dos Santos and Hallewell 2002; Ribeiro 1995, p. 224; Telles 2002). As a result, some elites, such as Silvio Romero, believed that *pardo* people, having lighter features than those in the *preto* category, could be seen a positive sign of whitening in Brazil (Osorio 2003).

Ultimately, the attempts to whiten Brazil were quite successful, particularly in the south of the country. The white proportion of Brazil’s population increased from 38% in 1872 to 63% in 1940 (Ribeiro 1995, p. 229). When focusing just on the south, where the vast majority of these European immigrants went, the numbers are more extreme. The province of São Paulo went from 51.8% white in 1872 to 88.8% white in 1950. Massive amounts of European immigration to Rio de Janeiro also resulted in large racial shifts in the population as well (dos

Santos and Hallewell 2002). However, as European immigration to Brazil died down, the country's white population began to decrease as well, suggesting that the second stage of the whitening plan, racial mixing, did not result in the elimination of black Brazilians that the elites expected. Brazil's white population fell to 55% in 1990 and reached 47.7% in the 2010 census, where the black population once again formed the majority in Brazil (Ribeiro 1995, p.229; *Instituto Brasileiro de Geografia e Estatística* 2010a).

In conclusion, Brazil's history has been filled with racial mixing from its early days as a colony through the slavery period and beyond. The social situations caused by this racial mixing and the dynamics of having both free and enslaved people of African descent led to a variety of labels for racial and social categorization which have been unclear, and which have gone through changes in meaning over time. By the time slavery was abolished in Brazil, all of these factors had laid a foundation for today's complex concept of race that differs starkly from that of the United States. Let's explore this concept next.

3.2. The current demographic situation in Brazil

3.2.1. Differences between race in Brazil and race in the United States

The academic consensus on the difference between the concept of race in Brazil in comparison to that of the United States is that Brazil bases race (also referred to as "color") on one's physical appearance, while the United States bases race on one's ancestry. The earliest scholarly work that appears to describe this distinction is Nogueira (1955), which has frequently, but not always, been cited in work since then which comments on this difference. Notably, there were no works found in the background research for this dissertation which disputed this distinction given by Nogueira (1955). Harris and Kottak (1963) describe racial prejudice in Brazil

as based on the marking race (physical appearance) from Nogueira while they describe prejudice in the United States as based on origin race (ancestry). Kottak (1967) reiterates this difference, stating that in the United States, an individual with one ancestor in the subordinate group (i.e., African American), no matter how remote, is automatically assigned to that group at birth. This was even written into laws in the United States. Meanwhile, in Brazil, the factors which affect racial classification include phenotype, economic status, and the person making the classification, not the presence of a singular African ancestor. Additionally, Schwartzman (1999) also cites Nogueira, and his distinction of racism based on origin in the United States and racism based on color in Brazil.

In a similar vein to Nogueira, Ribeiro (1995, p. 225) states that the “distinctive characteristic” of Brazilian racism is that it is based on the color of one’s skin, not one’s origin. He also distinguishes Anglo-Saxon prejudice of race from Brazilian prejudice of color (p. 236). Pena et al. (2011) also describes Brazilian racial classification as being based on phenotype, rather than ancestry (Harris 1964), and gives skin pigmentation, hair pigmentation and type, eye melanization, and facial features such as nose and lip shape as characteristics which contribute to racial classification in Brazil (Telles 2002).

Further supporting the notion of different concepts of race in different countries, Nobles (2003) states that the definitions of the terms “race,” “ethnicity,” and “color” are nation-specific and depend largely on historical circumstance. She also describes the “one-drop rule,” which was applied in the United States but not Brazil. The one-drop rule is the idea that an individual who can be shown to have any degree of non-white ancestry, regardless of how small the percentage or how white the individual may appear to be, is automatically considered to be

non-white. Also supporting the distinction between ancestry and racial classification, Osorio (2003) and Pena et al. (2000) refer to the fact that it has been demonstrated that white people in Brazil have African and indigenous ancestry as well.

Meanwhile, Nishida (2003, p. 153) describes Brazilian racial classification as one based on physical appearance and socioeconomic standing. Additionally, she describes a situation where two full siblings are classified into different races and how this has resulted in different treatment within society and even within their own family (Patai 1988). Harris and Kottak (1963) and Kottak (1967) state that full siblings can be classified into different racial groups as well. This is also corroborated in the sample of participants from Salvador in this dissertation, where within a group of full siblings, one identified as *preto*, one as *pardo*, and one as white. Furthermore, this sample contains a mother-son pair where the mother identified as *pardo* and the son identified as white.

It should also be noted that this Brazilian concept of race applies to all socioeconomic classes, not exclusively to those in the lower or middle class. For example, the fact that Brazilian elites wanted to encourage European immigration to the country in order to mix with the existing population and make future generations whiter (dos Santos and Hallewell 2002; Telles 2002) can be seen as an indication that all levels of Brazilian society believe in a classification of race based more on physical appearance than on ancestry.

With regards to terminology, Telles (2002) states that the Brazilian term *cor* “color” captures the equivalent of the English term “race.” This has been the term used by many in Brazil because the word color captures the continuous nature of Brazil’s racial concepts (Telles 2002; Guimarães 1999). However, Telles argues that given the concept of race is a social

construct, the term “race” is still applicable in Brazil, and he prefers to use it in his paper.

Additionally, many Afro-Brazilian activists feel that the term “color” suggests a hierarchy and activists have criticized the *IBGE* for using the term “color” instead of “race” (Telles 2002; Nobles 2000). When speaking with the informant from Salvador for this project, he indicated a preference for the term *raça* “race.” Keeping these things in mind, this dissertation will use the term “race” rather than “color” to describe the situation in Brazil.

3.2.2. The variety of racial terms used in Brazil

While in recent decades (starting in 1991), the *IBGE* has had five race categories: *amarelo* “yellow” (Asian), *branco* “white,” *indígena* “indigenous,” *pardo* “mixed race” or “brown,” and *preto* “black,” far more terms have been used colloquially over the course of Brazil’s history, particularly for people of African descent. This is largely due to the phenotypic approach to racial classification in Brazil. People have a wide variety of physical appearances, and it is not clear how much of a difference in appearance warrants a new racial category. As a result, countless terms meant to distinguish one person from another developed over time. Looking at just how many terms came into use, the answer appears to be well over one hundred, though the number of terms varies by publication. At the lower end of the range, approximately 13 terms along a black to white spectrum were given in some publications in the 1940s and 1950s (Harris and Kottak 1963; Hutchinson 1952; Pierson 1942; Zimmerman 1952). Meanwhile, they elicited 40 different terms from 100 people in the same Bahian village from just 9 drawings along a phenotypic spectrum (i.e., 40 terms from 900 elicitations). Looking at the higher end of the range, Osorio (2003) states that studies on racial identification since the 1950s have revealed anywhere from five to forty different racial categories. However, he also

discusses a survey done in 1976 which revealed 136 different responses. Nevertheless, the majority of respondents used very few terms. 57% of them used the *IBGE* terms and another 38% used only three additional terms: *morena* (ambiguous term which can mean “brunette” or brown skin), *morena-clara*, and *clara* (literally “clear”). Therefore, despite a large range of terms used in the population as a whole, over 90% of people only use a small subset of the terms. Similarly, Schwartzman (1999) found with an *IBGE* survey of over 34 million Brazilians that more than 130 different terms were used. However, 14 of these terms comprised over 99% of responses, with 69.8% of them being the *IBGE* terms and 20.77% being the term *moreno*, which was larger than *pardo* (10.33%).

Despite the consistency of the racial categories of the census in the last few decades, historically, the census in Brazil has had some variation in the options given, and many censuses did not track race at all. Looking at previous censuses, the 1900, 1920, and 1970 censuses did not ask for race and there were no censuses conducted in 1880, 1910, or 1930. Approximately every ten years beginning in 1890 there was a census in Brazil. Every census which asked about race had the categories *branco* “white” and *preto* “black.” *Pardo* was used in the first census but was replaced by *mestiço* (mestizo) in 1890. From 1940, when race returned to the Brazilian census, onward, *pardo* was used as the middle ground term between “black” and “white.” For indigenous people, the term used in 1872 and 1890 was *caboclo*, followed by *índio* in 1960. Beginning in 1991, the term *indígena* “indigenous” was used. In all other census, there was no indigenous category given. Lastly, an Asian category, *amarelo* “yellow,” was added in 1940 and continues through today (*Instituto Brasileiro de Geografia e Estatística* 2023a; Nobles 2000; Telles 2002).

3.2.3. Self-identification vs. classification by others

As one might guess, the large number of terms used to describe race is too much for people to keep track of and it leads to a lack of sociological meaning. Kottak (1967) found in the Bahian fishing village of Areembepe that participants disagreed over the ordering of various color terms when arranging them from lightest to darkest. However, there was complete agreement that *branco* “white” was on the lightest end and *preto* “black” on the darkest end. Inconsistency like this is why the *IBGE* uses few terms in comparison. However, the approach done by the *IBGE* takes this variety and consolidates all intermediate positions between “black” and “white” into *pardo*. There also are no defined criteria for these racial categories, though it is fair to assume that people identifying as *pardo* have both African and European ancestors due to its status as an intermediate category (Ososio 2003; Telles 2002). Ultimately, the wide range captured by the term *pardo* creates ambiguity within the term. Combine this information with the facts that Portuguese language dictionaries are not consistent in their definitions of the word, calling it either “gray” or “brown,” and the infrequency of the term *pardo* in everyday speech (Nobles 2000), and it is no surprise that it has been found that racial categorization terms are used inconsistently throughout Brazil.

To complicate matters further, Brazilians are not consistent with how they classify themselves or each other. Bastos et al. (2009) examined the difference between self-identification of race and the race assigned by female interviewers in southern Brazil among respondents aged 40 and over. They found that respondents were more likely to identify as *pardo* and less likely to identify as white or *preto* when speaking with black interviewers as opposed to white interviewers. In other words, there was a tendency to avoid the ends of the

spectrum among respondents with black interviewers. Additionally, Bastos et al. (2009) found that black interviewers were less likely than white interviewers to classify men as black, suggesting a greater degree of whitening by interviewers than respondents. Regarding younger respondents, Santos et al. (2009) found that high school students near Rio de Janeiro tended to whiten their responses when asked to self-identify their race within the *IBGE* options in comparison to their responses in an open-ended question. The students also tended to place their classmates in lighter color categories than the classmates put themselves into. Thus, these two studies found a greater tendency for whitening when others classify one's race than one's self-identification. On the other hand, Lima-Costa et al. (2015) found that participants from Salvador tended to whiten their self-identification more than participants from cities in the south and southeast of Brazil. Thus, there is no consistent tendency in previous research regarding whether whitening is more likely in self-identification versus classification by others. However, there may be some differences in how whitening in racial categorization patterns in the south of Brazil versus the north, based on the results of the aforementioned studies.

Taking into consideration some further examples of individuals changing their own identification, in Sansone (2007), respondents in Salvador and Camaçari, Bahia, described themselves with multiple racial terms (not necessarily just *IBGE* terms) over the course of a single interview. Similarly, because of inconsistency in racial categorization both from person to person and from the same individual, Kottak (1967) avoided using categories from participants when examining the correlation between skin color and wealth in Arembépe, Bahia. Furthermore, people have been known to switch their racial identification to match that of their spouse (Nishida 2003, p. 131). Given these facts, it is clear that the racial categories in Brazil are

not clearly defined, and it is not reasonable to expect everyone to agree on how an individual can be classified.

To add a final layer of complexity, factors such as education level and socioeconomic status affect both self-identification and racial classification by others. “Money whitens” is a phrase that is often used to describe this effect in Brazil. The association between lighter skin color and being better off socioeconomically can be traced back several centuries and it continues to this day. People who are perceived as “black” are often thought of as being in the lowest socioeconomic position (Nishida 2003, p. 130, 154). Thus, the two variables of race and socioeconomic status are so linked in the minds of Brazilians that it adds another factor into racial classification beyond the phenotype. Harris and Kottak (1963) state that racial identification is influenced by the educational and economic statuses of both an interviewer and a respondent being classified, and Kottak (1967) explains that there is a positive correlation between light skin color and wealth in Brazil. Osorio (2003) found that wealthier people were more likely to identify as white despite their physical features. At the same time, he found that poorer people were more likely to identify as *pardo* or *preto* in comparison to categorization by outside observers. Ribeiro (1995, p. 233-234) also states that *preto* and *pardo* people are worse off socioeconomically than white people. Lastly, Telles (2002) found that interviewers tended to whiten more highly educated respondents while they darkened less educated respondents. He also found that interviewers tended to whiten respondents more than respondents whitened themselves, similar to Santos et al. (2009) and Bastos et al. (2009). It should be noted, however, that Santos et al. (2009) suggests that the phrase “money whitens” in Brazil has become less

true and that in the decade leading up to the publication of their paper, a growing number of young, educated Brazilians were identifying as black (*negro*) rather than switching to white.

3.2.4. Afro-Brazilian movements and their effect on racial identity

The increased usage of the term *negro* among the Brazilian population is the result of Afro-Brazilian movements aiming to unify those identified as both *preto* and *pardo*, i.e., people with “visible” African ancestry, and promote the freedoms of black Brazilians. Such movements began to take hold in the 1930s and gained prominence over the second half of the 20th century onward (Nishida 2003, p. 154-155). While both *preto* and *negro* translate to “black” in English, the term *negro* is preferred by many because they believe *preto* is a term that is used to describe objects while *negro* is a term used to describe people. This unification of *preto* and *pardo* people is further warranted by the fact that both groups are more or less socioeconomically equal, with the difference lying between them and the white population of Brazil (Levine 1999, p. 14; Nobles 2000; Oliveira et al 1985; Osorio 2003; Ribeiro 1995, p. 234; Santos et al 2009; Schwartzman 1999; Telles 2002). Such social movements make up what is known as the *movimento negro* or “black movement” and there are various organizations that have risen to prominence as part of it such as *Olodum*, *Ilê Aiyê*, and the *Movimento Negro Unificado (MNU)* (Nishida 2003, p. 154-155; Dixon 2016, Ch. 3-4). Activists also use the unification of *pretos* and *pardos* as means to fight back against the hierarchy suggested by the Brazilian color categories (Telles 2002). In many ways, this movement to unify black Brazilians creates a racial situation more similar to that of the United States, where all who physically appear to have any amount of African ancestry are considered black. The population referred to with the term *negro* in Brazil would be equivalent to what is generally referred to as “black” in

the United States (Levine 1999, p. 12-13). The sole difference here would be that people who have African ancestry but are phenotypically white would be considered white in Brazil, while they could still be considered black in the United States.

While the term *negro* is preferred by many in Brazil, it is still not an option when it comes to the census. This forces people to choose between either *preto* or *pardo*. It appears that as the *movimento negro* gains popularity and influence in Brazil, more and more people are specifically identifying as *negro*. Without the ability to select this preferred option in the census, people are likely to default to the next closest option, *preto*, since this also means “black.” In order to ensure consistency in racial identification and understanding of the racial category options by participants, the demographic survey for this dissertation gave the same options as the *IBGE* has in the Brazilian census. As shown in the racial identification data in table 3.1 below, the participants in the current study are much more likely to identify as *preto* than *pardo* when compared to the census data. However, the percentage of *preto* + *pardo* (i.e., *negro*) is similar in both, as is the percentage of people identifying as white. The indigenous and Asian populations of Salvador are quite small and among the participants, only one person identified as indigenous and none as Asian.

Table 3.1: The number and percentage of participants identifying with each *IBGE* category compared with the percentage of each category in the 2022 and 2010 censuses in Salvador².

Racial identification	Participants	2022 census	2010 census
<i>Preto</i>	60% (21)	34.1%	27.8%
<i>Pardo</i>	17.1% (6)	49.1%	51.7%
<i>Preto + Pardo</i> (i.e., <i>Negro</i>)	77.1% (27)	83.2%	79.5%
White	20% (7)	16.5%	18.9%
Indigenous	2.9% (1)	0.2%	0.3%
Asian	0% (0)	0.1%	1.3%

² Instituto Brasileiro de Geografia e Estatística (2010b, 2023a)

Although there is a discrepancy in the proportions of participants identifying as *preto* and *pardo* when compared to the census data, the aforementioned evidence that both groups can be combined in the category *negro* indicates that the sample in this dissertation is likely still racially representative of Salvador's population. One reason for the gap between these categories could be a social shift in identification over time, particularly since the proportion of people identifying as *preto* increased about 6.3% from the 2010 census to the 2022 census. At the same time, the proportion of people identifying as *pardo* went down about 2.6%. Given that the last census there is data for in Brazil was thirteen years ago (2010), over a decade of social changes in the reference population are missing, making such a shift plausible. Additionally, other studies have found that self-identification skewed more toward *preto* and less toward *pardo* than the census. In their genetic study, Lima-Costa et al. (2015) found that 49.3% of the 876 participants from Salvador identified as *preto* while 43.3% identified as *pardo*. Similarly, Telles (2002) found that a 1995 nationwide survey had more *preto* and less *pardo* respondents than the 1991 census but the sum of these two categories (i.e., *negro*) was about the same. Thus, it appears that self-identifications, like the data in this dissertation, will have more identifying as *preto* and less as *pardo* than the census. Lastly, the Brazilian census is completed by interviewers completing forms about residents rather than residents filling out the forms themselves. The interviewers are supposed to ask about race, but they often do not, whether it be because they think they know the correct response, they feel uncomfortable asking the question, or they are rushing through the interview. On top of this, data for an entire household comes from only one respondent who may make their own decisions regarding how to identify other members of the household (Telles 2002; Pinto 1996). Combine this information with

aforementioned tendency for outside observers to categorize people into a lighter category than they self-identify, and it makes sense that the census would have a larger proportion of people categorized as *pardo* than participants who picked their own category. Overall, there are many factors at play that can explain why so many more participants identified as *preto* than in the 2010 census. These factors ultimately suggest that despite the difference from the census data, the participants in this dissertation likely are racially representative of the population of Salvador.

3.3. Genetic studies in Brazil and their relation to demographics

In his book *Patterns of Race in the Americas* (1964, p. 1-2), Marvin Harris states that Latin America, as a whole, consists of about equal portions of African, European, and indigenous American DNA. However, he states that this DNA is not equally distributed across Latin America and that the eastern coast of South America (where the majority of Brazil's population is situated) is dominated more by African and European DNA than indigenous American DNA. Although these claims are based on historical information and he does not have any genetic tests supporting these claims, genetic tests that have since been done corroborate this claim: Brazilians have predominantly African and European DNA, with small portions of indigenous American DNA.

Biologists, especially geneticists, have held the view that race is insignificant while social scientists have maintained that it is indeed significant (Santos et al. 2009; REGWG 2005; Templeton 1999). Race in Brazil is an excellent example of this dichotomy. Genetic studies show that extensive racial mixing has made the already subtle genetic distinctions between races even less clear in Brazil. Meanwhile, race has had a profound effect in the lives of Brazilians. As

has been mentioned, a huge economic divide exists between white and black people in Brazil. Additionally, there is a significant difference in the deletion rate of the rhotic phoneme by race. Therefore, it is important to understand exactly to what extent genetic studies show similarities across racial groups in Brazil in order to know how the differences by race in this dissertation are socially driven.

Kehdy et al. (2015) examined the DNA of 1,309 people from Salvador along with over 5,000 people from Pelotas in the south and Bambuí in the southeast of Brazil for proportions of African, European, and indigenous DNA as well as where within Africa and Europe the DNA likely originated from. Salvador had the highest proportion of African ancestry on average at 50.8%, along with 42.9% European and 6.4% indigenous DNA. By contrast, the other two cities were over 75% European, with about 15% African and 6-8% indigenous DNA. Regarding more specific origins of the African and European DNA, the DNA in Salvador was found to be more non-Bantu/West African (particularly Yoruba and Mandinka) and Iberian while the DNA in the south was found to be more Bantu/East African and more broadly European than the Salvador DNA. Additionally, they concluded that the European colonization of Salvador occurred earlier than in the south. Across all three locations, they found evidence that paternal lines tended to be European in origin while maternal lines were African or indigenous in origin. This is consistent with the claims made in historical references that the colonists in Brazil were predominantly male and had children with indigenous and African women.

Looking at the same three cities, Lima-Costa et al. (2015) examined the proportions of African, European, and indigenous DNA in over 5,800 Brazilians by racial identification: *preto*, *pardo*, and white. Once again, Salvador had the highest proportion of African ancestry at 50.5%.

The proportions of African and European in DNA in Salvador showed a lot of overlap across members of the racial groups, especially compared to the other cities. Overall, the white participants from Salvador had the most African DNA among white participants from the three sites, averaging about 40%, while the *preto* participants from Salvador did not have the highest proportion of African DNA. However, among *preto* participants, Salvador had about the same proportion of African DNA as Pelotas, averaging about 55%. *Preto* participants from Bambuí had the highest proportion of African DNA at about 75% on average. In terms of African DNA among *pardo* participants, the average percentage of African DNA was between the percentages of African DNA for *preto* and white participants of the same city, with *pardo* participants from Salvador having the highest average proportion at about 50%. Once again, the proportion of indigenous ancestry was low, at around 5-6% for all the cities. These results are summarized in table 3.2 below. The relative uniformity across racial groups among Salvador's participants demonstrates the high degree of mixing over several centuries that has resulted in a phenotypically variable population coming from the same ancestral populations. It also suggests that this mixing has been at play longer in Salvador than the other two locations. Ultimately, Lima-Costa et al. (2015) found a very low, though significant, association between ancestry and racial classification in Salvador: a pseudo r^2 value of 0.13 and they concluded that self-identification of race is not completely random relative to ancestry, but it is certainly not strongly determinative, either.

Table 3.2: Approximate percentage of median genetic ancestry by city and racial identification found in Lima-Costa et al. (2015).³

City	Genetic Ancestry	Racial Category			City-Wide
		<i>Preto</i>	<i>Pardo</i>	White	
Salvador	African	58%	45%	38%	50.5%
	European	37%	50%	55%	42.4%
	Indigenous	5%	6%	7%	5.8%
Pelotas (South)	African	55%	27%	5%	6.6%
	European	40%	65%	90%	85.3%
	Indigenous	6%	8%	6%	6.3%
Bambu� (Southeast)	African	75%	15%	5%	9.6%
	European	20%	77%	90%	83.8%
	Indigenous	4%	7%	5%	5.4%

Similar to Lima-Costa et al. (2015), Silva et al. (2015) examined the proportions of African, European, and indigenous DNA in over 500 people from Salvador and over 600 people from Fortaleza (also in the northeast of Brazil). In terms of African ancestry in Salvador, they found a much larger difference between those identifying as white (20.0% African DNA on average) and those identifying as *preto* (56.4% African DNA on average). Ultimately, they found a significant correlation between African ancestry and racial identification of $r=0.585$. While racial mixing began nearly 500 years ago in Brazil, the authors believe this association between phenotype and ancestry could be due to assortative mating and the fact that the European and African source populations were nearly continuously reintroduced into Brazil's overall population. Regardless of the association, it should be noted that it is still the norm for people who identify as white in Brazil, particularly in Salvador, to have a sizeable portion of African DNA

³ Note that the percentages for a given city and racial identification may not add up to 100 percent, as they are approximated from graphs in the paper. The city-wide percentages are taken directly from a table in the paper and only add up to 98.2 to 98.8 percent.

and that there is overlap in the proportion of African DNA across the three racial categories of *preto*, *pardo*, and white, meaning that not everyone who is white automatically has less African ancestry than everyone who is *preto* or *pardo*.

Looking at a study outside of Salvador, Santos et al. (2009) examined the DNA of students at a school in the metropolitan area of Rio de Janeiro for proportions of African, European, and indigenous DNA. When the results of the tests were revealed to the students, many *preto* and *pardo* students were surprised to hear how much European DNA they had (in many instances much more European DNA than African DNA). However, the students ultimately concluded that appearance mattered more than genetic results. Thus, it is important to keep in mind with these DNA tests that despite the large amounts of shared ancestry, Brazilians still hold a major social importance on their racial identity based on appearance.

Overall, genetic studies in Brazil have confirmed what has been claimed in historical sources: Brazilians, regardless of racial identification, have a mix of African, European, and Indigenous ancestry. Additionally, the relatively high proportion of African DNA in Salvador corroborates the claim that Bahia was among the largest recipients of enslaved people from Africa. Despite this shared heritage, Brazilians consider the racial categories based on their differences in appearance to hold social significance. Therefore, genetic studies in Brazil serve as a means of confirming the fact that race is biologically insignificant but socially significant. In the next section, the social significance of race with regards to rhotic deletion will be explored.

3.4. Race in Brazil and how it relates to rhotics in Salvador

All of this information is fascinating to learn about but what exactly does it have to do with this dissertation? In a nutshell, what we have seen is that people in Brazil come from a mix

of African, European, and indigenous DNA regardless of racial identification (*preto*, *pardo*, and white) to the point that these ancestries no longer play a role in determining one's identification. It is a purely social marker based on physical appearance (and socioeconomic status to some extent).

It has been established in previous research that rhotic deletion was one of many features that were the result of substratal influences of African languages on Brazilian Portuguese. Given that Brazil received more enslaved people from Africa than any other country and that one of the most common destinations in Brazil for enslaved people was Bahia, it would make sense that the dialects that developed there would have strong African influences. Furthermore, Salvador's status as the capital of Bahia and a port city meant that, in addition to the people enslaved there, countless Africans were brought through the city before being sent elsewhere. Combine these with the fact that today, about 80% of Salvador's population identifies as black, these African influences are likely to remain strong in the standard dialect of Salvador. Add in the fact that white people in Salvador are generally expected to have African ancestry and that it is not unusual for them to have black parents or full siblings, it would make sense for the white residents of the city to be constantly exposed to and influenced by the African-derived features of the Salvador dialect.

With these things in mind, it would not be surprising if there were not a difference between black and white speakers in Salvador. Thus, the question is which makes African linguistic influences stronger in a modern-day Soteropolitan's speech: having African ancestry or identifying as a racial category associated with African phenotypical features? If the answer is ancestry, there would be no difference between those identifying as black and those identifying

as white. However, if the answer is racial identification, there will be a difference between those identifying as black and those identifying as white, despite all the shared heritage.

When taking into consideration differences that may exist between the northeast of Brazil and the south, São Paulo did not have the extensive history of slavery that existed in Salvador. The slave trade played a major role in Salvador's early history and the population was predominantly African shortly after its founding. São Paulo, being in the south of Brazil, did not have a prominent role in Brazilian slavery until the later days of the slavery era. Additionally, São Paulo was home to the massive wave of European immigration that was part of the whitening plan for Brazil. On the other hand, Salvador received very few European immigrants by comparison and remained a predominantly black city. Thus, the demographics in the two cities have historically been quite different and remain different today. Given all these differences, it is possible that rhotics would pattern differently in the two cities as well.

Ultimately, the complex history surrounding demographics in Brazil is playing a major role in the deletion rates of rhotics among the participants from Salvador. When looking at the phenomenon of rhotic deletion, the history and demographic difference are crucial to understanding the bigger picture. As the data is presented in upcoming chapters, it is important to keep this information in mind and use it to contextualize the data.

4. Questions and hypotheses

Firstly, this dissertation aims to establish how the rhotic phoneme surfaces in Salvador, and how frequently deletion occurs on the phoneme. Additionally, there is the question of how deletion in one environment can affect deletion in another environment. This results in the following research questions:

1. What is the inventory of allophones of the non-flap rhotic in the Salvador variety?
2. At what rate does the Salvador variety delete word-final rhotics, both in isolation and in context?
3. At what rate does the rhotic delete in other environments (syllable-final, word-initial, intervocalic) in the Salvador variety?
4. When rhotics are not deleted in the Salvador variety, to what extent have they undergone lenition?
5. To what extent is deletion in one environment related to deletion in another environment?

Other than Rennicke's (2015) expectation that the rhotic in Bahia would be a "back fricative" ([χ], [ʁ], [h], [ɦ]), there is little information in the existing literature indicating which allophones of the rhotic phoneme exist in Salvador. Consequently, no precise hypothesis can be made regarding this first question. In terms of deletion rate in the word-final position, it is hypothesized that speakers of the Salvador variety will delete at a higher rate than speakers of other varieties both in isolation and in context. The basis for this hypothesis is the historical influence of African languages on the variety in Salvador and the connection between rhotic deletion and African influences that has been made in previous work. For deletion in

environments outside of the word-final position, it is hypothesized that all other positions will delete less. Among the other positions, it is hypothesized that rhotics in syllable-final position will delete more than in the other two positions due to it being the position most similar to word-final (i.e., they are both coda positions). However, how deletion rates between word-initial and intervocalic tokens will compare with each other is not something that can be hypothesized. When not deleted, it is hypothesized that the rhotic phoneme will be undergoing high rates of lenition in all environments and that this lenition will largely take the form of the glottal fricatives [h] and [ɦ]. If not glottal, it is hypothesized that this lenition will result in the surface forms of the rhotic being predominantly fricatives, with the flapping phenomenon on word-final tokens before a vowel comprising the majority of non-fricative productions. Finally, regarding the relationship between deletion rates across environments, it is hypothesized that participants who delete more in one environment will also delete more in another environment. In other words, it is hypothesized that an individual's tendency to delete the phoneme would not be limited to one particular environment, but rather exist in all environments.

With regard to the demographic differences that may exist in Salvador, there is the question of to what extent demographic groups have been affected by the African influence of deletion. Additionally, there are the questions of whether different demographic groups have adopted different degrees of lenition and if the non-lenited surface forms of the rhotic vary by demographic variables. This results in the following research questions:

1. Do black participants delete the rhotic more frequently than white participants?
2. Are there differences in deletion rate that exist by other demographic variables?

3. Do the various demographic groups produce non-lenited (i.e., oral or non-glottal) surface forms at different rates?
4. Do the non-lenited surface forms of the rhotic vary by demographic variables?

Regarding differences in deletion rate by race, it is hypothesized that there will not be a difference between the black and white participants. This is because both black and white residents of Salvador have been shown to come from the same ancestral backgrounds, as shown in chapter 3, and therefore, they have been exposed to the same historical African influences that would lead to deletion of the rhotic. Likewise, it is hypothesized that there will not be any differences in deletion rate by any other demographic variables. This is due to the lack of anything in the existing literature that attributes rhotic deletion to any other demographic variable. Thus, there is no basis for a hypothesis that deletion rates will vary by any other demographic variable. Regarding the rate and distribution of non-lenited surface forms, it is hypothesized that no differences will exist by demographic variable, as there is no indication in the existing literature that oral surface forms will vary by demographics that can serve as a basis for any hypothesis that a difference will exist.

The final set of questions and hypotheses is related to a second dataset collected from the city of São Paulo. Existing work (Reed and Leite 1947; Silva 1998) suggests that the rhotic would surface as an alveolar trill [r] or approximant [ɹ], rather than the “back fricatives” for Bahia suggested by Rennicke (2015). Furthermore, as was shown in chapters 1 and 3, the proportion of white residents in São Paulo is higher than in Salvador, and the historical circumstances surrounding the city are different as well. This suggests that African influences would be weaker in São Paulo than in Salvador. This results in the following research questions:

1. Do speakers from Salvador delete the rhotic more frequently than speakers from São Paulo?
2. Do speakers from São Paulo produce non-lenited surface forms more frequently than speakers from Salvador?
3. Is there a difference between the surface forms of the rhotic in São Paulo and Salvador?

Given the differences between the history and demographics of the cities, it is hypothesized that speakers from Salvador delete the rhotic more than speakers from São Paulo. It is highly likely considering the historical and demographic situation that African influences are weaker in São Paulo than in Salvador. If African influences do indeed play a role in deletion, then it would follow that deletion would occur less frequently in São Paulo. Considering the surface forms of São Paulo cited in the existing literature in comparison to what was postulated for Bahia by Rennie (2015), it is hypothesized that non-lenited surface forms will be more common in São Paulo than in Salvador. This has two aspects: the rate of oral production in surface forms, as well as the type of oral surface forms. Firstly, it is hypothesized that glottal fricatives will comprise a smaller proportion of the surface forms in São Paulo than in Salvador. Secondly, it is hypothesized that oral fricatives will comprise a smaller proportion of oral surface forms of the rhotic in São Paulo as well, with surface forms primarily being fortified as the alveolar trill [r] or approximant [ɹ]. The basis for this is that they are cited in existing work (Reed and Leite 1947; Silva 1998) as surface forms of the rhotic in São Paulo, thus suggesting that fricatives are not expected as surface forms in the city. This second aspect also suggests that the surface forms of the rhotic will vary between the two cities. Thus, in response to the third research question, it is similarly hypothesized that a difference in the distribution of surface

forms exists between the two cities. It is hypothesized that this difference will manifest as glottal and oral fricative surface forms in Salvador and the alveolar trill [r] and approximant [ɹ] as surface forms in São Paulo. Consequently, it is hypothesized that the surface forms between the two cities will overlap little, if at all.

5. Methodology

5.1. Participants and their demographics

5.1.1. Participants from Salvador

The primary group of participants in this study are native speakers of the variety of Brazilian Portuguese of Salvador, Bahia. Data were collected from thirty-six participants from Salvador, but one was discovered as not having been raised in Salvador and was, therefore, eliminated from the sample. The demographic characteristics of these participants are given in table A1.1 in appendix A1.

Among the participants from Salvador, twenty-one identified as *preto*, six identified as *pardo*, seven identified as white, and one identified as indigenous. Of the seven white participants, four have black parents and siblings, meaning that thirty-one of the thirty-five participants are known to be of Afro-Brazilian descent, though only twenty-seven would typically be considered Afro-Brazilian. Neighborhoods in which the participants have lived at some point in their lives include Itapuã, Roma, Calabar, Pau Miúdo, Pirajá, Barris, Ribeira, Paripe, Tororó, Barbalho, São Rafael, Nordeste de Amaralina, Brotas, Mirantes de Periperi, Rio Vermelho, Garcia, Stiep, and Candeal. The participants range in age from 20 to 61 (approximate birth years 1960 to 2002) with a median approximate birth year of 1993. Fourteen participants identified as male, twenty as female, and one as trans non-binary. Twenty-one participants identified as lower class, and fourteen identified as middle class. No participants identified as upper class. Lastly, twenty-eight of the thirty-five participants have lived in Salvador their whole lives, while seven have lived elsewhere (in Brazil or abroad) at some point in their lives.

Regarding how the demographics of the participants compare with the demographics of Salvador in the 2022 census, tables 5.1 and 5.2 are given below with the data for age and gender respectively. The data for race will be discussed in chapter 3. It is not possible to use the census data for the other demographic variables. Although the *IBGE* does give data pertaining to income (*Instituto Brasileiro de Geografia e Estatística 2023b*), there are several brackets, and it is not clear how they would map to the classifications of lower, middle, and upper class.

Table 5.1: Comparison of the proportion of the participants per census age group with the proportion of residents of Salvador (aged 20-64) in each age group⁴.

Age group	Participants	2022 census
20-24 years old	20% (7)	11.1%
25-29 years old	37.1% (13)	10.9%
30-34 years old	14.3% (5)	10.9%
35-39 years old	2.9% (1)	13.3%
40-44 years old	0% (0)	14.1%
45-49 years old	2.9% (1)	11.5%
50-54 years old	2.9% (1)	10.5%
55-59 years old	14.3% (5)	9.5%
60-64 years old	5.7% (2)	8.1%

In table 5.1, it can be seen that there are several differences in the proportion of each age group. Ultimately, the sample skews in favor of the younger groups aged 20-34, as well as the 55-59 age group. There are no participants in the 40-44 age group and the remaining groups are underrepresented.

Table 5.2: Comparison of the gender of the participants (male and female only) and the gender of Salvador in the 2022 census⁵.

Gender	Participants	2022 census
Male	41.2% (14)	45.6%
Female	58.8% (20)	54.4%

⁴ (*Instituto Brasileiro de Geografia e Estatística 2023a*). Participants who were recorded in 2021 had their reported ages increased by 1 for this table.

⁵ (*Instituto Brasileiro de Geografia e Estatística 2023a*). The trans non-binary participant is not included here due to the census data only having male and female categories.

In table 5.2, there is a slight overrepresentation of females in comparison to males, but it is very close to the census data. A shift of two participants from female to male would result in the males being overrepresented, indicating the gender balance of the sample is quite close to the gender balance of Salvador.

Approximately two years for remote participants and one year for in-person participants after their initial participation, participants were recontacted and asked what they considered their heritage to be as well as where their parents and grandparents were from. However, one participant was not able to be contacted. This person identified as *preto*. Of the thirty-four contacted, twenty-nine responded. Of those who did not respond, four identified as *preto* and one identified as *pardo*. One of the five *pardo* respondents identified their heritage as Afro-Brazilian, while the remaining four identified as Brazilian. Nine of the sixteen *preto* respondents identified as having African, Afro-Brazilian, or black heritage, while six identified as Brazilian and one simply gave a definition of “a line of generations of a family.” Of the seven white respondents, one identified as having African heritage (alongside indigenous and Portuguese heritage), while the remaining six identified as Brazilian, Brazilian and Italian, or Arab and Portuguese. With regard to the origin of the parents and grandparents, sixteen of the twenty-nine respondents had at least one parent from the interior of Bahia and twenty-four of the twenty-nine had at least one grandparent from the interior of Bahia.

5.1.2. Participants from São Paulo

From São Paulo, nineteen participants were recorded. Of these nineteen, ten were selected to be demographically matched with a subset of the Salvador participants. The participants were matched for race, gender, age, and socioeconomic class as could be best

accomplished with the available participants. The demographics of all nineteen participants from São Paulo can be found in table A1.2 in appendix A1. Among the ten participants from São Paulo selected for analysis, one identified as *preto*, three as *pardo*, and six as white. None of the white participants have black parents or siblings, though one indicated doing a DNA test that showed he had African ancestry, and one reported having a great-grandmother who was black. Therefore, although none of the white participants have black parents or siblings, it is not unheard of for white residents of São Paulo to have African ancestry. The participants range in age from 19 to 56 (approximate birth years 1967 to 2004), with a median approximate birth year of 1991.5. Six participants identified as female and four as male. One participant identified as lower class, and nine identified as middle class. No participants identified as upper class. Six of the ten participants have lived in São Paulo their whole lives, while four have lived elsewhere (in Brazil or abroad) at some point in their lives. The participants selected from Salvador for the comparison were participants 4, 5, 11, 12, 13, 21, 22, 23, 30 and 33. These participants are indicated with asterisks (*) in the table of the demographics of the participants from Salvador (table A1.1 in appendix A1).

5.2. Data collection

Ten remote and twenty-five in-person participants were recorded for the sample from Salvador. Remote participants completed the project between June and August 2021, and in-person participants completed the project in May and June 2022. Remote participants were recruited through a consultant who provided contact information for potential participants. These people were then contacted via email and/or *WhatsApp* and asked whether they were interested in participating. If they indicated interest, a consent form was sent to them.

Throughout this process, in communications with participants, the researcher indicated a desire to know how the people of Salvador speak, in hopes of avoiding them speaking in a manner they perceived as standardized. In-person participants were recruited through the same consultant. With these participants, a time and place to meet was arranged. During the meeting, participants were presented with the consent form and instructed to indicate if they had any questions.

After they granted consent, remote participants were emailed slides and were asked to record themselves into a .wav reading the text on the slides. Remote participants used their own personal device to complete the recordings. In-person participants were recorded with the author present using earbuds with a microphone plugged into the author's computer, a Windows Surface Laptop 3. In-person participants were recorded with a stereo .wav file using the program *Audacity* with a sampling rate of 44100 Hz. Each slide contained only one token. Both the isolated tokens and the sentences were in the same PowerPoint file. The slides began with the first half of a set of isolated tokens, then the first half of a set of sentences, followed by the second half of the set of isolated tokens and finally, the second half of the set of sentences. This division was made in order to avoid making the task feel too repetitive for the speaker. In total, there were 113 slides, 38 with isolated tokens and 75 with sentences. Among the isolated tokens, 32 had at least one instance of the rhotic phoneme, while 6 were distractors. Among the sentences, 65 contained at least one instance of the rhotic phoneme, while 10 were distractors. The tokens were not identical between the isolated and context tokens however, there was significant overlap between the two groups. In total, the corpus consists of 4964

instances of the rhotic phoneme from Salvador. The stimuli along with their translations into English can be found in appendix A3.

Two in-person and eight remote participants were recorded for the sample from São Paulo. All participants completed the project between March and August 2023. The procedure used to record the participants from São Paulo was the same as what was used for the participants from Salvador. In total, the corpus consists of 1419 instances of the rhotic phoneme from São Paulo.

5.3. Data analysis

The data were analyzed with Praat (Boersma & Weenink 2021). Using a method from Rennie (2015), spectrograms were examined for formant changes indicating the presence of a segment other than a vowel in the syllable. To do this, the values of the formants in the vowels adjacent to the rhotic were measured in Praat and compared with the formants present in the potential production of the rhotic phoneme. When the formant values were substantially different, this indicated a movement to production in a specific place of articulation, with the change in formants depending on the vowel quality and where the rhotic was articulated. When the formants indicated movement to a specific place of articulation, the production was classified by referring to the relation between formant frequencies and position of the constriction given in Stevens (1972), as well as impressionistically. This was done for all classifications regardless of how front or back the place of constriction was. If the formant values did not change, the spectrogram was examined for changes in intensity. If the intensity decreased, this indicated a glottal fricative. If the intensity remained the same, this indicated that no consonant was present and the rhotic had been deleted. If movement to a specific place

of articulation was found, the waveform and spectrogram were examined for signs of frication. If there were no signs of frication, other manners of articulation were considered. Rounding on velar fricatives was determined impressionistically.

After determination of the place of the rhotic, voicing was determined by examining the spectrogram for a voicing bar, as well as glottal pulses. Note that it is possible that in some circumstances, an echo overlapping with a voiceless rhotic production was misinterpreted as a voiced production. In the case of flaps and trills, the waveform and spectrogram were examined for occlusions. One occlusion indicated a flap, and two or more occlusions indicated a trill. Flap + fricative variants were determined by the presence of a single occlusion along with a fricative that was determined using the method described above. Uvular trills [ʀ] were classified based on movement in the spectrogram like that of a uvular fricative with the addition of occlusions on the spectrogram as well. Productions which contained neither frication nor occlusions, but had a dip in the third formant were classified as [ɹ]. Additionally, some rhotic productions in coda position contained an epenthesized vowel. This was classified by the presence of a high-intensity segment in the spectrogram located between the rhotic and the expected following segment. Classification of the productions was done by the author, and when the author was unsure how to classify a specific production, a phonetic consultant aided in making the classifications. To eliminate the possibility of mistaking one production of the rhotic for another, word-final rhotics and word-initial rhotics were not placed adjacent to each other in the stimuli.

Figures 5.1-5.3 below exemplify the classifications when there was not movement to another place of articulation: [h], [ɦ], and deletion. Figures 5.4-5.5 below give two examples of classifications where there was an indication of movement to a place of articulation with the

fricatives [x] and [χ]. Figures 5.6-5.8 below exemplify various possible productions involving coronal occlusions: [r], the flap + fricative variant [rʰ], and the flap [ɾ]. Figure 5.9 shows an example production of [ɻ], and figure 5.10 exemplifies a production with an epenthesized vowel. Figures 5.1-5.3 and 5.8 are from a 27-year-old female speaker from Salvador. Figures 5.4-5.5 are from a 21-year-old female speaker from Salvador. Figures 5.6-5.7 are from a 19-year-old male speaker from São Paulo. Figures 5.9 and 5.10 are from a 19-year-old female speaker from São Paulo.

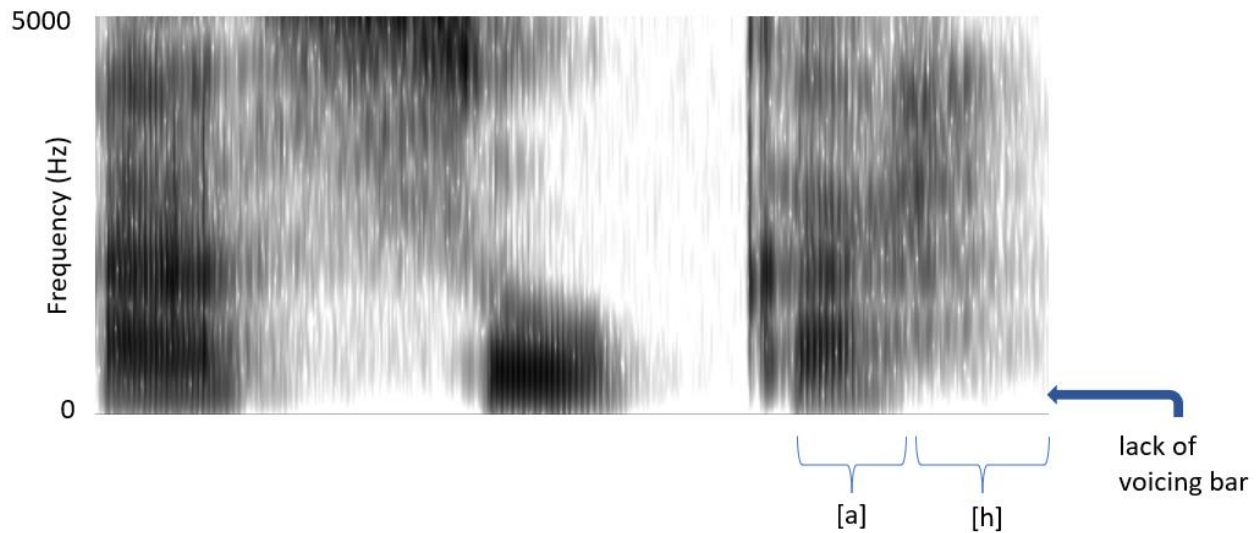


Figure 5.1: Example of the rhotic produced as [h] in word-final position on the token *açúcar* “sugar” by a 27-year-old female speaker from Salvador.

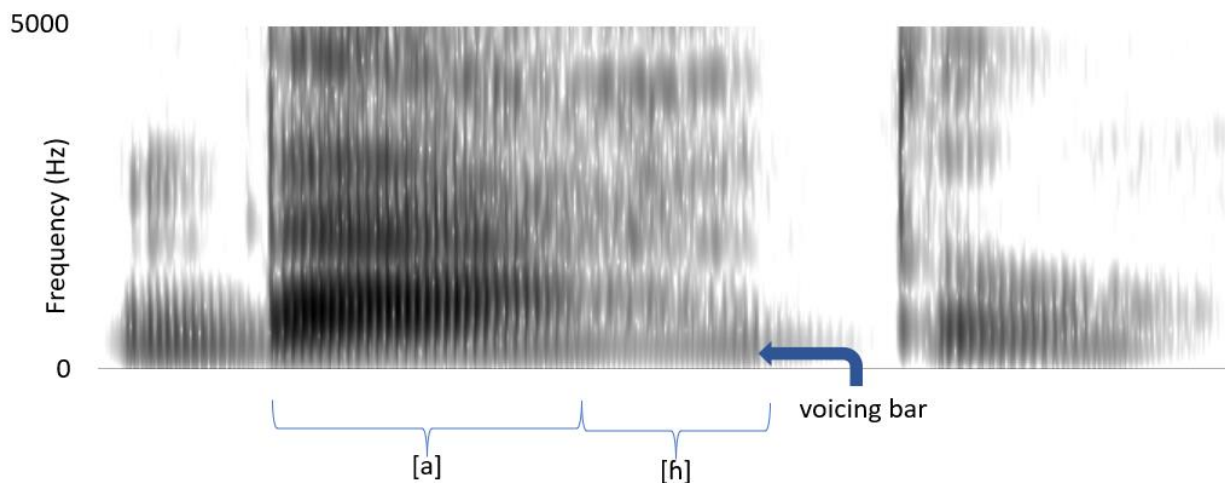


Figure 5.2: Example of the rhotic produced as [h] in syllable-final position on the token *dardo* "dart" by a 27-year-old female speaker from Salvador.

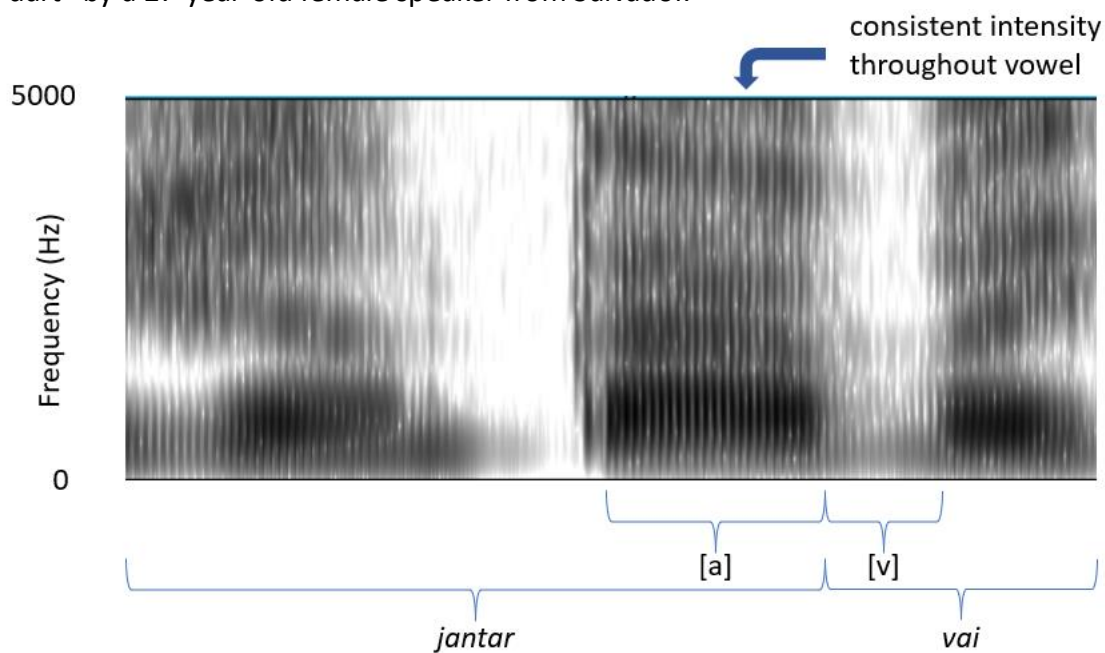


Figure 5.3: Example of the rhotic deleted in word-final position on the token *jantar* "dinner" before the word *vai* "goes." This was produced by a 27-year-old female speaker from Salvador.

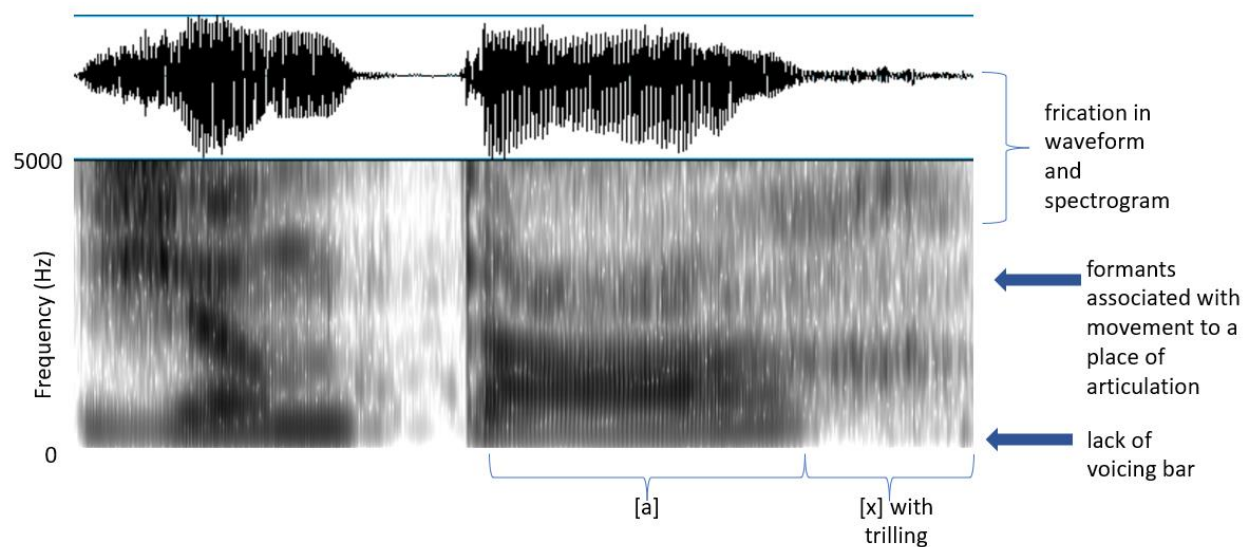


Figure 5.4: Example of the rhotic produced as [x] in word-final position on the token *jantar* “dinner” by a 21-year-old female speaker from Salvador.

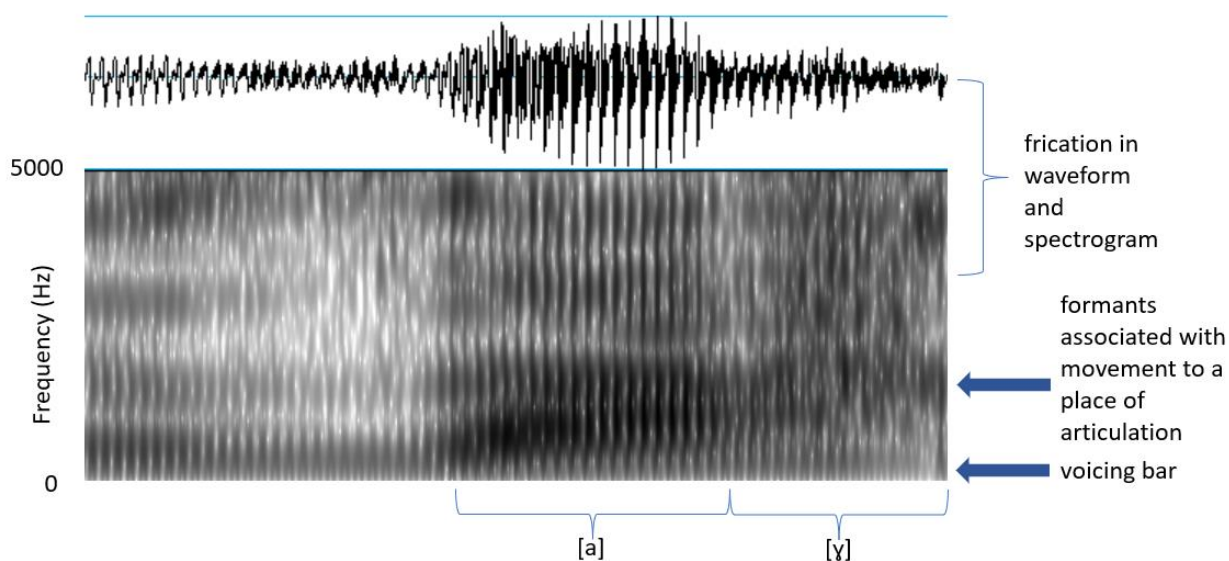


Figure 5.5: Example of the rhotic produced as [ɣ] in word-final position on the token *usar* “to use” by a 21-year-old female speaker from Salvador.

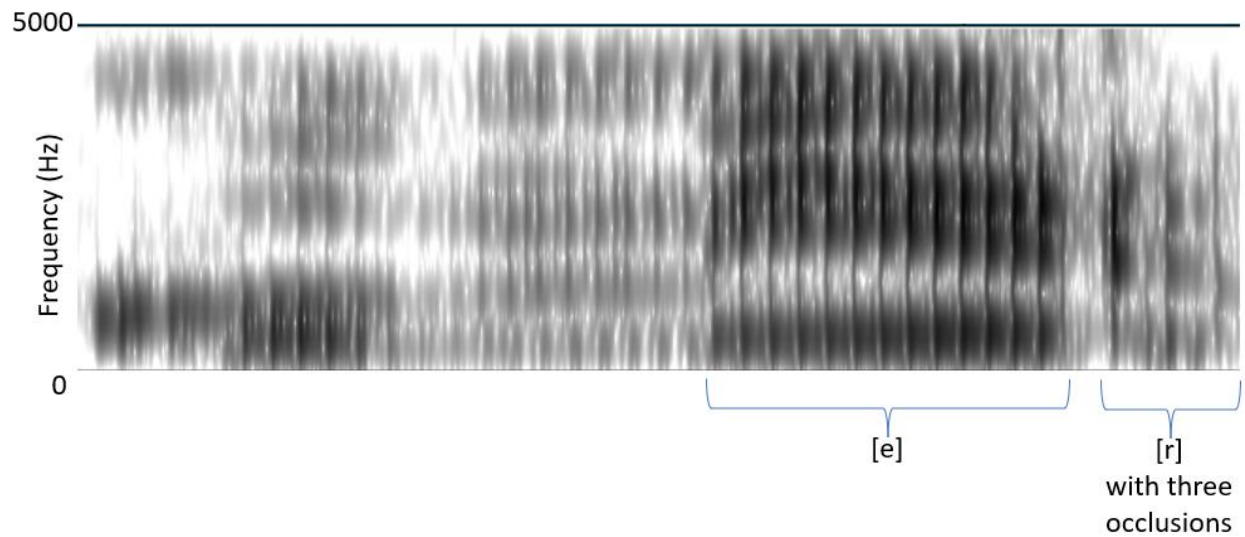


Figure 5.6: Example of the rhotic produced as [r] in word-final position on the token *comer* “to eat” by a 19-year-old male speaker from São Paulo.

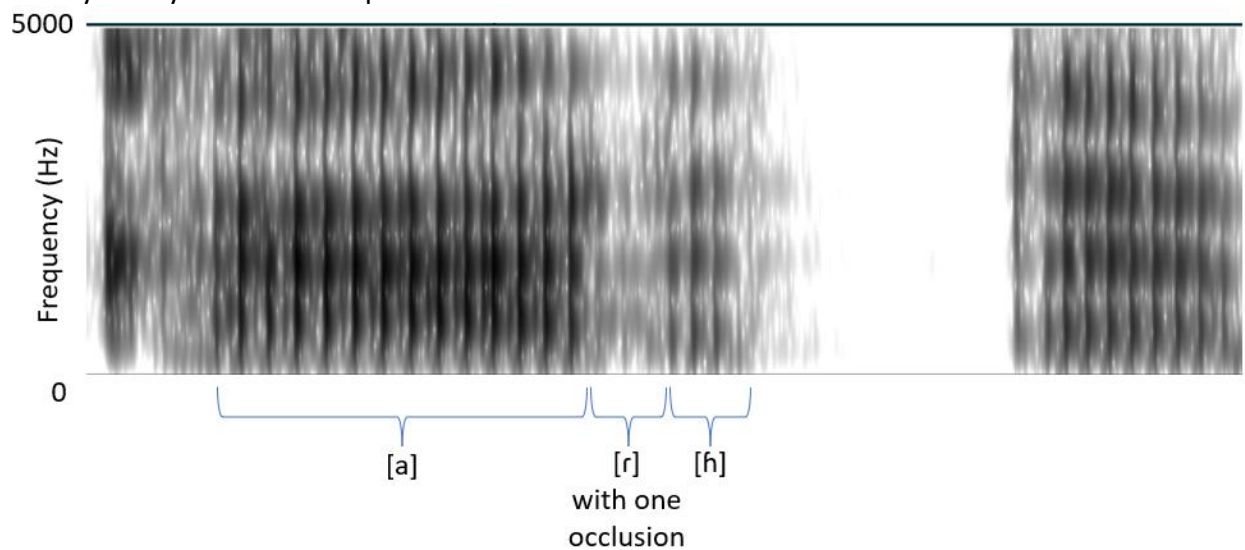


Figure 5.7: Example of the rhotic produced as the flap + fricative variant [rh] in syllable-final position on the token *carta* “letter” by a 19-year-old male speaker from São Paulo.

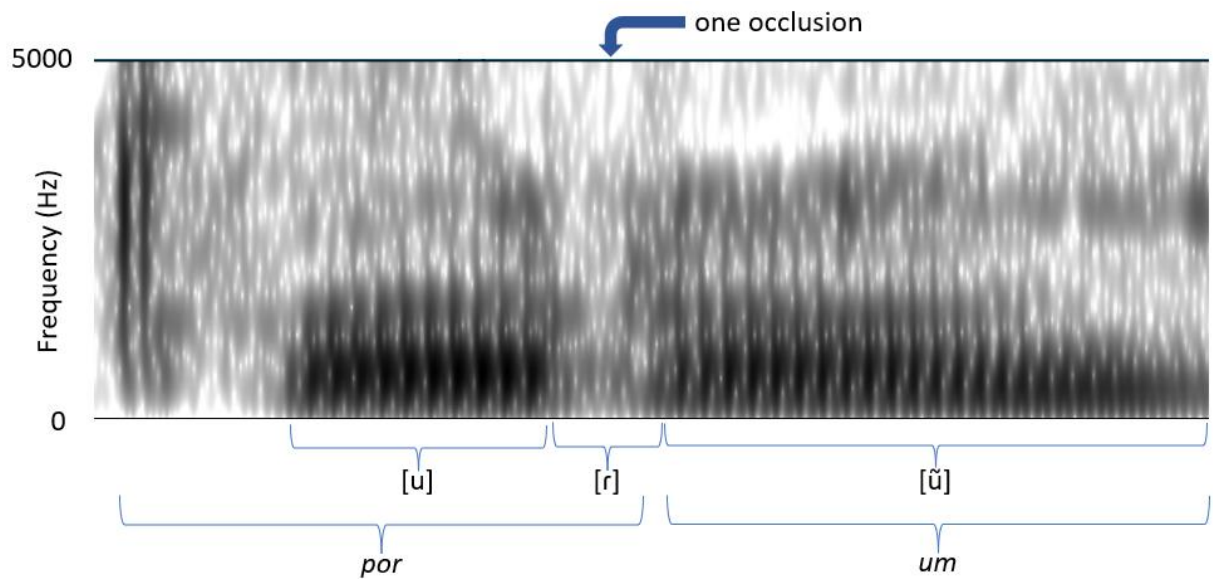


Figure 5.8: Example of the rhotic produced as a flap [r] in word-final position before a vowel on the token *por* “by” before the word *um* “a.” This was produced by a 27-year-old female speaker from Salvador.

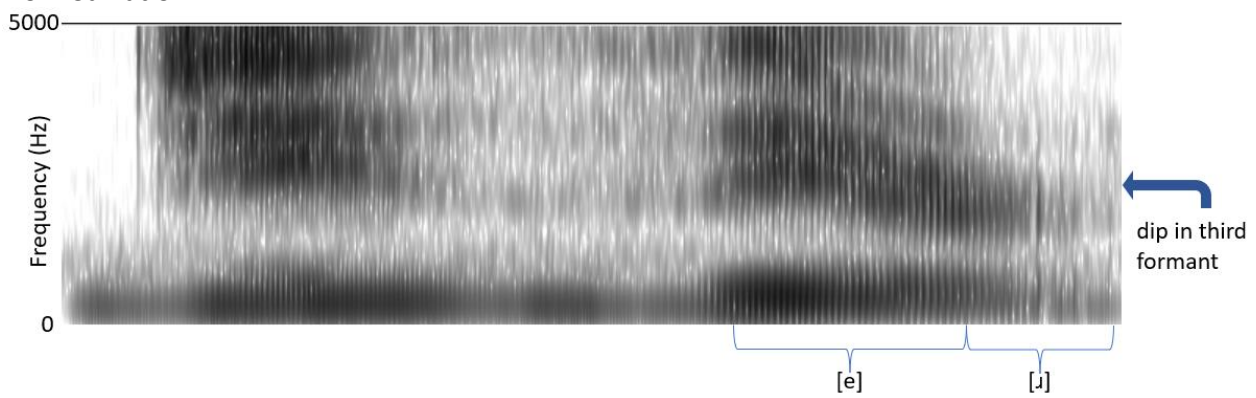


Figure 5.9: Example of the rhotic produced as [ɻ] in word-final position on the token *dizer* “to say” by a 19-year-old female speaker from São Paulo.

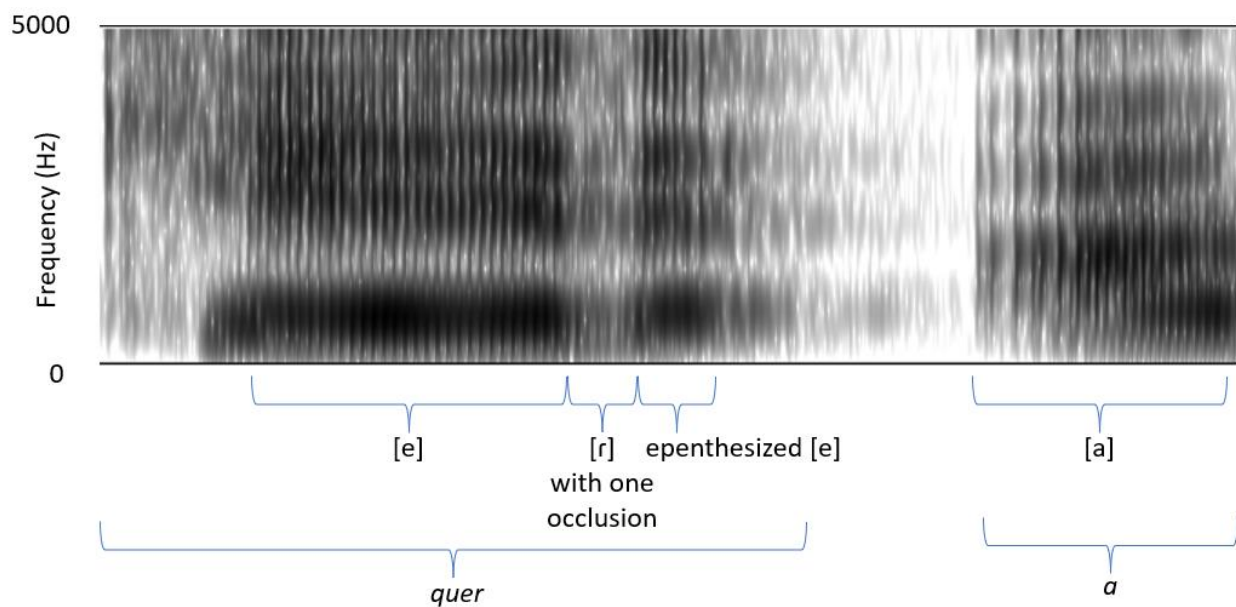


Figure 5.10: Example of the rhotic produced as a flap [r] with an epenthesis vowel in word-final position on the word *quer* “wants” before the word *a* “the.” This was produced by a 19-year-old female speaker from São Paulo.

After classifying all the productions, various *Python* scripts were written in order to tabulate the data. The output of these scripts included counts of each surface form (including deletions) by environment, isolated vs. context, adjacent segments, stress, etc., as well as deletion rates and rates of oral surface forms by environments and other variables. The counts from the *Python* scripts were then used to conduct various statistical tests in both *RStudio* and *Microsoft Excel*. The data and associated statistical tests are presented in chapters 6, 8, and 9.

6. Data from Salvador

This chapter presents the data of the study and the various tests conducted on the deletion rates in the corpus in order to determine what factors affect deletion rates and the surface forms of the rhotic, as well as what sort of relationships exist between the deletion rates in different environments. The tests in this chapter only look at the sample from Salvador as a whole and they do not look at the differences between the demographic subgroups from Salvador, nor do they look at the differences between Salvador and São Paulo. The tests for those differences are in chapters 8 and 9 respectively. Section 1 of this chapter presents the various surface forms of the data and the tests to evaluate where deletion occurs most and least frequently. Section 2 presents the correlations between two different environments to determine to what extent deletion in one environment can predict deletion in another environment.

6.1. Data for the study

This section presents the various surface forms of the rhotic by environment as well as the deletion rates that occur in each environment. It also explores the various factors affecting the surface forms and the deletion rate of the rhotic.

The statistical tests of this section are paired *t*-tests (two dependent samples) of the arcsine transformed deletion rates of two different categories within the data. The arcsine transformed deletion rate is calculated as the arcsine of the square root of the deletion rate:

$$\text{transformed deletion rate} = \sin^{-1} \sqrt{\text{deletion rate}}$$

This transformation was used because it stabilizes the variance of proportions by removing its reliance on the proportion, and can help make the data more normally distributed. This

happens by pulling out the ends of the distribution (Sokal and Rohlf, 1995; Zar, 1999). Note that this transformation is also known as the arcsine square root transformation and the angular transformation. This transformation is also used in section 2 of this chapter and chapters 8 and 9 of this dissertation.

This paired t -test was chosen because it is based on the difference in deletion rate between two environments by each individual participant. This allows for insight into the factors affecting deletion across individuals in the sample. The results of the t -tests are integrated in the presentation of the data at relevant points. All the t -tests concern the deletion rate for each subset of the data being compared, paired by participant. Since there are 35 participants, each t -test has 34 degrees of freedom. The null hypothesis for all the t -tests is $\mu_d = 0$, where μ_d represents the true mean difference between each subset's deletion rate and $\mu_d = (\text{first subset}) - (\text{second subset})$. The t -tests are two-tailed with the alternative hypothesis " $\mu_d \neq 0$," unless otherwise indicated. The direction of a one-tailed test is based on claims made in previous research or when the data indicated that one category appeared to be deleted more than the other, but its significance needed to be tested. The significance threshold is $\alpha = .05$. Significant tests are noted with an asterisk (*) and bolded. The results of these tests will be used to propose a deletion hierarchy in chapter 7.

6.1.1. Word-final productions and deletions

Table 6.1 below gives the count of each surface form of the rhotic for word-final tokens in isolation and in context, with the final column combining both isolation and context tokens. Examples of word-final tokens include *por* "by," *pinta* "to paint," and *açúcar* "sugar." The count

of deleted tokens is included as well and is denoted as “null.” Phonetic symbols followed by a capital V (e.g., “hV”) indicate vowel epenthesis after the given phone.

Table 6.1: Counts of the surface forms of the rhotic in word-final environment.

Surface form	Isolation	Context	Overall
null	127	945	1072
r	0	316	316
h	281	175	456
ɦ	62	259	321
hV	0	2	2
ɦV	0	14	14
χ	1	5	6
ʁ	0	1	1
x	49	34	83
x ^w	6	21	27
x ^w V	0	1	1
ɣ	0	2	2
ɣV	0	1	1
ç	28	26	54
ɟ	1	12	13
r	3	0	3
φ	1	3	4
ɦ	1	1	2
Total	560	1818	2378

In Table 6.1, it can be seen that a fair number of tokens were deleted and that tokens were much more likely to be deleted in context than in isolation. About 22.7% (127/560) of tokens in isolation were deleted, while about 52.0% (945/1818) of tokens in context were deleted. Context tokens were found to delete significantly more than isolation tokens ($\mu_d > 0$, $p = 3.23 \times 10^{-12}$, $t = 10.233$). When produced, the word-final rhotic appears to surface primarily as a glottal fricative or an alveolar flap, with palatal and velar fricatives being somewhat common as well. Additionally, there are three instances of the alveolar trill [r] in isolation. These were all produced by a single speaker. It is unclear why this speaker produced this form; however since they are the only person to produce alveolar trills from Salvador, and there are so few instances

of alveolar trills, it is likely that these are not typical surface forms in Salvador. Furthermore, Gomes (2016) found that Bahian learners of Spanish struggled to acquire the alveolar trill in Spanish, suggesting that the trill is not a typical surface form of the rhotic in Salvador. On the other hand, while the voiceless bilabial fricative [ɸ] only occurred four times word-finally, it occurred with a different participant each time. Each instance of [ɸ] was preceded by the rounded vowel [o] and the one instance that was followed by a consonant was followed by [p]. Thus, each instance of [ɸ] had at least one adjacent labial element, suggesting that this is likely a result of coarticulation. This distribution of the surface forms is given in figure 6.1, which shows the proportions of each surface form in the word-final environment.

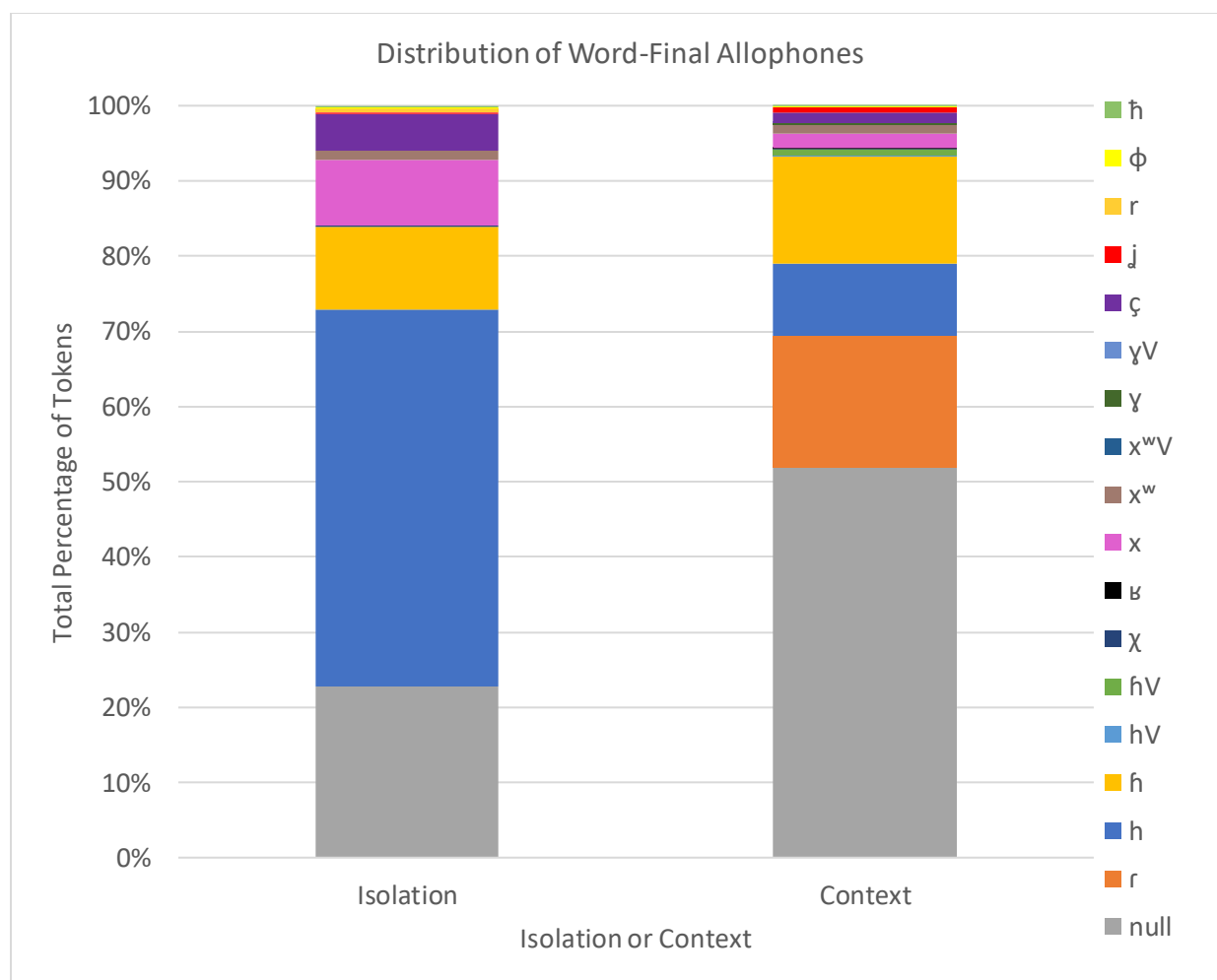


Figure 6.1: Proportion of surface forms of the rhotic in word-final environment

In Figure 6.1, it can be seen that the voiceless glottal fricative [h] predominated in the isolated productions while deletion predominated in context. The alveolar flap [r] was quite common in context but it only occurred in word-final tokens followed by a vowel-initial word (e.g., *limpar as janelas* “to clean the windows”). In this respect, regardless of whether the token was in context or in isolation, the glottal fricatives were by far the preferred surface forms among the participants. However, the voiced glottal fricative [ɦ] is more common in context, while the voiceless glottal fricative [h] is more common in isolation. The other surface forms appear to comprise over 15% of tokens in isolation, but less than 10% of tokens in context.

Another thing to note is that the participants themselves demonstrated a range of deletion rates of word-final context tokens from 3.8% (two deletions) to 96.2%. It should be noted that only three participants had deletion rates lower than 20% (3.8%, 5.8%, 15.4%), with the next lowest participant having a deletion rate of about 23.1%, suggesting that these speakers might form a different group from the others. Meanwhile, seven participants had deletion rates above 80%, including two over 90%, with the next highest deletion rate at about 78.8%, indicating that speakers from Salvador in the corpus are more likely to be at the upper end of the deletion range than at the lower end.

In order to determine how the segment type following the word-final rhotic affected its surface form and deletion, word-final tokens in context were separated according to following environment. Table 6.2 and figure 6.2 below give the count and distribution of each surface form of the rhotic for word-final context tokens separated by following environment.

Table 6.2: Counts of the surface forms of the rhotic on word-final context tokens separated by following environment.

Surface form	Consonant	Pause	Vowel
null	484	100	361
r	0	0	316
h	51	96	28
ɦ	204	31	24
hV	1	0	1
ɦV	14	0	0
χ	0	3	2
ʁ	0	0	1
x	15	13	6
x ^w	7	14	0
x ^w V	1	0	0
ʏ	2	0	0
ʏV	1	0	0
ç	11	6	9
ɟ	8	2	2
φ	1	2	0
ħ	0	1	0
Total	800	268	750

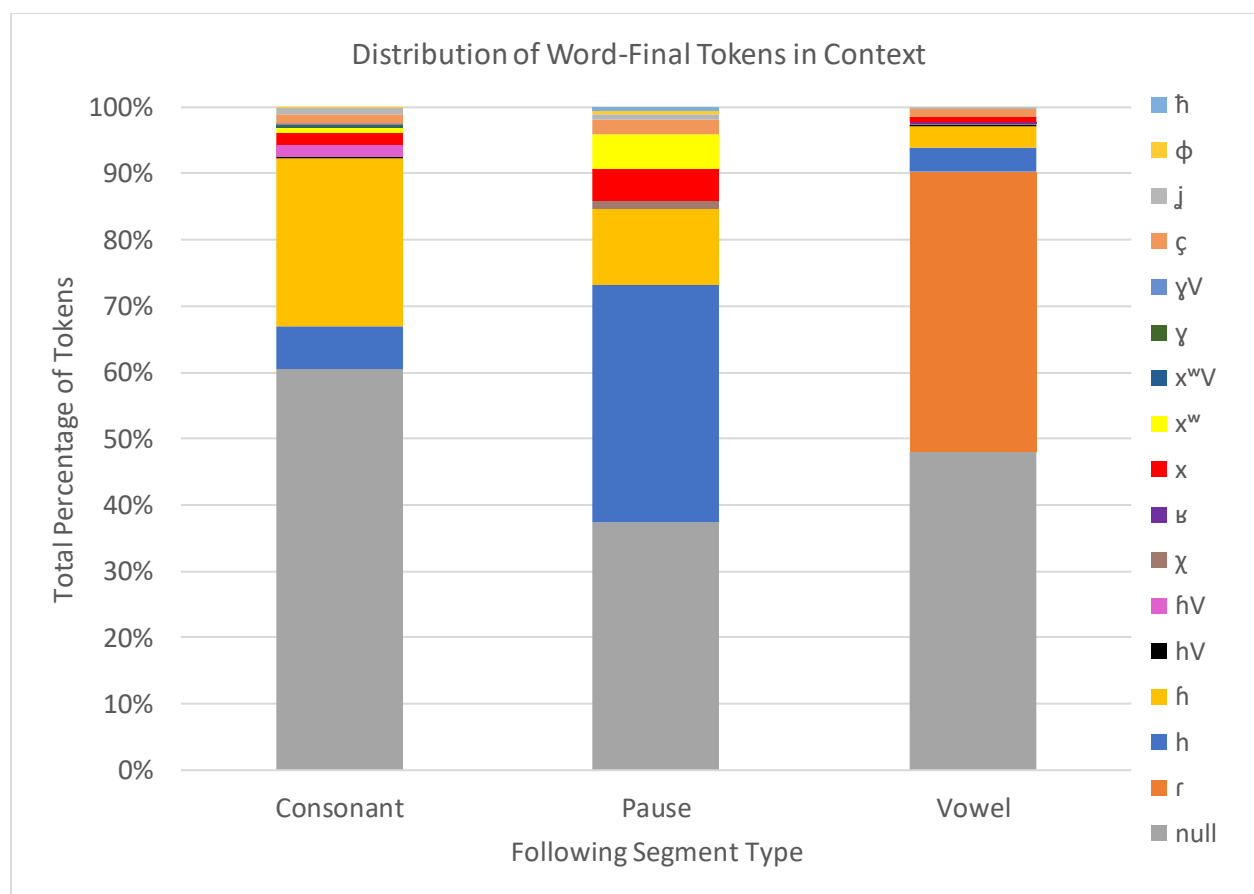


Figure 6.2: Proportion of surface forms of the word-final rhotic by following segment type.

From the table and figure above, it can be seen that tokens before a consonant (e.g., *por gato* “per cat”) have the highest deletion rate at 60.5% (484/800), followed by tokens before a vowel (e.g., *limpar as janelas* “to clean the windows”) at about 48.1% (361/750) and tokens before a pause (*comida com açúcar##* “food with sugar”) at about 37.3% (100/268). These differences are statistically significant between tokens before a consonant and tokens before a vowel ($\mu_d > 0$, $p = 6.03 \times 10^{-4}$, $t = 3.533$) as well as between tokens before a vowel and tokens before a pause ($\mu_d > 0$, $p = 0.019$, $t = 2.454$). Thus, it is the case that the following segment type is playing a major role in the deletion rate of the rhotic in word-final position. Although tokens before a pause are deleting less often than the tokens in other contexts, they are still deleting more often than tokens in isolation ($\mu_d > 0$, $p = 1.56 \times 10^{-4}$, $t = 4.012$). With this, there is

a four-step hierarchy among the deletion rates of word-final tokens with tokens before a consonant in the highest position and isolation tokens in the lowest position.

Regarding surface form variability by following segment type, the most common variant is different for each segment type as well. While glottal fricatives are most common before both consonants and pauses, they are usually voiced before consonants and voiceless before pauses. Meanwhile, alveolar flaps [ɾ] are the most common surface form before vowels. Given that this is the only place where flaps occurred in Salvador, this suggests that the word-final environment before vowels induces a sort of neutralization or realignment of the rhotic and flap phonemes. Further discussion of this phenomenon is given in subsection 6.1.1.2.

With the presence of vowels following the word-final rhotic resulting in a neutralization with the flap and a reduction in the deletion rate, it was natural to wonder whether having the same vowel both precede and follow the rhotic would delete at a different rate than having a different vowel on each side. In the corpus, deletion occurred about 45.3% of the time when the preceding and following vowels were the same, while it occurred about 49.4% of the time when the preceding and following vowels were different. This difference was not statistically significant ($p = 0.459$, $t = 0.748$), indicating that there is effectively no difference in deletion rates based on whether or not the surrounding vowels are the same.

Other contextual effects on rhotic deletion were also tested, based on claims in previous literature (Oliveira 1983; Brandão 1995; Cardoso 1999; Serra & Callou 2013; Rennie 2015): a stressed syllable is less likely to have the rhotic delete (e.g. *trabalhador* “worker” less likely to delete than *açúcar* “sugar”) and verbs are more likely to have a word-final rhotic delete (e.g., *cantar* “to sing” more likely to delete than *cantar* “singer”). These factors were taken into

consideration when designing stimuli to ensure representation of all possibilities. However, because word-final stress and verbs comprise the majority of words with word-final rhotics in Portuguese, the number of tokens in each stress and morphosyntactic category is not equal. Neither effect was statistically significant in this corpus. In fact, both of these tests yielded p -values greater than 0.5 because in the data, the opposite of these claims turned out to be the case. Stressed syllables (47.2%) did not delete less than unstressed syllables (29.1%) ($\mu_d < 0$, $p > 0.999$, $t = 5.341$), and verbs (43.9%) did not delete more than non-verbs (47.2%) ($\mu_d > 0$, $p = 0.88$, $t = -1.185$). However, most verbs with word-final rhotics are infinitives with word-final stress, creating a paradox for some tokens. If these results were tested in the opposite direction, the test for verbs versus non-verbs would not be significant, but the test for stressed versus unstressed syllables would be. However, given that only 8 unstressed tokens occurred per speaker and that word-final tokens have a significant deletion difference based on following segment type, there is no need to separate word-final tokens by stress in the deletion hierarchy proposed in chapter 7.

6.1.1.1. Factors affecting the voicing and place of word-final surface forms

Lastly, it is important to consider the possibility of the surface form of the rhotic being affected by the voicing and place of articulation of adjacent segments. In terms of the word-final context, all tokens are preceded by a voiced segment since they are all preceded by vowels. Because this is consistent for all word-final tokens, the voicing of the preceding segment cannot be considered a factor affecting the surface form of the rhotic. However, in the more variable case of word-final tokens which are followed by a segment, 491 out of 550 (89.3%) instances of the rhotic which are followed by a voiced segment (*por gato* “per cat” or *fizer uma estátua* “to

make a statue”) are also voiced. Meanwhile, only 74 out of 155 (47.7%) word-final tokens followed by a voiceless segment (*comer comida* “to eat food”) are also voiceless. Therefore, regardless of the voicing of the following segment, the rhotic tends to surface as voiced. However, a voiceless segment following the rhotic increases the likelihood of the rhotic being voiceless. This is statistically significant using a Fisher’s exact test ($p < 2.2 \times 10^{-16}$ *, **odds ratio = 7.752**). The voicing of an adjacent segment does not mean that the rhotic will match in voicing, as voiceless rhotics still occasionally occurred before voiced segments and voiced rhotics still occurred before voiceless segments. At the same time, if there is no segment following the word-final rhotic, as in isolation or before a pause, there is a preference for the rhotic to surface as voiceless (366 out of 433 (84.5%) in isolation, 134 out of 168 (79.8%) in context before a pause). These tendencies can also be seen in Figures 6.1 and 6.2. This suggests that utterance-final productions tend to be voiceless.

With regard to the place of articulation for adjacent segments, the predominance of the glottal fricatives and flap (for tokens followed by a vowel) indicates that generally, the surface form of the rhotic is unaffected by place of articulation of an adjacent segment. However, if the rhotic surfaces as something other than a glottal fricative or a flap, to what extent does it match the neighboring place of articulation? Where the vowels preceding word-final productions are concerned, 102 out of 197 (51.8%) non-glottal, non-flap productions matched the place of constriction of the vowel. Of these, 2 were pharyngeal fricatives after [a], 50 were palatal fricatives after [i], 16 were palatal fricatives after [e], 3 were velar fricatives after [u], 27 were rounded velar fricatives after [o], and 4 were bilabial fricatives after [o]. Thus, when the rhotic and preceding segment match in place, they are typically matching at the hard palate or labially

with rounding. Where the following segment is concerned, only 21 out of 66 (31.8%) non-glottal, non-flap productions matched in place of articulation. Of these, 11 were velar fricatives preceding [k] or [g], 1 was the bilabial fricative preceding [p], 6 were rounded velar fricatives preceding [p], 2 were velar fricatives preceding [u], and 1 was a palatal fricative preceding [e]. Thus, when the rhotic and following segment match in place, they are typically matching at the velum or labially with rounding. When both the preceding and following segments are considered, it appears that palatal vowels before the rhotic, velar consonants after the rhotic, and a labial element before or after the rhotic tend to be the main drivers that cause a match in place of articulation between a word-final rhotic and an adjacent segment. However, there is a greater tendency for the rhotic to match a preceding vowel than a following consonant. Nevertheless, a surface form that diverges from a glottal fricative is not necessarily going to match the place of an adjacent segment. Tables 6.3-6.5 below show the non-glottal, non-flap surface forms of the rhotic for each preceding and following segment in the word-final environment. Note that the counts for preceding segments and following segments are not the same as some word-final tokens are not followed by a segment (those in isolation and those followed by a pause).

Table 6.3: Preceding segments and non-glottal, non-flap surface forms of the rhotic in the word-final environment.

Preceding segment	Surface forms											
	ç	j	x	x ^w	x ^w V	ɣ	ɣV	χ	ʁ	r	ħ	φ
a	0	0	34	0	0	1	0	4	0	3	2	0
i	43	7	12	0	0	0	0	1	0	0	0	0
e	10	6	29	0	0	0	0	0	1	0	0	0
u	0	0	1	0	1	1	0	1	0	0	0	0
o	1	0	7	27	0	0	1	0	0	0	0	4

Table 6.4: Following consonants and non-glottal, non-flap surface forms of the rhotic in the word-final environment.

Following segment	Surface forms							
	ç	j	x	x ^w	x ^w V	ɣ	ɣV	ϕ
k	6	6	7	0	0	1	0	0
g	0	0	1	0	1	1	0	0
s	0	0	3	0	0	0	0	0
p	5	2	0	6	0	0	0	1
f	0	0	2	0	0	0	0	0
d	0	0	1	1	0	0	1	0
t	0	0	1	0	0	0	0	0

Table 6.5: Following vowels and non-glottal, non-flap surface forms of the rhotic in the word-final environment.

Following segment	Surface forms				
	ç	j	x	χ	ʁ
ũ	2	0	0	1	0
a	1	0	2	0	1
u	5	2	2	1	0
e	1	0	0	0	0
ĩ	0	0	2	0	0

Overall, it is clear that in the word-final environment, the rhotic has a strong tendency to delete. When not deleted, it typically either surfaces as a glottal fricative ([h], [ɦ]) or, if in context and followed by a vowel, an alveolar flap [r]. This flapping leads to a major reduction in the deletion rate, compared to context tokens followed by a consonant, but the deletion rate remains higher than context tokens followed by a pause. A variety of other surface forms occur in small numbers as well. In terms of using the voicing or place of articulation of an adjacent segment to help predict the surface form, it appears that some tendencies exist but none gives a very certain means of prediction. For voicing, it appears that the better predictor is whether or not the rhotic is followed by anything. For place on non-glottal and non-flap productions, it seems likely that the preceding or following segment can have an influence on the surface form of the rhotic; however, they are no guarantee as to how it will surface. As a whole, the data in

the corpus indicate that the word-final rhotic is undergoing large amounts of lenition both in the form of deletion and in the form of glottal fricatives. The corpus also indicates that the previously reported inventory of allophones ([r], [r̥], [ɹ], [x], [χ], [ɣ], [ʁ], [h], [ɦ]) for the rhotic phoneme in Brazilian Portuguese does not contain all the surface forms found in Salvador (e.g., [ç], [j]).

6.1.1.2. Flapping phenomenon on word-final tokens before a vowel

While it has been stated earlier in this dissertation that the alveolar flap /r/ and the rhotic discussed here are separate phonemes, the fact that the most common allophone of the rhotic word-finally before a vowel is the flap blurs this line substantially. Firstly, the two phonemes have a marginal contrast, only being distinguished in word-medial, intervocalic position, and secondly, the flap is occurring in the word-final environment, which is supposed to have the rhotic that does not surface as a flap. Notably, the word-final context before a vowel is also intervocalic, like the environment where the two phonemes are supposed to contrast with one another. Therefore, is it correct still to consider these separate phonemes? The answer to this would be yes, as in the corpus (for both cities), there are no instances of flaps occurring in the environment where the two are said to contrast: word-medial, intervocalic position. In this position, rhotics with the orthography “r” are produced as flaps and all rhotics with the orthography “rr” are not produced as a flap.

On the surface, this appears to be a sort of merger, but in reality, this resembles what is referred to in historical linguistics as a primary split. A primary split is where an allophone of one phoneme switches to resemble another phoneme, but both phonemes still remain (Campbell 2013, p. 23; Millar and Trask 2023, p. 75). What is happening here is that word-final

rhotics are conditioned to surface as one phoneme or the other depending on the following context. If the following context is a vowel, the rhotic is generally conditioned to surface as a flap, although some productions still surfaced as the other phoneme instead. In all other following contexts, the rhotic is conditioned to surface as the non-flap phoneme. Various Portuguese language phonologies indicate that the coda rhotic in Portugal is the flap, rather than the rhotic phoneme discussed in this dissertation, giving additional evidence supporting the notion that word-final rhotics in Portuguese can be the flap phoneme (Barbosa 1983, 1984; Silva 1998; Mateus and d'Andrade 2000). In fact, Mateus and d'Andrade (2000) specifically distinguish varieties spoken in Portugal from those spoken in Brazil by the way coda rhotics surface. Thus, it is not unusual for the coda rhotic in Portuguese to surface as a flap. It just happens to be the case in Salvador that the rhotic will only do so when it is word-final before a vowel. Otherwise, the rhotic will surface as the rhotic phoneme.

Moreover, on nouns with word-final rhotics, the plural is formed with an epenthesis vowel [i] and an [s]. When this suffix is added, the rhotic changes from the “strong r” phoneme to the flap. For example, the singular *sabor* [saboɾ] “flavor” becomes *sabores* [saboris] “flavors” in the plural. Thus, the addition of the plural suffix on nouns with word-final rhotics conditions a change to the flap phoneme. Given the similarities between this alternation and flapping on word-final rhotics before a vowel, this further suggests that the following segment tends to condition which phoneme a word-final rhotic will surface as.

One final thing to note regarding the flapping phenomenon is that the root to which suffixes are added for the future tense and conditional mood in Portuguese is identical to the infinitive, which ends with a word-final rhotic, for many verbs. The suffixes added to the root for

these conjugations all begin with vowels (Mateus and d'Andrade 2000). Consequently, this means that the root will have the rhotic surface as a flap in all of these conjugations, as it is a single “r” in the orthography intervocalically. The existence of these verb conjugations may also influence the ability of the word-final rhotic to surface as a flap before vowels.

6.1.2. Syllable-final productions and deletions

Table 6.6 gives the count of each surface form of the rhotic for syllable-final tokens in isolation and in context, with the final column combining both isolation and context tokens. Examples of syllable-final tokens include *carta* “letter” and *árvores* “trees.” Thus, these are word-medial, syllable final tokens, as opposed to word-final, syllable-final tokens in the preceding subsection. There are a few additional surface forms in the syllable-final environment that did not occur word-finally and some forms which did occur word-finally did not occur syllable-finally ([ʀ], [ɣʷ], [ɬ], [θ], and [ɕ]).

Table 6.6: Counts of the surface forms of the rhotic in the word-medial, syllable-final environment

Surface form	Isolation	Context	Overall
null	12	200	212
h	45	99	144
ɦ	171	564	735
hV	3	0	3
ɦV	22	7	29
χ	1	1	2
ʁ	3	2	5
R	1	0	1
x	11	27	38
χ ^w	4	25	29
ʁ	2	6	8
ʁ ^w	1	5	6
ç	0	26	26
ɟ	0	13	13
r	3	0	3
ɾ	0	1	1
θ	0	1	1
ç	0	1	1
φ	0	1	1
Total	279	979	1258

In Table 6.6, it can be seen that a fair number of syllable-final tokens were deleted in context, though at a deletion rate of 20.4% (200/979), it is noticeably lower than word-final tokens. Additionally, only 12 out of 279 tokens (4.3%) in isolation were deleted. Once again, tokens in context delete the rhotic more frequently than tokens in isolation ($\mu_d > 0$, $p = 4.81 \times 10^{-12}$, $t = 10.076$). Similarly, the glottal fricatives were again the preferred surface form in both isolation and in context. However, unlike the word-final environment, the voiced glottal fricative [ɦ] and its variants were preferred over voiceless [h] in both isolation and in context. Palatal and velar fricatives were common as well, and there were a few uvular productions. The bilabial fricative [φ] occurred after the rounded vowel [o], as was the case for all the word-final productions of this form. Once again, there were three instances of the alveolar trill [r] in

isolation. These came from the same participant as the word-final trills. There were also three unusual productions: the voiceless alveolar lateral fricative [ɬ], the voiceless interdental fricative [θ], and the voiceless alveolo-palatal fricative [ç]. Interestingly, all three of these occurred with the word *porta* “door” in a sentence. It is not clear what could have driven this, but the [t] following the rhotic on this token and the fact that these are all coronal productions suggest that it may be a coarticulation effect. The proportions of the surface forms in the syllable-final environment are given in Figure 6.3.

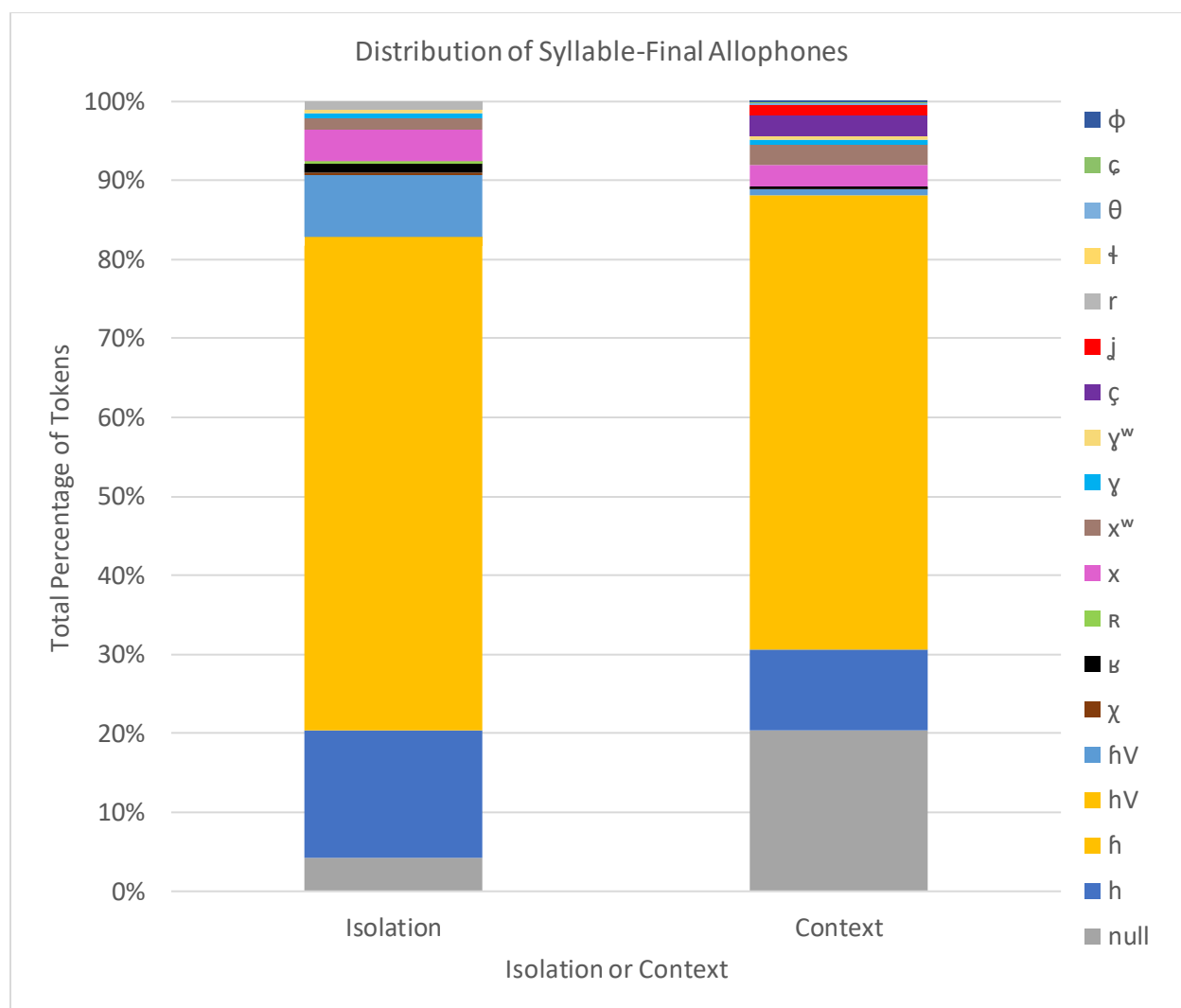


Figure 6.3: Proportion of surface forms of the rhotic in the word-medial, syllable-final environment

In Figure 6.3, it can be seen how deletion is not nearly as predominant with syllable-final tokens as it was with word-final tokens, particularly in the case of isolation. The bulk of each consists of the voiced glottal fricative [h], with many of the remaining instances being either the voiceless glottal fricative [h] or other glottal variants such as [hV]. In total, other places of articulation comprise about 10% of the isolation tokens and about 12% of the context tokens.

Regarding stress, the deletion on unstressed syllables (15.8%) (e.g., *informações* “information”) was not significantly higher than on stressed syllables (17.8%) (e.g., *porta*

“door”) ($\mu_d > 0$, $p = 0.926$, $t = -1.478$). There is no need to conduct a test on deletion for verbs versus non-verbs, as the existing literature only mentions its significance word-finally.

6.1.2.1. *Factors affecting the voicing and place of syllable-final surface forms*

Regarding the voicing of the adjacent segments to the syllable-final rhotic, there is a similar situation to that of the word-final rhotic. Like their word-final counterpart, all syllable-final rhotics are preceded by vowels and therefore, preceded by voiced segments. However, unlike the word-final environment, the following segment can only be a consonant, which may be voiced or voiceless.

When considering the more variable following environment, among syllable-final rhotics followed by voiced consonants (e.g., *árvores* “trees”), 506 out of 526 (96.2%) are voiced. Meanwhile, for those followed by a voiceless consonant (e.g., *carta* “letter”), 226 out 520 (43.5%) are voiceless. Thus, while it is the case that most of the syllable-final rhotics followed by voiceless consonants are still voiced, there is a noticeably larger proportion of them that are voiceless. This is statistically significant using a Fisher’s exact test ($p < 2.2 \times 10^{-16}$ *, **odds ratio = 19.401**). However, since it is the case that there is a strong preference for voiced variants in the syllable-final position, the voicing on the preceding vowel likely has a stronger influence over the surface form of the syllable-final rhotic than the following consonant. Nevertheless, a voiceless consonant following a syllable-final rhotic drastically increases the likelihood of the rhotic surfacing as a voiceless consonant.

With regard to the place of articulation for adjacent segments, the predominance of the glottal fricatives indicates that generally, the surface form of the rhotic is unaffected by an adjacent segment’s place of articulation, as was the case word-finally. However, there are some

non-glottal surface forms worth considering. In the case of the following consonant, 55 out of 135 (40.7%) non-glottal productions matched the place of articulation. Of these, 2 were rounded velar fricatives before [m], 1 was the bilabial fricative [ɸ] before [m], 1 was the aforementioned [ɬ] before [t], 1 was the alveolar trill before [d], 49 were velar fricatives before [k] and [g], and 1 was the rounded velar fricative [ɣ^w] before [v]. Where the place of articulation for the preceding vowel is concerned, 77 out of 135 (57.0%) non-glottal productions of the syllable-final rhotic matched. Of these, 7 were rounded velar fricatives after [o], 3 were the voiced uvular fricative [ʁ] after [o], 1 was the bilabial fricative [ɸ] after [o], 39 were palatal fricatives before [e] or [i], 27 were velar fricatives before [u]. Therefore, it appears that place of articulation of the adjacent segments syllable-finally offer a fair indication as to where a non-glottal rhotic may be articulated. While adjacent segment place can help predict the place of articulation, the adjacent segment only offers a general idea of how the rhotic may surface, rather than a clear assimilation rule. Notably, for both word-final and syllable-final tokens, the preceding vowel did a better job at predicting the place of a non-glottal surface form than the following consonant did. Tables 6.7 and 6.8 below show the non-glottal surface forms of the rhotic for each preceding and following segment in the syllable-final environment.

Table 6.7: Preceding segments and non-glottal surface forms of the rhotic in the syllable-final environment.

Preceding segment	Surface forms													
	ç	j	x	x ^w	ɣ	ɣ ^w	χ	ʁ	R	r	ɬ	θ	ç	ɸ
ɔ	0	0	12	1	1	0	1	2	1	1	1	1	1	0
o	0	0	6	5	2	2	0	3	0	1	0	0	0	1
a	0	0	6	0	1	0	1	0	0	1	0	0	0	0
e	7	4	12	0	4	0	0	0	0	0	0	0	0	0
u	0	0	0	23	0	4	0	0	0	0	0	0	0	0
i	19	9	2	0	0	0	0	0	0	0	0	0	0	0

Table 6.8: Following segments and non-glottal surface forms of the rhotic in the syllable-final environment.

Following segment	Surface forms													
	ç	j	x	x ^w	ɣ	ɣ ^w	χ	ʁ	R	r	ɾ	θ	ç	φ
b	0	0	0	0	0	0	0	1	1	1	0	0	0	0
m	0	1	1	2	0	0	0	2	0	1	0	0	0	1
t	0	0	17	3	2	1	1	1	0	0	1	1	1	0
d	0	0	0	0	0	0	0	0	0	1	0	0	0	0
n	0	0	2	1	1	1	0	0	0	0	0	0	0	0
k	25	12	18	23	4	3	1	0	0	0	0	0	0	0
v	0	0	0	0	0	1	0	0	0	0	0	0	0	0
g	0	0	0	0	1	0	0	1	0	0	0	0	0	0
f	1	0	0	0	0	0	0	0	0	0	0	0	0	0

Overall, the syllable-final rhotic is undergoing a fair amount of lenition, although it is less than the word-final environment. Since this lenition is not nearly as progressed, it surfaces much more as a glottal fricative than it does as deletion. Nevertheless, deletion of the syllable-final rhotic is present and cannot be ignored. Several other surface forms besides the glottal fricatives exist in this environment as well. While the syllable-final environment appears to prefer voiced surface forms over voiceless ones, it still seems that a voiceless consonant following the rhotic makes it more likely for the rhotic to surface as voiceless. In the case of the place of articulation of non-glottal surface forms, both the preceding vowel and the following consonant appear to give a fair amount of insight into what the surface form may look like. As a whole, the syllable-final forms indicate that the pattern of lenition on the rhotic is not limited to the word-final environment. Additionally, they further corroborate the notion that this phoneme surfaces with a large variety of forms in the Salvador variety.

6.1.2.2. Vowel epenthesis on rhotics in coda position

It is worth noting that there were some cases both word-finally and syllable-finally where a vowel was epenthesised after the rhotic. Ultimately, this phenomenon makes up a

small portion of the corpus; however it is a topic that warrants brief discussion. Only 18 productions of the 1818 word-final context tokens had vowel epenthesis, and all but one occurred before a consonant. The one other production occurred before a vowel. Meanwhile, 32 of the 1258 syllable-final productions had vowel epenthesis, with 25 of these being in isolation and 7 of these being in context. Additionally, instances of vowel epenthesis in coda position were found in São Paulo as well. This suggests that a consonant following a rhotic in coda position may trigger occasional epenthesis. Furthermore, this could be linked to the tendency to maintain a CV syllable structure that was discussed in chapter 2, though with so few instances of vowel epenthesis in the corpus, it is not possible to explore this phenomenon in more detail.

6.1.3. Word-initial productions and deletions

Table 6.9 gives the count of each surface form of the rhotic for word-initial tokens in isolation and in context, with the final column combining both isolation and context tokens. Examples of word-initial tokens include *rato* “rat” and *rapidamente* “quickly.” The word-initial environment showed less variation and therefore, there are fewer forms in table 6.4 and the corresponding figure 6.4.

Table 6.9: Counts of the surface forms of the rhotic in the word-initial environment

Surface form	Isolation	Context	Overall
null	6	39	45
h	161	99	260
ɦ	41	377	418
x	0	3	3
ç	0	2	2
ç̥	1	1	2
ç̞	1	2	3
ɾ	0	1	1
Total	210	524	734

In table 6.9, it is clear that word-initial tokens were far less likely to delete than syllable-final tokens in context and all word-final tokens. With only 39 deletions out of 524 tokens in context, this is a deletion rate of about 7.4%. In isolation, only 6 of 210 tokens (2.9%) were deleted. Despite the fact that both of these deletion rates are quite low, tokens in context were still found to delete significantly more than tokens in isolation ($\mu_d > 0$, $p = 2.47 \times 10^{-4*}$, $t = 3.852$). As with the previous environments, the majority of productions appear to be glottal fricatives, with only ten productions not being glottal. In terms of unusual surface forms, there are the voiced epiglottal fricative [ʕ] and the voiced pharyngeal fricative [ʁ]. Additionally, a voiceless alveolar trill [r] occurred in context with same participant that produced the other alveolar trills.

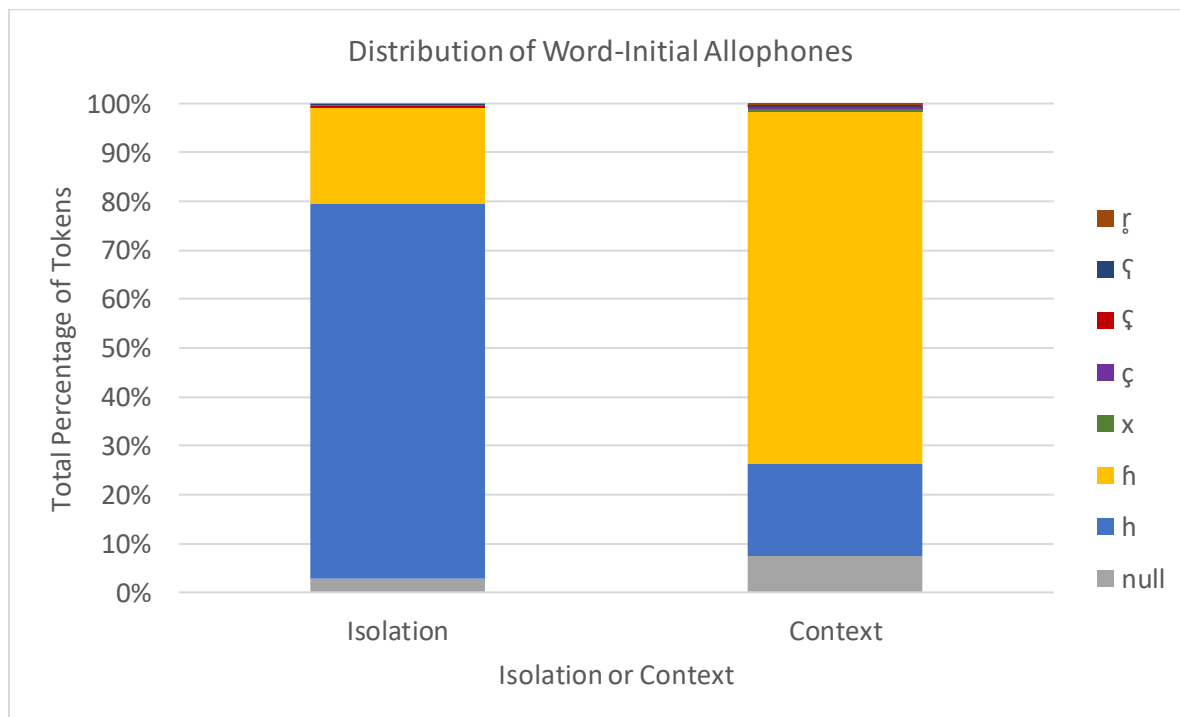


Figure 6.4: Proportion of surface forms of the rhotic in word-initial environment

In Figure 6.4, it is clear that the glottal fricatives predominate once again, with the voiceless glottal fricative [h] preferred in isolation and the voiced glottal fricative [ɦ] preferred in context. This is reminiscent of what was seen with the word-final rhotic earlier, where the voiceless glottal fricative was preferred in isolation and before a pause while the voiced glottal fricative was preferred before consonants and vowels. Deletion still exists but is noticeably reduced compared to the previous two environments. Other surface forms only appear in context and comprise well under 10% of productions. Because the preferences for glottal fricative voicing in isolation versus in context resemble the situation with the word-final environment, it is necessary to split up the context tokens by preceding environment to make some comparisons. Table 6.10 and figure 6.5 give this information.

Table 6.10: Counts of the surface forms of the rhotic on word-initial context tokens by preceding segment type.

Surface form	Consonant	Vowel	Pause
null	27	7	5
h	15	12	72
ɦ	96	258	23
x	1	1	1
ç	0	0	2
ɕ	0	0	1
ʃ	0	1	1
ʒ	0	1	0
Total	139	280	105

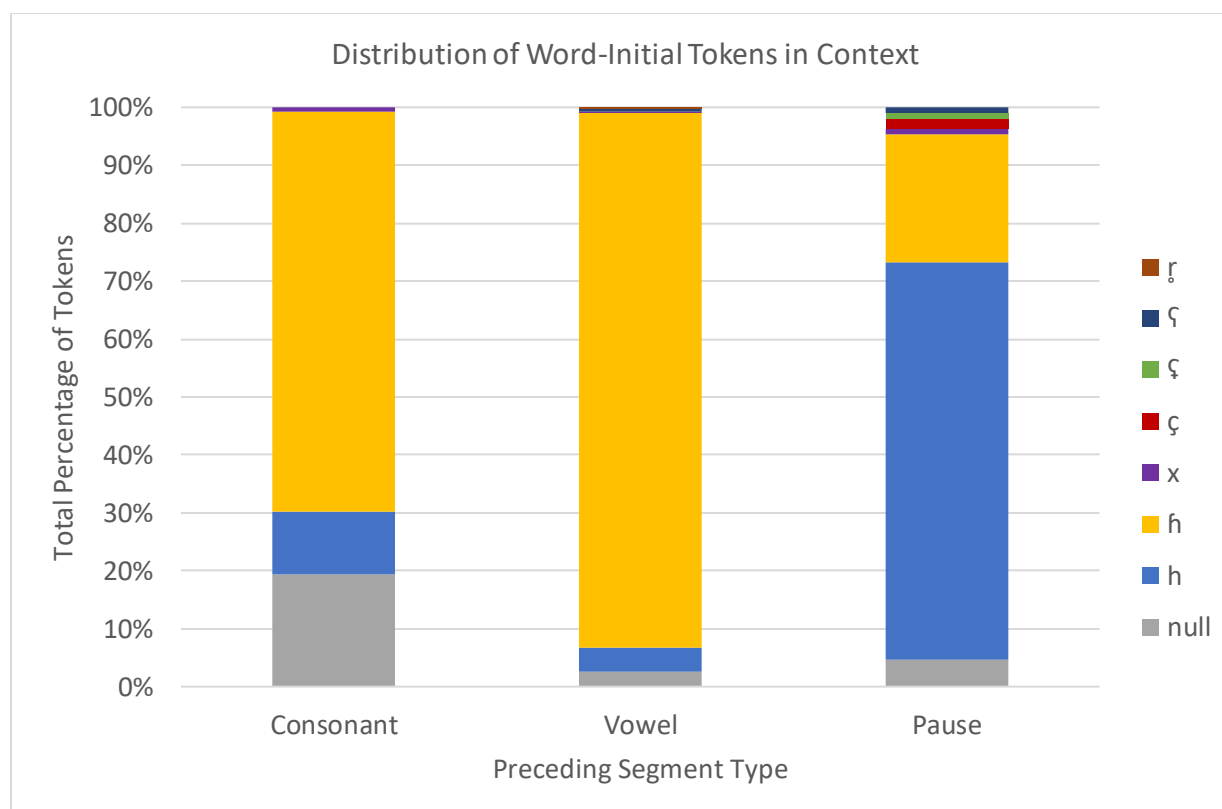


Figure 6.5: Proportion of surface forms of the word-initial rhotic by preceding segment type

In table 6.10 and figure 6.5, it can be seen that for both a consonant and a vowel preceding the word-initial rhotic, the predominant surface form is the voiced glottal fricative [h]. Meanwhile, when the rhotic is preceded by a pause (i.e. at the beginning of the sentence), the predominant surface form is voiceless [h̥]. As already discussed, [h̥] is also the preferred surface form in isolation for the participants. This indicates that the word-initial rhotic tends to be voiceless when utterance-initial. Given this parallel with the word-final rhotic before pauses and in isolation, it appears that the rhotic at the beginning or end of an utterance tends to be voiceless and that otherwise, it tends to be voiced. In terms of deletion, post consonantal rhotics deleted about 19.3% of the time on average, post-vowel rhotics deleted about 2.5% of time, and post-pausal rhotics deleted about 4.8% of the time.

In terms of deletion rates in context by preceding environment, post-consonantal rhotics (e.g., *muitas regiões* “many regions”) delete more than post-vowel (e.g., *eu recebi* “I received”) ($\mu_d > 0$, $p = 2.77 \times 10^{-5}$, $t = 4.605$) and post-pausal rhotics (e.g., *##Rimas* “rhymes”) ($\mu_d > 0$, $p = 0.021$, $t = 2.107$). However, post-pausal rhotics do not delete significantly more than post-vowel rhotics ($\mu_d > 0$, $p = 0.749$, $t = 0.323$). Similarly, post-pausal and post-vowel rhotics as a combined group do not delete at a different rate than word-initial isolated tokens ($p = 0.297$, $t = 1.060$). Additionally, unlike the previous two environments, stressed tokens (e.g., *rato* “rat”) were found to delete less (2.1% vs. 16.5%) than unstressed tokens (e.g., *refeição* “meal”) ($\mu_d < 0$, $p = 1.55 \times 10^{-9}$, $t = -5.582$).

Among the 10 non-glottal word-initial productions, 3 were pharyngeal fricatives matching the place of the following vowel [a] and 2 were palatal fricatives also matching the place of the following vowel [i]. One of the pharyngeal fricatives was also preceded by [a]. With so few non-glottal productions here, it is hard to say much about them, but it appears that the following vowel may have more influence than the preceding segment on the place of articulation of non-glottal word-initial rhotics. Tables 6.11 and 6.12 below show the non-glottal surface forms of the rhotic for each preceding and following segment in the word-initial environment. Note that the total counts are not the same in each table since some word-initial tokens did not have a preceding segment (those in isolation and those following a pause).

Table 6.11: Preceding segments and non-glottal surface forms of the rhotic in the word-initial environment.

Preceding segment	Surface forms		
	x	ç	ʁ
a	0	1	1
o	1	0	0
s	1	0	0

Table 6.12: Following segments and non-glottal surface forms of the rhotic in the word-initial environment.

Following segment	Surface forms				
	ç	x	ʃ	ʒ	ɹ
a	0	2	3	0	0
e	0	0	0	0	1
o	0	1	0	0	0
i	2	0	0	1	0

Regarding voicing of the rhotic in word-initial position, all word-initial rhotics are followed by vowels, but they can be preceded by a vowel or the alveolar fricative /s/ in the corpus. Ultimately, 259 of the 273 (94.9%) of the word-initial rhotics following a vowel were voiced, while 96 of the 112 (85.7%) word-initial rhotics following /s/ were voiced. Thus, there is a slight decrease in the proportion of voiced tokens after /s/ compared to after a vowel. This is statistically significant using a Fisher's exact test ($p = 0.005^*$, **odds ratio = 3.073**). Thus, it is likely that the voicelessness of the /s/ is causing the rhotic to surface as a voiceless segment more frequently, though the effect is weaker than that of being utterance-initial.

As a whole, the word-initial environment is certainly undergoing lenition as well, but it is to an even lesser extent than the syllable-final environment. While there were some cases of deletion, the glottal fricatives were the predominant form of lenition in this case. Additionally, with only ten non-glottal productions, other surface forms were not very common in this environment. Within the glottal fricatives, there is a preference for voiceless [h] when in isolation or preceded by a pause (i.e. the beginning of an utterance). Meanwhile, it typically surfaces as voiced [ɦ] in all other situations. This factor appears to be the main determinator in the voicing of the rhotic, rather than the voicing of an adjacent segment. Overall, the word-

initial rhotic shows its own pattern of lenition while also showing a voicing pattern which relates to the voicing pattern found in the word-final environment.

6.1.4. Intervocalic productions and deletions

The final environment to review is the intervocalic environment. Table 6.13 gives the count of each surface form of the rhotic for intervocalic tokens in isolation and in context, with the final column combining both isolation and context tokens. Examples of intervocalic tokens include *terra* “land” and *carro* “car.”

Surface form	Isolation	Context	Overall
null	0	42	42
h	14	13	27
ɦ	160	355	515
ʁ	1	1	2
x	0	1	1
ɣ	0	1	1
ç	0	2	2
j	0	3	3
ħ	0	1	1
ʕ	0	1	1
Total	175	420	595

Table 6.13 Counts of the surface forms of the rhotic in the intervocalic environment

In Table 6.13, it can be seen that deletion only occurred on tokens in context. Among them, 10% (42/420) were deleted. Given the lack of deletion in isolation, it does not take much for the deletion rate in context to be significantly higher, and accordingly, there is a significant difference ($\mu_d > 0, p = 2.87 \times 10^{-11}, t = 9.387$). When the productions are considered, the vast majority of forms are glottal and most of the glottal fricatives are voiced as well. In total there are only 11 non-glottal productions. In terms of unusual productions, there are two pharyngeal productions, one voiceless and one voiced. Interestingly, both of these occurred with the token

amarrado “tied.” The proportions of the surface forms in the intervocalic environment are given

Figure 6.6.

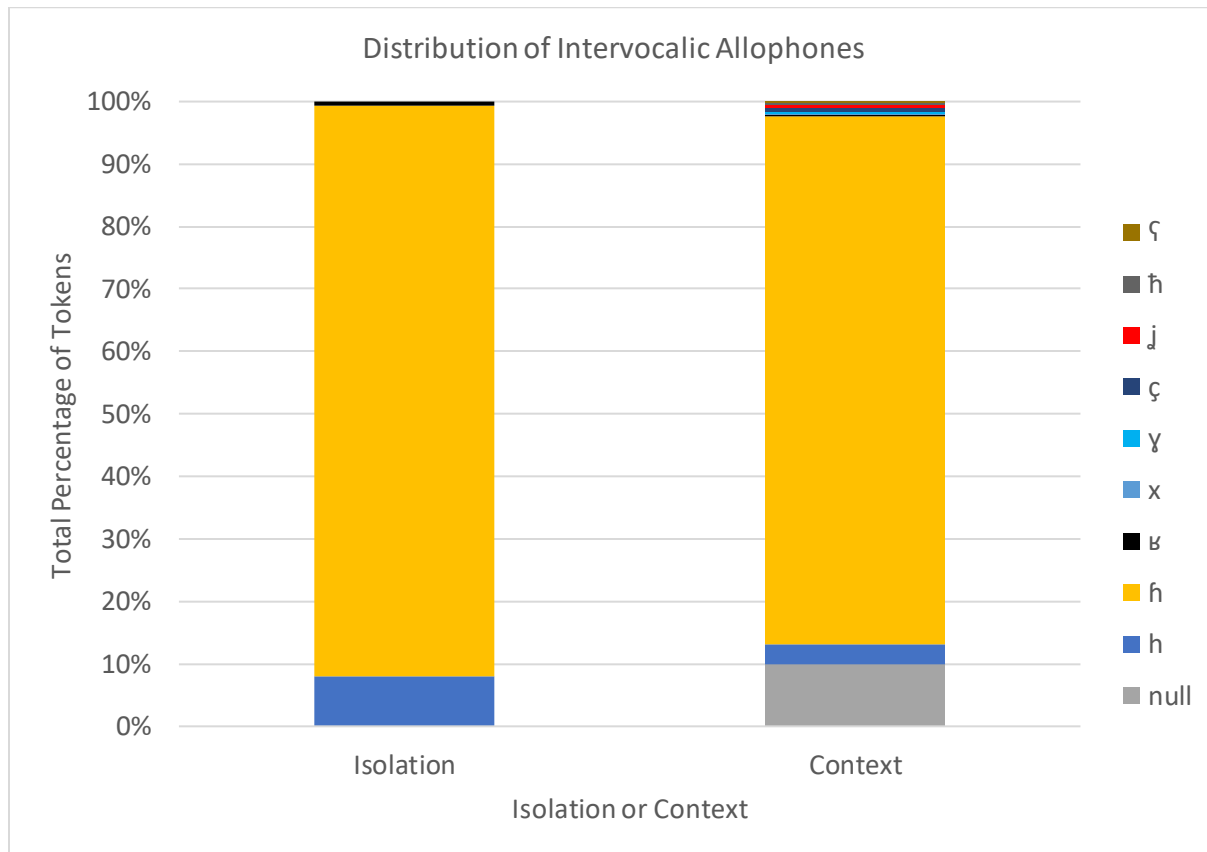


Figure 6.6: Proportion of surface forms of the rhotic in intervocalic environment

In Figure 6.6, it can be seen that the voiced glottal fricative [h] predominates both in isolation and in context. There are some instances of [h], with very few other productions. Deletion is certainly noticeable but is relatively minor, as was the case with word-initial tokens. The fact that the voiced glottal fricative is predominating in this position is not surprising, given that it is surrounded by vowels (voiced segments). With only 11 non-glottal productions, there is not much to say on the place of articulation of the rhotic versus the surrounding vowels. Among the 11 non-glottal productions, 2 were the aforementioned pharyngeal fricatives matching the surrounding vowels [a], all 5 palatal productions were followed by palatal vowels [i] or [e] and 2

of these were also preceded by [i], and 1 velar production matched a preceding [u] while 1 velar production matched a following [u]. The remaining 2 non-glottal productions were uvular and did not match an adjacent vowel in place. Since intervocalic tokens are surrounded by vowels, it is not possible to check for any devoicing effect caused by adjacent voiceless segments. Tables 6.14 and 6.15 below show the non-glottal surface forms of the rhotic for each preceding and following segment in the intervocalic environment.

Table 6.14: Preceding segments and non-glottal surface forms of the rhotic in the intervocalic environment.

Preceding segment	Surface forms						
	ç	j	x	ɣ	ʁ	ʕ	ħ
a	1	0	1	0	2	1	1
o	0	2	0	0	0	0	0
u	0	0	0	1	0	0	0
i	1	1	0	0	0	0	0

Table 6.15: Following segments and non-glottal surface forms of the rhotic in the intervocalic environment.

Following segment	Surface forms						
	ç	j	x	ɣ	ʁ	ʕ	ħ
a	0	0	0	0	0	1	1
u	0	0	1	0	2	0	0
e	0	2	0	0	0	0	0
i	2	1	0	1	0	0	0

Regarding the effect of stress on deletion rate, there is a more similar situation to that of the word-initial environment. When the stress is located after the rhotic (this is equivalent to a stressed syllable, e.g., *currículo* “curriculum”), the rhotic is less likely to delete than when stress is not located after the rhotic (equivalent to an unstressed syllable, e.g., *terra* “land”) (2.0% vs. 21.1%) ($\mu_d < 0$, $p = 6.91 \times 10^{-9*}$, $t = -7.405$). This test was only conducted using context tokens, since no intervocalic isolation tokens were deleted.

Overall, the lenition situation intervocalically is similar to that occurring word-initially. There is some deletion occurring but, glottal fricatives are the much more common form of lenition. Other surface forms are not common in this environment either, though they do occur. Being surrounded by vowels, it is not surprising that the rhotic in this context is overwhelmingly voiced. As a whole, after reviewing the other environments first, the intervocalic environment does not come with many surprises. However, this lack of surprises helps strengthen and reinforce the understanding and findings regarding the rhotic in the other environments.

6.1.5. Comparing allophones and deletions by environment

Lastly, before moving on to the regression analysis in the next subsection, it is important to compare the distribution of allophones of the rhotic across all four environments (word-final = WF, syllable-final = SF, word-initial = WI, intervocalic = IV). This is shown in Figure 6.7.

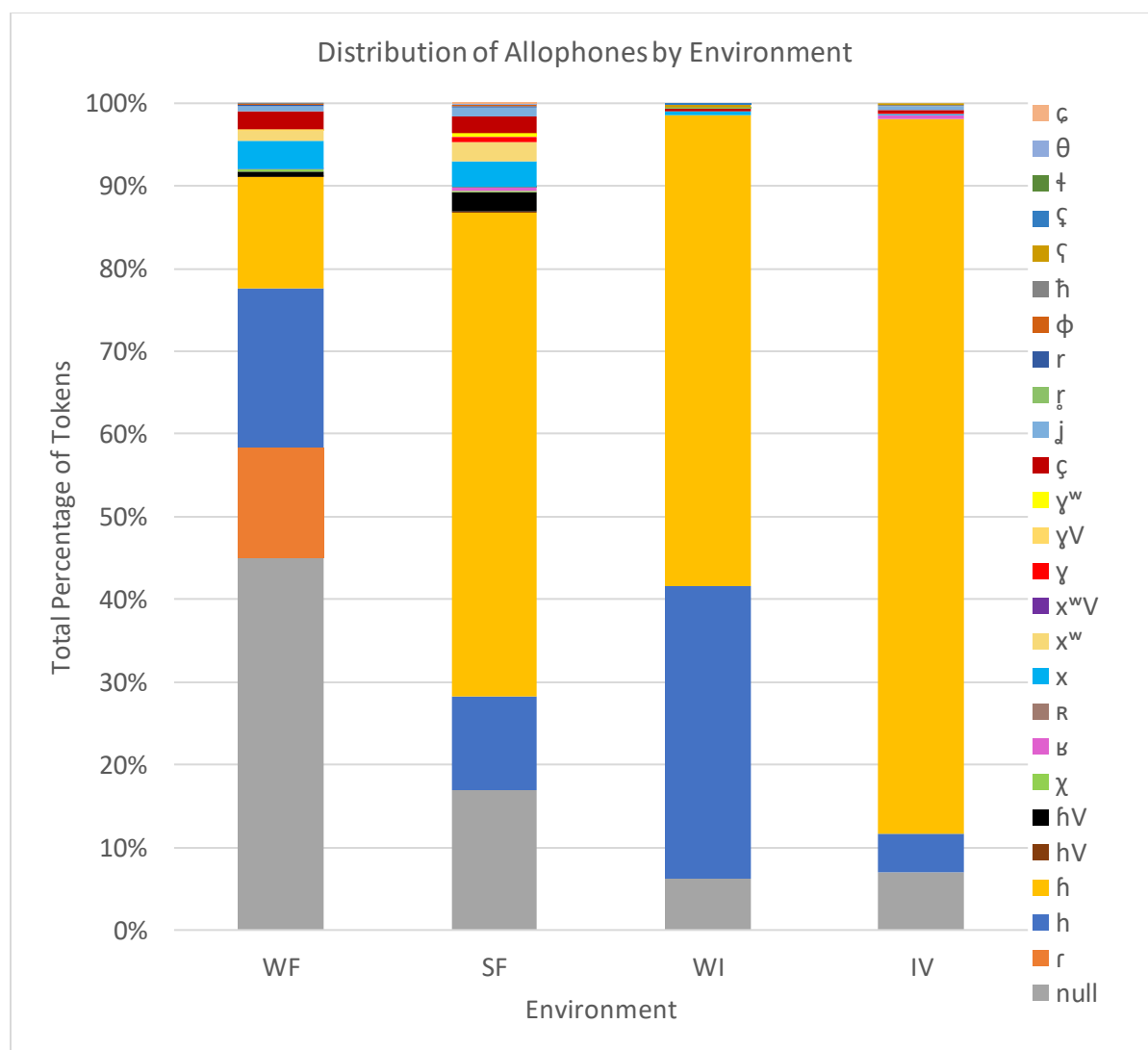


Figure 6.7: Proportion of surface forms of the rhotic by environment

Figure 6.7 demonstrates the tendency for deletion in the word-final environment and how much more common it is there than the other environments. Following word-final, syllable-final has the next highest deletion rate, then intervocalic, and lastly, word-initial. As previously mentioned, the alveolar flap [ɾ] is present only in the word-final environment. Beyond deletions and the flap, the glottal fricatives are the most common surface forms. Word-finally, [h] is more common than [h̥]. This tendency for voicelessness word-finally is largely driven by deletion and flapping instead of [h̥], as well as devoicing on the isolated tokens and

context tokens before a pause. Meanwhile, [h] is more common in the other environments. The remaining surface forms are the most common syllable-finally, at about 15%, while they comprise about 10% of word-final tokens, and 5% or less of word-initial and intervocalic tokens. As a whole, seeing these distributions side by side demonstrates that there are differences by environment, but they have more in common than not.

The remaining *t*-tests involve either all environments pooled together to examine the significance of a particular factor on deletion as a whole, or they compare some element of two different environments in an attempt to establish a hierarchy of deletion rates across the corpus. The finalized hierarchy will be presented in chapter 7.

Regarding stress across all environments, the result is the opposite of what the literature predicted: stressed tokens are not less likely to delete than unstressed tokens (37.1% vs. 8.5%) ($\mu_d < 0$, $p > 0.999$, $t = 12.410$). With stress only being significant in two environments (significant for word-initial and intervocalic) and the fact that tokens with the rhotic in coda position comprise most of the corpus (3636/4965), this is not surprising, but it is good to verify with the test. If the directionality were switched with this test, this would be a significant result indicating that stressed tokens delete more than unstressed tokens. The second test pooling together all environments considers whether the following segment is a consonant or not. Within the consonant category would be all syllable-final tokens and word-final tokens followed by a consonant. All other tokens fall into the non-consonant category. The result of this test is that rhotics which are followed by consonants are significantly more likely to delete than those which are not followed by consonants (38.5% vs. 27.6%) ($\mu_d > 0$, $p = 2.54 \times 10^{-6*}$, $t = 5.407$). This

test result confirms the significance of the higher deletion rates on syllable-final tokens and word-final tokens followed by consonants presented earlier.

All other *t*-tests in this chapter compare two different environments, with some of these being subsets of each environment. The first test is to see if all word-final categories delete more than syllable-final context tokens (20.4%). Since the category of word-final tokens with the lowest deletion rate was isolation tokens (22.7%), it is the category used in this test. The result of this test was not significant ($\mu_d > 0$, $p = 0.279$, $t = 0.591$), indicating that only the word-final context tokens delete more than syllable-final context tokens. Thus, word-final isolation tokens and syllable-final context tokens are on the same level in the hierarchy. Intervocalic context tokens with stress not on the syllable after the rhotic deleted at a similar rate as well (21%). This was not significantly different from either syllable-final context tokens ($p = 0.358$, $t = 0.931$) or word-final isolation tokens ($p = 0.933$, $t = 0.084$). Similarly, word-initial context tokens after a consonant were also deleted at a similar rate (19.3%) as these three sets of tokens. There was not a significant difference between the deletion rates for word-initial context tokens after a consonant and any of the three sets of tokens. The test with syllable-final context tokens had the lowest *p*-value ($p = 0.052$, $t = 2.018$). This tests with word-final isolation tokens ($p = 0.329$, $t = 0.991$) and intervocalic context tokens with stress not on the syllable after the rhotic ($p = 0.330$, $t = 0.989$) had similar *p*-values. Thus, word-final isolation tokens, syllable-final context tokens, intervocalic context tokens with stress not on the syllable after the rhotic, and word-initial context tokens after a consonant all are on the same level of the deletion hierarchy.

Since word-initial context tokens after a vowel or a pause (the remaining word-initial context tokens) were shown to delete less frequently than word-initial context tokens after a

consonant in subsection 3, the next step is to test these tokens for differences in deletion rate with the remaining sets of tokens (syllable-final isolation and intervocalic context tokens with post-rhotic stress). Post-pausal and post-vowel rhotics as a combined group do not delete at a different rate than syllable-final isolation tokens (4.3%) ($p = 0.846$, $t = 0.196$) or intervocalic context tokens with post-rhotic stress (2.0%) ($p = 0.103$, $t = 1.677$). Since post-pausal and post-vowel rhotics as a combined group were shown in subsection 3 to delete at the same rate as word-initial isolation tokens, these environments combined form the second-lowest spot in the deletion hierarchy, with intervocalic isolation tokens (which had no deleted rhotics) occupying the lowest spot. This hierarchy is presented in chapter 7.

6.2. Individual variability in deletion rate and its relationship to environments

Due to the fact that deletion is occurring across environments and speakers appear to be demonstrating a general trend of deleting word-final rhotics most, followed by syllable-final, and lastly, word-initial and intervocalic, it is worth checking if a correlation exists between the deletion rates for these environments. In other words, given a participant's deletion rate for one environment, how accurately can one predict the same person's deletion rate in another environment? This subsection explores this question. The regression tests here were conducted on the arcsine transformed deletion rates for each set of tokens used in the test. The results of the regression tests are integrated into the paragraphs following each figure. Note that all the regression tests in this section have 33 degrees of freedom and that the significance threshold is $\alpha = .05$. Significant tests are noted with an asterisk (*) and bolded.

In order to ensure that graphs are intuitive, all figures in this section display the untransformed deletion rates. However, the curve on each graph is the arcsine transformed regression line back transformed to fit the original data.

Since only 12 deletions occurred on syllable-final isolated tokens and only 6 deletions occurred on word-initial tokens, these sets of tokens are excluded from the regression analyses, as having a value of 0 for the majority of the participants for a variable in the regression weakens the regression's ability to be meaningful. Likewise, there were no deletions on intervocalic isolated tokens, meaning that it is not possible to do regression analyses with this set of tokens. However, given that the word-final isolation tokens were deleted about 22.7% of the time, this is sufficient to work with for a regression. Because of this, figure 6.8 below gives the scatterplot and regression curve for the regression between word-final (WF) context tokens and word-final isolation tokens.

WF Context vs. WF Isolation Deletion Rates

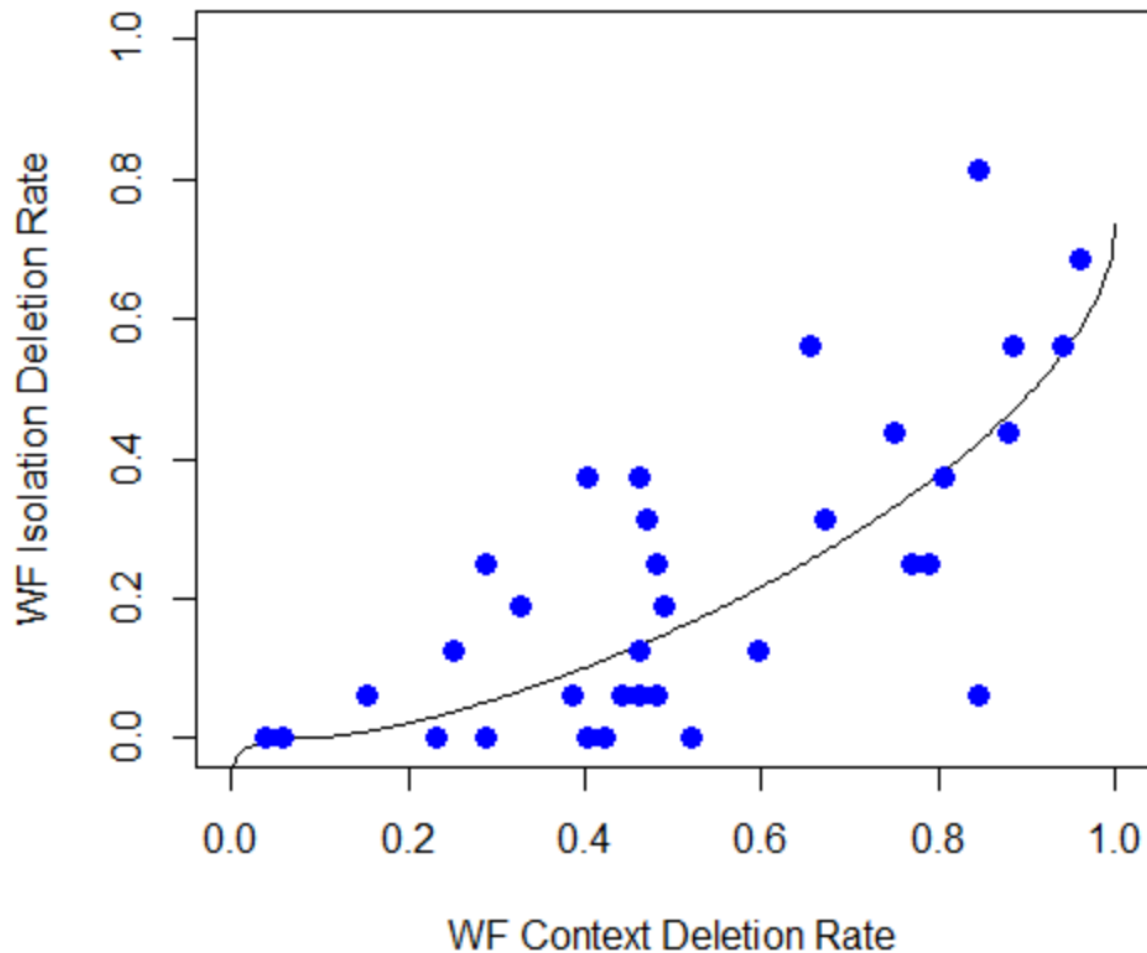


Figure 6.8: Scatterplot and regression curve between word-final (WF) context and word-final isolation tokens.

The regression shown in figure 6.8 yielded a significant result with a strong, positive correlation between word-final isolation and context deletion rates. ($r = 0.732$, $r^2 = 0.536$, $p = 5.71 \times 10^{-7}$, $t = 6.179$). Thus, around 53.6% of the variation in the word-final isolation deletion rate can be accounted for by the variation in the word-final context deletion rate. Given the amount that the word-final environment is driving deletion in the corpus, it makes sense that

those who delete in context more frequently are also going to delete more frequently in isolation. Next, the regressions between different environments (context tokens only) will be shown.

Beginning with regression between the word-final and syllable final-environments (correlation between tokens like *canta*_r “to sing” and *porta* “door”), the following results are presented in Figure 6.9:

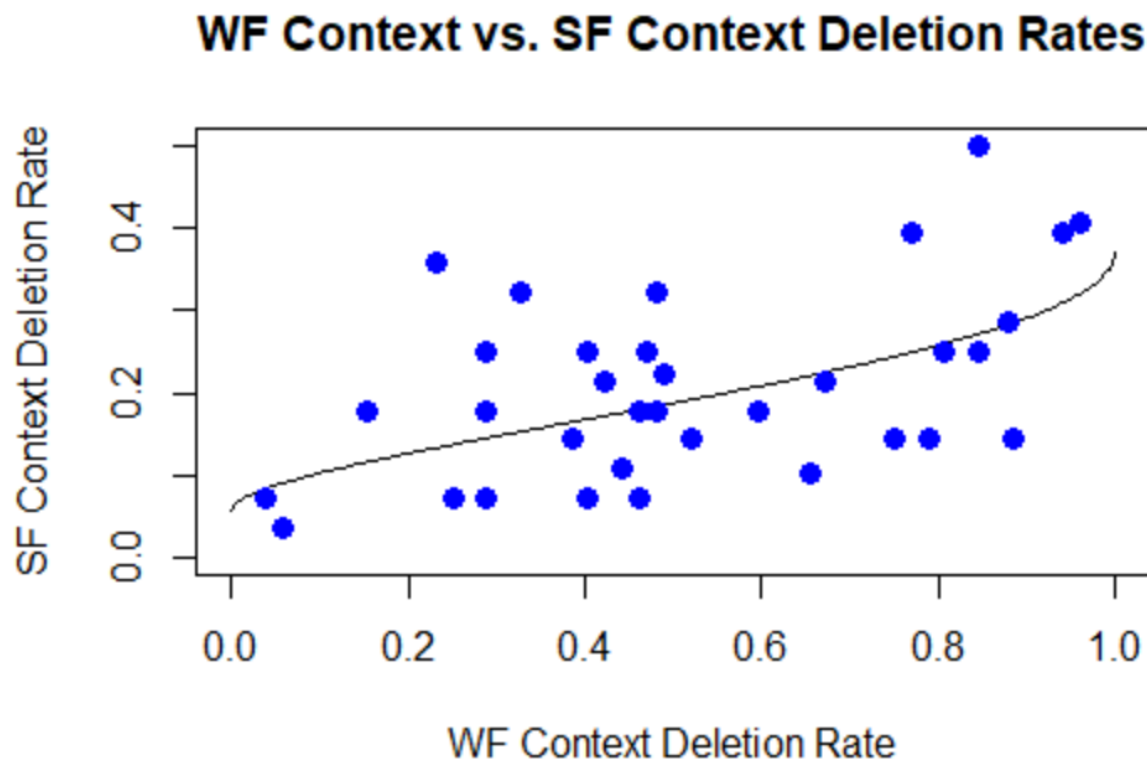


Figure 6.9: Scatterplot and regression curve between word-final (WF) and syllable-final (SF) context environments.

The regression shown in Figure 6.9 yielded a significant result, with a moderate, positive correlation between word-final and syllable-final deletion rates ($r = 0.527$, $r^2 = 0.278$, $p = 0.001^*$, $t = 3.563$). Thus, about 27.8% of the variation in syllable-final deletion rates can be

explained by the variation in word-final deletion rates. This is a fair indicator; however, given the parallels between word-final tokens followed by consonants and syllable-final tokens, if these two factors are compared, how can the prediction of syllable-final deletions be improved?

Next, Figure 6.10 gives the results of a regression test between word-final tokens followed by a consonant and syllable-final context tokens (correlation between tokens like *por**g**ato* “per cat” and *por**t**a* “door”).

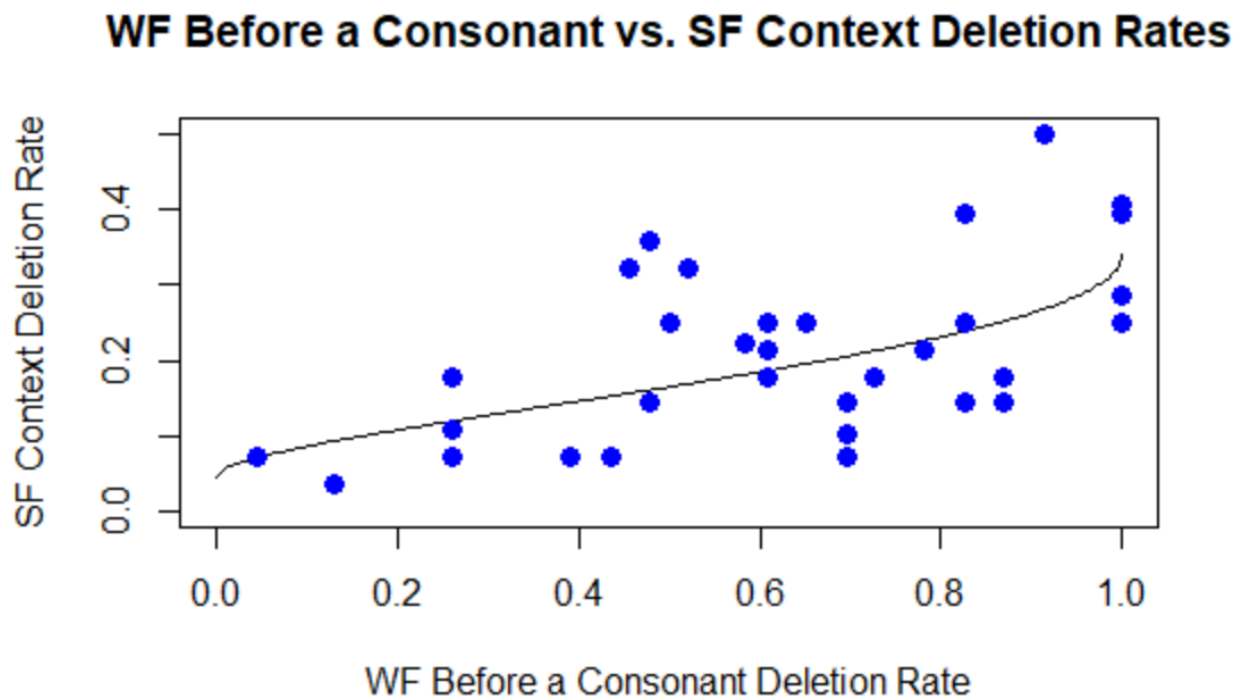


Figure 6.10: Scatterplot and regression line between word-final before a consonant and syllable-final context environments.

The regression shown in Figure 6.10 yielded a significant result, with a slightly stronger correlation than the regression shown in figure 6.9 ($r = 0.603$, $r^2 = 0.364$, $p = 1.24 \times 10^{-4*}$, $t = 4.347$). Thus, about 36.4% of the variation in syllable-final deletion rates can be explained by the variation in word-final before a consonant deletion rates, an improvement of about 8.6%. Thus,

this is a better way to predict syllable-final deletion than word-final context tokens as a whole.

From the results of these two regression tests, it is clear that there is a relationship between word-final deletion rates and syllable-final deletion rates.

With regard to the possibility of relationships between word-final deletion and deletion in the remaining two environments, Figure 6.11 presents the results of a regression test between word-final context deletion and word-initial context deletion (correlation between tokens like *canta*r “to sing” and *ra*to “rat”).

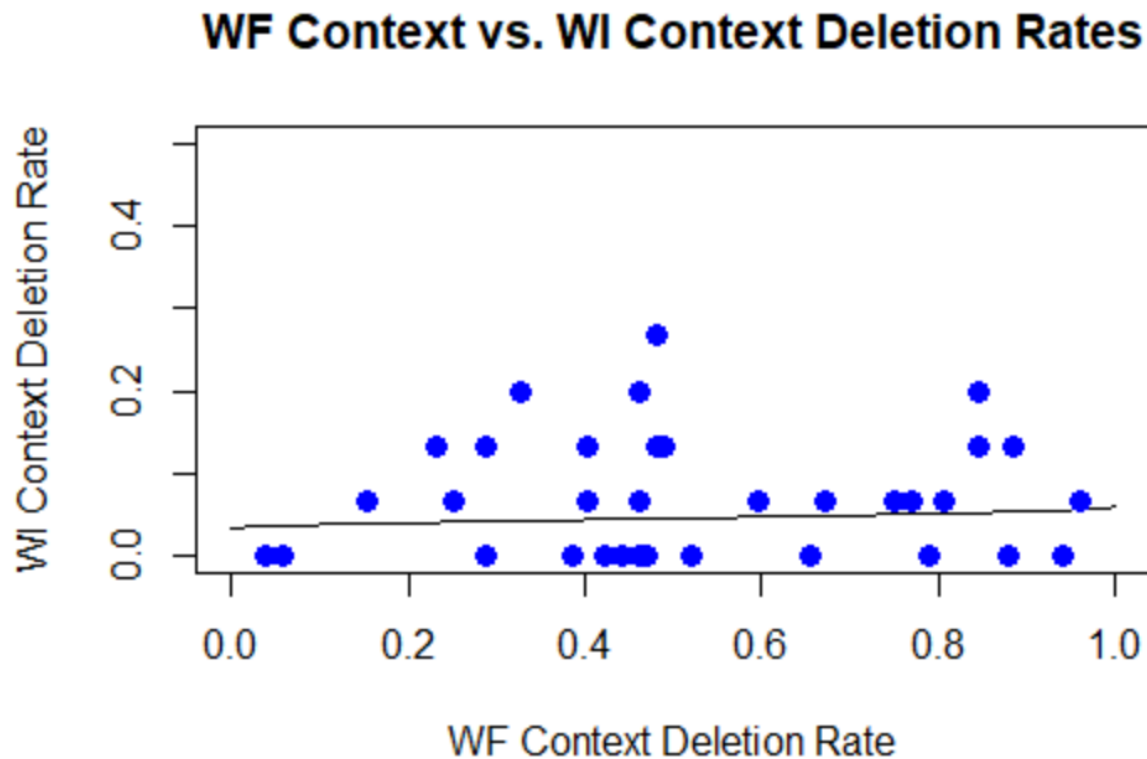


Figure 6.11: Scatterplot and regression line between word-final and word-initial (WI) context environments.

The regression shown in figure 6.11 did not yield a significant result ($r = -0.067$, $r^2 = 0.005$, $p = 0.700$, $t = 0.388$), suggesting that there is not a relationship between word-final and

word-initial deletion rates. This is possibly driven by the fact that deletion word-initially is far less frequent than word-finally or syllable-finally. Additionally, there is the fact that the word-initial environment is an onset environment, while the word-final environment is a coda environment. The previous regressions so far in this section compared coda environments with each other. Thus, there could be a connection between coda environment deletion rates that does not appear to carry over to onset environment deletion rates.

Next, the regression test between the word-final and intervocalic context tokens (correlation between tokens like *cantar* “to sing” and *carro* “car”) is presented in Figure 6.12.

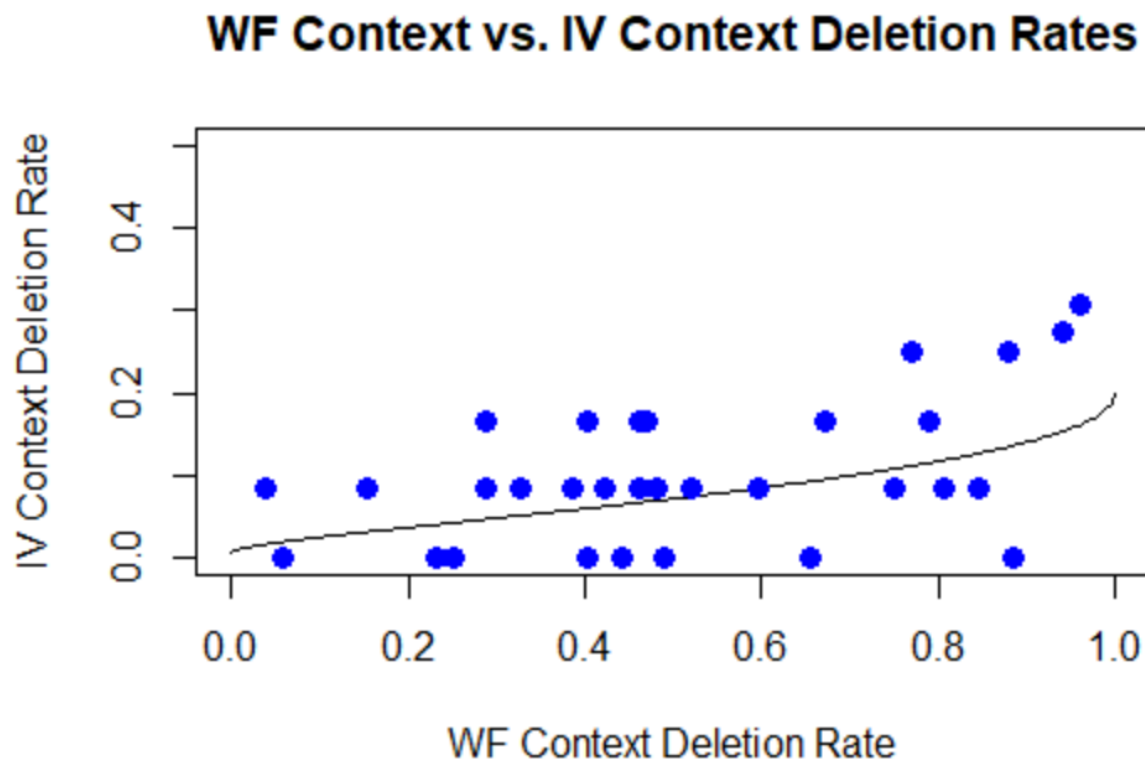


Figure 6.12: Scatterplot and regression line between word-final and intervocalic (IV) context environments.

The regression shown in Figure 6.12 yielded a significant result, with a moderate, positive correlation between word-final and intervocalic deletion rates ($r = 0.405$, $r^2 = 0.164$, $p = 0.016^*$, $t = 2.545$). Thus, about 16.4% of the variation in intervocalic deletion rates can be explained by the variation in word-final deletion rates. This correlation is weaker than what was found between the coda environments, but it does indicate that word-final deletion rates can be linked to more than just coda environments.

The remaining regression tests do not involve the word-final environment. Figure 6.13 demonstrates the correlation between the syllable-final and word-initial context environments (correlation between tokens like *porta* “door” and *rato* “rat”).

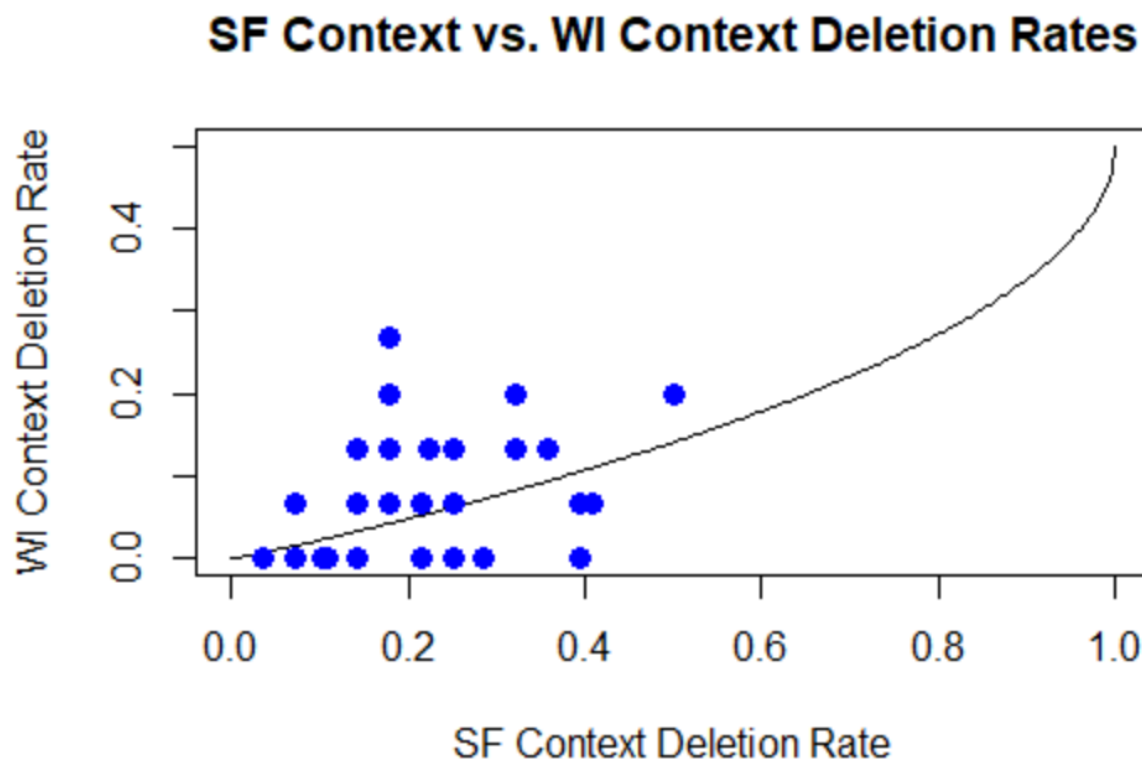


Figure 6.13: Scatterplot and regression line between syllable-final and word-initial context environments.

The regression shown in Figure 6.13 yielded a significant result, with a moderate, positive correlation between word-final and intervocalic deletion rates ($r = 0.407$, $r^2 = 0.165$, $p = 0.015^*$, $t = 2.556$). Thus, about 16.5% of the variation in word-initial deletion rates can be explained by the variation in syllable-final deletion rates.

Next, the regression between the syllable-final and intervocalic context environments (correlation between tokens like *porta* “door” and *carro* “car”) is presented in Figure 6.14.

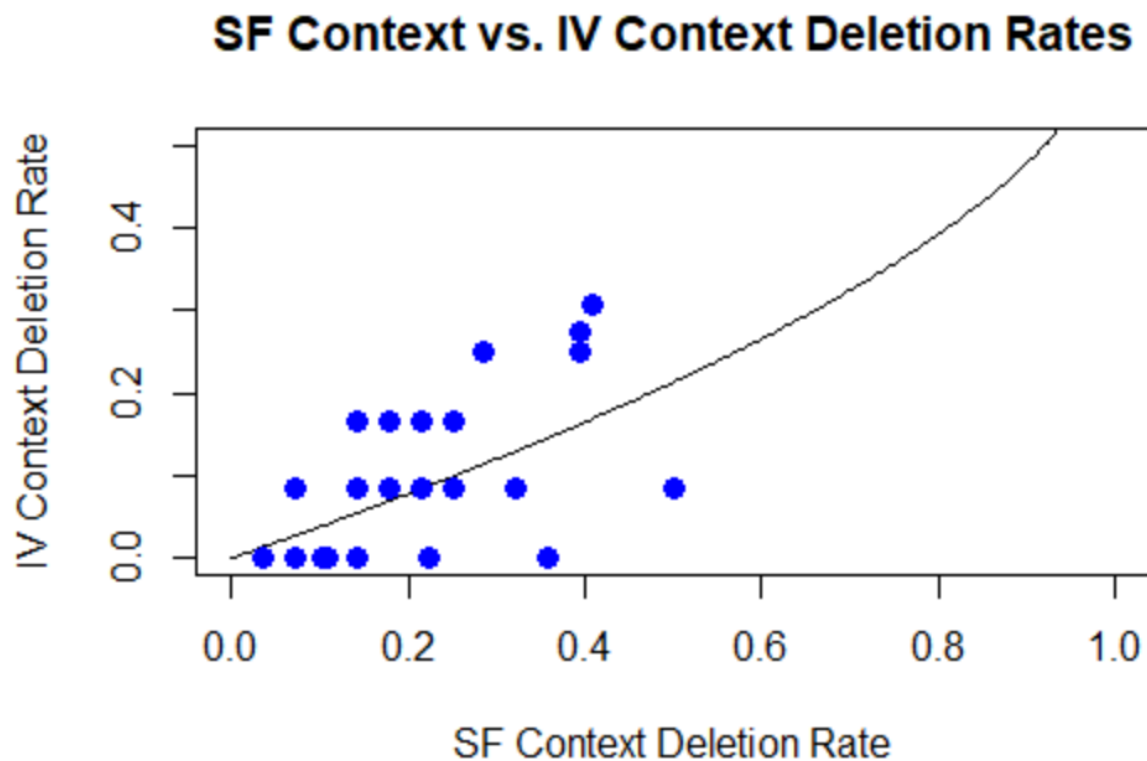


Figure 6.14: Scatterplot and regression line between syllable-final and intervocalic environments.

The regression shown in Figure 6.14 yielded a significant result, with a moderate, positive correlation between word-final and intervocalic deletion rates ($r = 0.508$, $r^2 = 0.258$, $p =$

0.002*, $t = 3.387$). Thus, about 25.8% of the variation in intervocalic deletion rates can be explained by the variation in syllable-final deletion rates.

The final pair of environments to test for a significant correlation is word-initial and intervocalic context deletions (correlation between tokens like rato “rat” and carro “car”), presented in Figure 6.15.

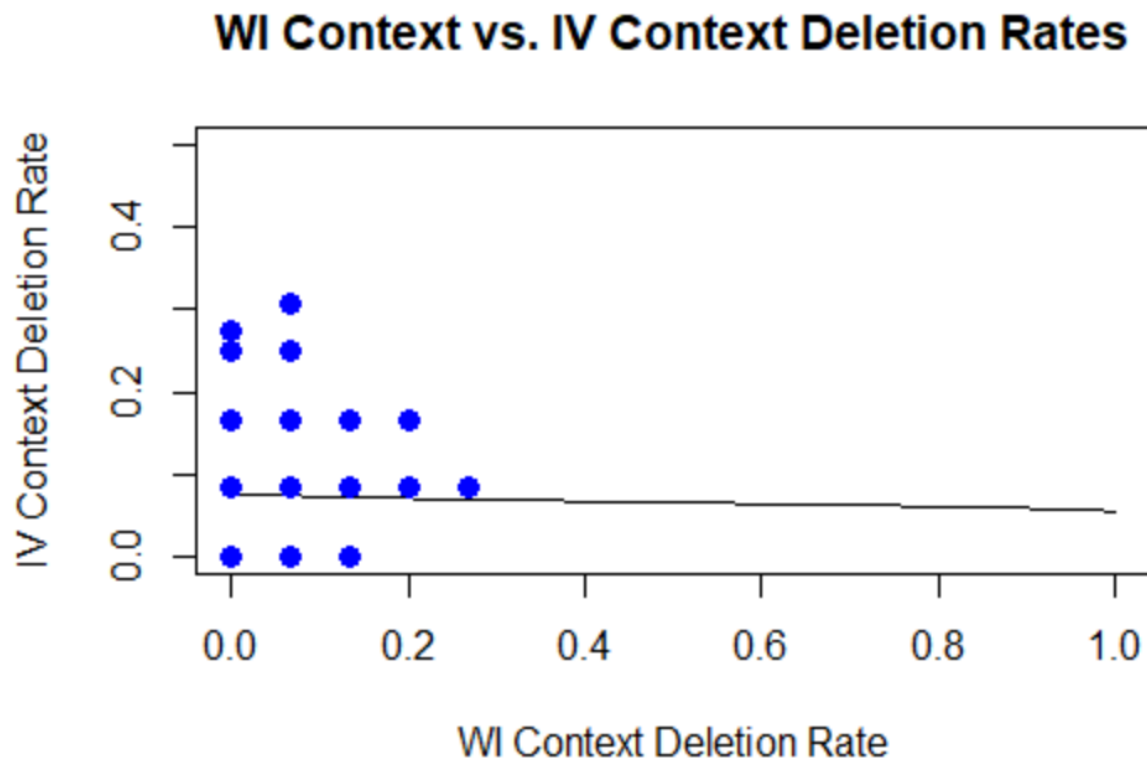


Figure 6.15: Scatterplot and regression line between word-initial and intervocalic context environments.

The regression shown in figure 6.15 did not yield a significant result ($r = -0.032$, $r^2 = 0.001$, $p = 0.854$, $t = 0.186$), suggesting that there is not a relationship between word-initial and intervocalic deletion rates. This is possibly driven by the fact that deletion word-initially and

intervocally is far less frequent than word-finally or syllable-finally. However, it is difficult to be certain.

Overall, these regression tests have revealed that while some deletion rates correlate with one another, others do not show any significant correlation. Generally, the deletion rates did correlate with each other; however deletion on word-initial context tokens did not correlate with deletion on word-final or intervocalic context tokens. Word-initial context tokens did delete at the lowest rate of the four environments, suggesting that this likely played a role in the lack of significance of these tests. While the discussion of figure 6.11 suggested the possibility that deletion in coda position and deletion in onset position may not be linked to each other, the other three tests (word-final vs. intervocalic, syllable-final vs. intervocalic, syllable-final vs. word-initial) making such a comparison were all significant. Thus, it is more likely that the low deletion rate was causing this correlation not to be significant than the fact it was between a coda environment and an onset environment.

As a whole, these tests point to the notion that deletion of the rhotic phoneme is more of a widespread phenomenon than one specific to each environment. However, given that most of the significant correlations were only moderate and had r^2 values well under 50%, the fact that one participant deletes more than another participant in a certain environment does not guarantee that this participant will also delete more in another environment.

7. Discussion of data and statistical test results from Salvador

The counts of the various surface forms and deletions within the data demonstrate just how variable the rhotic phoneme can be in the Salvador variety of Brazilian Portuguese. Nearly every possible surface form reported in chapter 2 was found in the corpus, with [ɹ] being the only exception. Furthermore, there were additional surface forms which were not cited in previous research. Despite not being cited as a surface form in the previous research, it is suspected that the palatal fricatives found in the corpus ([ç], [j]) are not so unusual. Both occurred frequently in the data (84 for [ç] and 29 for [j]) and they can fall under the generalization of a “back fricative” that Rennie (2015) stated was expected in Bahia. Similarly, the rounded variants of the uvular fricatives can be classified this way. The remaining surface forms that were not previously cited are unusual in the corpus ([ʈ], [θ], [ɕ], [ɸ], [ħ], [ʕ], and [ʕ̣]), being present only in low numbers. [ħ], [ʕ], and [ʕ̣] would qualify as back fricatives, and [ɸ], [ħ], and [ʕ] all exhibited place matching with adjacent segments, suggesting that perhaps these surface forms can be considered typical for the rhotic despite their low frequencies, though they are not included in the following discussion as part of the inventory of allophones in Salvador. On the other hand, [ʈ], [θ], and [ɕ] each only occurred once and are more likely to be misarticulations than rare but typical surface forms of the rhotic. Finally, although [r] and [ɹ] are cited as typical productions of the phoneme in previous research, the fact that only one participant produced these forms and the fact that the existing literature does not suggest that the rhotic would surface as alveolar trills in Salvador indicate that these productions can be considered exceptional, rather than typical.

It is also worth noting that rhotics are generally considered to be sonorants and as such, would not be expected to be voiceless, yet there were several voiceless productions in the corpus. While there is no directly provable way to show why this is the case, it is possible that the lenition of the rhotic phoneme which resulted in it being produced primarily as fricatives opened the door to this possibility. Since fricatives are not sonorants, this may cause speakers to devoice the rhotic phoneme. On a related note, it was shown in chapter 6 that devoicing tends to occur utterance-initially and utterance-finally, suggesting that transitioning in or out of an utterance plays a large role in whether the phoneme is devoiced. Similarly, the phoneme has higher likelihood of devoicing when adjacent to a voiceless consonant, although most productions are still voiced in this environment. Thus, the main factors behind voiceless surface forms of the rhotic are possibly linked to the fact that the surface typically surfaces as an obstruent and articulatory effects. For the remaining surface forms that are voiceless but do not fall into the aforementioned categories (e.g., intervocalic rhotics), the only factor that seems to make this possible is the fact that the rhotic typically surfaces as a fricative. Other than this, there is no clear indicator as to how a typically sonorant sound, a rhotic, can have voiceless surface forms.

Based on the results of the t-tests, a deletion hierarchy has been developed. When more than one category is on the same line, this indicates that there was not a significant difference in deletion rate between the categories. The hierarchy, starting with the highest deletion rate, is as follows:

1. Word-final preceding a consonant
2. Word-final preceding a vowel
3. Word-final preceding a pause
4. Word-final isolation, syllable final context, word-initial following a consonant, intervocalic context with non-post-rhotic stress
5. Intervocalic context with post-rhotic stress, word-initial following a vowel or a pause, syllable-final isolation, word-initial isolation
6. Intervocalic isolation

Thus, it is clear that word-final context tokens, regardless of what they precede, delete the most. Among these, word-final tokens before a consonant deleted at the highest rate, followed by those before a vowel, and finally, those before a pause. Next in the hierarchy is a variety of context tokens and word-final isolated tokens, which all deleted around 20% of the time. The next level is for any other set of tokens in which deletion occurred. These sets have deletion rates varying from about 2% to 5%. Lastly, intervocalic isolation tokens, which were the only tokens to have no deletions, are at the bottom of the hierarchy.

Notably, both word-final tokens before a consonant and word-initial tokens after a consonant deleted at the highest rate among all subsets of their respective environments. This suggests that a tendency for CV syllable structure, which was listed as a possible influence on deletion in chapter 2, is likely a factor affecting deletion in the corpus. Furthermore, coda environments tend to delete at a higher rate than onset environments, suggesting that there is a preference to delete codas over onsets, indicating a tendency for CV syllables rather than VC syllables.

It is also the case that isolated tokens are generally in a lower position in the deletion hierarchy than context tokens in the same environment. The sole exception to this is that word-initial isolated tokens are on the same level as word-initial context tokens following a vowel or a pause. However, word-initial isolated tokens still delete less than word-initial context tokens following a consonant. Thus, tokens in context can generally be considered to be deleted more frequently than tokens in isolation.

While it is true that rhotics in coda position deleted more than those in onset position, the hierarchy indicates that the coda factor can be split into two factors: being word-final and preceding a consonant. Moreover, the similar deletion rate between syllable-final context tokens and word-initial tokens following a consonant can be accounted for by a factor of being adjacent to a consonant. Thus, both the factors of preceding a consonant and following a consonant can be combined into a single “consonant-adjacent” category. With this, the factors that lead to a higher likelihood of deletion are word-final position, being in context rather than in isolation, being consonant-adjacent, and for intervocalic tokens, not having post-rhotic stress. Based on the fact that all non-word-final token types do not delete significantly more than any word-final token type, it can be presumed that the factor of word-final position is stronger than the other factors. Meanwhile, the other factors appear to have about the same effect on the deletion rate of the rhotic phoneme. Keeping this in mind, if a weight of 2 is given to the word-final factor and a weight of 1 is given to the other factors, the result is table 7.1 below.

Table 7.1: Deletion hierarchy and the factors affecting deletion.

Level in hierarchy	Token type	Word-final	Context	Consonant-adjacent	Intervocalic non-post-rhotic stress	Total weight
1	Word-final preceding a consonant	2	1	1		4
2	Word-final preceding a vowel	2	1			3
3	Word-final preceding a pause	2	1			3
4	Word-final isolation	2				2
	Syllable-final context		1	1		2
	Word-initial following a consonant		1	1		2
	Intervocalic context with non-post-rhotic stress		1		1	2
5	Intervocalic context with post-rhotic stress		1			1
	Word-initial following a vowel or a pause		1			1
	Syllable-final isolation			1		1
	Word-initial isolation					0
6	Intervocalic isolation					0

From table 7.1, it is clear that the total weight of the factors affecting deletion has a strong influence on where specific token types are in the deletion hierarchy. Nevertheless, these

factors cannot explain why word-final tokens preceding a vowel and word-final tokens preceding a pause are on different levels of the hierarchy, or why word-initial isolation tokens and intervocalic isolation tokens are not on the same level. Notably, word-final tokens before a pause have no segments following them, as is the case for word-final isolation tokens. This may be a driver behind the lower deletion rate for word-final tokens preceding a pause when compared to word-final tokens preceding a vowel. However, no such parallel exists to potentially explain why word-initial isolation tokens delete more than intervocalic isolation tokens, despite both token types having a total weight of 0. Furthermore, word-initial isolation rhotics are pause-adjacent, yet they are deleting more than intervocalic isolation rhotics which are surrounded by vowels. In the end, although not perfect, these factors generally do a good job of explaining the deletion rates in the data and why various unrelated tokens types can end up on the same level in the deletion hierarchy.

The foregoing discussion of the results permits the following answers to the first set of research questions posed in chapter 4:

1. What is the inventory of allophones of the non-flap rhotic in the Salvador variety?

The inventory of allophones in the Salvador variety consists of the voiceless and voiced fricatives in the palatal, velar (with possible labialization), uvular, and glottal places of articulation, as well as the uvular trill. If the rhotic is word-final followed by a vowel, it may also surface as an alveolar flap. This is summarized in Table 7.2:

Table 7.2: Inventory of allophones of the rhotic phoneme in the Salvador variety.

Place of articulation	Possible Productions
Alveolar	[r] (exclusively word-finally before vowels)
Palatal	[ç], [j]
Velar	[x ^(w)], [ɣ ^(w)]
Uvular	[χ], [ʁ], [R]
Glottal	[h], [ɦ]

2. At what rate does the Salvador variety delete word-final rhotics, both in isolation and in context?

As shown in Table 6.1 in chapter 6, 127 of 560 (22.7%) isolated tokens were deleted, and 945 of 1818 (52.0%) context tokens were deleted. This is reflective of a general trend in the data where isolated tokens tend to delete less than context tokens. It is likely that this difference is driven by the fact that isolated tokens present the speakers with an opportunity to fixate on their articulation of a word, as opposed to when the tokens are in context. The additional fixation results in a decreased deletion rate in isolation. Likewise, the tokens in context present an opportunity to fixate more on articulation than in an informal interview. Thus, it is quite likely that context tokens would have a lower deletion rate than tokens in an informal interview as well.

Within the word-final context tokens, it was determined that the following segment type had an effect on deletion rate, with those preceding a consonant deleting at the highest rate, followed by those preceding a vowel, and finally, those preceding a pause deleted at the lowest rate. The factors behind this are discussed along with the presentation of the deletion hierarchy above. In total, 484 of 800 (60.5%) word-final tokens before a consonant, 361 of 750 (48.1%)

before a vowel, and 100 of 268 before a pause (37.3%) were deleted. Given these numbers, the deletion rates for word-final tokens are estimated in Table 7.3. The estimations are based on the average deletion rates among the participants with a buffer space on each side.

Table 7.3: Estimated deletion rates for word-final tokens.

Type of Token	Estimated Deletion Rate
Word-final isolation	20-25%
Word-final context	50-55%
Word-final context before a consonant	58-63%
Word-final context before a vowel	45-50%
Word-final context before a pause	35-40%

Compared to previous research on other varieties, Reis (2015) found that speakers from Belo Horizonte (in the state of Minas Gerais in the southeast region) deleted the word-final rhotic 0.8% of the time in isolation and 9.2% of the time in context. Additionally, he found that 34.9% of tokens in informal interviews has the word-final rhotic deleted. Ultimately, the word-final deletion is much more common in Salvador than in Belo Horizonte, with the least common word-final context environment in Salvador (before a pause) being comparable in deletion rate to the informal interviews in Belo Horizonte. Meanwhile, Cardoso (1999) found that speakers from Belém (in the state of Pará in the north region) deleted the word-final rhotic 2% of the time in isolation and 25% of the time in context. In a task similar to the tokens in context here but with blanks on the target words to be filled in by the participants, Rennie (2015) found that 13.4% of word-final rhotics in speakers from the state of Minas Gerais were deleted. Thus, as hypothesized, participants from Salvador are deleting word-final rhotics in isolation and in

context at a higher rate than speakers of other varieties. This suggests a strong possibility that historical African influences are stronger in Salvador than in other Brazilian cities.

With regards to how these results compare to informal interviews, Serra and Callou (2013) found that speakers from Rio de Janeiro (southeast region) deleted the word-final rhotic in interviews about 76% of the time. Similarly, Cardoso (1999) found that word-final rhotic deletion occurred about 77% of the time in informal interviews. Rennie (2015) had an even higher deletion rate, finding that 81.1% of word-final rhotics in informal interviews were deleted, and Vieira (2021) found that 81% of word-final rhotics in interviews were deleted. While the informal interviews from Vieira (2021), Rennie (2015), Serra and Callou (2013), and Cardoso (1999) ultimately had higher deletion rates than any set of tokens in the corpus from Salvador, the isolation and context tokens from Cardoso (1999) and Reis (2015) are notably lower in deletion rate than the sample here from Salvador. Additionally, of all the cities in Callou et al. (1996), the speakers from Salvador deleted the most word-finally in interviews at 62%, compared with 50% for Recife (in the state of Pernambuco in the northeast region), 49% for São Paulo, 47% for Rio de Janeiro, and 37% for Porto Alegre (in the state of Rio Grande do Sul in the south region). Given these facts, it is reasonable to believe that Salvador could produce higher word-final deletion rates in informal interviews than speakers from other cities as well.

3. At what rate does the rhotic delete in other environments (syllable-final, word-initial, intervocalic) in the Salvador variety?

From the data presented in chapter 6, for the syllable-final environment, 12 of 219 (4.3%) isolated tokens were deleted and 200 of 979 (20.4%) context tokens were deleted. For word-initial rhotics, 6 of 210 (2.9%) isolated tokens were deleted and 39 of 524 (7.4%) in

context were deleted. There was a significant difference between word-initial context tokens preceded by a consonant (27 of 139 or 19.4% deleted) and word-initial context tokens preceded by a vowel or pause (12 of 385 or 3.1% deleted). Lastly, intervocalically, no tokens were deleted in isolation and 42 of 420 (10%) in context were deleted. There was also a significant difference by stress, with 2.0% (5 of 245) of tokens with post-rhotic stress deleted and 21% (37 of 175) of tokens without post-rhotic stress deleted. Given this information, along with the statistical tests for significance and the deletion hierarchy, Table 7.4 presents the estimations for the deletion rates. The estimations are based on the average deletion rates among the participants with a buffer space on each side. Although no intervocalic isolation tokens were deleted in this sample, the estimate accounts for the possibility that someone may delete in this circumstance.

Table 7.4: Estimated deletion rates for syllable-final, word-initial, and intervocalic environments.

Type of Token	Estimated Deletion Rate
Syllable-final isolation	2-6%
Syllable-final context	18-22%
Word-initial isolation	2-4%
Word-initial context	5-10%
Word-initial context after a consonant	18-22%
Word-initial context after a vowel	2-4%
Word-initial context after a pause	2-4%
Intervocalic isolation	0-1%
Intervocalic context	8-12%
Intervocalic context with post-rhotic stress	1-3%
Intervocalic context without post-rhotic stress	19-23%

Given the results in table 7.4, the hypothesis that word-final rhotics would delete more than the other environments was correct. However, the hypothesis that syllable-final rhotics would delete more than the other two positions was only partially correct. While the syllable-final isolated tokens did not delete more than word-initial isolated tokens, both deleted more than intervocalic isolated tokens, as none of those were deleted. With regards to the context tokens, the syllable-final context tokens deleted more than the broad categories of word-initial and intervocalic context tokens, but when these two groups were split up into their subsets, the word-initial tokens following a consonant and the intervocalic tokens without post-rhotic stress deleted at about the same rate as the syllable-final tokens. Thus, there is a general tendency

that syllable-final tokens will delete more than word-initial and intervocalic tokens, but certain factors can negate this difference.

4. When rhotics are not deleted in the Salvador variety, to what extent have they undergone lenition?

The vast majority of non-deleted rhotics in the Salvador variety have lenited to the point of surfacing as glottal fricatives. The sole environment where glottal fricatives do not dominate the surface forms of the rhotic is that of word-final tokens preceding a vowel. In this circumstance, the rhotic predominantly neutralizes with the alveolar flap phoneme. Thus, the hypothesis that the phoneme is undergoing high rates of lenition, and that this lenition would be mostly yielding glottal fricatives, followed by other fricatives, was correct.

5. To what extent is deletion in one environment related to deletion in another environment?

From the results of the regression tests, it is clear that deletion rates across environments tend to be associated with each other, though it is not always the case. Specifically, word-initial context tokens did not have a significant correlation with either word-final or intervocalic context tokens. The remaining correlations between context environments were moderate and had r^2 values well under 50%. However, the correlation between word-final isolation and word-final context tokens was strong, with an r^2 value of 0.536, indicating that participants who deleted frequently on word-final context tokens also deleted frequently on word-final isolation tokens. Therefore, while participants who delete more in one environment tend to delete more in another environment as well, it is far from guaranteed that this will be

the case. Consequently, the hypothesis that participants who delete more in one environment will also delete more in another environment turned out to be only partially correct.

Finally, the high degree of variation in the rhotic production and deletion of Salvador may be linked to the complex social and demographic history of the city. Given the focus on morphosyntactic phenomena in the existing Brazilian Portuguese contact literature, it is important to consider the potential phonetic implications of this contact history as well. This will be explored in the next chapter.

8. Demographic differences in Salvador

This chapter breaks down the differences in deletion rate, rate of oral surface forms, and distribution of oral surface forms by demographic variable. The demographic variables tested were race, age, gender, socioeconomic class, Afro-Brazilian identity, whether or not a participant has lived their whole life in Salvador, and data collection method (remote versus in-person). The environments tested for differences in deletion rate were word-final in isolation and in context; syllable-final, word-initial, and intervocalic in context; and overall deletion rate across all tokens. Syllable-final (12 deletions), word-initial (6 deletions), and intervocalic (0 deletions) isolation tokens were excluded due to a lack of enough deletions to give the tests meaning. Word-final context and word-initial context tokens were also subdivided into the three possible preceding and following segment type categories (vowel, consonant, and pause) and intervocalic context tokens were subdivided based on whether or not the syllable following the rhotic was stressed. However, these subsets are only considered if the test for all the tokens in a given environment is significant. Intervocalic isolation tokens were not considered for demographic differences, as no participants deleted any of them. The environments tested for differences in oral production rate and distribution of oral surface forms were the same as for deletion rate, except that they exclude the word-initial and intervocalic environments, as only 10 word-initial and 11 intervocalic productions in the corpus from Salvador were oral. The significance level for all the tests is $\alpha=0.05$. Significant p -values are indicated with an asterisk (*) and the significant tests are bolded in their tables.

This chapter is organized as follows: Section 1 reviews the differences in deletion rate by demographic variable. Section 2 examines the oral production forms by demographic variable.

Section 3 addresses the concern that demographic variables may be associated with one another and may confound the results of this chapter. Lastly, section 4 discusses the findings of this chapter.

8.1. Differences in deletion rate by demographic variables

All tests for differences in deletion rate by all demographic variables except age were two independent sample *t*-tests on the arcsine transformed deletion rates of individual participants split into their respective categories as the two samples. All *t*-tests were two-tailed unless otherwise indicated. To determine whether to use the equal variance (Student's *t*-test) or unequal variance (Welch's *t*-test) assumption for a particular test, an *F*-test was conducted. If the *F*-test gave a significant result ($p \leq 0.05$), the unequal variance assumption was used. Note that if no participants of a particular demographic group deleted the rhotic in a certain environment, the *F*-statistic could not be calculated due to a variance of zero for that group. In this case, the unequal variance assumption was used as well. Otherwise, the equal variance assumption was used. Tests which used the unequal variance assumption are indicated with footnotes that give the *p*-value of the *F*-test and the degrees of freedom of the *t*-test. To evaluate the connection between deletion rate and age, a regression test was conducted on the deletion rates for a given environment and the approximate birth year of the participants. The approximate birth year for a participant was calculated as the year of data collection minus the participant's age at the time of data collection.

The tests for socioeconomic class, age, whether or not a participant has lived their whole life in Salvador, and data collection method (remote versus in-person) contain all 35 participants. The tests for gender contain 34 participants, as they do not include the one

participant who identified as trans non-binary since one person is not sufficient to form a group. The tests for race contain 34 participants when *preto*, *pardo*, and white participants are considered, and 27 participants when only *preto* and *pardo* participants are considered. The indigenous participant was not included in the tests for differences by race. Finally, the tests for Afro-Brazilian identity contain the 21 participants who identified as *preto* and *pardo* and responded to the extra questions asked in summer 2023.

8.1.1. Differences in deletion rate by race

The first step in evaluating differences in deletion rate by race was to conduct *t*-tests to determine if any differences exist between the deletion rates of the *preto* and *pardo* participants. This was done to evaluate the validity of the claims discussed in chapter 3 that the census groups *preto* and *pardo* should be combined into one category, *negro*, and determine to what extent the groups can be combined into single group for analyses comparing the deletion rates of black and white participants.

The average deletion rates, *p*-values, and *t*-scores of the tests conducted are given in table 8.1 below. The *t*-scores given represent the *preto* mean minus the *pardo* mean for a given environment.

Table 8.1: Average deletion rates, p -values, and t -scores for the difference between the race categories of *preto* and *pardo* by environment.

Environment	<i>Preto</i> average	<i>Pardo</i> average	p -value	t -score
Word-final isolation	0.280	0.198	$p = 0.545$	$t = 0.614$
Word-final context	0.560	0.584	$p = 0.846$	$t = -0.196$
Syllable-final context	0.234	0.198	$p = 0.602$	$t = 0.529$
Word-initial context	0.076	0.089	$p = 0.494$	$t = -0.694$
Intervocalic context	0.119	0.083	$p = 0.216$	$t = 1.271$
Overall deletion rate	0.305	0.296	$p = 0.947$	$t = 0.067$

From the table above, it is clear that there are no significant differences in deletion rates between the *preto* and *pardo* participants in any environment. Because of this, these groups are pooled together in the following tests which compare the deletion rates of the black participants (*preto* + *pardo*) with those of the white participants.

Because of the existing claims in the literature discussed in chapter 2 that deletion was largely driven by African influences on Brazilian Portuguese, the tests for differences in deletion rate between black and white participants were one-tailed t -tests with the alternative hypothesis that the average deletion rate for black participants was higher than the average deletion rate for white participants ($\mu_{\text{black}} > \mu_{\text{white}}$). Table 8.2 below gives the results of these tests along with the average deletion rate for each group. The positive t -scores indicate that black participants deleted more in each particular environment. Rows which are indented are subsets of the nearest preceding non-indented row.

Table 8.2: Average deletion rates, p -values, and t -scores for the difference between black and white participants by environment.

Environment	Black average	White average	p -value	t -score
Word-final isolation	0.262	0.125	$p = 0.071$	$t = 1.507$
Word-final context	0.565	0.346	$p = 0.021^*$	$t = 2.117$
Word-final before a consonant	0.650	0.452	$p = 0.040^*$	$t = 1.810$
Word-final before a vowel	0.531	0.271	$p = 0.020^*$	$t = 2.139$
Word-final before a pause	0.410	0.242	$p = 0.083$	$t = 1.420$
Syllable-final context	0.226	0.132	$p = 0.024^*$	$t = 2.059$
Word-initial context	0.079	0.067	$p = 0.329$	$t = 0.446$
Intervocalic context	0.111	0.060	$p = 0.027^*$	$t = 2.002$
Intervocalic context followed by stress	0.021	0.020	$p = 0.463$	$t = 0.093$
Intervocalic context not followed by stress	0.235	0.114	$p = 0.015^*$	$t = 2.267$
Overall deletion rate	0.303	0.180	$p = 0.012^*$	$t = 2.385$

From the table above, it is clear that the black participants are deleting the rhotic more than the white participants in many environments. These differences tend to be in coda position in context; however, intervocalic context tokens (particularly those not followed by stress) were significant as well. Additionally, the subset of word-final context tokens which were followed by a pause did not have a significant difference in deletion rate ($p = 0.083$, $t = 1.420$). The overall deletion rate in the corpus was also significantly different by race. Given that the corpus is

predominated by rhotics in the coda environment (104/142 per participant), this difference is not surprising, and it is likely driven by the coda tokens.

As discussed in chapter 2, rhotic deletion in coda position has been historically linked to African influences in Brazilian Portuguese. Given this, the difference by race in coda environments makes sense if one is to expect greater African influences in the black residents of Salvador. However, in the intervocalic environment, an onset position, there is no historical connection in the literature between deletion and race. Thus, it is unclear what could be driving this difference, but the difference in this environment does not match the pattern of everything else. The implications of the differences in deletion rate by race are broken down in section 4 of this chapter.

8.1.2. Differences in deletion rate by socioeconomic class

In the post-recording survey, participants only identified as either lower or middle class, with none selecting the upper-class option. No other choices were given in the survey. Consequently, there are only two groups to compare. The average deletion rates, *p*-values, and *t*-scores of the tests conducted are given in table 8.3 below. The *t*-scores given represent the lower-class mean minus the middle-class mean for a given environment.

Table 8.3: Average deletion rates, p -values, and t -scores of the various environments for difference in deletion rate by socioeconomic class.

Environment	Lower class average	Middle class average	p -value	t -score
Word-final isolation	0.211	0.250	$p = 0.531$	$t = -0.633$
Word-final context	0.515	0.527	$p = 0.836$	$t = -0.209$
Syllable-final context	0.181	0.240	$p = 0.158$	$t = -1.446$
Word-initial context	0.057	0.1	$p = 0.041^*$	$t = -2.131$
Word-initial after a consonant	0.131	0.286	$p = 0.020^*$	$t = -2.442$
Word-initial after a vowel	0.024	0.027	$p = 0.868$	$t = -0.168$
Word-initial after a pause	0.048	0.048	$p = 1$	$t = 0$
Intervocalic context	0.102	0.097	$p = 0.474$	$t = 0.723$
Overall deletion rate	0.266	0.292	$p = 0.527$	$t = -0.640$

Given the table above, it is clear that there is no consistent trend for a difference in deletion rate by socioeconomic class. However, word-initial context tokens and their subset of tokens after a consonant had middle class participants deleting at a higher rate than lower class participants. In fact, it is the case in most environments; the middle class group is deleting more than the lower class group. This is somewhat surprising given that five of the seven white participants identified as middle class and therefore, they make up over one-third of the middle class group. Nevertheless, this is not driving the middle class group to delete less than the lower class group. Ultimately, it is not clear what would be causing the word-initial tests to be significant. Given that there is not a consistent difference in deletion rate between the two

groups that can be attributed to anything, it is not likely that socioeconomic class is affecting deletion rate.

8.1.3. Differences in deletion rate by gender

For the tests for differences in deletion rate by gender, participants were split into male and female categories, with the one person who identified as trans non-binary excluded from the analyses since this was not sufficient to form a group. The average deletion rates, p -values, and t -scores of the tests conducted are given in table 8.4 below. The t -scores given represent the female mean minus the male mean for a given environment.

Table 8.4: Average deletion rates, p -values, and t -scores of the various environments for difference in deletion rate by gender.

Environment	Female average	Male average	p -value	t -score
Word-final isolation	0.281	0.125	$p = 0.019^*$	$t = 2.468$
Word-final context	0.525	0.487	$p = 0.620$	$t = 0.500$
Syllable-final context	0.183	0.240	$p = 0.105$	$t = -1.668$
Word-initial context	0.063	0.086	$p = 0.328$	$t = -0.993$
Intervocalic context	0.087	0.125	$p = 0.106$	$t = -1.662$
Overall deletion rate	0.276	0.265	$p = 0.822$	$t = 0.227$

From the table above, it is clear there is not a consistent trend for a significant difference in deletion rate by gender, with the only significant test being word-final isolation tokens. Given that there is no consistent trend in a deletion rate difference between the two groups, it is not certain what could be driving this difference.

8.1.4. Differences in deletion rate by places lived

For the tests by places lived, participants were placed into two categories: those who have only lived in Salvador and those who have lived elsewhere at some point in their lives. The average deletion rates, p -values, and t -scores of the tests conducted are given in table 8.5 below. The t -scores given represent the mean of those who have only lived in Salvador minus the mean of those who have lived outside the city.

Table 8.5: Average deletion rates, p -values, and t -scores of the various environments for difference in deletion rate by whether or not participants lived their entire lives in Salvador.

Environment	Salvador only average	Has lived elsewhere average	p -value	t -score
Word-final isolation	0.196	0.348	$p = 0.139$	$t = -1.515$
Word-final context	0.481	0.676	$p = 0.065$	$t = -1.906$
Syllable-final context	0.203	0.211	$p = 0.903$	$t = -0.123$
Word-initial context	0.081	0.048	$p = 0.423$	$t = 0.812$
Intervocalic context	0.096	0.115	$p = 0.850$	$t = -0.196^6$
Overall deletion rate	0.259	0.345	$p = 0.142$	$t = -1.504$

From the table above, it is clear that there is not a significant difference in deletion rate between participants who have lived their whole lives in Salvador and those who have not. Given that there is no significant deletion rate difference between the two groups, there is no evidence that living elsewhere, along with the potential influence of another variety of Portuguese, is causing a difference in deletion rate in natives of Salvador.

⁶ F -test p -value: 0.050. Unequal variance test degrees of freedom: 8.058.

8.1.5. Differences in deletion rate by Afro-Brazilian identity

When participants were recontacted in the summer of 2023, one of the questions asked was “What would you consider your heritage to be? (e.g., Brazilian, African, Portuguese, Japanese, etc.).” The responses given by the black participants were then split into those which identified as having African or Afro-Brazilian heritage and those who did not. In total, 10 black participants identified as African or Afro-Brazilian, while 11 did not, giving a total of 21 participants for the tests conducted in this subsection. The average deletion rates, p -values, and t -scores of the tests conducted are given in table 8.6 below. The t -scores given represent the mean of those who identified as having an Afro-Brazilian or African heritage minus those who did not indicate such a heritage.

Table 8.6: Average deletion rates, p -values, and t -scores of the various environments for difference in deletion rate by whether or not participants identified as having Afro-Brazilian or African heritage.

Environment	Identified as Afro-Brazilian average	Did not identify as Afro-Brazilian average	p -value	t -score
Word-final isolation	0.244	0.295	$p = 0.471$	$t = -0.735$
Word-final context	0.427	0.698	$p = 0.015^*$	$t = -2.664$
Word-final before a consonant	0.541	0.776	$p = 0.026^*$	$t = -2.416$
Word-final before a vowel	0.358	0.654	$p = 0.044^*$	$t = -2.157$
Word-final before a pause	0.284	0.580	$p = 0.010^*$	$t = -2.846$
Syllable-final context	0.236	0.223	$p = 0.930$	$t = 0.090$
Word-initial context	0.1	0.061	$p = 0.246$	$t = 1.197$
Intervocalic context	0.075	0.126	$p = 0.335$	$t = -0.990$
Overall deletion rate	0.251	0.352	$p = 0.075$	$t = -0.631$

From the table above, it is clear that the only significant difference between the groups is word-final context tokens, and this significance is maintained in all three subsets of the word-final context tokens when they are divided up. Interestingly, all of these have negative t -scores, indicating that those who did not indicate themselves as having Afro-Brazilian or African heritage were deleting these tokens more than those who did identify as such. Given that deletion has been connected to African influences and there is a greater deletion rate for black participants than white participants, it would be reasonable to think that it is more likely for those who identify with an African heritage to delete more, yet that is not the case. It is also important to note that only 21 participants are included here, as the question of heritage was

only asked when participants were recontacted in 2023 and not all participants responded, so it is not representative of the entire sample collected. These tests only include the black participants who answered the heritage question. In the end, it is not certain what could be causing this difference, but it does appear to offer a bit of contrast to the difference in deletions rates by race which occurred.

8.1.6. Differences in deletion rate by data collection method

To test for possible differences by data collection method, participants were split into whether they completed the recording remotely or in-person. All 10 remote participants completed the recording in summer 2021, and all 25 in-person participants completed the recording in summer 2022. In both instances, they communicated with the author, who is not a native speaker of Portuguese, throughout the participation process. The average deletion rates, *p*-values, and *t*-scores of the tests conducted are given in table 8.7 below. The *t*-scores given represent the mean of those who participated remotely minus those who participated in-person.

Table 8.7: Average deletion rates, p -values and t -scores of the various environments for difference in deletion rate by data collection method.

Environment	Remote average	In-person average	p -value	t -score
Word-final isolation	0.219	0.23	$p = 0.675$	$t = -0.422$
Word-final context	0.533	0.515	$p = 0.796$	$t = 0.261$
Syllable-final context	0.233	0.193	$p = 0.336$	$t = 0.977$
Word-initial context	0.073	0.075	$p = 0.661$	$t = -0.443$
Intervocalic context	0.133	0.087	$p = 0.150$	$t = 1.473$
Overall deletion rate	0.288	0.272	$p = 0.821$	$t = 0.228$

From the table above, it is clear that data collection method did not play a role in deletion rate among the participants, as none of the tests were significant. Because of this, we can be fairly certain that differences in deletion rate within the sample are not due to the data collection method.

8.1.7. Differences in deletion rate by age

To test differences in deletion rate by age, regressions were conducted between the various deletion rates by participant for each environment along with their approximate birth year (calculated as year of data collection minus age at data collection). Table 8.8 below gives the correlation coefficient (r), r^2 , p -values, and t -scores for the various regression tests conducted. Positive r values indicate a positively sloped regression line (deletion rate increasing as birth year increases) while negative r values indicate a negatively sloped regression line (deletion rate decreasing as birth year increases).

Table 8.8: Correlation coefficient (r), r^2 , p -values and t -scores of the various environments for difference in deletion rate by age.

Environment	r	r^2	p -value	t -score
Word-final isolation	$r = -0.263$	$r^2 = 0.069$	$p = 0.127$	$t = -1.565$
Word-final context	$r = -0.105$	$r^2 = 0.011$	$p = 0.550$	$t = -0.060$
Syllable-final context	$r = -0.047$	$r^2 = 0.002$	$p = 0.790$	$t = -0.269$
Word-initial context	$r = -0.282$	$r^2 = 0.079$	$p = 0.101$	$t = -1.688$
Intervocalic context	$r = 0.032$	$r^2 = 0.001$	$p = 0.856$	$t = 0.183$
Overall deletion rate	$r = -0.145$	$r^2 = 0.021$	$p = 0.407$	$t = -0.840$

From the table above, it is clear that there is no detectable connection between deletion rate and age, as the tests are all insignificant and the r^2 values are all very low, indicating that very little variation in deletion rate can be explained by the variation in approximate birth year. To help visualize these regressions, figure 8.1 below plots approximate birth year against the overall deletion rates of the participants. The curve on the plot is the arcsine transformed regression line back transformed to fit the original data.

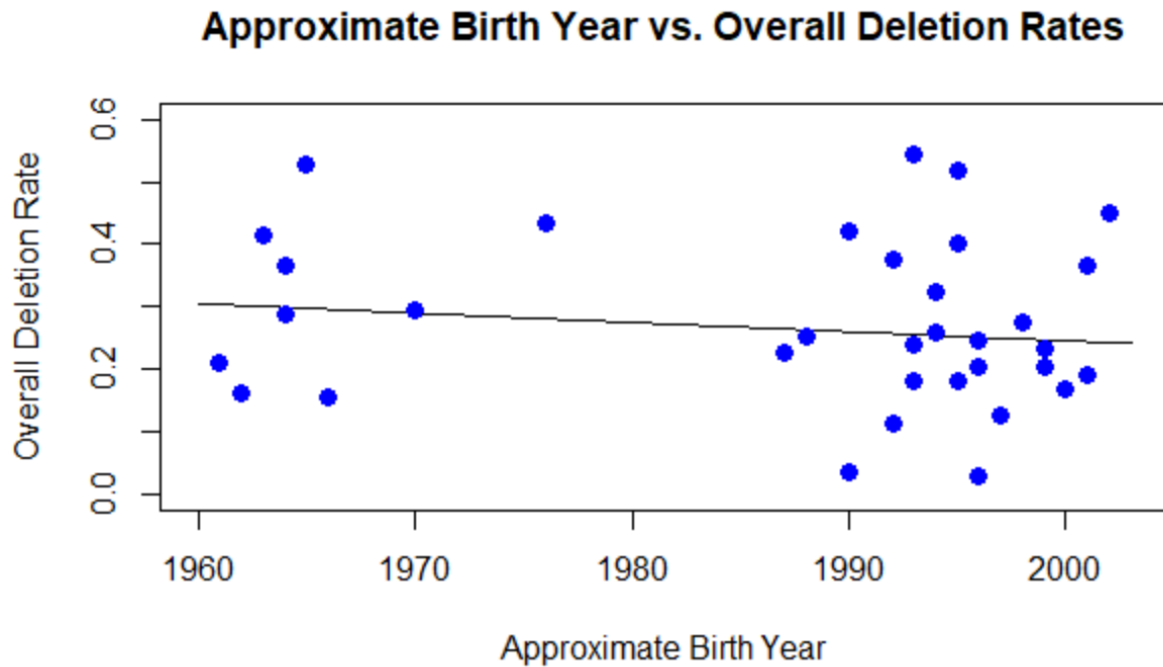


Figure 8.1: Regression graph of the approximate birth year of participants from Salvador and their overall deletion rates in the corpus.

From the figure above, it can be seen that there is no clear pattern between approximate birth year and overall deletion rate, further solidifying the results of the regression tests.

8.1.8. Conclusions regarding differences in deletion rate by demographic variables

In conclusion, there is no strong evidence to suggest that any demographic variables besides race are contributing to a difference in deletion rate among the participants. Additionally, there is no evidence that data collection method led to a difference in deletion rate either. While some of the tests did turn out to be significant, these were not systematic. Considering the large number of comparisons, it is not clear what could be causing these tests

to be significant. Therefore, it is quite likely that race is the main driver for individual differences in deletion rate.

8.2. Oral production forms by demographic variable

In addition to establishing the differences in deletion rate by demographic variable, it is also important with such a variable phoneme to consider how frequently and which oral production forms (i.e., non-glottal) are used. By establishing differences in oral production across demographics, we can have a deeper understanding of how the lenition processes may be occurring across populations. In terms of potential differences in production of oral forms by demographic variable, there are two things to consider: the frequency of oral forms and the way the oral forms surface. Consequently, the potential demographic differences were evaluated with two sets of tests. First, *t*-tests and regression tests were conducted on the rate of oral production forms in the same manner as in section 1 of this chapter. Second, the distribution of the oral forms produced by each demographic variable were tested with chi-squared and Fisher's exact tests. Subsection 1 presents the tests on the rates of oral forms, and subsection 2 presents the tests on the distribution of the oral surface forms. Furthermore, these tests will focus only on the rhotics in coda position, as only 10 word-initial and 11 intervocalic productions were oral in the entire corpus, meaning that less than one-third of participants produced oral forms in either environment.

8.2.1. Rates of oral production forms by demographic variables

The tests for differences in the rate of oral production forms were conducted in a similar manner to the tests for deletion rate. The tests for all demographic variables except age were two-tailed independent two-sample *t*-tests on the arcsine transformed oral production rates of

individual participants split into their respective categories. The *t*-tests were two-tailed because the existing literature does not claim that one particular demographic would diverge from glottal forms more than another. For age, a regression test was conducted on the deletion rates for a given environment and the approximate birth year of the participants.

8.2.1.1. Differences in oral production rate by race

Beginning with tests between the *preto* and *pardo* groups, the results of the tests for differences in oral production rates between the two groups are given in table 8.9 below. The negative *t*-scores indicate that the *pardo* participants had higher rates of oral production in each environment.

Table 8.9: Average rates of oral productions, *p*-values, and *t*-scores for the difference between the race categories *preto* and *pardo*.

Environment	<i>Preto</i> average	<i>Pardo</i> average	<i>p</i> -value	<i>t</i> -score
Word-final isolation	0.083	0.385	<i>p</i> = 0.001*	<i>t</i> = -3.830
Word-final context	0.189	0.246	<i>p</i> = 0.388	<i>t</i> = -0.879
Syllable-final isolation	0.072	0.188	<i>p</i> = 0.015*	<i>t</i> = -2.609
Syllable-final context	0.106	0.132	<i>p</i> = 0.460	<i>t</i> = -0.750
Overall oral production rate	0.108	0.174	<i>p</i> = 0.104	<i>t</i> = -1.689

From the table above, we can see that the isolation tokens for both the word-final and syllable-final environments are significantly different, with the *pardo* group producing more oral forms in each case. However, the context tokens are not significant, and neither is the overall oral production rate, which consists predominantly of context productions (35/142 per speaker). It is not clear what could be causing this difference between oral production rates of *preto* and *pardo* participants.

When combining the *preto* and *pardo* groups to create a comparison between the black and white participants, the following results are obtained, as shown in table 8.10. The *t*-scores given represent the mean for black participants minus the mean for white participants for a given environment.

Table 8.10: Average rates of oral productions, *p*-values, and *t*-scores for the difference between black and white participants by environment.

Environment	Black average	White average	<i>p</i> -value	<i>t</i> -score
Word-final isolation	0.150	0.188	<i>p</i> = 0.428	<i>t</i> = -0.803
Word-final context	0.202	0.354	<i>p</i> = 0.020*	<i>t</i> = -2.447
Word-final before a consonant	0.046	0.100	<i>p</i> = 0.016*	<i>t</i> = -2.557
Word-final before a vowel	0.398	0.645	<i>p</i> = 0.067	<i>t</i> = -1.897
Word-final before a pause	0.117	0.293	<i>p</i> = 0.024*	<i>t</i> = -2.377
Syllable-final isolation	0.098	0.071	<i>p</i> = 0.500	<i>t</i> = 0.682
Syllable-final context	0.112	0.102	<i>p</i> = 0.810	<i>t</i> = 0.242
Overall oral production rate	0.122	0.179	<i>p</i> = 0.063	<i>t</i> = -1.929

From the table above, we can see that there is a significant difference on word-final context tokens, particularly in the subsets of word-final tokens followed by a consonant or a pause, with the white participants producing more oral forms in these cases. While it is not certain what is causing this difference, it is worth noting that deletion takes away from the total possible number of tokens that can be produced with an oral surface form. In other words, lower deletion rates open the door for higher oral production rates. Additionally, if glottal surface forms are considered the standard in Salvador, deletion and oral production offer two opposite ways to diverge from the norm, with deletion serving as lenition and oral production

serving as fortition. Since it is the case that black participants are deleting the phoneme more (i.e., showing signs of more advanced lenition), it is plausible that white participants would produce more oral surface forms as a feature of having less lenited speech, though there is not sufficient evidence to determine this with any degree of certainty.

8.2.1.2. Differences in oral production rate by socioeconomic class

The results of the tests for differences in oral production rates by socioeconomic class are given in table 8.11 below. The *t*-scores given represent the lower-class mean minus the middle-class mean for a given environment.

Table 8.11: Average rates of oral productions, *p*-values, and *t*-scores for the difference by socioeconomic class.

Environment	Lower class average	Middle class average	<i>p</i> -value	<i>t</i> -score
Word-final isolation	0.158	0.165	<i>p</i> = 0.657	<i>t</i> = -0.449
Word-final context	0.207	0.271	<i>p</i> = 0.294	<i>t</i> = -1.067
Syllable-final isolation	0.083	0.108	<i>p</i> = 0.715	<i>t</i> = -0.368
Syllable-final context	0.104	0.122	<i>p</i> = 0.447	<i>t</i> = -0.77
Overall oral production rate	0.124	0.151	<i>p</i> = 0.382	<i>t</i> = -0.886

From the table above, we can see that there were no tests which were significant for differences in oral production rates by socioeconomic class, suggesting that this variable does not play a role in oral production rate.

8.2.1.3. Differences in oral production rate by gender

The results of the tests for differences in oral production rates by gender are given in table 8.12 below. The *t*-scores given represent the female mean minus the male mean for a given environment.

Table 8.12: Average rates of oral productions, p -values, and t -scores for the difference by gender.

Environment	Female average	Male average	p -value	t -score
Word-final isolation	0.209	0.102	$p = 0.204$	$t = 1.295$
Word-final context	0.214	0.272	$p = 0.323$	$t = -1.004$
Syllable-final isolation	0.119	0.064	$p = 0.205$	$t = 1.293$
Syllable-final context	0.117	0.110	$p = 0.569$	$t = 0.575$
Overall oral production rate	0.136	0.141	$p = 0.847$	$t = -0.194$

From the table above, we can see that there were no tests which were significant for differences in oral production rates by gender, suggesting that this variable does not play a role in oral production rate.

8.2.1.4. Differences in oral production rate by places lived

The results of the tests for differences in oral production rates by places lived (whether or not a participant has lived their entire life in Salvador) are given in table 8.13 below. The t -scores given represent the mean of those who have only lived in Salvador minus the mean of those who have lived outside the city.

Table 8.13: Average rates of oral productions, p -values, and t -scores for the difference by whether a participant has lived in Salvador their entire life.

Environment	Salvador only average	Has lived elsewhere average	p -value	t -score
Word-final isolation	0.189	0.044	$p = 0.020^*$	$t = 2.455$
Word-final context	0.250	0.162	$p = 0.141$	$t = 1.507$
Syllable-final isolation	0.094	0.089	$p = 0.673$	$t = 0.426$
Syllable-final context	0.119	0.082	$p = 0.121$	$t = 1.592$
Overall oral production rate	0.146	0.088	$p = 0.055$	$t = 1.989$

From the table above, we can see that the only environment which was significant for differences in oral production rates by places lived is word-final isolation. It is not clear what could be causing this difference by places lived.

8.2.1.5. Differences in oral production rate by Afro-Brazilian identity

The results of the tests for differences in oral production rates by Afro-Brazilian identity are given in table 8.14 below. The t -scores given represent the mean of those who identified as having an Afro-Brazilian or African heritage minus those who did not indicate such a heritage.

Table 8.14: Average rates of oral productions, p -values, and t -scores for the difference by whether or not participants identified as having Afro-Brazilian or African heritage.

Environment	Identified as Afro-Brazilian average	Did not identify as Afro-Brazilian average	p -value	t -score
Word-final isolation	0.119	0.119	$p = 0.819$	$t = -0.233$
Word-final context	0.276	0.145	$p = 0.068$	$t = 1.934$
Syllable-final isolation	0.1	0.091	$p = 0.974$	$t = 0.033$
Syllable-final context	0.121	0.092	$p = 0.227$	$t = 1.249$
Overall oral production rate	0.15	0.093	$p = 0.107$	$t = 1.694$

From the table above, we can see that there were no tests which were significant for differences in oral production rates by Afro-Brazilian identity, suggesting that this variable does not play a role in oral production rate.

8.2.1.6. Differences in oral production rate by data collection method

The results of the tests for differences in oral production rates by data collection method (remote versus in-person) are given in table 8.15 below. The t -scores given represent the mean of those who participated remotely minus those who participated in-person.

Table 8.15: Average rates of oral productions, p -values, and t -scores for the difference by data collection method (remote versus in-person).

Environment	Remote average	In-person average	p -value	t -score
Word-final isolation	0.05	0.205	$p = 0.017^*$	$t = -2.504$
Word-final context	0.256	0.223	$p = 0.570$	$t = 0.574$
Syllable-final isolation	0.088	0.096	$p = 0.997$	$t = -0.004$
Syllable-final context	0.122	0.107	$p = 0.473$	$t = 0.726$
Overall oral production rate	0.134	0.135	$p = 0.891$	$t = -0.138$

From the table above, we can see that the only environment which was significant for differences in oral production rates by data collection method is word-final isolation. It is not clear what could be causing this difference, though it might indicate that detecting oral constriction in isolated tokens is sensitive to the quality of the recording.

8.2.1.7. Differences in oral production rate by age

The results of the regression tests for oral production rates by age are given in table 8.15 below. Positive correlation coefficients (r) indicate a positively sloped regression line (oral production rate increasing as birth year increases) while negative r values indicate a negatively sloped regression line (oral production rate decreasing as birth year increases).

Table 8.16: Correlation coefficient (r), r^2 , p -values and t -scores of the various environments for difference in oral production rate by age.

Environment	r	r^2	p -value	t -score
Word-final isolation	$r = -0.164$	$r^2 = 0.027$	$p = 0.347$	$t = -0.953$
Word-final context	$r = 0.024$	$r^2 = 0.001$	$p = 0.889$	$t = 0.140$
Syllable-final isolation	$r = 0.115$	$r^2 = 0.013$	$p = 0.511$	$t = 0.665$
Syllable-final context	$r = -0.215$	$r^2 = 0.046$	$p = 0.214$	$t = -1.266$
Overall deletion rate	$r = -0.033$	$r^2 = 0.001$	$p = 0.851$	$t = -0.189$

From the table above, it is clear that there is no connection between oral production rate and age, as the tests are all insignificant and the r^2 values are all very low, indicating that very little variation in deletion rate can be explained by the variation in approximate birth year. To help visualize these regressions, figure 8.2 below plots approximate birth year against the overall oral production rates of the participants. The curve on the plot is the arcsine transformed regression line back transformed to fit the original data.

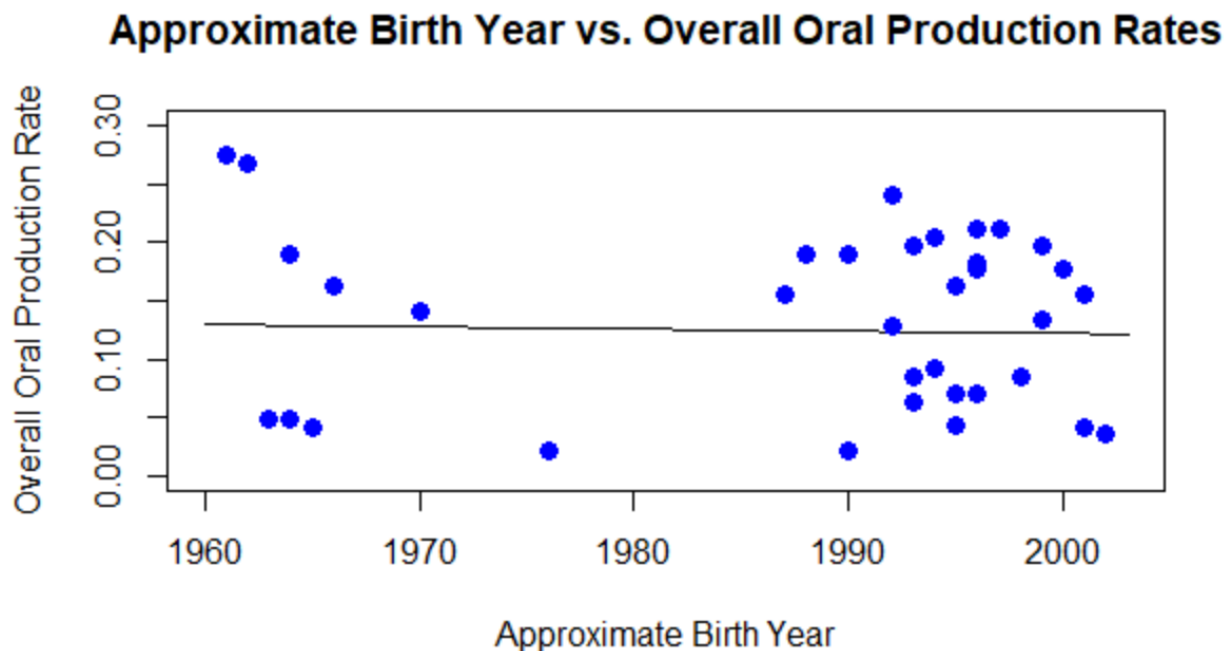


Figure 8.2: Regression graph of the approximate birth year of participants from Salvador and their overall oral production rates in the corpus.

From the figure above, it can be seen that there is no clear pattern between approximate birth year and overall oral production rate, further solidifying the results of the regression tests.

8.2.1.8. Conclusions regarding differences in oral production rate by demographic variable

In conclusion, there is not a strong likelihood that any demographic variable is causing a difference in the rate of oral production among the participants. While some of the tests did turn out to be significant, there is no clear indication as to what could be driving these differences. However, it is possible that the significant differences between black and white participants on word-final context tokens could be driven by the fact that deletion is more common in the black participants, leaving a larger proportion of the productions available to be produced with oral surface forms. Nevertheless, these tests do not address the possibility that when the rhotic is oral, the surface form produced may vary by demographics. The next subsection addresses this concern.

8.2.2. Distribution of oral surface forms by demographic variables

The tests conducted for distribution of oral surface forms by demographic variables were chi-squared and Fisher's exact tests. Due to low counts for many surface forms, the tests here group the surface forms into places of articulation. Even with this grouping, many places still had low counts, meaning that some of these significant results should be taken with a grain of salt. To help mitigate the effects of low counts, additional tests were conducted with the places of articulation restricted to the most common oral places: palatal and velar, as well as alveolar for the tests which included word-final context tokens before a vowel (word-final context, word-final before a vowel, overall corpus), as the alveolar flap was the most common surface form in this environment.

Most of the tests conducted were chi-squared tests; however, in the case that a test was conducted on a 2x2 contingency table, a Fisher's exact test was used instead as a means of

mitigating the low counts that existed on some variables. Note that age is not included here due to the fact it is a continuous variable and cannot be split into groups for these analyses. In the tables which present the results of the tests, the test used can be determined by the test statistic given. The test statistic for chi-squared tests is χ^2 , and the test statistic for Fisher's exact tests the odds ratio.

8.2.2.1. Differences in oral surface forms by race

The results of the tests for differences in oral surface forms by race are given in tables 8.17 and 8.18 below. Table 8.17 gives the results for tests where the *preto*, *pardo*, and white participants are all in separate groups while table 8.18 gives the results for test between black (*preto* + *pardo*) and white participants. The contingency tables for these tests can be found in tables A2.1 to A2.8 appendix A2.

Table 8.17: *p*-values and test statistics for the tests on differences in the distribution of oral surface forms by race with the *preto*, *pardo*, and white participants split into separate groups.

All places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	<i>p</i> = 0.143	$\chi^2 = 14.703$
Word-final context	<i>p</i> = 0.005*	$\chi^2 = 25.338$
Word-final before a consonant	<i>p</i> = 0.328	$\chi^2 = 4.629$
Word-final before a vowel	<i>p</i> = 0.040*	$\chi^2 = 13.195$
Word-final before a pause	<i>p</i> = 0.425	$\chi^2 = 8.086$
Syllable-final isolation	<i>p</i> = 0.079	$\chi^2 = 8.366$
Syllable-final context	<i>p</i> = 0.731	$\chi^2 = 8.667$
Overall corpus	<i>p</i> = 0.048*	$\chi^2 = 26.454$
High frequency places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	<i>p</i> = 0.056	$\chi^2 = 5.778$
Word-final context	<i>p</i> = 0.008*	$\chi^2 = 13.733$
Word-final before a consonant	<i>p</i> = 0.244	$\chi^2 = 2.824$
Word-final before a vowel	<i>p</i> = 0.027*	$\chi^2 = 10.981$
Word-final before a pause	<i>p</i> = 0.559	$\chi^2 = 1.165$
Syllable-final isolation	n/a	Test not conducted due to a lack of palatal syllable-final isolated productions
Syllable-final context	<i>p</i> = 0.496	$\chi^2 = 1.402$
Overall corpus	<i>p</i> = 0.007*	$\chi^2 = 14.1$

From the table above, it can be seen that word-final context tokens and their subset of tokens preceding vowels have significant differences in distributions of oral surface forms, as well as the overall corpus. With regards to their respective contingency tables in appendix A2, it appears that this is possibly driven by a larger proportion of alveolar surface forms (flaps) for the *preto* participants in these tests.

Combining the *preto* and *pardo* participants into a single group gives the following results:

Table 8.18: *p*-values and test statistics for the tests on differences in the distribution of oral surface forms between black and white participants.

All places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	<i>p</i> = 0.836	$\chi^2 = 2.093$
Word-final context	<i>p</i> = 0.026*	$\chi^2 = 12.715$
Word-final before a consonant	<i>p</i> = 0.222	$\chi^2 = 3.014$
Word-final before a vowel	<i>p</i> = 0.191	$\chi^2 = 4.745$
Word-final before a pause	<i>p</i> = 0.299	$\chi^2 = 4.891$
Syllable-final isolation	<i>p</i> = 0.326	$\chi^2 = 2.241$
Syllable-final context	<i>p</i> = 0.902	$\chi^2 = 2.189$
Overall corpus	<i>p</i> = 0.729	$\chi^2 = 5.268$
High frequency places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	<i>p</i> = 1	Odds ratio = 0.951
Word-final context	<i>p</i> = 0.358	$\chi^2 = 2.0522$
Syllable-final isolation	n/a	Test not conducted due to a lack of palatal syllable-final isolated productions
Syllable-final context	<i>p</i> = 0.606	Odds ratio = 1.532
Overall corpus	<i>p</i> = 0.685	$\chi^2 = 0.757$

From the table above, it can be seen that the only test which was significant for differences in the distribution of oral surface forms between black and white participants was word-final context tokens taking all places of articulation into consideration. However, when splitting this environment into its various subsets based on the following segment type, or only looking at high frequency places of articulation, the tests are no longer significant. This suggests that the significant result of this test is due to the low counts of surface forms in the bilabial, uvular, and pharyngeal places of articulation. Therefore, it is not likely that there is a difference in the distribution of oral surface forms between black and white participants.

8.2.2.2. Differences in oral surface forms by socioeconomic class

The results of the tests for differences in oral surface forms by socioeconomic class are given in 8.19 below. The contingency tables for these tests can be found in tables A2.9 to A2.13 appendix A2.

Table 8.19: *p*-values and test statistics for the tests on differences in the distribution of oral surface forms by socioeconomic class.

All places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	<i>p</i> = 0.131	$\chi^2 = 8.487$
Word-final context	<i>p</i> = 0.093	$\chi^2 = 9.427$
Syllable-final isolation	<i>p</i> = 0.086	$\chi^2 = 4.897$
Syllable-final context	<i>p</i> = 0.300	$\chi^2 = 8.235$
Overall corpus	<i>p</i> = 0.002*	$\chi^2 = 24.103$
High frequency places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	<i>p</i> = 0.348	Odds ratio = 1.709
Word-final context	<i>p</i> = 0.096	$\chi^2 = 4.691$
Syllable-final isolation	n/a	Test not conducted due to a lack of palatal syllable-final isolated productions
Syllable-final context	<i>p</i> = 0.103	Odds ratio = 2.050
Overall corpus	<i>p</i> = 0.002*	$\chi^2 = 12.689$

From the table above, it can be seen that the various environments did not have significant differences in distributions of oral surface forms, however the overall corpus did have a difference. Based on table A2.13 in appendix A2, this difference is likely driven by the fact that alveolar productions (flaps) proportionally comprise more of the middle-class group's oral surface forms. However, when the tokens are split into the various environments, this difference is no longer statistically significant.

8.2.2.3. Differences in oral surface forms by gender

The results of the tests for differences in oral surface forms by gender are given in 8.20 below. The contingency tables for these tests can be found in tables A2.14 to A2.21 appendix A2.

Table 8.20: *p*-values and test statistics for the tests on differences in the distribution of oral surface forms by gender.

All places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	<i>p</i> = 0.441	$\chi^2 = 4.798$
Word-final context	<i>p</i> = 0.007*	$\chi^2 = 15.828$
Word-final before a consonant	<i>p</i> = 0.7365	$\chi^2 = 0.612$
Word-final before a vowel	<i>p</i> = 0.012*	$\chi^2 = 10.988$
Word-final before a pause	<i>p</i> = 0.132	$\chi^2 = 8.070$
Syllable-final isolation	<i>p</i> = 0.130	$\chi^2 = 4.088$
Syllable-final context	<i>p</i> = 0.597	$\chi^2 = 4.595$
Overall corpus	<i>p</i> = 0.002*	$\chi^2 = 24.999$
High frequency places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	<i>p</i> = 0.130	Odds ratio = 2.168
Word-final context	<i>p</i> = 0.002*	$\chi^2 = 12.216$
Word-final before a consonant	<i>p</i> = 1	Odds ratio = 0.935
Word-final before a vowel	<i>p</i> = 0.005	$\chi^2 = 10.652$
Word-final before a pause	<i>p</i> = 0.402	Odds ratio = 2.312
Syllable-final isolation	n/a	Test not conducted due to a lack of palatal syllable-final isolated productions
Syllable-final context	<i>p</i> = 0.681	Odds ratio = 0.832
Overall corpus	<i>p</i> = 2.068x10⁻⁵*	$\chi^2 = 21.573$

From the table above, it can be seen that word-final context tokens and their subset of tokens preceding vowels have significant differences in distributions of oral surface forms, as well as the overall corpus. With regards to their respective contingency tables in appendix A2, it

appears that this is likely driven by a larger proportion of alveolar surface forms (flaps) for the male participants in these tests.

8.2.2.4. Differences in oral surface forms by places lived

The results of the tests for differences in oral surface forms by whether or not a participant has lived their entire life in Salvador are given in 8.21 below. The contingency tables for these tests can be found in tables A2.22 to A2.26 appendix A2.

Table 8.21: *p*-values and test statistics for the tests on differences in the distribution of oral surface forms by whether or not a participant has lived their entire life in Salvador.

All places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	$p = 0.848$	$\chi^2 = 2.007$
Word-final context	$p = 0.211$	$\chi^2 = 8.137$
Syllable-final isolation	$p = 0.253$	$\chi^2 = 2.751$
Syllable-final context	$p = 0.676$	$\chi^2 = 4.002$
Overall corpus	$p = 0.477$	$\chi^2 = 8.562$
High frequency places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	$p = 0.335$	Odds ratio = 0.332
Word-final context	$p = 0.068$	$\chi^2 = 5.369$
Syllable-final isolation	n/a	Test not conducted due to a lack of palatal syllable-final isolated productions
Syllable-final context	$p = 0.154$	Odds ratio = 2.798
Overall corpus	$p = 0.118$	$\chi^2 = 4.281$

From the table above, it can be seen that no differences were found between the distribution of oral surface forms and whether or not a participant has their entire life Salvador, suggesting that this variable does not affect a speaker's surface forms.

8.2.2.5. Differences in oral surface forms by Afro-Brazilian identity

The results of the tests for differences in oral surface forms by Afro-Brazilian identity are given in 8.22 below. The contingency tables for these tests can be found in tables A2.27 to A2.31 appendix A2.

Table 8.22: *p*-values and test statistics for the tests on differences in the distribution of oral surface forms by whether a participant identified as having Afro-Brazilian or African heritage.

All places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	<i>p</i> = 0.144	$\chi^2 = 5.414$
Word-final context	n/a	Only high frequency places (alveolar, palatal, and velar) were present. See high frequency test.
Syllable-final isolation	<i>p</i> = 0.085	$\chi^2 = 4.933$
Syllable-final context	<i>p</i> = 0.730	$\chi^2 = 2.806$
Overall corpus	<i>p</i> = 0.037*	$\chi^2 = 14.903$
High frequency places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	<i>p</i> = 0.721	Odds ratio = 0.600
Word-final context	<i>p</i> = 0.115	$\chi^2 = 4.333$
Syllable-final isolation	n/a	Test not conducted due to a lack of palatal syllable-final isolated productions
Syllable-final context	<i>p</i> = 0.790	Odds ratio = 1.267
Overall corpus	<i>p</i> = 0.011*	$\chi^2 = 9.088$

From the table above, it can be seen that the various environments did not have significant differences in distributions of oral surface forms, however the overall corpus did have a difference. Based on table A2.31 in appendix A2, this difference is likely driven by the fact that velar productions proportionally comprise more productions of those who did not identify as Afro-Brazilian and alveolar (flap) productions comprise more productions of those who identified as Afro-Brazilian. However, when the tokens are split into the various environments, this difference is no longer statistically significant.

8.2.2.6. Differences in oral surface forms by data collection method

The results of the tests for differences in oral surface forms by data collection method are given in 8.23 below. The contingency tables for these tests can be found in tables A2.32 to A2.39 appendix A2.

Table 8.23: *p*-values and test statistics for the tests on differences in the distribution of oral surface forms by data collection method (remote versus in-person).

All places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	<i>p</i> = 0.985	$\chi^2 = 0.664$
Word-final context	<i>p</i> = 0.002*	$\chi^2 = 18.833$
Word-final before a consonant	<i>p</i> = 0.372	$\chi^2 = 1.976$
Word-final before a vowel	<i>p</i> = 0.037*	$\chi^2 = 8.513$
Word-final before a pause	<i>p</i> = 0.456	$\chi^2 = 3.644$
Syllable-final isolation	<i>p</i> = 0.462	$\chi^2 = 1.547$
Syllable-final context	<i>p</i> = 0.421	$\chi^2 = 6.019$
Overall corpus	<i>p</i> = 0.002	$\chi^2 = 24.666$
High frequency places of articulation		
Environment	<i>p</i> -value	Test statistic
Word-final isolation	<i>p</i> = 1	Odds ratio = 1.152
Word-final context	<i>p</i> = 4.46x10⁻⁴*	$\chi^2 = 15.43$
Word-final before a consonant	<i>p</i> = 0.281	Odds ratio = 2.484
Word-final before a vowel	<i>p</i> = 0.032*	$\chi^2 = 6.867$
Word-final before a pause	<i>p</i> = 0.568	Odds ratio = 2.579
Syllable-final isolation	n/a	Test not conducted due to a lack of palatal syllable-final isolated productions
Syllable-final context	<i>p</i> = 0.508	Odds ratio = 0.692
Overall corpus	<i>p</i> = 1.098x10⁻⁴*	$\chi^2 = 18.233$

From the table above, it can be seen that word-final context tokens and their subset of tokens preceding vowels have significant differences in distributions of oral surface forms, as well as the overall corpus. With regards to their respective contingency tables in appendix A2, it appears that this is likely driven by a larger proportion alveolar surface forms (flaps) for the remote participants in these tests.

8.2.2.7. Conclusions regarding differences in oral surface forms by demographic variables

Among the various demographic variables tested, the only tests which were ever significant for a difference in distribution of oral surface forms were word-final context tokens,

the subset of word-final tokens before a vowel, and the overall corpus. Given the predominance of the alveolar flap in the word-final before a vowel environment, this is certainly the main driver of differences by demographic variables. In other words, the differences in distribution of surface forms are generally being driven by the prominence of the phenomenon of the rhotic merging with the flap when in word-final position before a vowel. The sole exception to this was word-final context tokens between the black and white participants, where the significance was driven by the low count surface forms and the test was no longer significant when restricted to the high frequency places of palatal, velar, and alveolar.

8.3. Addressing potential associations between demographic variables

To fully contextualize these demographic differences found in the data, it is important to establish whether the demographic variables are independent from each other. To test for independence of the three race categories (*preto*, *pardo*, and white) and other demographic variables, chi-squared tests were conducted. For the other tests, Fisher's exact tests were conducted on each pair of demographic variables at a time (excluding age, as it is continuous and not categorical). This test was chosen over the chi-squared test because it is better for 2x2 contingency tables with low counts. The demographic categories with only one person (indigenous and trans non-binary) were excluded from these tests. Table 8.24 below presents the results of these tests. The upper entry for each test gives the *p*-value, and the lower entry gives the test statistic. The contingency tables used in these tests can be found in tables A2.40 to A2.54 in appendix A2.

Table 8.24: p -values and test statistics of the tests for independence between demographic variables.

	Data collection method	Afro-Brazilian Identity	Places lived	Gender	Class
Race (preto, pardo, and white separate)	$p = 0.731$	$p = 0.311$ (<i>preto</i> and <i>pardo</i> only)	$p = 0.375$	$p = 0.937$	$p = 0.189$
	$\chi^2 = 0.627$	Odds ratio = 4.769	$\chi^2 = 1.962$	$\chi^2 = 0.131$	$\chi^2 = 3.331$
Race (black vs. white)	$p = 1$	n/a	$p = 0.615$	$p = 1$	$p = 0.097$
	Odds ratio = 1.051	n/a	Odds ratio = 1.728	Odds ratio = 0.838	Odds ratio = 4.5751
Class	$p = 0.704$	$p = 1$	$p = 1$	$p = 0.296$	
	Odds ratio = 1.803	Odds ratio = 0.760	Odds ratio = 1.154	Odds ratio = 0.440	
Gender	$p = 0.252$	$p = 1$	$p = 1$		
	Odds ratio = 2.898	Odds ratio = 1.470	Odds ratio = 1.483		
Places lived	$p = 0.381$	$p = 0.035^*$			
	Odds ratio = 0.456	Odds ratio = infinity			
Afro-Brazilian Identity	$p = 0.659$				
	Odds ratio = 1.729				

From the table above, it can be seen that the only test which was significant for a relationship between demographic variables was the test between Afro-Brazilian identity and places lived. The contingency table from appendix A2 used for this test (table A2.53) is replicated below as table 8.25.

Table 8.25: Counts for the variables of Afro-Brazilian identity and whether a participant has lived their whole life in Salvador.

	Identified as Afro-Brazilian	Identified as something else
Salvador only	10	6
Has lived elsewhere	0	5

As shown in the table above, no participants who lived outside of Salvador at some point in their lives identified as having an Afro-Brazilian or African heritage. This is clearly the driver of

the significance of this test. However, it is hard to be certain whether living elsewhere has contributed to identifying as having something other than Afro-Brazilian or African, as only five people in this test have lived outside Salvador. This small amount is likely skewing the results. Additionally, only 21 of the participants are included here. Because of these factors, it is unlikely that the significant test here indicates that the variables of Afro-Brazilian identity of whether a participant has lived their whole life in Salvador are confounding any results in this dissertation. Nevertheless, this may indicate some sort of dissociation between Afro-Brazilian identity and leaving the city of Salvador.

At the same time, there is one non-significant test worth noting. As was discussed in chapter 3, socioeconomic class has been linked to race in Brazil. While the test here was insignificant ($p = 0.189$ with *preto* and *pardo* as separate categories, $p = 0.096$ with *preto* and *pardo* combined into one category), it is likely that the lack of significant result here was driven by low counts, particularly for the white participants. Take into consideration tables 8.26 and 8.27 below.

Table 8.26: Counts for the variables of race and socioeconomic class.

	<i>Preto</i>	<i>Pardo</i>	White
Lower class	14	4	2
Middle class	7	2	5

In the table above, the proportions of lower- and middle-class participants in the categories of *preto* and *pardo* are identical. This can be seen as further evidence supporting the notion that the two categories should be combined in the category *negro*, as was discussed in the literature review of chapter 3. When keeping the three categories separate, this same proportion between *preto* and *pardo* participants certainly drives down the significance.

However, even combining the two groups into one category still does not yield a significant result.

Table 8.27: Counts for the variables of race and socioeconomic class.

	Black	White
Lower class	18	2
Middle class	9	5

In the table above, two-thirds of the black participants identified as lower class (66.7%), while five of the seven white participants identified as middle class (71.4%). Clearly, this is a major difference, yet it is not statistically significant. Undoubtedly, if these proportions were maintained with a higher sample size, the test would become significant. It is also reasonable to believe that such proportions would be maintained in a larger sample size when considering the existing literature on the connection between race and socioeconomic class in Brazil discussed in section 2 of chapter 3. Therefore, while this test was not significant, it is highly likely that race and socioeconomic class are indeed dependent on each other in Salvador. However, given the reality that there is no strong indication of differences in the data by socioeconomic class, it is not likely to be confounding the significant results for differences in deletion rate by race.

In conclusion, while the test between Afro-Brazilian identity and whether a participant has lived their whole life in Salvador was significant, it is unlikely that these two variables are confounding anything in the data. At the same time, the non-significant test between race and socioeconomic class is likely being affected by low counts, and the two variables are most likely connected, especially when considering the connection between these variables established in the existing literature. However, due to the lack of evidence that socioeconomic class is driving any demographic differences in the data, it is not likely to be confounding the results of the tests for differences by race. Therefore, it is fair to conclude that the higher deletion rates by

race in the data are indeed driven by the difference in racial identification and not by other variables.

8.4. Discussion of demographic differences

In this chapter, the results of various tests for differences in the data by demographic variable were presented, as well as an evaluation of whether the demographic variables were associated with each other in any way that could confound the findings of this dissertation. With this analysis, it is now possible to answer the second set of research questions posed in chapter 4:

1. Do black participants delete the rhotic more frequently than white participants?
2. Are there differences in deletion rate that exist by other demographic variables?

There was a consistently higher deletion rate among the black participants than among the white participants on context tokens in coda position. There were some tests which were significant for other demographic variables as well, though these differences were not as widespread as the differences that existed by race. Additionally, there is no indication as to what could be driving these differences. Thus, the hypothesis that deletion rates would not vary by race was not correct. This suggests that although both the black and white residents of Salvador generally come from the same ancestral background, there are likely still stronger African influences in the speech of those who identify as *preto* or *pardo*. Regarding the hypothesis that differences in deletion rate would not exist by other demographic variables, this hypothesis was also not correct, as there were some significant tests for these demographic variables.

3. Do the various demographic groups produce non-lenited (i.e., oral or non-glottal) surface forms at different rates?

4. Do the non-lenited surface forms of the rhotic vary by demographic variables?

With regards to oral production rate, there were various significant tests, though there is no clear indication as to what could be causing this. However, the significant tests between black and white participants here may be driven by the fact that white participants deleted the rhotic less, which could then open the door to the possibility of greater proportion of oral surface forms. In terms of the distribution of oral surface forms, the differences here tend to be driven by variation in the flapping phenomenon on word-final context tokens before a vowel, and in the high frequency oral places of articulation: palatal and velar. Thus, the hypotheses regarding oral production rate and oral surface forms were also not correct, as there were significant tests here as well.

In addition to reviewing the deletion rates, oral production rates, and oral surface forms by demographic variable, this chapter checked for independence of demographic variables. Ultimately, it was shown that the only demographic variables associated with each other are Afro-Brazilian identity and whether a participant has lived their entire life in Salvador. However, the fact that only three-fifths of the participants (21) were included in this test and the fact that only five of these participants have lived outside Salvador are likely skewing the results of the independence test for these variables. Therefore, it is unlikely that the connection between these variables is confounding any results. Also of note is the fact that socioeconomic class and race are probably connected to one another, despite the fact that the test on the contingency table 8.27 is insignificant. This is because of the relative proportion of white participants who identified as middle class and the connection between race and socioeconomic class in Brazil discussed in section 2 of chapter 3. Just as the significance of the test between Afro-Brazilian

identity and whether a participant has lived their entire life in Salvador was likely driven by low counts, the low number (7) of white participants is likely causing the test for independence between race and socioeconomic class to be insignificant. Regardless, the fact that the significant tests for differences in deletion rate by race and socioeconomic class do not overlap eliminates any concern that the differences in deletion rate by race are being confounded by socioeconomic class.

With regards to the social significance of the difference in deletion rate by race in coda environments, it is important to take into consideration the information presented in chapter 3. In particular, the main takeaway to apply here is that race in Brazil is a purely social marker based on physical appearance (and socioeconomic status to some extent); it is not something directly determined by ancestry. It is known that white residents of Salvador typically have African ancestry and four of the seven white participants in this study have black parents or siblings. Nevertheless, the race variable is creating a significant difference in deletion rates of the rhotic among participants. At surface level, such a social difference may not be surprising, but when we dig deeper, it begins to be clear that this social difference is not very straightforward, and it reveals the differences between the concept of race in the United States and the concept of race in Brazil that were discussed in chapter 3.

Ultimately, it is important to ask whether having African ancestry or identifying as a racial category associated with African phenotypical features (i.e., *preto* or *pardo*) plays a larger role in strengthening African linguistic influences in a modern speaker. If the answer were ancestry, it would be reasonable to believe that this would lead to the deletion rates being the same for both black and white speakers. This is plausible because, as was established in the

genetic studies discussed in section 3 of chapter 3, all three racial categories *preto*, *pardo*, and white have a mix of African, European, and indigenous American DNA, and all three tend to have European paternal lines and African or indigenous maternal lines. This indicates that despite the varied appearances, all three groups are descended from the same ancestral populations in Salvador. Consequently, it would be reasonable to believe that all three groups speak varieties which have been subject to the same historical influences. However, despite all this shared heritage, there is still a difference in deletion rate by race in the city. Therefore, it appears that racial identification plays a larger role.

While this may seem like something obvious at first glance, this is overlooking the complexity of the race situation in Brazil and Salvador. As seen in section 2 of chapter 3, it has been shown that people are not consistent with their own identification and that their identification also often differs from what racial category is assigned by an outside observer. For example, people have even been known to switch their identification to white when becoming wealthier. Thus, identifying as *preto* or *pardo* is not necessarily a permanent status that will socially condition someone to speak a certain way indefinitely, nor does identifying as such mean that others in society will perceive and treat them that way. Additionally, the fact that white people in Salvador are often known to have black parents or full siblings (the case for at least two of the seven white participants from Salvador) means that they are often raised alongside each other and have a lot of shared early experiences.

On top of this, as discussed in section 1 of chapter 3, though it was somewhat uncommon to have a relationship between two people with different skin colors, interracial relationships were not discouraged or forbidden among the lower classes in Brazil and such

relationships were far from unheard of. Given the fact that Salvador's population developed over several centuries, occasional interracial relationships would still ensure regular social contact between those identified as black and those identified as white, at least among non-elites. Furthermore, none of the participants in the study identified as being upper class, meaning that such mixing over the centuries likely applies to them. Moreover, genetic studies have confirmed the shared heritage of Salvador's residents regardless of racial identification, suggesting that the racial categories are little more than labels based on appearance. It is also important to note the fact that there was no significant difference in deletion rate between those who identified as *preto* and those who identified as *pardo*. This can be seen as further justification for the combination of the two categories into one: *negro*.

Considering all of the above, the difference in rhotic deletion rates by racial identification must be purely driven by whether a person identifies as black or white, rather than by other social factors hidden within the racial identifications themselves. Additionally, the lack of a consistent difference in deletion rate by other demographic variables or data collection method (remote versus in-person) and the fact that no relationship between race and another demographic variable was found show that there is not a confounding variable at play that could also be driving the deletion. What separates this from a similar racial difference in the United States would be the fact that people in the United States with any non-white ancestor would be expected to identify as non-white regardless of their appearance, as discussed in section 2 of chapter 3. Because of this, one would not expect to have participants identifying as white and non-white while coming from the same ancestral populations in the United States. However, this is the norm in Salvador. What it boils down to is this: the social differences by race

in Salvador come down to divisions primarily driven by individual identification while social differences in the United States would be rooted in deeper, historical social divisions between black and white people that simply did not exist in Salvador. Thus, it appears that the difference in deletion rates in Salvador is driven by racial identification alone and not by other social variables that are often connected to race.

Given all these results presented in this chapter, it is safe to say that identifying as *preto* or *pardo* is a driving factor that increases deletion rate the most among speakers from Salvador. While the links of African influences on Brazilian Portuguese to rhotic deletion established in the literature may make this seem unsurprising, the historical and sociolinguistic context established in chapter 3 demonstrates that white speakers in Salvador have been exposed to the same African influences as black speakers, both through their ancestry and with the present-day reality that many white residents of Salvador have black parents or siblings. Thus, this difference between the black and white participants in the data must be driven specifically by identity, rather than by ancestry or social separation of the two racial groups.

9. Data from São Paulo and its comparison to Salvador

This chapter presents the data collected from the São Paulo participants, as well as the various tests conducted to compare the deletion rates, oral production rates, and surface form distribution of the São Paulo and Salvador samples. Section 1 of this chapter presents the São Paulo data. Section 2 compares the data of the two cities. Lastly, section 3 discusses the findings of this chapter.

9.1. São Paulo data

9.1.1. São Paulo word-final productions and deletions

Table 9.1 below gives the count of each surface form of the rhotic for word-final tokens in isolation and in context, with the final column combining both isolation and context tokens. The count of deleted tokens is included as well and is denoted as ‘null’.

Table 9.1: Counts of the surface forms of the rhotic in the word-final environment for the São Paulo participants.

Surface form	Isolation	Context	Overall
null	3	48	51
r	1	173	174
h	10	1	11
ɦ	9	71	80
rh	2	0	2
rɦ	17	50	67
rç	1	0	1
rj	0	1	1
rʂ	9	7	16
rʐ	1	1	2
rs	0	1	1
rθ	1	0	1
rV	2	32	34
ɾ	28	15	43
r	2	15	17
ɹ	44	88	132
ɹV	0	1	1
x	3	1	4
x ^w	0	1	1
ç	1	1	2
j	0	2	2
ʂ	19	7	26
ʐ	6	2	8
z	0	1	1
l	0	1	1
Total	159	520	679

From the table above, it can be seen that, while a fair number of tokens were deleted, many surface forms were more prominent than deletion, such as flaps, flap + fricative variants, glottal fricatives, and the approximant [ɹ]. In total, about 9.2% (48/520) of the context tokens were deleted while only three isolation tokens were deleted. Figure 9.1 below shows the proportion of surface forms in the word-final environment.

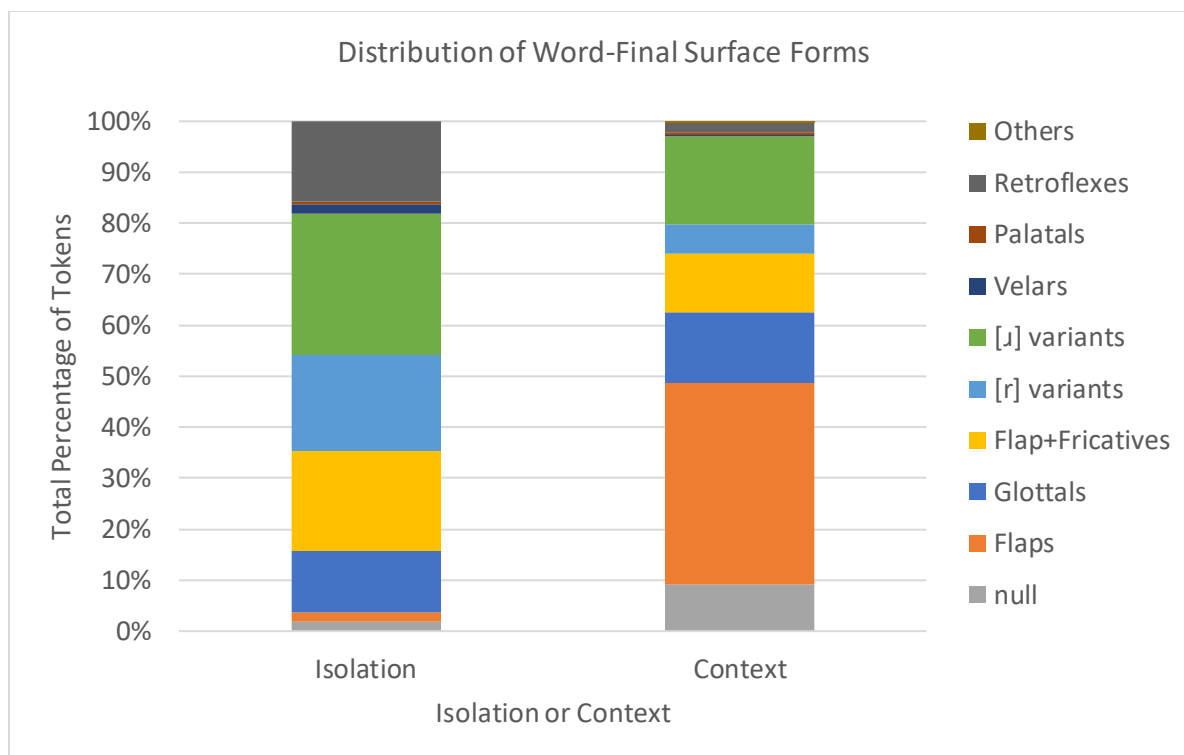


Figure 9.1: Proportion of surface forms in the word-final environment for the São Paulo participants.

In figure 9.1, it can be seen that there is quite a bit of variation in the surface forms, with flaps being the most common in context and [ɹ] variants being the most common in isolation. However, alveolar trill [r] variants, flap + fricative variants, and glottal fricatives all compromise notable portions of the surface forms, as well as retroflexes for the isolation tokens.

Table 9.2 and figure 9.2 below give the count and distribution of each surface form of the rhotic for word-final context tokens separated by following environment.

Table 9.2: Counts of the surface forms of the rhotic on word-final context tokens separated by following environment for the São Paulo participants.

Surface form	Consonant	Pause	Vowel
null	29	9	10
r	8	1	164
h	0	1	0
ɦ	61	7	3
rɦ	41	7	2
rj	1	0	0
rʃ	0	7	0
rʒ	1	0	0
rs	0	1	0
rV	24	2	6
ɾ	1	14	0
r	10	0	5
ɹ	48	20	20
ɹV	1	0	0
x	0	1	0
x ^w	1	0	0
ç	0	1	0
j	2	0	0
ʃ	0	7	0
ʒ	1	1	0
z	1	0	0
l	0	0	1
Total	230	79	211

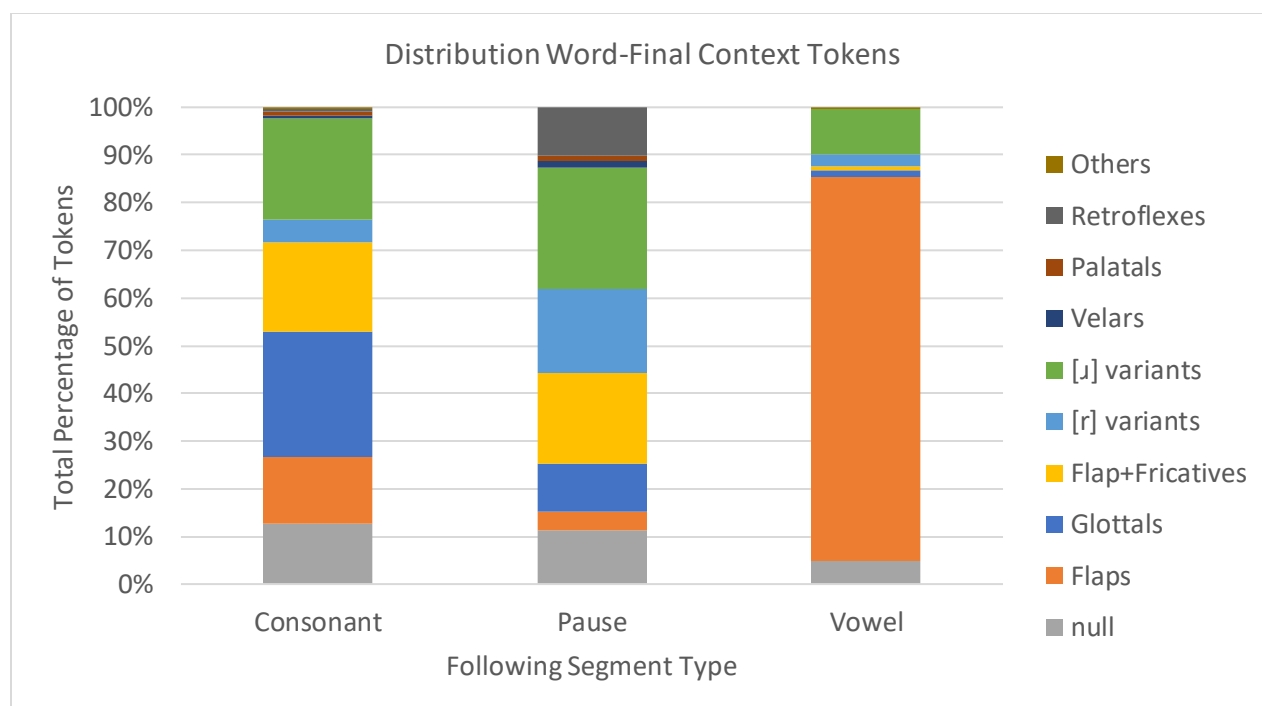


Figure 9.2: Proportion of word-final context surface forms separated by following segment type for the São Paulo participants.

In the table and figure above, it can be seen that flaps occurred before all three segment types, but were most frequent before vowels, comprising the majority of productions. Deletion also occurred before all three types, about 12.6% (29/230) of the time before consonants, 11.4% (9/79) of the time before pauses, and 4.7% (10/211) of the time before vowels. [ɹ] variants frequently occurred before all three following segment types as well. Before consonants and pauses, glottal fricatives, flap + fricative variants, and alveolar trill [r] variants were common, with all trills before pauses being devoiced. Lastly, retroflex fricatives were common before pauses as well. This similarity with the isolated tokens suggests that being utterance-final plays a role in the rhotic surfacing as a retroflex.

9.1.2. São Paulo syllable-final productions and deletions

Table 9.3 below gives the count of each surface form of the rhotic for syllable-final tokens in isolation and in context, with the third column for each surface form combining both isolation and context tokens.

Table 9.3: Counts of the surface forms of the rhotic in the syllable-final environment for the São Paulo participants.

Surface form	Isolation	Context	Overall
null	3	22	25
r	2	15	17
ɾ	0	2	2
h	1	3	4
ɦ	4	59	63
ɦV	0	1	1
rh	0	3	3
rɦ	18	56	74
rç	0	1	1
rs	0	1	1
rV	12	33	45
ɹ	1	4	5
r	17	9	26
ɹ	21	57	78
ɹV	0	1	1
x	0	2	2
x ^w	0	1	1
j	0	6	6
z	2	0	2
zV	0	1	1
θ	0	1	1
ɥ	0	1	1
Total	81	279	360

From the table above, it can be seen that deletion occurred in the syllable-final environment as well, occurring about 7.9% (22/279) of the time in context. Additionally, only three tokens had the rhotic deleted in isolation. There is quite a bit of variation in this

environment as well, though there are slightly less surface forms here than word-finally. Figure 9.3 below shows the proportion of surface forms in the syllable-final environment.

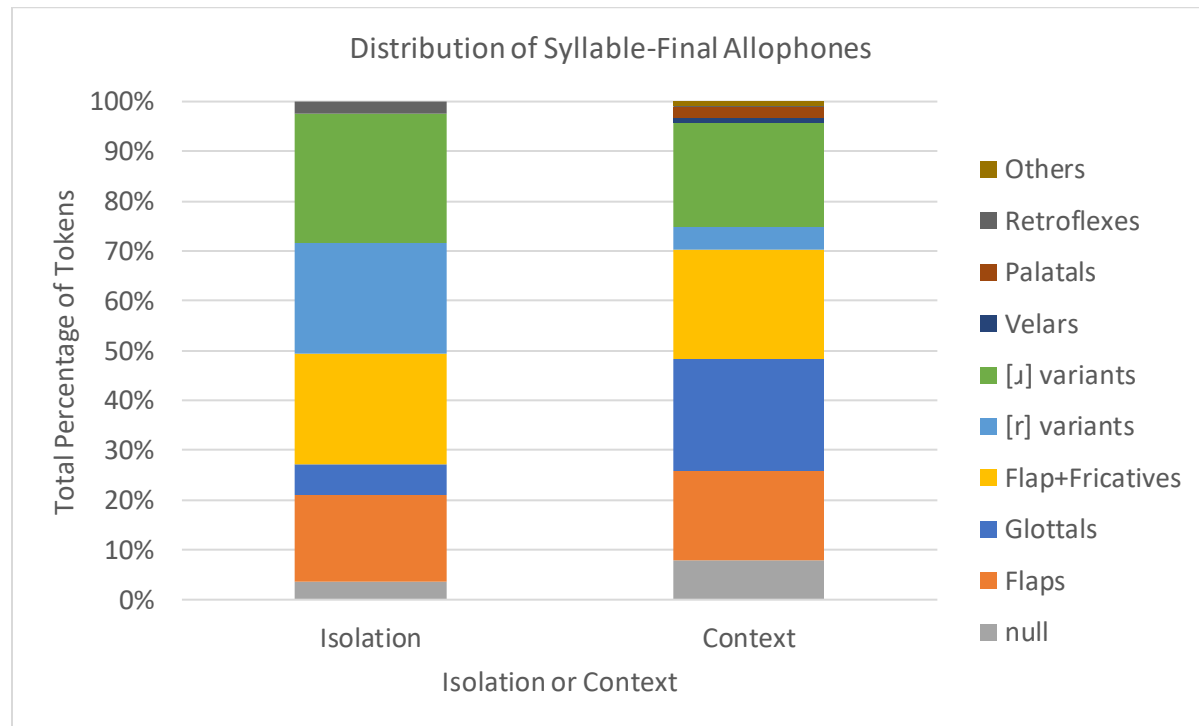


Figure 9.3: Proportion of surface forms in the syllable-final environment for the São Paulo participants.

In figure 9.3, it can be seen that the surface forms tend to be dominated by flaps, glottal fricatives, flap + fricative variants, [r] variants, and [ɹ] variants, with few other surface forms. Unlike with the word-final tokens, there does not appear to be much of a difference between isolation and context tokens here, although the isolation tokens do have less deletion and fewer glottal fricatives while context tokens have fewer trill [r] variants.

9.1.3. São Paulo word-initial productions and deletions

Table 9.4 below gives the count of each surface form of the rhotic for word-initial tokens in isolation and in context, with the third column for each surface form combining both isolation and context tokens.

Table 9.4: Counts of the surface forms of the rhotic in the word-initial environment for the São Paulo participants.

Surface form	Isolation	Context	Overall
null	0	7	7
h	49	30	79
ɦ	4	96	100
χ	0	1	1
ʁ	0	1	1
x	5	7	12
ç	0	6	6
ʃ	1	2	3
ʒ	1	0	1
Total	60	150	210

From the table above, it can be seen that deletion was not common word-initially, occurring only 7 times in context and not occurring at all in isolation. Furthermore, there is significantly reduced variation in this environment, with the surface forms being restricted to fricatives and no presence of the coronal consonants which were predominant in coda position. Figure 9.4 below shows the proportion of surface forms in the word-initial environment.

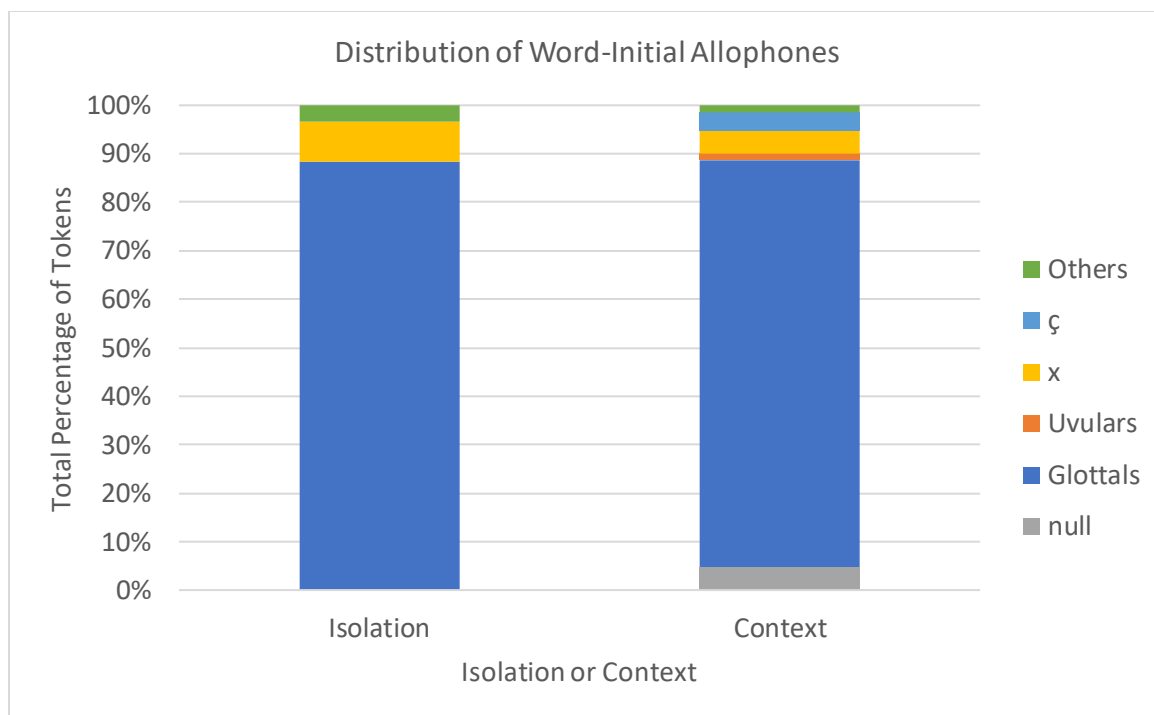


Figure 9.4: Proportion of surface forms in the word-initial environment for the São Paulo participants.

In figure 9.4, it can be seen that the surface forms here are quite different from what is in the coda environments. Glottal fricatives comprise the vast majority of productions word-initially both in isolation and in context. The remaining surface forms combined comprise about 12% of word-initial tokens.

Table 9.5 and figure 9.5 below give the count and distribution of each surface form of the rhotic for word-initial context tokens separated by preceding environment.

Table 9.5: Counts of the surface forms of the rhotic on word-initial context tokens separated by preceding environment for the São Paulo participants.

Surface form	Consonant	Pause	Vowel
null	6	0	1
h	8	19	3
ɦ	24	2	70
χ	1	0	0
ʁ	0	1	0
x	1	3	3
ç	0	6	0
ʃ	0	0	2
Total	40	31	79

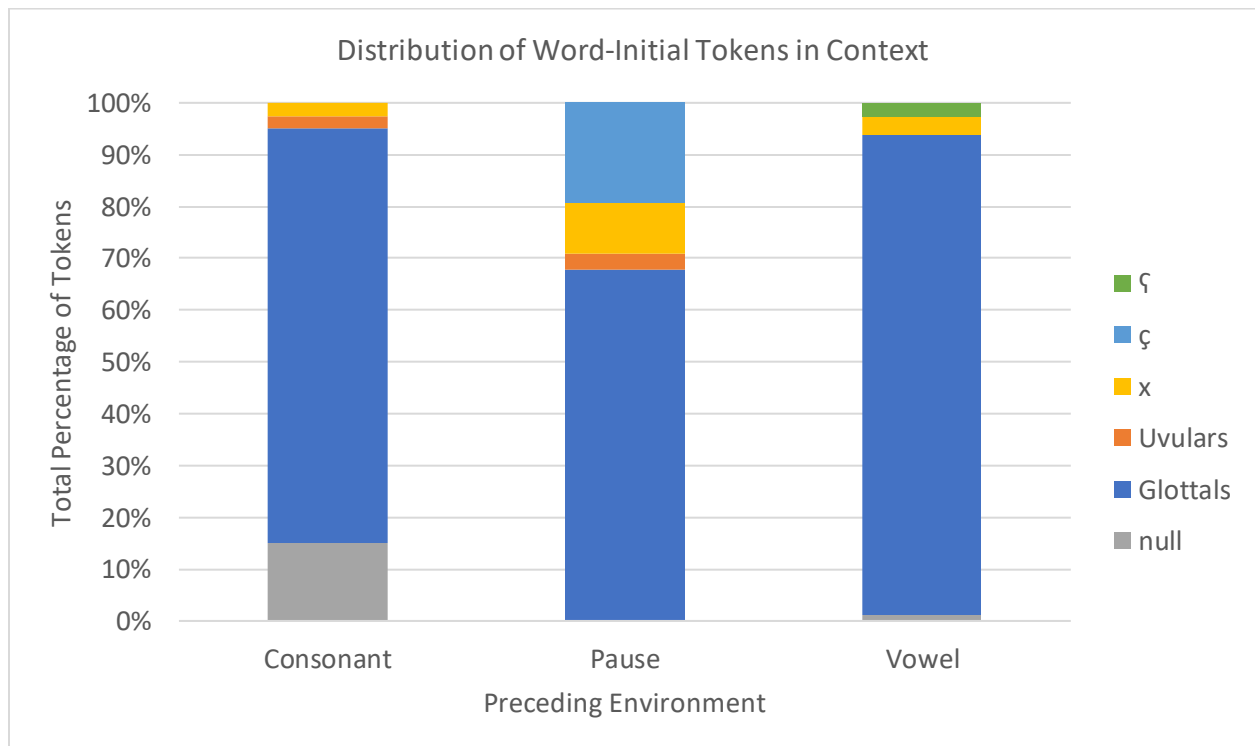


Figure 9.5: Proportion of word-initial context surface forms separated by preceding segment type for the São Paulo participants.

In the table and figure above, it can be seen that the glottal fricatives were the most common in word-initial context tokens regardless of the preceding segment type; however over 30% of tokens preceded by a pause surfaced as another fricative, with the voiceless palatal

fricative [ç] being the most common. Additionally, 6 of the 7 deleted tokens occurred after a consonant and one occurred after a vowel, with none occurring after a pause.

9.1.4. São Paulo intervocalic productions and deletions

Table 9.6 below gives the count of each surface form of the rhotic for intervocalic tokens in isolation and in context, with the third row under each surface form combining both isolation and context tokens.

Table 9.6: Counts of the surface forms of the rhotic in the intervocalic environment for the São Paulo participants.

Surface form	Isolation	Context	Overall
null	0	3	3
h	5	3	8
ɦ	30	103	133
ʁ	1	1	2
x	7	3	10
x ^w	1	0	1
ɣ	4	2	6
ɣ ^w	1	0	1
ç	1	1	2
j	0	3	3
ɥ	0	1	1
Total	50	120	170

From the table above, it can be seen that deletion was not common intervocalically, occurring only 3 times in context and not occurring at all in isolation. As was the case word-initially, there is significantly reduced variation in this environment, with no presence of the coronal consonants which were predominant in coda position. This suggests that the rhotic in onset position tends to be a fricative, particularly a glottal fricative, similar to what is found in Salvador, while coda position tends to have a coronal element, though other surface forms occur there as well. Additionally, there is greater variation in coda position than in onset

position in São Paulo, something that occurred with the Salvador participants as well. Figure 9.6 below shows the proportion of surface forms in the intervocalic environment.

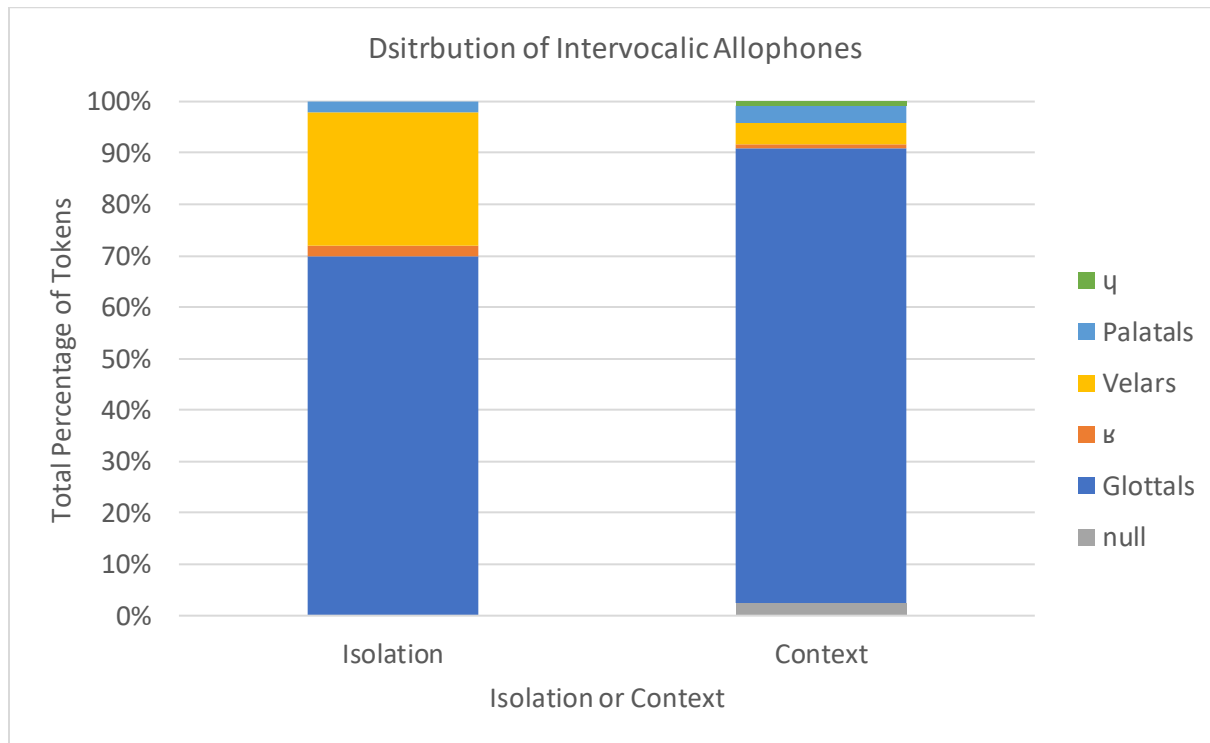


Figure 9.6: Proportion of surface forms in the intervocalic environment for the São Paulo participants.

In figure 9.6, it can be seen that glottal fricatives comprise the majority of surface forms in isolation and in context, however 30% of the isolated tokens were non-glottal, with velar fricatives being the predominant forms. By contrast, only about 10% of the context tokens were non-glottal. It is worth noting here that while variation is usually greater in context than in isolation, the opposite is occurring here: there is more variation in isolation than in context. Typically, the greater variation in context is due to the fact that there are many more factors occurring that can result in alterations in the surface form. This may be an exceptional circumstance, but further research would be needed to confirm this is the case.

9.1.5. Comparing São Paulo productions and deletions by environment

Lastly, figure 9.7 below compares the distribution of the surface forms across all four environments.

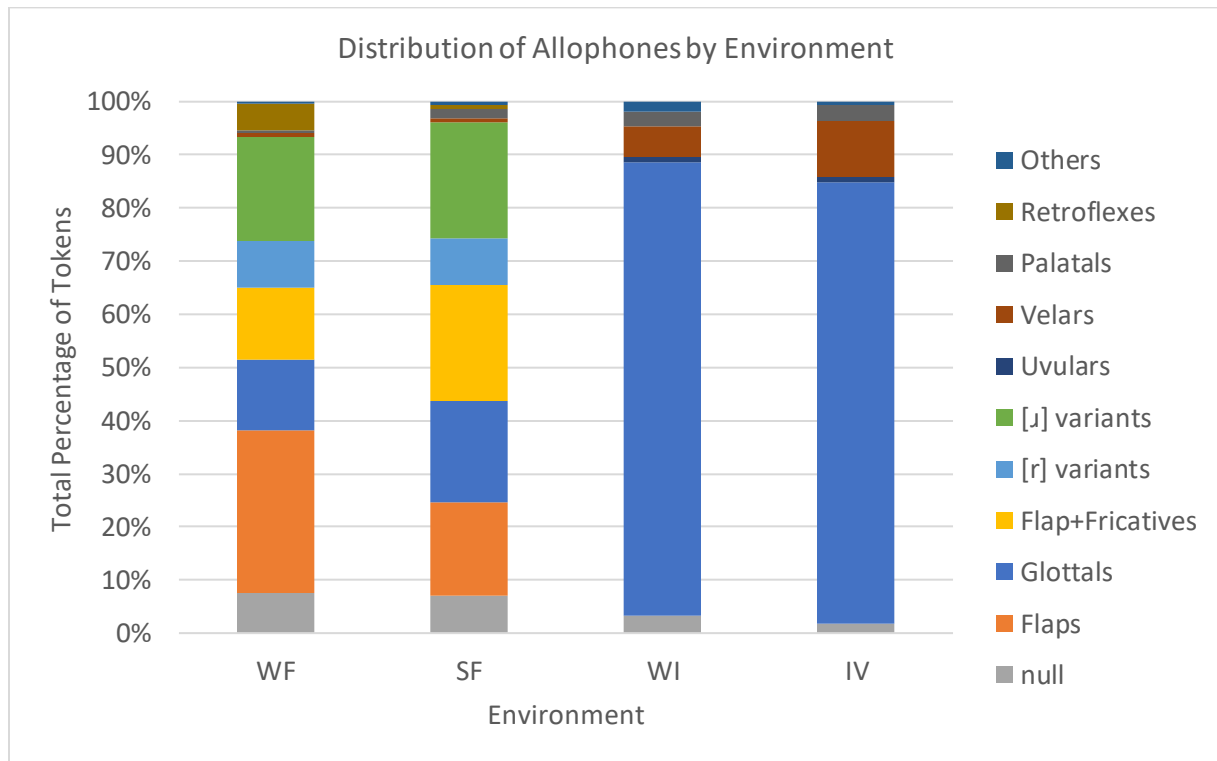


Figure 9.7: Proportion of surface forms of the rhotic by environment for the São Paulo participants.

Figure 9.7 confirms that glottal fricatives are the predominant form in onset position, while there is extensive variation, with a tendency for coronal consonants in coda position. Additionally, deletion appears to be a bit higher in coda position than in onset position, though this difference is not nearly as pronounced as it was in Salvador.

In conclusion, at first glance, the distributions of surface forms in São Paulo do not appear to differ greatly from Salvador in onset position (word-initial and intervocalic environments), while the distributions in coda position (word-final and syllable-final

environments) appear to have quite a few differences. Furthermore, it appears that deletion is substantially less common in São Paulo than in Salvador. Section 2 presents the statistical tests that determine to what extent these differences are significant.

9.2. Comparing São Paulo and Salvador data

This section breaks down the differences in deletion rate, oral production rate, and distribution of surface forms between the São Paulo and Salvador participants. The environments tested for differences in deletion rate were word-final and word-initial in isolation and in context, syllable-final and intervocalic in context, and overall deletion rate across all tokens. Word-final context and word-initial context tokens were also subdivided into the three possible preceding and following segment type categories (vowel, consonant, and pause) and intervocalic context tokens were subdivided based on whether or not the syllable following the rhotic was stressed. However, these subsets are only considered if the test for all the tokens in a given environment is significant. Intervocalic isolation tokens were not considered for differences, as no participants deleted any of them. Syllable-final (12 in Salvador, 3 in São Paulo) and word-initial (6 in Salvador, 0 in São Paulo) isolation tokens were not considered for differences, as they had the majority of participants in both cities not delete in any tokens, making it difficult for tests on these deletion rates to be meaningful. The same environments were considered for oral production rate and distribution of surface forms as for deletion, with the addition of the syllable-final, word-initial, and intervocalic isolation tokens. The isolation tokens were added in here because of the high number of oral surface forms relative to the number of participants in comparison to Salvador, suggesting that differences likely exist. The

significance level for all the tests is $\alpha=0.05$. Significant p -values are indicated with an asterisk (*) and the significant tests are bolded.

This section is organized as follows: Subsection 1 reviews the differences in deletion rate between the two cities. Subsection 2 examines the differences in the oral production rates. Subsection 3 evaluates the differences in surface form distributions. Lastly, subsection 4 discusses the differences found between the São Paulo and Salvador participants.

9.2.1. Comparing São Paulo and Salvador deletion rates

All tests for differences in deletion rate between São Paulo and Salvador were two independent sample t -tests on the arcsine transformed deletion rates of individual participants split into their respective categories as the two samples. All t -tests were one-tailed with the alternative hypothesis that the participants from Salvador deleted more. The basis for this is that given the history of the two cities and their current demographic situations discussed in chapters 1 and 3, African influences are expected to be stronger in Salvador than in São Paulo. Since rhotic deletion has been historically linked to African influences in Brazilian Portuguese, as discussed in chapter 2, the hypothesis is that deletion is stronger in Salvador than São Paulo, particularly in coda positions.

To determine whether to use the equal variance (Student's t -test) or unequal variance (Welch's t -test) assumption for a particular test, an F -test was conducted. If the F -test gave a significant result ($p \leq 0.05$), the unequal variance assumption was used. Note that if no participants of a particular demographic group deleted the rhotic in a certain environment, the F -statistic could not be calculated due to a variance of zero for that group. In this case, the unequal variance assumption was used as well. Otherwise, the equal variance assumption was

used. Tests which used the unequal variance assumption are indicated with footnotes that give the p -value of the F -test and the degrees of freedom of the t -test. The significance level for all the tests is $\alpha=0.05$. Significant p -values are indicated with an asterisk (*) and the significant tests are bolded in their tables.

Due to the significant differences in deletion rate by race among the Salvador participants, a subset of ten participants from Salvador were used for the deletion rate comparison with São Paulo. The participants selected were those which allowed for the best possible demographic match between the datasets. The flexibility in selecting participants was limited by the fact that only 4 participants from São Paulo identified as black and only one participant from São Paulo identified as lower class, both of which were characteristics of the majority of participants from Salvador. The demographic characteristics of each sample are given in table 9.7 below.

Table 9.7: Demographic characteristics of the São Paulo participants and the subset of Salvador participants used in these deletion rate tests.

Demographic Characteristic	São Paulo	Salvador
Race	4 black (3 <i>pardo</i> , 1 <i>preto</i>), 6 white	4 black (1 <i>pardo</i> , 3 <i>preto</i>), 6 white
Socioeconomic class	1 lower class, 9 middle class	1 lower class, 9 middle class
Gender	4 male, 6 female	4 male, 6 female
Average approximate birth year	1990.1	1989.5
Median approximate birth year	1991	1994.5
Youngest approximate birth year	2004	2001
Oldest approximate birth year	1967	1964

From the table above, it can be seen that the gender and socioeconomic class variables were identical between the two groups. In order to keep the variables balanced, the black

participants have different compositions of *preto* and *pardo* participants. However, as shown in chapter 7, since there were no differences in deletion rate between *preto* and *pardo* participants from Salvador, this should not confound any of the results in this section. Lastly, while the average approximate birth year for the two cities is about the same, the range of years for São Paulo is about three years younger than Salvador on each end, as is the median age. Overall, combining these similar demographics with the findings of section 1 of chapter 7, the demographic variables should not be confounding any differences in deletion rate found in this chapter, and the differences can be attributed to the geographic difference alone. The demographics of the participants used in these comparisons can be found in tables A1.1 and A1.2 in appendix A1.

The average deletion rates, *p*-values, and *t*-scores of the tests conducted are given in table 9.8 below. The *t*-scores given represent the Salvador mean minus the São Paulo mean for a given environment.

Table 9.8: Average deletion rates, p -values, and t -scores for the difference between Salvador and São Paulo participants by environment.

Environment	Salvador average	São Paulo average	p -value	t -score
Word-final isolation	0.269	0.019	$p = 0.002^*$	$t = 3.528^7$
Word-final context	0.542	0.092	$p = 1.62 \times 10^{-5}^*$	$t = 5.494$
Word-final before a consonant	0.660	0.126	$p = 1.79 \times 10^{-5}^*$	$t = 5.445$
Word-final before a vowel	0.484	0.047	$p = 1.93 \times 10^{-4}^*$	$t = 4.869^8$
Word-final before a pause	0.361	0.114	$p = 0.010^*$	$t = 2.527$
Syllable-final context	0.217	0.079	$p = 0.003^*$	$t = 3.121$
Word-initial context	0.08	0.047	$p = 0.150$	$t = 1.069$
Intervocalic context	0.102	0.025	$p = 0.016^*$	$t = 2.324$
Intervocalic context followed by stress	0.043	0	$p = 0.088$	$t = 1.485^9$
Intervocalic context not followed by stress	0.185	0.06	$p = 0.024^*$	$t = 2.114$
Overall deletion rate	0.294	0.061	$p = 1.87 \times 10^{-5}^*$	$t = 5.424$

From table 9.8 above, it can be seen that the Salvador participants are deleting the rhotic significantly more than the São Paulo participants in the majority of context environments, word-final, syllable-final, and intervocalic, as well as on word-final isolation

⁷ F -test p -value: 0.004. Unequal variance test degrees of freedom: 11.143.

⁸ F -test p -value: 0.033. Unequal variance test degrees of freedom: 12.726.

⁹ F -test could not be done due to no deletion on intervocalic context tokens followed by stress by São Paulo participants. Unequal variance test degrees of freedom: 9

tokens. Regarding the subsets of the word-final and intervocalic context tokens, the Salvador participants delete significantly more in all subsets of the word-final tokens, but only in the subset of intervocalic tokens where the rhotic is not followed by stress. Furthermore, the Salvador participants are deleting significantly more in the overall corpus. Given that the majority of the tests were significant, this is unsurprising. This consistent difference between the two groups, even when controlled for demographics, suggests that deletion is indeed more common in Salvador than in São Paulo and that African influences may possibly be stronger in Salvador as well.

9.2.2. Comparing São Paulo and Salvador oral production rates

The tests for differences in the rate of oral production forms were conducted in a similar manner to the tests for deletion rate. All tests for differences in oral production rate between São Paulo and Salvador were two independent sample *t*-tests on the arcsine transformed oral production rates of individual participants split into their respective categories as the two samples. All the *t*-tests were one-tailed, as the distributions of the coda environments for São Paulo presented in section 1 of this chapter demonstrated a high tendency for non-glottal forms. With this, it is expected that the oral production rate will be higher in coda position in São Paulo than in Salvador. Additionally, although glottal surface forms were the most common in onset position (word-initial and intervocalic) in both cities, the ten participants from São Paulo produced more oral forms in these environments than the entire corpus of Salvador, suggesting that the São Paulo participants likely have a higher rate of oral production in these environments as well. Note that these tests included all participants from Salvador, as there was no indication in chapter 8 that demographic variables played a major role in oral production

rate. Furthermore, when conducting these tests with only the subset used on the deletion rate tests, the significant tests were the same as with the entire sample.

The average oral production rates, p -values, and t -scores of the tests conducted are given in table 9.9 below. The t -scores given represent the Salvador mean minus the São Paulo mean for a given environment.

Table 9.9: Average oral production rates, p -values, and t -scores for the difference between Salvador and São Paulo participants by environment.

Environment	Salvador average	São Paulo average	p -value	t -score
Word-final isolation	0.161	0.863	$p = 5.35 \times 10^{-12*}$	$t = -15.169$
Word-final context	0.232	0.770	$p = 1.69 \times 10^{-11*}$	$t = -14.159$
Word-final before a consonant	0.058	0.609	$p = 2.38 \times 10^{-6*}$	$t = -9.862^{10}$
Word-final before a vowel	0.448	0.939	$p = 1.16 \times 10^{-10*}$	$t = -9.460^{11}$
Word-final before a pause	0.153	0.786	$p = 3.77 \times 10^{-10*}$	$t = -11.705$
Syllable-final isolation	0.093	0.9	$p = 3.44 \times 10^{-7*}$	$t = -10.072^{12}$
Syllable-final context	0.111	0.694	$p = 1.64 \times 10^{-6*}$	$t = -10.103^{13}$
Word-initial isolation	0.005	0.117	$p = 0.076$	$t = -1.564^{14}$
Word-initial context	0.017	0.113	$p = 4.94 \times 10^{-6*}$	$t = -6.064$
Word-initial after a consonant	0.007	0.05	$p = 0.223$	$t = -0.798^{15}$
Word-initial after a vowel	0.011	0.063	$p = 0.045^*$	$t = -1.872^{16}$
Word-initial after a pause	0.048	0.325	$p = 0.003^*$	$t = -3.514^{17}$
Intervocalic isolation	0.006	0.3	$p = 0.010^*$	$t = -2.799^{18}$
Intervocalic context	0.024	0.092	$p = 0.053$	$t = -1.774^{19}$
Overall oral rate	0.135	0.602	$p = 2.12 \times 10^{-16*}$	$t = -27.297$

¹⁰ F -test p -value: 0.033. Unequal variance test degrees of freedom: 10.953.

¹¹ F -test p -value: 0.004. Unequal variance test degrees of freedom: 39.203.

¹² F -test p -value: 0.044. Unequal variance test degrees of freedom: 11.065.

¹³ F -test p -value: 2.32×10^{-4} . Unequal variance test degrees of freedom: 9.953.

¹⁴ F -test p -value: 6.17×10^{-13} . Unequal variance test degrees of freedom: 9.183.

¹⁵ F -test p -value: 7.21×10^{-6} . Unequal variance test degrees of freedom: 9.662.

¹⁶ F -test p -value: 0.002. Unequal variance test degrees of freedom: 10.225.

¹⁷ F -test p -value: 0.004. Unequal variance test degrees of freedom: 10.361.

¹⁸ F -test p -value: 1.8×10^{-16} . Unequal variance test degrees of freedom: 9.107.

¹⁹ F -test p -value: 0.021. Unequal variance test degrees of freedom: 10.796.

From table 9.9 above, it can be seen that the São Paulo participants are producing oral surface forms at a significantly higher rate in the majority of context environments, word-final, syllable-final, and word-initial, as well as on word-final, syllable-final, and intervocalic isolation tokens. Regarding the subsets of the word-final and word-initial context tokens, the São Paulo participants produce oral surface forms at a significantly higher rate on all subsets of word-final tokens, but only on the subsets of word-initial tokens that occur after vowels and pauses. Additionally, the São Paulo participants are producing oral forms at a significantly higher rate in the overall corpus. From the distributions seen in section 1 of this chapter, the high number of significant tests here is not surprising. Furthermore, the lower deletion rates of the São Paulo participants open the door for a larger proportion of oral surface forms to be produced as well. This result, combined with the result from the previous subsection, suggest that there is significantly less lenition occurring with the rhotic phoneme in São Paulo than in Salvador.

9.2.3. Comparing São Paulo and Salvador surface form distributions

While the last two subsections addressed the fact that deletion is more frequent in Salvador and oral surface forms are more frequent in São Paulo, they do not address whether the surface forms have different distributions in the two cities. This subsection tests for differences in distributions using chi-squared tests. For the tests, surface forms were combined into categories similar to those in section 1 of this chapter in order to help keep the counts in each box of the contingency table at a more reasonable number. The contingency tables used for these tests are tables A2.55 to A2.71 in appendix A2. Note that these tests included all participants from Salvador.

9.2.3.1. Comparison of word-final surface form distributions of São Paulo and Salvador

Figure 9.8 below compares the proportion of surface forms in the word-final environment for the São Paulo and Salvador participants.

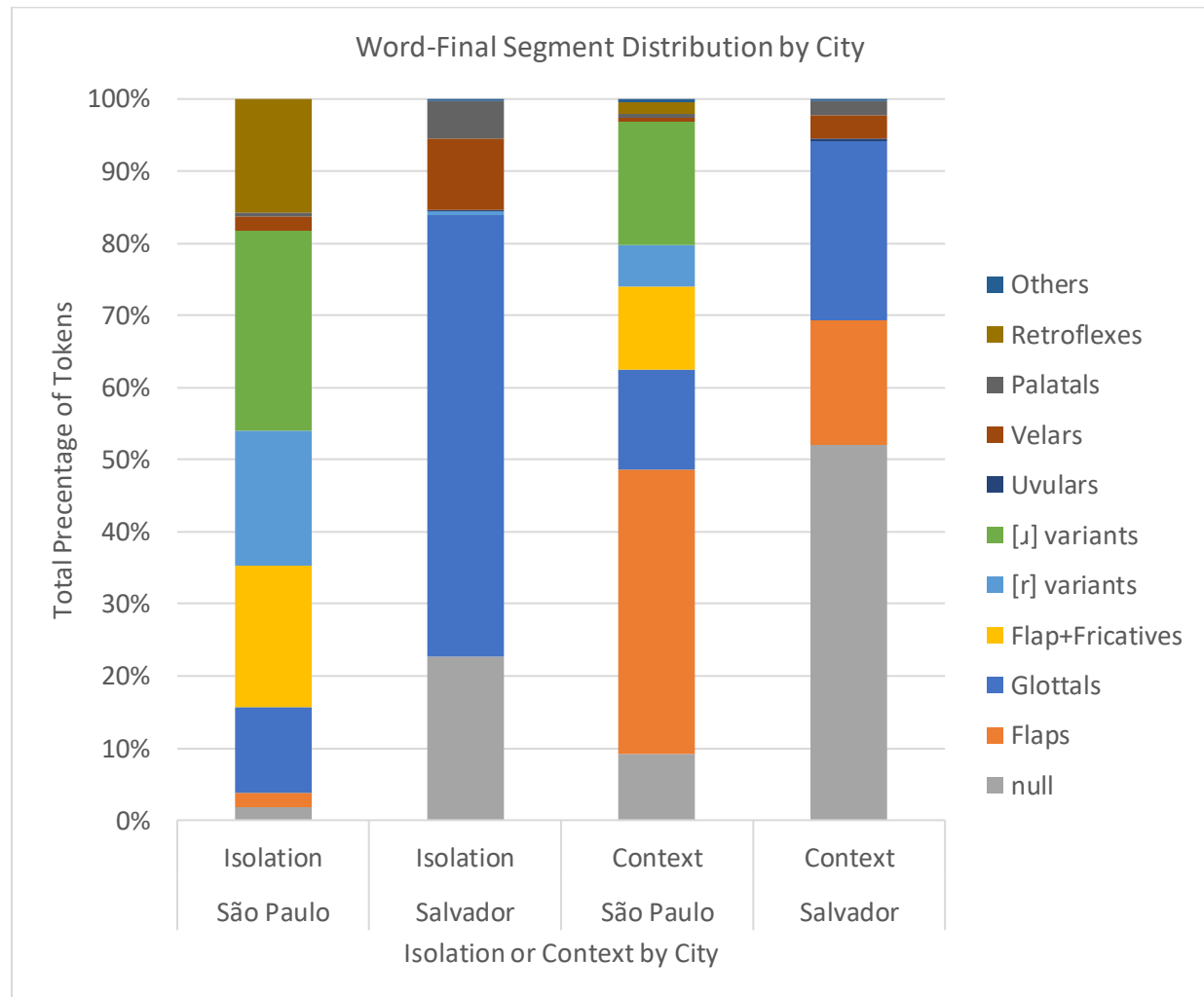


Figure 9.8: Proportion of surface forms in the word-final environment for the São Paulo and Salvador participants.

From the figure above, it can be seen that deletion and glottal fricatives are most common in word-final isolation tokens in Salvador while flap + fricative variants, alveolar trill [r] variants, [ɹ] variants, and retroflexes are most common in São Paulo. Regarding the context tokens, flaps are common in both, but flaps comprise a much higher proportion of the tokens in

São Paulo than in Salvador. Furthermore, Salvador has more deletion and glottal fricatives in context while São Paulo has flap + fricative variants, alveolar trill [r] variants, and [ɹ] variants in addition to deletion and glottal fricatives.

Both the distributions of isolation ($p < 2.2 \times 10^{-16}$, $\chi^2 = 462.96$) and context ($p < 2.2 \times 10^{-16}$, $\chi^2 = 500.08$) tokens in the word-final environment are significantly different. This is a strong indication that the distribution of surface forms is different both in isolation and in context. Given the prominence of flap + fricative variants, alveolar trill [r] variants, and [ɹ] variants in coda position, as well as retroflexes in isolation tokens in São Paulo, this result is not surprising. Of these categories, only [r] variants occurred in Salvador, and only one participant produced this surface form a total of three times in isolation. The contingency tables for these tests are A2.55 and A2.56 in appendix A2.

Given the significance of the word-final context token surface form distributions, it is important to break this environment down by following segment type. Figure 9.9 below compares the proportion of surface forms in the word-final environment for the São Paulo and Salvador participants.

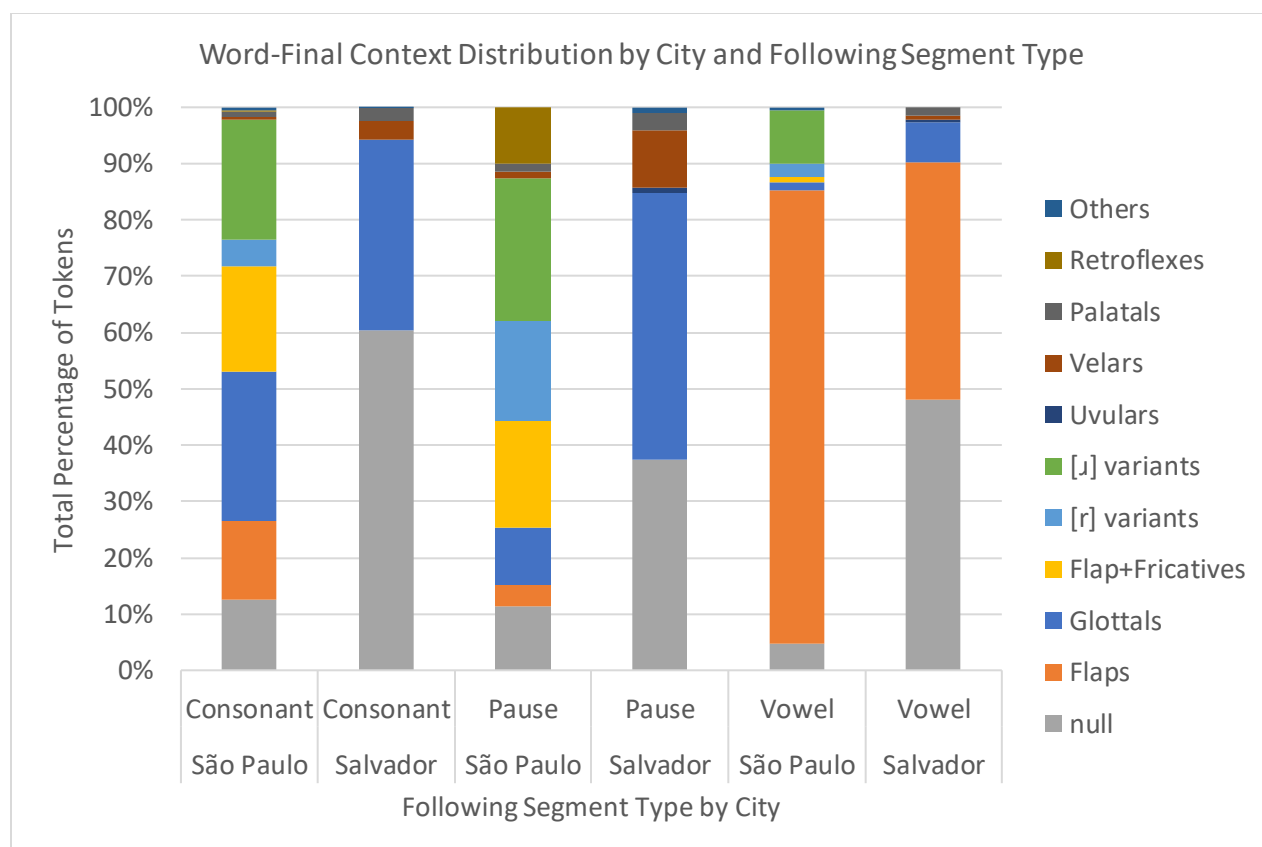


Figure 9.9: Proportion of surface forms on word-final context tokens for the São Paulo and Salvador participants separated by following segment type.

From the figure above, it can be seen Salvador participants tend to delete the rhotic or produce a glottal fricative before consonants and pauses while they tend to delete or produce a flap before a vowel. While flaps are much more common before vowels in São Paulo than in Salvador, the total percentage of tokens that are either flaps or deleted is similar. Before consonants and pauses, São Paulo participants demonstrate more variation with glottal fricatives, flaps, flap + fricative variants, alveolar trill [r] variants, and [ɹ] variants common. Additionally, retroflexes are common before pauses in the São Paulo participants.

The distributions were significantly different in all subsets of word-final context tokens (before a consonant: $p < 2.2 \times 10^{-16}$, $\chi^2 = 293.83$; before a pause: $p < 2.2 \times 10^{-16}$, $\chi^2 = 192.82$;

before a vowel: $p = 4.26 \times 10^{-15}$ *, $\chi^2 = 85.254$). This is a strong indication that the distribution of surface forms is different in word-final context tokens regardless of the following segment type. Given the apparent differences between the two cities in figure 9.9, these results are not surprising. Therefore, it can be concluded that a clear difference in the distribution of surface forms between São Paulo and Salvador exists in the word-final environment regardless of whether the token is in isolation or in context or what segment type follows the rhotic. The contingency tables for these tests are A2.57-A2.59 in appendix A2.

9.2.3.2. Comparison of syllable-final surface form distributions of São Paulo and Salvador

Figure 9.10 below compares the proportion of surface forms in the syllable-final environment for the São Paulo and Salvador participants.

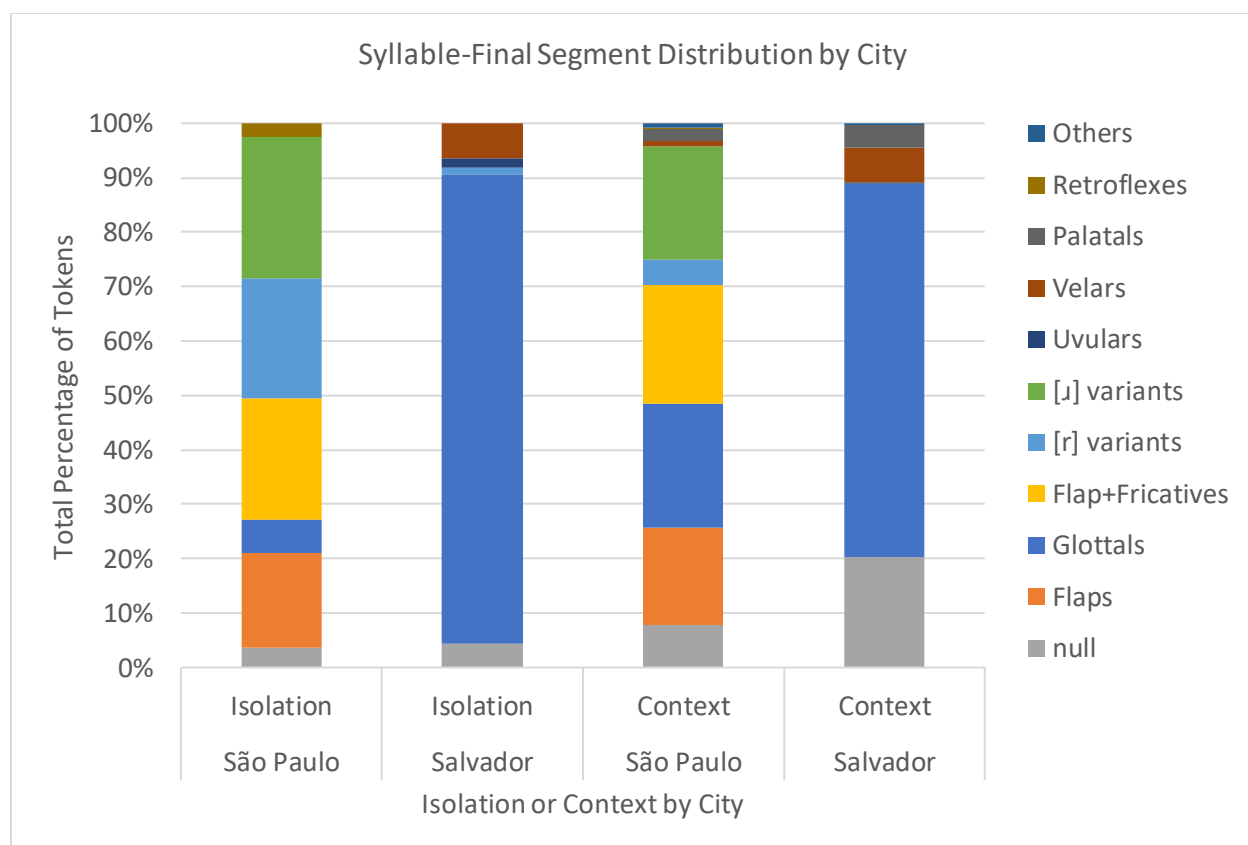


Figure 9.10: Proportion of surface forms in the syllable-final environment for the São Paulo and Salvador participants.

From the figure above, it can be seen that while glottal fricatives comprise the bulk of syllable-final allophones in Salvador, they comprise relatively few productions in São Paulo. As was the case in the word-final environment, there is extensive variation in São Paulo, with flaps, flap + fricative variants, alveolar trill [r] variants, and [ɺ] variants, as well as glottal fricatives.

Both the distributions of isolation ($p < 2.2 \times 10^{-16}$, $\chi^2 = 302.31$) and context ($p < 2.2 \times 10^{-16}$, $\chi^2 = 676.91$) tokens in the syllable-final environment are significantly different. This is a strong indication that the distribution of surface forms is different both in isolation and in context. Once again, with the variety of surface forms seen from the São Paulo participants relative to the Salvador participants, this is not particularly surprising. Furthermore, no

productions from Salvador fell into the categories of flaps, flap + fricative variants, alveolar trill [r] variants, or [ɹ] variants. The contingency tables for these tests are A2.60 and A2.61 in appendix A2.

9.2.3.3. Comparison of word-initial surface form distributions of São Paulo and Salvador

Figure 9.11 below compares the proportion of surface forms in the word-initial environment for the São Paulo and Salvador participants.

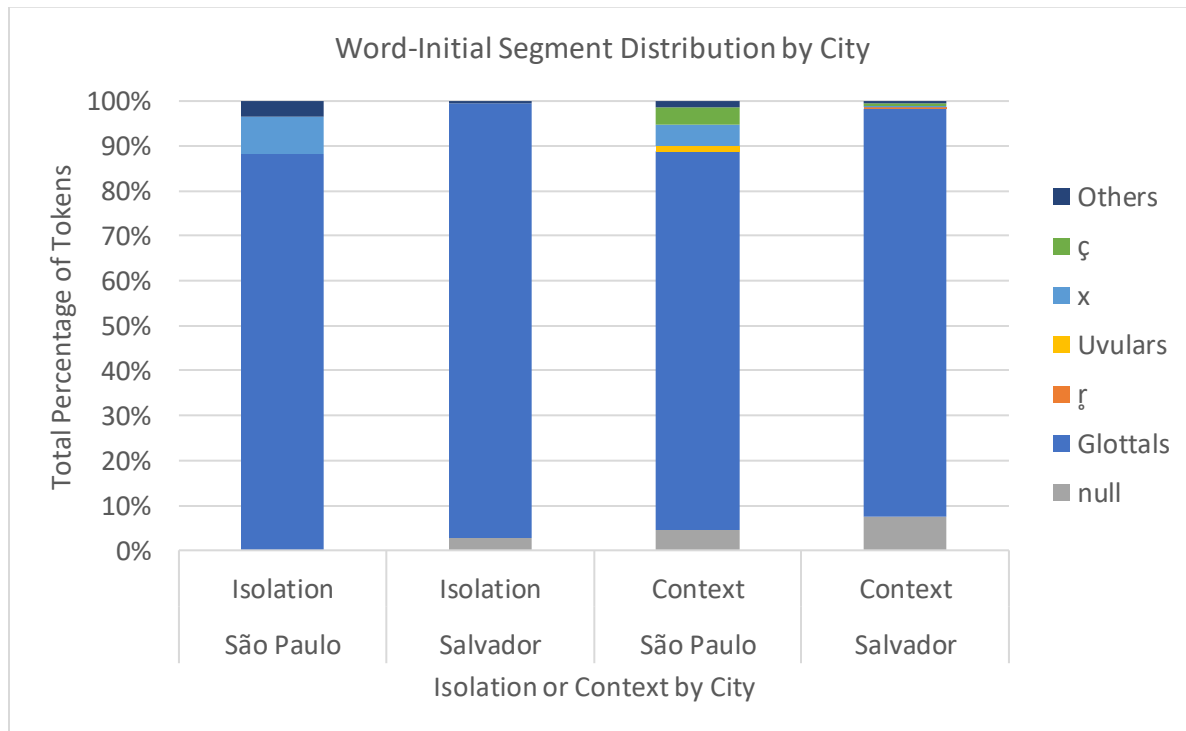


Figure 9.11: Proportion of surface forms in the word-initial environment for the São Paulo and Salvador participants.

From the figure above, it can be seen that regardless of the city or whether the token is in isolation or in context, the preferred form in the word-initial environment is a glottal fricative. In fact, only one production in isolation for Salvador was not a glottal fricative. Deletion occurred in isolation and in context for Salvador, but it only occurred in context for São Paulo.

Additionally, there is a larger proportion of oral surface forms in São Paulo both in isolation and in context.

Both the distributions of isolation ($p = 3.06 \times 10^{-5}$, $\chi^2 = 20.789$) and context ($p = 2.20 \times 10^{-6}$, $\chi^2 = 34.176$) tokens in the syllable-final environment are significantly different. This is a strong indication that the distribution of surface forms is different both in isolation and in context. Given the preference for glottal forms regardless of the city, this result is not exactly something expected just by looking at figure 9.11. However, it should also be noted that although the proportion of oral surface forms is clearly higher in São Paulo, only 7 oral surface forms occurred in isolation and 17 occurred in context for São Paulo participants. Therefore, although these tests did take all surface forms into consideration, the lack of forms other than glottal means the results of these tests should be taken with a grain of salt. The contingency tables for these tests are A2.62 and A2.63 in appendix A2. Nevertheless, given the significant result on word-initial context tokens, the tokens were separated by preceding segment type and tested for differences between the two cities. Figure 9.12 below compares the proportion of surface forms in the word-initial environment for the São Paulo and Salvador participants separated by preceding segment type.

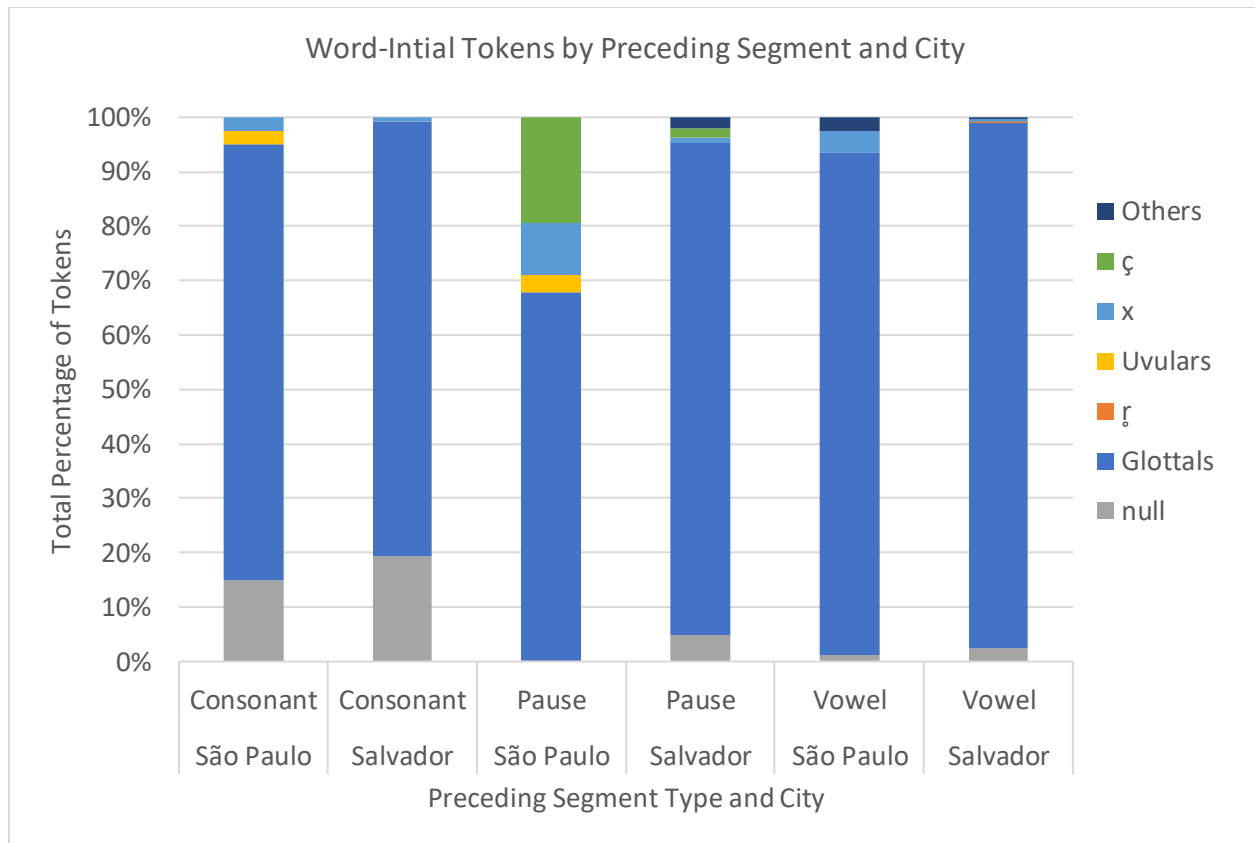


Figure 9.12: Proportion of surface forms on word-initial context tokens for the São Paulo and Salvador participants separated by preceding segment type.

From the figure above, it can be seen that over 30% of tokens preceded by a vowel are oral surface forms in São Paulo, while Salvador has a much lower percentage at about 5%. It is also the case that there appear to be a larger proportion of oral surface forms in São Paulo on tokens preceded by a consonant and a vowel than in Salvador, though the difference is much less pronounced.

The tests for tokens preceded by a pause ($p = 1.08 \times 10^{-4}$ *, $\chi^2 = 23.338$) and tokens preceded by a vowel ($p = 0.016$ *, $\chi^2 = 10.336$) were significant. However, the test for tokens preceded by a consonant was not ($p = 0.125$, $\chi^2 = 4.159$). Therefore, the difference between the distributions of surface forms in the word-initial environment is not as prominent as in the coda

environments. Furthermore, given the relatively small portion of oral surface forms the significance of these tests may be overstated. The contingency tables for these tests are A2.64-A2.66 In appendix A2.

9.2.3.4. Comparison of intervocalic surface form distributions of São Paulo and Salvador

Figure 9.13 below compares the proportion of surface forms in the intervocalic environment for the São Paulo and Salvador participants.

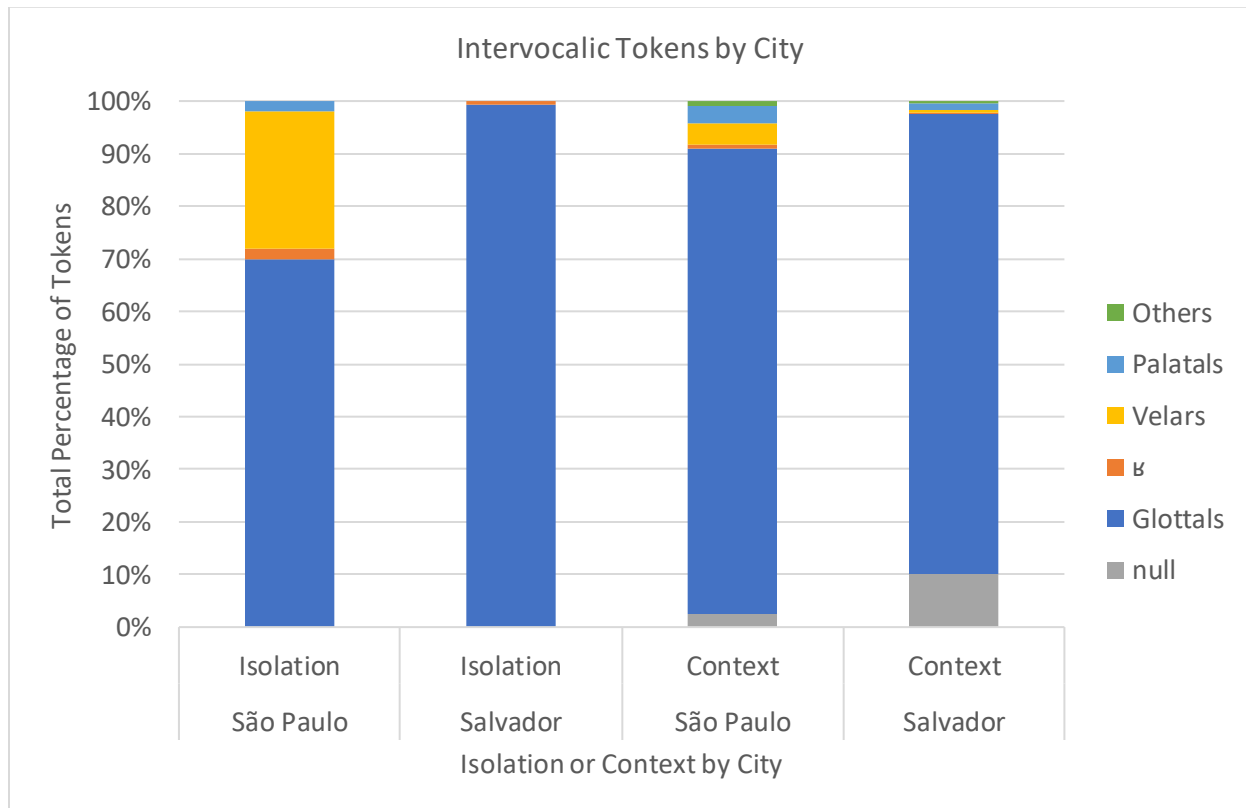


Figure 9.13: Proportion of surface forms in the intervocalic environment for the São Paulo and Salvador participants.

From the figure above, it can be seen that glottal fricatives are the most common surface form intervocalically for both cities whether in isolation or in context. Individuals from neither city deleted in isolation, though some deletion occurred in context for individuals from both

cities. There were very few oral surface forms in Salvador, while oral surface forms comprised nearly 10% of tokens in context and about 30% of tokens in isolation in São Paulo.

Both the distributions of isolation ($p = 1.42 \times 10^{-11}^*$, $\chi^2 = 53.519$) and context ($p = 0.015^*$, $\chi^2 = 12.369$) tokens in the intervocalic environment are significantly different. Given the large proportion of oral surface forms in isolation in São Paulo, the result for isolated tokens is not surprising. However, there were less obvious differences between the cities on context tokens in figure 9.13, meaning the result of this test was not necessarily expected by looking at the figure alone. However, it should also be noted that although the proportion of oral surface forms is clearly higher in São Paulo, only 15 oral surface forms occurred in isolation and 11 occurred in context for São Paulo participants. Therefore, although these tests did take all surface forms into consideration, the lack of forms other than glottal means the results of these tests should be taken with a grain of salt. The contingency tables for these tests are A2.67 and A2.68 in the appendix A2. Nevertheless, given the significant result on intervocalic context tokens, the tokens were separated by whether or not stress occurred after the rhotic and tested for differences between the two cities. Figure 9.14 below compares the proportion of surface forms in the intervocalic environment for the São Paulo and Salvador participants separated by whether or not stress occurred post-rhotic.

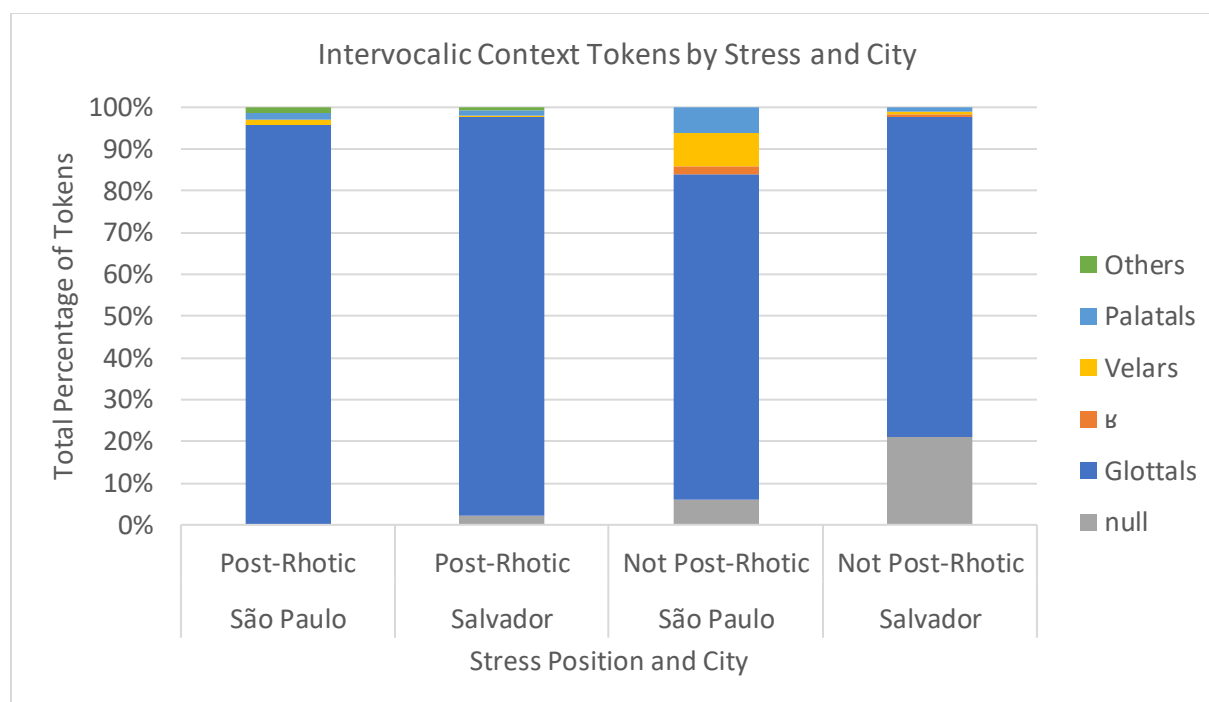


Figure 9.14: Proportion of surface forms on intervocalic context tokens for the São Paulo and Salvador participants separated by whether or not stress occurred post-rhotic.

From the figure above, it can be seen that very few surface forms, around 5% or less, are oral productions for Salvador and for tokens with post-rhotic stress in São Paulo. However, the proportion of oral productions on tokens without post-rhotic stress in São Paulo is about 15%.

The test on tokens where stress occurred post-rhotic was not significant ($p = 0.780$, $\chi^2 = 1.090$), while the test on tokens where stress did not occur post-rhotic was significant ($p = 0.006^*$, $\chi^2 = 12.408$). Therefore, as was the case word-initially, the difference between the distributions of surface forms in the intervocalic environment is not as prominent as in the coda environments. Furthermore, given the relatively small portion of oral surface forms the significance of the test on tokens without post-rhotic stress may be overstated. The contingency tables for these tests are A2.69 and A2.70 in appendix A2.

9.2.3.5. Comparison of overall corpus surface form distributions of São Paulo and Salvador

Figure 9.15 below compares the proportion of surface forms in the overall corpus for the São Paulo and Salvador participants.

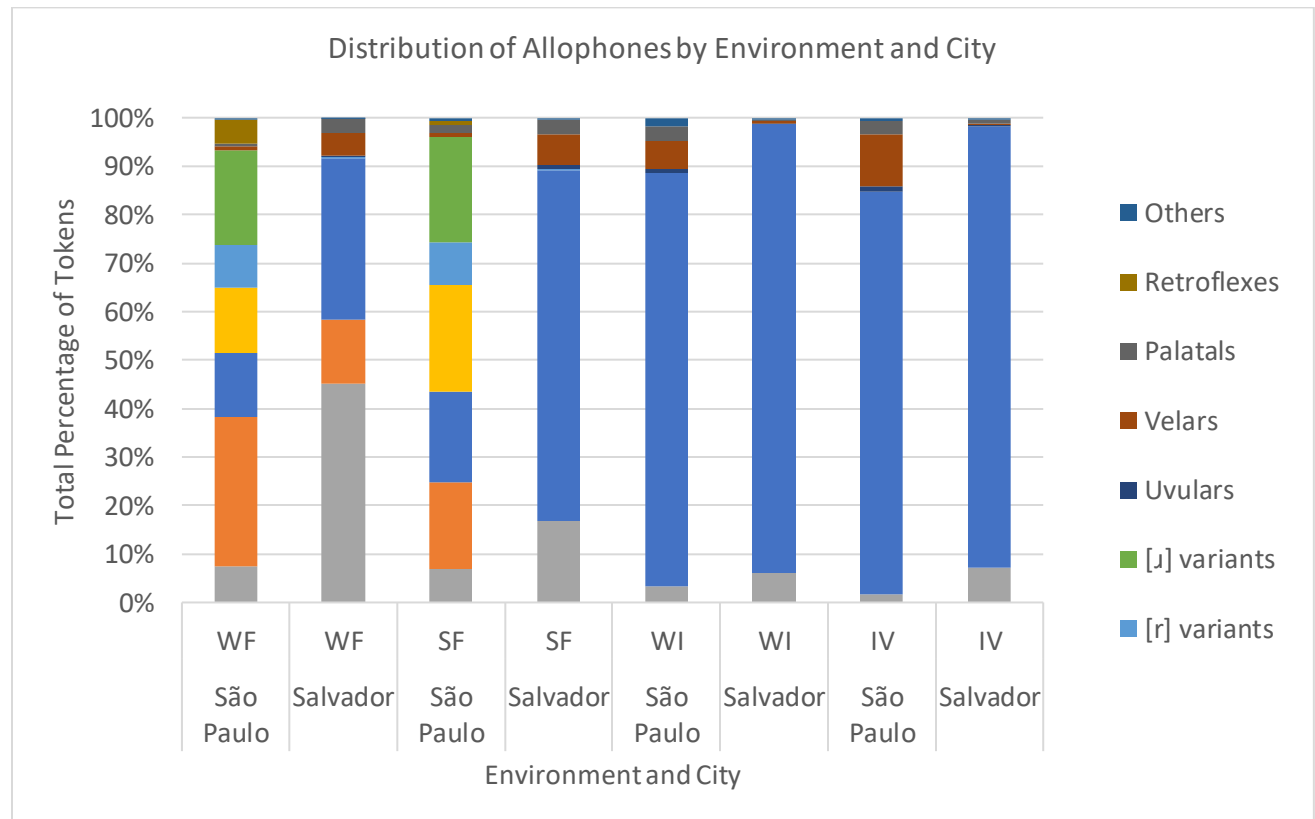


Figure 9.15: Proportion of surface forms in the corpus for the São Paulo and Salvador participants.

From the figure above, it can be seen that, as the tests indicated, the distributions in the word-initial and intervocalic environments are more similar between the two cities than in the word-final and syllable-final environments. Thus, the samples from the two cities have a fair amount in common in terms of rhotic production in onset position, but they differ greatly in coda position. A final chi-squared test was conducted by summing the productions across all environments in both cities into the contingency table given in table A2.71 in appendix A2. This test was significant ($p < 2.2 \times 10^{-16}$, $\chi^2 = 1770.2$). Considering that most of the tests presented in

this section were significant, this result was expected. Additionally, considering the fact that coda tokens are more common in the corpus (104/142 per participant), it would make sense for these significant differences to carry over into the overall corpus.

As a final consideration, figure 9.16 below gives the distribution of surface forms of the overall corpus separated by race (black versus white) and city. This figure helps visualize the effects of racial identification versus geography in the corpus.

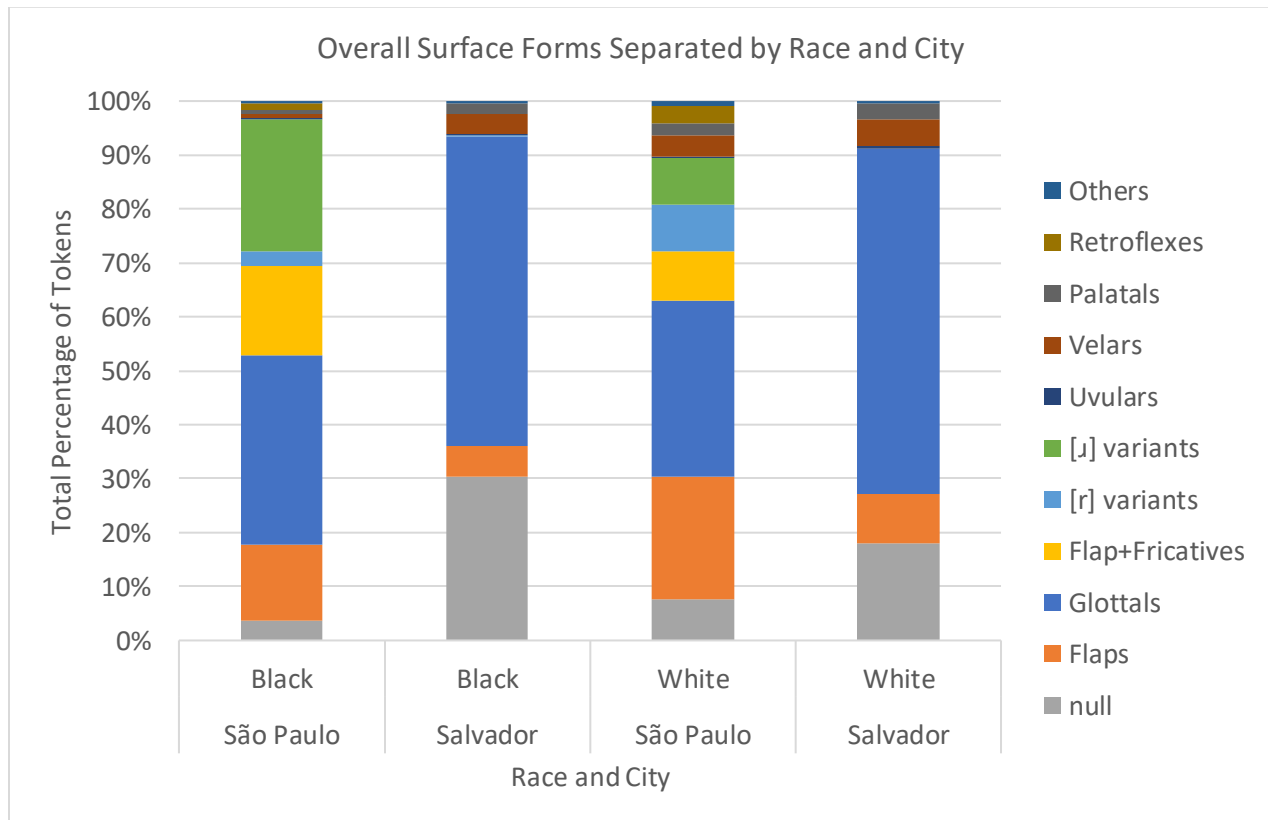


Figure 9.16: Surface forms in the overall corpus separated by race and city.

From the figure above, it can be seen that regardless of race, participants from Salvador are deleting more than participants from São Paulo. When looking at the surface forms above the null portion from Salvador, the surface forms look nearly identical. Meanwhile, the surface forms appear mostly similar between the two racial groups in São Paulo, but there do appear to

be some differences, such as more [ɹ] variants and fewer [r] variants produced by the black participants. Interestingly, the deletion rate appears to be lower for black participants from São Paulo than white participants. However, whether this difference is significant was not tested. Overall, it appears that the only noticeable difference between black and white participants from Salvador is deletion rate, while black and white participants from São Paulo tend to have more differences from each other. When comparing black participants from both cities to each other, they do not have very much in common, and the same can be said for the white participants. They have more in common with others from their own city than with others of their own race in the other city.

9.3. Discussion of the São Paulo data and its differences from Salvador

In this chapter, the data from the São Paulo participants was presented along with various statistical tests to see what differences exist between the São Paulo and Salvador data. Ultimately, the majority of tests were significant, indicating that differences exist between the two cities in deletion rate, oral production rate, and distribution of surface forms.

When it comes to the data from São Paulo, the rhotic has an extreme amount of variation. While this variation includes the trills and alveolar approximant [ɹ] that the previous literature discussed in chapter 2 cites as allophones of the phoneme in the city, several additional surface forms were found as well. Furthermore, the previously cited allophones are limited to coda position, but many other surface forms were frequently found in coda position as well, such as the flap + fricative variants, flaps that are not in the word-final position preceding a vowel context, and retroflex fricatives. These surface forms suggest that the trill is undergoing lenition in the form of a reduction in the number of occlusions and frication. Similar

reductions in occlusions and “pre-aspirated trills” ([hr] rather than the [rh] found here) have been cited in varieties of Spanish in the Dominican Republic and Spain (Willis, 2006, 2007; Willis and Bradley, 2008; Henriksen and Willis, 2010).

Additionally, glottal fricatives were quite common in coda position as well, which further suggests the lenition of the phoneme in São Paulo. However, glottal fricatives were more common in onset position than coda position. This suggests that the phoneme may have undergone more lenition in onsets than in codas in São Paulo, but deletion is more common in codas than in onsets, which would suggest the opposite. Nevertheless, it is clear that the phoneme is undergoing lenition in São Paulo as well and that the variation in São Paulo is likely even more extreme than in Salvador.

Given the various forms of lenition found in the two cities, there appears to be a progression of lenition from trills being the least lenited, to flap + fricative variants, to oral fricatives, to glottal fricatives, and finally deletion being the most lenited. There is indication that something similar has occurred in Puerto Rican Spanish, where the trill also has lenited into velar, uvular and glottal fricatives (Delgado-Diaz and Galarza, 2016; Delgado-Diaz et al., 2021). Moreover, the latter half of this lenition progression is similar to a lenition process that has happened in some dialects of Spanish with coda /s/, which has [h] as an intermediate form of lenition between [s] and deletion (File-Muriel and Brown, 2011; Gradoville et al., 2022).

Regarding the differences between São Paulo and Salvador, this chapter found the answers to the third set of research questions given in chapter 4:

1. Do speakers from Salvador delete the rhotic more frequently than speakers from São Paulo?

Deletion is significantly higher in Salvador in coda position for all environments of tokens in context, as well as with intervocalic context tokens and word-final isolation tokens. Additionally, the overall deletion rate across the corpus was significantly higher with the Salvador participants. Given the attribution in the literature of African influences to rhotic deletion in coda position in Brazilian Portuguese discussed in chapter 2 and the demographic differences between the cities discussed in chapters 1 and 3, this is evidence for the notion that African influences have been much stronger in Salvador than in São Paulo. This evidence is further strengthened by the fact that the tests for deletion rate were conducted with a subset of participants from Salvador who were demographically matched with the sample from São Paulo. Thus, the fact that black participants from Salvador deleted the rhotic more frequently than white participants from Salvador was mitigated by the subset used in the comparison. Taking this into consideration, it is fair to say that the variety of Salvador has remaining traces of African influences which are affecting today's speakers. Thus, the hypothesis that deletion rates would be higher in Salvador was correct.

2. Do speakers from São Paulo produce non-lenited surface forms more frequently than speakers from Salvador?

Regarding differences in oral production rate between the two cities, oral surface forms occurred at a significantly higher rate in São Paulo in word-final isolation and context tokens, syllable-final isolation and context tokens, word-initial context tokens, intervocalic isolation tokens, and the overall corpus. Many of these environments were also significant for deletion rate, suggesting that the more a population deletes the rhotic in a given environment, the less it will produce oral surface forms. Since deletion reduces the number of opportunities for oral

surface forms, this is a plausible explanation. However, it is clearly the case when taking into consideration figures 9.8 and 9.10 that oral surface forms are simply more widespread in São Paulo than Salvador, even without the influence of deletion. This fact, combined with the higher deletion rates in Salvador, suggests that Salvador has undergone substantially more lenition on the rhotic phoneme than São Paulo. Thus, the hypothesis that non-lenited surface forms would be more common in São Paulo than Salvador was correct. Additionally, the hypotheses that glottal and oral fricatives would comprise smaller proportions of the surface forms in São Paulo than in Salvador were mostly correct, although glottal fricatives did comprise a substantial proportion of productions in onset position in São Paulo, and fricatives occurred at lower frequencies in coda position as well. Thus, fricatives were more common as surface forms than hypothesized. Additionally, the lenition of the trill [r] to a single occlusion flap [ɾ] or flap + fricative variant was not hypothesized.

3. Is there a difference between the surface forms of the rhotic in São Paulo and Salvador?

In terms of differences in the distribution of surface forms of the rhotic, almost all environments were significantly different between the two cities, with word-initial tokens preceded by a consonant and intervocalic context tokens with post-rhotic stress being the only tests which were not significant. These results further confirm what was found on the tests for deletion rate and oral production rate: the rhotic phoneme surfaces very differently in São Paulo from how it surfaces in Salvador. In São Paulo, the rhotic in coda position typically, though not always, involves a coronal element. However, in Salvador, there is very rarely a coronal element to the rhotic. In onset position, the two cities are quite similar in the sense that glottal fricatives predominate. Nevertheless, the higher proportion of oral surface forms was still enough to

make most of the tests in these environments significant. The fact that oral surface forms occurred in low numbers for both cities in onset position means that these significant results should be taken with a grain of salt, but given the clear tendency across the board for São Paulo speakers to produce oral surface forms more frequently than Salvador speakers, these tests are worth taking into consideration. Thus, the hypothesis that a difference in surface forms exists between the two cities was correct. However, the difference did not manifest exactly as hypothesized, as glottal and oral fricative productions were found in São Paulo and one participant in Salvador did produce a few alveolar trills [r].

Finally, based on the separation of the overall corpus by race and city, it is clear that despite the significant differences in deletion rate by race in Salvador, the participants have more in common with those from the same city but a different racial group than with those who are from the same racial group but a different city. This further suggests that if African influences are prevalent in the black speakers of Salvador, they must be present in the white speakers of Salvador as well. Therefore, if the higher deletion rate on the black participants in Salvador is due to greater African influences, there is a spectrum of degree of African influence which ranges from black speakers from predominantly black cities such as Salvador to white speakers from predominantly black cities to speakers from cities that are not predominantly black.

In conclusion, there are extensive differences between São Paulo and Salvador when it comes to the rhotic phoneme. Salvador deletes the phoneme more, while São Paulo has a greater oral production rate and a wider variety of surface forms. These differences between the cities are maintained across racial groups, meaning that the differences between the cities

are due to geography rather than race. Ultimately, this comparison between the two cities reveals just how complex and variable the “strong r” rhotic phoneme in Brazilian Portuguese is.

10. Conclusions and future work

The data obtained from Salvador indicate that the rhotic phoneme exhibits substantial variation in addition to its extensive lenition phenomenon. This lenition primarily manifests as either a glottal fricative or deletion. When a production of the phoneme contains an oral element, it is typically a palatal, velar, or uvular fricative, though some additional variants were also found in the corpus. Which variant the rhotic surfaces as cannot be determined with a set of phonological rules; however, tendencies do exist. For example, utterance-initial and utterance-final rhotics tend to be voiceless. Additionally, the presence of a voiceless consonant adjacent to the rhotic increases the likelihood that the rhotic will be voiceless, but most productions are still voiced. Furthermore, a rhotic that is surrounded by voiced segments will not necessarily be voiced. With regards to the place of the rhotic when it surfaces as a non-glottal form, the preceding vowel had a greater influence on the surface form of rhotics in coda position than the following segment did. In onset position, there were far fewer non-glottal productions, but more oral surface forms did tend to match the place of following vowels more than of preceding segments. Thus, an oral surface form of the rhotic tends to match the place of the vowel serving as the nucleus of the syllable containing the rhotic more than the place of an adjacent consonant or of an adjacent vowel in another syllable.

The word-final environment, as expected, was the most heavily lenited, with the highest rate of deletion. When it comes to studies which also used isolated tokens and sentences, previous work in other cities demonstrated far lower deletion rates. Unfortunately, there is no way to know with certainty how this deletion rate would compare to informal interviews in other studies; however, given the higher deletion rates that occurred with predetermined

stimuli, it is reasonable to believe that speakers from Salvador could produce higher word-final deletion rates in informal interviews than speakers in other cities. It is also important to note that word-final context tokens before vowels demonstrate that the following segment type may have an influence over which rhotic phoneme the word-final rhotic surfaces as. In terms of lenition in other environments, all other environments demonstrate high rates of lenition as well, with all environments exhibiting deletion, albeit at lower rates than the word-final environment, and the glottal fricatives being the predominant surface form as well.

Previous research, as discussed in chapter 2, has proposed that African influences played a role in rhotic deletion in Brazilian Portuguese, and if this is the case, this influence would have been strong in Salvador due to its history of slavery and the resulting demographics of the city discussed in the chapter 3. While slavery played a major role in the history of Brazil, it played an even larger role in Salvador than the rest of the country. Furthermore, while places like São Paulo underwent significant demographic changes due to European immigration in the late 19th and 20th centuries, Salvador did not have such a demographic shift. Consequently, any African influences on Brazilian Portuguese would have a greater likelihood of survival into the modern variety of Salvador than the modern varieties of cities like São Paulo. Given this, it is reasonable to believe that the variety of Salvador would delete the rhotic more often than other varieties of Brazilian Portuguese. Additionally, as discussed in chapter 2, deletion in Brazilian Portuguese has been attributed to a general tendency to maintain a CV syllable structure, something that is somewhat a recurring theme in rhotic deletion literature. Therefore, it is possible that rhotic deletion in Salvador is driven by a tendency for a CV syllable structure (similar to deletion in

other languages), which could also be rooted in the African influences of Brazilian Portuguese (particularly prominent in Salvador with its demographic profile).

Because of the notably high deletion rates in this project, even with elicited stimuli, it is plausible that the African influences in Salvador were stronger and allowed for more widespread deletion than in other varieties of Brazilian Portuguese. Additionally, the correlation in deletion rates between certain environments suggests that some speakers have a greater tendency to delete than others. With this fact, one might ask what personal characteristics of the speakers might account for these differences and what connection to the historical demographics and African substrate influences of rhotic deletion this may indicate.

Ultimately, it was discovered in chapter 8 that race does indeed play a significant role in the deletion rates of participants from Salvador, with participants who identified as *preto* or *pardo* generally deleting at a higher rate than white participants. Thus, it is quite possible that African influences are stronger in the black participants from Salvador than the white participants from Salvador. At the same time, given the historical and genetic evidence discussed in chapter 3, the white residents of Salvador are generally expected to have African ancestry and to have been exposed to the same African influences. Therefore, it is likely the case that the white participants from Salvador still have a heavy amount of African influence in their speech, though it is less than the amount of African influence on the speech of the black participants from Salvador. This is demonstrated in chapter 9 when a subset of the Salvador participants was demographically matched with the São Paulo participants for race, age, gender, and socioeconomic class. The results in this chapter indicated that deletion is more common in Salvador than São Paulo regardless of race.

From the results obtained from the analysis of the data from the participants from both Salvador and São Paulo, it is clear that the rhotic phoneme is undergoing lenition in all environments in both cities. In Salvador, this lenition primarily manifests as either a glottal fricative or deletion. However, in São Paulo, deletion is far less common, and the lenition pattern of the rhotic phoneme is different. In onsets, the phoneme is usually lenited to a glottal fricative, as was the case in Salvador. In codas, on the other hand, the phoneme typically lenites to a flap or flap + fricative variant, though glottal fricatives occur here as well. This difference in coda lenition, combined with the higher deletion rates in Salvador and higher oral production rates in São Paulo, indicates that lenition as a whole is more advanced in Salvador than São Paulo. Lastly, while there is significant variation in production of the rhotic phoneme in both cities, the variation in São Paulo surpasses that in Salvador.

Throughout this dissertation, a variety of surface forms for the rhotic phoneme have been presented and discussed, but no form has ever been presented as the base form. For Salvador, it could be argued that the base form is /h/, as it comprised the bulk of productions and the surface form [h] can be reasonably well-predicted with optional rules that allow the phoneme to devoice utterance-initially, utterance-finally, and adjacent to voiceless consonants. A similar optional rule could help with the place of oral surface forms where the phoneme may match the place of the vowel that serves as the nucleus of the syllable. However, these rules would not account for voiceless forms which are surrounded by voiced segments or oral surface forms that do not match an adjacent segment, both of which occurred regularly in the corpus. Nevertheless, if a base form had to be selected, /h/ would be the best option for Salvador, as it

is the most frequent surface form, and there are ways to describe the phonological processes that can lead to it surfacing as a variety of other surface forms.

With regards to São Paulo, a base form of /h/ would not be the most effective way to represent the phoneme. Instead, the alveolar trill /r/ may serve as a better way to represent the base form. The other surface forms could then be described through lenition rules that turn the trill into a flap, flap + fricative, or a glottal fricative. However, as was the case with Salvador, these rules would be optional, as there is not a certain way to predict an exact surface form. Additionally, working the approximant [ɹ] into this set of rules would be logistically difficult.

Ultimately, if a base form of the rhotic phoneme had to be selected for both Salvador and São Paulo, or for Brazilian Portuguese as a whole, there is no great option. It would be easier to use phonological rules to turn the alveolar trill /r/ into the glottal fricatives than the other way around, but the lack of the trill in Salvador makes this option unappealing. At the end of the day, given the broader usage of the voiced glottal fricative and the inconsistency in the rhotic phoneme's production that makes phonological rules to predict the surface forms difficult, /h/ is probably the best way to represent the base form of the phoneme at least for the two varieties discussed here.

In conclusion, the non-flap rhotic phoneme in Brazilian Portuguese exhibits extensive variation and lenition phenomena that go beyond what is typically found in a language. This variation and lenition can be linked to the complex sociohistorical situation (particularly African influences) surrounding the development of Brazil as a country, and as a consequence, the development of Brazilian Portuguese.

Going forward, further analyses can be done with remaining data from São Paulo, as there were nine participants who were recorded but excluded for the purposes of demographic matching and timely completion of this dissertation. These additional participants may also strengthen some of the differences which exist between speakers from Salvador and speakers from São Paulo, as well as allow some insight into potential demographic differences that may exist in the data among the participants from São Paulo. Furthermore, the extensive variation and the significant differences in deletion rate by race indicate that perceptual work can help provide further insight and context to the production data presented here. A psycholinguistic study can explore how well individuals can distinguish the various allophones of the rhotic phoneme and how similar the allophones are to each other. Additionally, a sociolinguistic study can evaluate how well listeners can distinguish the various demographic characteristics of speakers, particularly race, given the significant differences in the data.

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A1. Participant Demographics

A1.1 Demographics of Participants from Salvador

Note: Participants 1-11 were recorded in 2022. Participants 12-36 were recorded in 2023. “TNB” stands for “trans non-binary.” “Ind.” Stands for “indigenous.” “All life” refers to whether or not a participant has lived their entire life in Salvador. “P-Int” refers to the number of a participant’s parents from the interior of Bahia, and “GP-Int” refers to the number of a participant’s grandparents from the interior of Bahia. Participants did not always know where parents or grandparents were from. In these cases, the parents and grandparents were not counted as being from the interior of Bahia. The final column indicates whether white participants have black parents or siblings. The columns to the right of “All life” represent the extra questions that participants were recontacted for in 2023. Footnotes indicate familial relationships between participants. An asterisk (*) indicates the participants that were included in the demographically matched comparison with São Paulo.

Table A1.1: Demographics of participants from Salvador

Participant	Age	Gender	Race	Class	All life	Heritage	P-Int	GP-Int	Black parents or siblings
1 ²⁰	25	Male	<i>Preto</i>	Middle	Yes	Afro-Brazilian	1	3	n/a
2	28	Female	<i>Preto</i>	Lower	No	Brazilian	0	2	n/a
3	28	Male	<i>Preto</i>	Lower	Yes	African	2	1	n/a
*4 ²¹	27	Female	White	Lower	No	Brazilian	0	3	Yes
*5	26	Female	<i>Preto</i>	Middle	Yes	No response			
6	26	Male	<i>Pardo</i>	Lower	Yes	Brazilian	2	0	n/a
7 ²²	22	Male	<i>Preto</i>	Lower	No	Brazilian	0	2	n/a
8	excluded due to not being raised in Salvador								
9 ²³	26	Female	<i>Preto</i>	Lower	Yes	African	2	2	n/a
10	31	Male	<i>Preto</i>	Lower	Yes	Black (<i>negro</i>)	1	1	n/a
*11	25	Male	White	Middle	Yes	Brazilian	1	2	No
*12 ²⁴	58	Female	<i>Pardo</i>	Middle	Yes	Brazilian	1	2	n/a
*13 ²⁵	57	Female	<i>Preto</i>	Middle	Yes	Brazilian, African, and Portuguese	2	2	n/a
14	59	Male	<i>Preto</i>	Middle	No	Brazilian	2	4	n/a
15	22	Male	<i>Preto</i>	Middle	Yes	African	0	2	n/a
16 ²⁶	60	Female	<i>Pardo</i>	Lower	Yes	No response			
17 ²⁷	61	Female	<i>Pardo</i>	Middle	Yes	Brazilian, Portuguese, and African	1	2	n/a
18 ²⁸	52	Female	<i>Preto</i>	Lower	Yes	African	0	0	n/a
19 ²⁹	58	Female	<i>Preto</i>	Lower	Yes	Brazilian	2	3	n/a

²⁰ Son of participant 17. Maternal nephew of participants 12, 13, and 16. Maternal first cousin of participant 30.

²¹ Sister of participants 20 and 24.

²² Son of participant 19.

²³ Sister of participant 34.

²⁴ Sister of participants 13, 16, and 17. Maternal aunt of participants 1 and 30.

²⁵ Sister of participants 12, 16, and 17. Maternal aunt of participants 1 and 30. This sister reported both parents being from the interior of Bahia, while the other two who responded to the extra questions indicated only their father as being from the interior of Bahia. The nephews both reported their maternal grandfather, but not their grandmother, as being from the interior of Bahia.

²⁶ Sister of participants 12, 13, and 17. Maternal aunt of participants 1 and 30.

²⁷ Sister of participants 12, 13, and 16. Mother of participant 1. Maternal aunt of participant 30.

²⁸ Sister of participant 32.

²⁹ Mother of participant 7.

20 ³⁰	29	Female	<i>Preto</i>	Lower	Yes	Brazilian, indigenous, and Spanish	0	3	n/a
*21	30	Male	White	Middle	Yes	Brazilian	1	3	No
*22	21	Female	White	Middle	Yes	Arab, Portuguese	0	0	Yes
*23	25	Female	White	Middle	No	Indigenous, Portuguese, and African	2	1	Yes
24 ³¹	20	Male	<i>Pardo</i>	Lower	Yes	Brazilian	0	3	n/a
25	35	Male	Ind.	Lower	Yes	Brazilian	0	4	n/a
26	21	Female	<i>Preto</i>	Lower	No	Brazilian	0	2	n/a
27	26	Female	<i>Preto</i>	Lower	Yes	No response			
28	24	Female	<i>Preto</i>	Lower	Yes	Unable to contact			
29	26	Female	White	Lower	Yes	Brazilian	0	0	No
*30 ³²	23	Male	White	Middle	Yes	Brazilian, Italian	0	1	Yes
31	46	TNB	<i>Preto</i>	Middle	No	Definition of ancestry	2	4	n/a
32 ³³	56	Female	<i>Preto</i>	Lower	Yes	Brazilian	0	0	n/a
*33	30	Male	<i>Preto</i>	Middle	Yes	No response			
34 ³⁴	28	Female	<i>Pardo</i>	Lower	Yes	Brazilian	1	1	n/a
35 ³⁵	32	Male	<i>Preto</i>	Lower	Yes	African	1	2	n/a
36 ³⁶	34	Female	<i>Preto</i>	Lower	Yes	No response			

³⁰ Sister of participants 4 and 24.

³¹ Brother of participants 4 and 20.

³² Son of participant 12. Maternal nephew of participants 13, 16, and 17. Maternal first cousin of participant 1.

³³ Sister of participant 18.

³⁴ Sister of participant 9. The discrepancy in the number of parents and grandparents from the interior is due to a lack of certainty for both participants.

³⁵ Brother of participant 36.

³⁶ Sister of participant 35.

A1.2 Demographics of Participants from São Paulo

Note: Participants from São Paulo were recorded in 2023. Of the 19 recorded, only 10 were included in order to demographically match a subset of the participants from Salvador as best as possible. Participants that were analyzed were assigned numbers. Those which were excluded have their original letter designation. None of the white participants had black parents or siblings. However, when possible, white participants who indicated something that indicated a known or likely African ancestry were selected over those who did not indicate such information.

Table A1.2: Demographics of participants from São Paulo

Participant	Age	Gender	Race	Class	Known or likely African ancestry indicated
Participants used in comparison					
SP1	29	Male	White	Middle	No
SP2	37	Female	White	Middle	Indicated having black great-grandmother
SP3	41	Male	White	Middle	Did DNA test that showed African DNA
SP4	56	Male	White	Middle	No
SP5 ³⁷	19	Female	White	Middle	All parents and grandparents from Bahia
SP6	19	Male	White	Middle	No
SP7 ³⁸	30	Female	<i>Pardo</i>	Middle	n/a
SP8 ³⁹	31	Female	<i>Pardo</i>	Middle	n/a
SP9 ⁴⁰	32	Female	<i>Pardo</i>	Lower	n/a
SP10	35	Female	<i>Preto</i>	Middle	n/a
Participants not analyzed					
SPA	34	Male	White	Middle	No
SPB	19	Female	White	Middle	No
SPC	29	Male	White	Middle	No
SPF	40	Female	White	Middle	No
SPH	38	Male	White	Middle	No
SPK	19	Female	White	Middle	All parents and grandparents from Bahia
SPL	19	Female	White	Middle	
SPM	31	Male	White	Middle	No
SPN	33	Female	White	Middle	No

³⁷ Triplet of participants SPK and SPL. In order to avoid repeating demographic variables of participants in the comparison sample, the triplet with the highest quality audio file was chosen.

³⁸ All parents and grandparents from Bahia.

³⁹ All parents and grandparents from Pernambuco, a state in the northeast adjacent to Bahia.

⁴⁰ One grandparent from Bahia and one grandparent from Pernambuco.

A2. Contingency tables for chi-squared and Fisher's exact tests

A2.1. Contingency tables for oral surface forms by demographic variables

A2.1.1. Contingency tables for oral surface forms by race

Note: The tests between black and white participants and their places of articulation were done by summing the columns of *preto* and *pardo* together and then using the summed column with the column for the white participants for the independence tests.

Table A2.1: Places of articulation of oral surface forms for word-final isolation tokens by race

Place	<i>Preto</i>	<i>Pardo</i>	White
Bilabial	0	1	0
Alveolar	0	3	0
Palatal	13	6	7
Velar	14	26	14
Uvular	0	1	0
Pharyngeal	1	0	0

Table A2.2: Places of articulation of oral surface forms for word-final context tokens by race.

Place	<i>Preto</i>	<i>Pardo</i>	White
Bilabial	0	0	3
Alveolar	166	51	91
Palatal	17	5	15
Velar	21	20	16
Uvular	1	1	4
Pharyngeal	1	0	0

Table A2.3: Places of articulation of oral surface forms for word-final context tokens before a consonant by race.

Place	<i>Preto</i>	<i>Pardo</i>	White
Bilabial	0	0	1
Palatal	8	2	8
Velar	10	8	7

Table A2.4: Places of articulation and oral surface forms for word-final context tokens before a vowel by race.

Place	<i>Preto</i>	<i>Pardo</i>	White
Alveolar	166	51	91
Palatal	5	2	4
Velar	2	4	0
Uvular	1	0	2

Table A2.5: Places of articulation and oral surface forms for word-final context tokens before a pause by race.

Place	<i>Preto</i>	<i>Pardo</i>	White
Bilabial	0	0	2
Palatal	4	1	3
Velar	9	8	9
Uvular	0	1	2
Pharyngeal	1	0	0

Table A2.6: Places of articulation and oral surface forms for syllable-final isolation tokens by race.

Place	<i>Preto</i>	<i>Pardo</i>	White
Alveolar	0	3	0
Velar	8	5	4
Uvular	4	1	0

Table A2.7: Places of articulation and oral surface forms for syllable-final context tokens by race.

Place	<i>Preto</i>	<i>Pardo</i>	White
Bilabial	0	1	0
Alveolar	1	0	0
Alveolopalatal	1	0	0
Palatal	21	10	6
Velar	36	11	14
Uvular	2	0	0
Interdental	1	0	0

Table A2.8: Places of articulation and oral surface forms for the overall corpus by race.

Place	<i>Preto</i>	<i>Pardo</i>	White
Bilabial	0	2	3
Alveolar	167	58	91
Alveolopalatal	1	0	0
Palatal	55	22	29
Velar	82	63	49
Uvular	8	3	5
Interdental	1	0	0
Pharyngeal	6	0	1
Epiglottal	1	0	0

A2.1.2. Contingency tables for oral surface forms by socioeconomic class

Table A2.9: Places of articulation and oral surface forms for word-final isolation tokens by socioeconomic class.

Place	Lower	Middle
Bilabial	0	1
Alveolar	0	3
Palatal	20	9
Velar	31	24
Uvular	1	0
Pharyngeal	1	0

Table A2.10: Places of articulation and oral surface forms for word-final context tokens by socioeconomic class.

Place	Lower	Middle
Bilabial	0	3
Alveolar	160	156
Palatal	25	13
Velar	36	23
Uvular	4	2
Pharyngeal	1	0

Table A2.11: Places of articulation and oral surface forms for syllable-final isolation tokens by socioeconomic class.

Place	Lower	Middle
Alveolar	0	3
Velar	10	8
Uvular	4	1

Table A2.12: Places of articulation and oral surface forms for syllable-final context tokens by socioeconomic class.

Place	Lower	Middle
Bilabial	0	1
Alveolar	1	0
Alveolopalatal	1	0
Palatal	26	13
Velar	31	32
Uvular	2	1
Interdental	0	1

Table A2.13: Places of articulation and oral surface forms for the overall corpus by socioeconomic class.

Place	Lower	Middle
Bilabial	0	5
Alveolar	161	163
Alveolopalatal	1	0
Palatal	78	35
Velar	111	89
Uvular	12	5
Interdental	0	1
Pharyngeal	5	2
Epiglottal	1	0

A2.1.3. Contingency tables for oral surface forms by gender

Table A2.14: Places of articulation and oral surface forms for word-final isolation tokens by gender.

Place	Male	Female
Bilabial	0	1
Alveolar	0	3
Palatal	11	18
Velar	12	43
Uvular	0	1
Pharyngeal	0	1

Table A2.15: Places of articulation and oral surface forms for word-final context tokens by gender.

Place	Male	Female
Bilabial	2	1
Alveolar	164	150
Palatal	12	26
Velar	19	40
Uvular	1	5
Pharyngeal	0	1

Table A2.16: Places of articulation and oral surface forms for word-final context tokens before a consonant by gender.

Place	Male	Female
Bilabial	0	1
Palatal	7	12
Velar	10	16

Table A2.17: Places of articulation and oral surface forms for word-final context tokens before a vowel by gender.

Place	Male	Female
Alveolar	164	150
Palatal	1	10
Velar	1	5
Uvular	1	2

Table A2.18: Places of articulation and oral surface forms for word-final context tokens before a pause by gender.

Place	Male	Female
Bilabial	2	0
Palatal	4	4
Velar	8	19
Uvular	0	3
Pharyngeal	0	1

Table A2.19: Places of articulation and oral surface forms for syllable-final isolation tokens by gender.

Place	Male	Female
Alveolar	0	3
Velar	4	14
Uvular	3	2

Table A2.20: Places of articulation and oral surface forms for syllable-final context tokens by gender.

Place	Male	Female
Bilabial	0	1
Alveolar	1	0
Alveolopalatal	0	1
Palatal	14	24
Velar	26	37
Uvular	2	1
Interdental	0	1

Table A2.21: Places of articulation and oral surface forms for the overall corpus by gender.

Place	Male	Female
Bilabial	2	3
Alveolar	165	157
Alveolopalatal	0	1
Palatal	41	71
Velar	63	137
Uvular	7	10
Interdental	0	1
Pharyngeal	2	5
Epiglottal	1	0

A2.1.4. Contingency tables for oral surface forms by places lived

Table A2.22: Places of articulation and oral surface forms for word-final isolation tokens by whether a participant has lived their entire life in Salvador.

Place	Salvador only	Has lived elsewhere
Bilabial	1	0
Alveolar	3	0
Palatal	26	3
Velar	53	2
Uvular	1	0
Pharyngeal	1	0

Table A2.23: Places of articulation and oral surface forms for word-final context tokens by whether a participant has lived their entire life in Salvador.

Place	Salvador only	Has lived elsewhere
Bilabial	3	0
Alveolar	264	52
Palatal	36	2
Velar	54	5
Uvular	6	0
Pharyngeal	1	0

Table A2.24: Places of articulation and oral surface forms for syllable-final isolation tokens by whether a participant has lived their entire life in Salvador.

Place	Salvador only	Has lived elsewhere
Alveolar	3	0
Velar	13	5
Uvular	5	0

Table A2.25: Places of articulation and oral surface forms for syllable-final context tokens by whether a participant has lived their entire life in Salvador.

Place	Salvador only	Has lived elsewhere
Bilabial	1	0
Alveolar	1	0
Alveolopalatal	1	0
Palatal	36	3
Velar	51	12
Uvular	2	1
Interdental	1	0

Table A2.26: Places of articulation and oral surface forms for the overall corpus by whether a participant has lived their entire life in Salvador.

Place	Salvador only	Has lived elsewhere
Bilabial	5	0
Alveolar	272	52
Alveolopalatal	1	0
Palatal	103	10
Velar	176	24
Uvular	16	1
Interdental	1	0
Pharyngeal	7	0
Epiglottal	1	0

A2.1.5. Contingency tables for oral surface forms by Afro-Brazilian identity

Table A2.27: Places of articulation and oral surface forms for word-final isolation tokens by whether a participant identified as having Afro-Brazilian or African heritage.

Place	Identified as Afro-Brazilian	Did not identify as Afro-Brazilian
Bilabial	1	0
Alveolar	3	0
Palatal	4	8
Velar	11	13

Table A2.28: Places of articulation and oral surface forms for word-final context tokens by whether a participant identified as having Afro-Brazilian or African heritage.

Place	Identified as Afro-Brazilian	Did not identify as Afro-Brazilian
Alveolar	124	70
Palatal	10	2
Velar	10	11

Table A2.29: Places of articulation and oral surface forms for syllable-final isolation tokens by whether a participant identified as having Afro-Brazilian or African heritage.

Place	Identified as Afro-Brazilian	Did not identify as Afro-Brazilian
Alveolar	3	0
Velar	3	7
Uvular	2	1

Table A2.30: Places of articulation and oral surface forms for syllable-final context tokens by whether a participant identified as having Afro-Brazilian or African heritage.

Place	Identified as Afro-Brazilian	Did not identify as Afro-Brazilian
Bilabial	1	0
Alveolar	1	0
Palatal	14	11
Velar	16	16
Uvular	1	1
Interdental	1	0

Table A2.31: Places of articulation and oral surface forms for the overall corpus by whether a participant identified as having Afro-Brazilian or African heritage.

Place	Identified as Afro-Brazilian	Did not identify as Afro-Brazilian
Bilabial	2	0
Alveolar	132	70
Palatal	31	23
Velar	42	48
Uvular	4	2
Interdental	1	0
Pharyngeal	0	2
Epiglottal	1	0

A2.1.6. Contingency tables for oral surface forms by data collection method

Table A2.32: Places of articulation and oral surface forms for word-final isolation tokens by gender.

Place	Remote	In-person
Bilabial	0	1
Alveolar	0	3
Palatal	3	26
Velar	5	50
Uvular	0	1
Pharyngeal	0	1

Table A2.33: Places of articulation and oral surface forms for word-final context tokens by gender.

Place	Remote	In-person
Bilabial	1	2
Alveolar	116	200
Palatal	9	29
Velar	7	52
Uvular	0	6
Pharyngeal	0	1

Table A2.34: Places of articulation and oral surface forms for word-final context tokens before a consonant by gender.

Place	Remote	In-person
Bilabial	0	1
Palatal	6	13
Velar	4	22

Table A2.35: Places of articulation and oral surface forms for word-final context tokens before a vowel by gender.

Place	Remote	In-person
Alveolar	116	200
Palatal	1	10
Velar	0	6
Uvular	0	3

Table A2.36: Places of articulation and oral surface forms for word-final context tokens before a pause by gender.

Place	Remote	In-person
Bilabial	1	1
Palatal	2	6
Velar	3	24
Uvular	0	3
Pharyngeal	0	1

Table A2.37: Places of articulation and oral surface forms for syllable-final isolation tokens by gender.

Place	Remote	In-person
Alveolar	0	3
Velar	5	13
Uvular	2	3

Table A2.38: Places of articulation and oral surface forms for syllable-final context tokens by gender.

Place	Remote	In-person
Bilabial	0	1
Alveolar	1	0
Alveolopalatal	0	1
Palatal	10	29
Velar	21	42
Uvular	2	1
Interdental	0	1

Table A2.39: Places of articulation and oral surface forms for the overall corpus by gender.

Place	Remote	In-person
Bilabial	1	4
Alveolar	117	207
Alveolopalatal	0	1
Palatal	27	86
Velar	39	161
Uvular	5	12
Interdental	0	1
Pharyngeal	0	7
Epiglottal	1	0

A2.2. Contingency tables for relationships between demographic variables

Note: The tests between black and white participants and their association with other variables were done by summing the columns of *preto* and *pardo* together and then using the summed column with the column for the white participants for the independence tests.

Table A2.40: Counts for the variables of race and data collection method.

	<i>Preto</i>	<i>Pardo</i>	White
Remote	7	1	2
In-person	14	5	5

Table A2.41: Counts for the variables of Afro-Brazilian identity and race.

	Identified as Afro-Brazilian	Identified as something else
<i>Preto</i>	9	7
<i>Pardo</i>	1	4

Table A2.42: Counts for the variables of race and whether or not a participant has lived their whole life in Salvador.

	<i>Preto</i>	<i>Pardo</i>	White
Salvador only	16	6	5
Elsewhere	5	0	2

Table A2.43: Counts for the variables of race and gender.

	<i>Preto</i>	<i>Pardo</i>	White
Male	8	2	3
Female	12	4	4

Table A2.44: Counts for the variables of race and socioeconomic class.

	<i>Preto</i>	<i>Pardo</i>	White
Lower class	14	4	2
Middle class	7	2	5

Table A2.45: Counts for the variables of socioeconomic class and data collection method.

	Lower class	Middle class
Remote	7	3
In-person	14	11

Table A2.46: Counts for the variables of Afro-Brazilian identity and socioeconomic class.

	Identified as Afro-Brazilian	Identified as something else
Lower class	7	7
Middle class	4	3

Table A2.47: Counts for the variables of socioeconomic class and whether or not a participant has lived their entire life in Salvador.

	Lower class	Middle class
Salvador only	17	11
Has lived elsewhere	4	3

Table A2.48: Counts for the variables of socioeconomic class and gender.

	Lower class	Middle class
Male	7	7
Female	14	6

Table A2.49: Counts for the variables of data collection method and gender.

	Remote	In-person
Male	6	8
Female	4	16

Table A2.50: Counts for the variables of Afro-Brazilian identity and gender.

	Identified as Afro-Brazilian	Identified as something else
Male	5	4
Female	5	6

Table A2.51: Counts for the variables of whether or not a participant has lived in Salvador their entire life and gender.

	Salvador only	Has lived elsewhere
Male	12	2
Female	16	4

Table A2.52: Counts for the variables of whether or not a participant has lived in Salvador their entire life and data collection method.

	Salvador only	Elsewhere
Remote	7	3
In-person	21	4

Table A2.53: Counts for the variables of Afro-Brazilian identity and whether a participant has lived their whole life in Salvador.

	Identified as Afro-Brazilian	Identified as something else
Salvador only	10	6
Has lived elsewhere	0	5

Table A2.54: Counts for the variables of Afro-Brazilian identity and data collection method.

	Identified as Afro-Brazilian	Identified as something else
Remote	4	3
In-person	6	8

A2.3. Contingency tables for surface forms differences between Salvador and São Paulo

Table A2.55: Counts for the various segment types produced on word-final isolation tokens by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Flaps	3	0
Glottals	19	343
Flap+Fricatives	31	0
[r] variants	30	3
[ɹ] variants	44	0
Uvulars	0	1
Velars	3	55
Palatals	1	29
Retroflexes	25	0
Others	0	2

Table A2.56: Counts for the various segment types produced on word-final context tokens by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Flaps	205	316
Glottals	72	450
Flap+Fricatives	60	0
[r] variants	30	0
[ɹ] variants	89	0
Uvulars	0	6
Velars	2	59
Palatals	3	38
Retroflexes	9	0
Others	2	4

Table A2.57: Counts for the various segment types produced on word-final context tokens before a consonant by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Flaps	32	0
Glottals	61	270
Flap+Fricatives	43	0
[r] variants	11	0
[ɹ] variants	49	0
Velars	1	26
Palatals	2	19
Retroflexes	1	0
Others	1	1

Table A2.58: Counts for the various segment types produced on word-final context tokens before a vowel by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Flaps	170	316
Glottals	3	53
Flap+Fricatives	2	0
[r] variants	5	0
[ɹ] variants	20	0
Uvulars	0	3
Velars	0	6
Palatals	0	11
Others	1	0

Table A2.59: Counts for the various segment types produced on word-final context tokens before a pause by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Flaps	3	0
Glottals	8	127
Flap+Fricatives	15	0
[r] variants	14	0
[ɹ] variants	20	0
Uvulars	0	3
Velars	1	27
Palatals	1	8
Retroflexes	8	0
Others	0	3

Table A2.60: Counts for the various segment types produced on syllable-final isolation tokens by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Flaps	14	0
Glottals	5	241
Flap+Fricatives	18	0
[r] variants	18	3
[ɹ] variants	21	0
Uvulars	0	5
Velars	0	18
Retroflexes	2	0

Table A2.61: Counts for the various segment types produced on syllable-final context tokens by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Flaps	50	0
Glottals	63	670
Flap+Fricatives	61	0
[r] variants	13	0
[ɹ] variants	58	0
Uvulars	0	3
Velars	3	63
Palatals	6	39
Retroflexes	1	0
Others	2	4

Table A2.62: Counts for the various segment types produced on word-initial isolation tokens by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Glottals	53	202
x	5	0
Others	2	1

Table A2.63: Counts for the various segment types produced on word-initial context tokens by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Glottals	126	476
ɾ	0	1
Uvulars	2	0
x	7	3
ç	6	2
Others	2	3

Table A2.64: Counts for the various segment types produced on word-initial context tokens after a consonant by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Glottals	32	111
Uvulars	1	0
x	1	1

Table A2.65: Counts for the various segment types produced on word-initial context tokens after a vowel by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Glottals	270	73
ɾ	1	0
x	1	3
Others	1	2

Table A2.66: Counts for the various segment types produced on word-initial context tokens after a pause by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Glottals	21	95
Uvulars	1	0
x	3	1
ç	6	2
Others	0	2

Table A2.67: Counts for the various segment types produced on intervocalic isolation tokens by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Glottals	35	174
ʁ	1	1
Velars	13	0
Palatals	1	0

Table A2.68: Counts for the various segment types produced on intervocalic context tokens by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Glottals	106	368
ʁ	1	1
Velars	5	2
Palatals	4	5
Others	1	2

Table A2.69: Counts for the various segment types produced on intervocalic context tokens with post-rhotic stress by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Glottals	67	234
Velars	1	1
Palatals	1	3
Others	1	2

Table A2.70: Counts for the various segment types produced on intervocalic context tokens with stress not post-rhotic by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Glottals	39	134
ʁ	1	1
Velars	4	1
Palatals	3	2

Table A2.71: Counts for the various segment types produced in the overall corpus by participants from São Paulo and Salvador.

Segment Type	São Paulo	Salvador
Flaps	272	316
Glottals	479	2924
Flap+Fricatives	170	0
[r] variants	91	7
[ɹ] variants	212	0
Uvulars	4	17
Velars	38	200
Palatals	21	113
Retroflexes	37	0
Others	9	16

A3. Stimuli with English translation

Note: The instances of the rhotic phoneme analyzed are underlined. For the sentences, the words containing instances of the rhotic phoneme analyzed are bolded. An orthographic “r” that is not underlined is the flap phoneme.

Stimulus	Translation
macaco	monkey
jantar <u>r</u>	dinner
or <u>r</u> be	orb
dormir <u>r</u>	sleep
açúcar <u>r</u>	sugar
terra <u>r</u>	dirt
<u>r</u> ato	rat
planta	plant
minutar <u>r</u>	draft
horta <u>r</u>	vegetable garden
dizer <u>r</u>	say
carro	car
cantar <u>r</u>	sing
dardo	dart
<u>r</u> etorno	return
sacada	balcony
admirar <u>r</u>	admire
colidir <u>r</u>	collide
borrado	blurred
Ele anda de bicicleta todos os dias.	He rides his bike everyday.
Ela tocou uma escala diatônica no piano.	She played a diatonic scale on the piano
João está no carro azul.	John is in the blue car.
Maria gosta de comer comida com açúcar .	Maria likes to eat food with sugar.
O jantar vai ser na terça-feira .	The dinner is going to be on Tuesday.
Eu fui ao supermercado ontem.	I went to the grocery store yesterday.
Eu compro um quilo de comida por gato a cada semana.	I buy a kilo of food per cat each week.
Ela encontrou uma tartaruga na praia.	She found a turtle at the beach.
A gente correu três quilômetros hoje.	We ran three kilometers today.
Ele espera plantar seis árvores este ano.	He hopes to plant six trees this year.
A refeição foi preparada por um chef.	The meal was made by a chef.

A toalha secou rapidamente .	The towel dried quickly.
Até você voltar para casa, vou esperar aqui.	Until you return home, I will wait here.
O orador falou por muito tempo.	The speaker talked for too long.
O currículo da escola está mudando no próximo ano.	The school's curriculum is changing next year.
O trabalhador viu um bando de pássaros voando acima dele.	The worker saw a flock of birds flying above him.
Ela não gosta de maçãs.	She doesn't like apples.
Ela quer consertar a luz antes de ir para a cama.	She wants to fix the light before going to bed.
Eles não sabem pintar o armário .	They don't know how to paint the closet.
A raquete de tênis tem um cabo azul.	The tennis racquet has a blue handle.
Eles estão viajando para Salvador para ver seus parentes.	They are travelling to Salvador to see their relatives.
Sofia trabalha em relações públicas.	Sofia works in public relations.
Ela derramou água na mesa.	She spilled water on the table.
Ele está aprendendo a dirigir um carro .	He is learning to drive a car.
Eu recebi uma carta pelos Correios .	I received a letter in the mail.
O cantor está no coral há sete anos.	The singer has been in the choir for seven years.
Nosso vizinho tem cinco gatos.	Our neighbor has five cats.
Ele ficará cansado até dormir .	He will be tired until he sleeps.
A banda conta com cinco percussionistas .	The band has five percussionists.
O computador possui uma tela retangular .	The computer has a rectangular screen.
A casa com a porta amarela é grande.	The house with the yellow door is big.
O artista esculpiu muitas esculturas.	The artist has built many sculptures.
Ela estuda a estrutura molecular de compostos orgânicos .	She studies the molecular structure of organic compounds.
A gente vai mostrar a eles como limpar as janelas.	We are going to show them how to clean the windows.
Graças à irrigação , o campo recebe bastante água.	Thanks to irrigation, the field receives plenty of water.
Se ele fizer uma estátua, a cidade a colocará no novo parque .	If he makes a statue, the city will put it in the new park.
Ela atende dois clientes por hora.	She sees two clients per hour.
O homem de sapatos azuis é seu tio.	The man with the blue shoes is her uncle.
batata	potato

caminhar <u>u</u>	walk
qualquer <u>u</u>	any
arte <u>u</u>	art
ramo <u>u</u>	branch
escalar <u>u</u>	climb
tomada	outlet
exigir <u>u</u>	demand
cordial	friendly
rolante	rolling
pesquisador <u>u</u>	researcher
parlamento	parliament
responsável	responsible
vender <u>u</u>	sell
janela	window
acontecer <u>u</u>	happen
marrom	brown
revestir <u>u</u>	coat
sorriso	smile
O lago estava congelado no mês passado.	The lake was frozen last month.
A piscina tem uma janela.	The pool has a window in it.
Ela vai vestir uma jaqueta.	She is going to put on a jacket.
A gente gosta de visitar a cidade.	We like to visit the city.
Muitas regiões têm essa planta.	Many regions have this plant.
O pai dele é ator.	His father is an actor.
O inseto foi preservado em âmbar .	The insect was preserved in amber.
A curva dessa estrada é muito acentuada.	The curve on this road is very sharp.
Ela passou sua carreira estudando insetos.	She has spent her career studying insects.
Meu celular tem uma tela grande.	My cell phone has a large screen.
Essa casa tem um lindo jardim .	That house has a beautiful garden.
Ela quer a mesa azul.	She wants the blue table.
O guitarrista dessa banda é muito talentoso.	The guitarist in this band is very talented.
Ele não sabe o que quer .	He doesn't know what he wants.
A gente gosta de ficar em casa quando está frio.	We like to stay at home when it's cold.
Ravioli é sua comida favorita.	Ravioli is his favorite food.
Ele vai esculpir o mármore .	He is going to sculpt the marble.
A gente assistiu a um filme em japonês.	We watched a movie in Japanese.

Até ele chegar , o cachorro vai ficar lá.	Until he arrives, the dog will stay there.
O andar inferior da biblioteca tem muitos romances famosos.	The lower level of the library has many famous novels.
Rimas são comuns em poemas.	Rhymes are common in poems.
Os escritórios estão localizados no andar superior da loja.	The offices are located on the upper level of the store.
Eles estão esperando a semente germinar .	They are waiting for the seed to germinate.
A gente vai trocar de trem na próxima parada.	We are going to transfer trains at the next stop.
Ela tem muitos retratos em sua casa.	She has many portraits in her house.
Ela vai verificar as informações na Internet .	She is going to verify the information on the Internet.
Reduzir a frequência com que ele come em restaurantes é importante para economizar dinheiro.	Reducing how often he eats at restaurants is important for him to save money.
A loja vende muitos rádios .	The store sells many radios.
Ele está bebendo chá gelado com o almoço.	He is drinking iced tea with his lunch.
Se ele quiser ir à loja, ele pode.	If he wants to go to the store, he can.
O tempo estava perfeito durante nossas férias.	The weather was perfect during our vacation.
Ela gosta de usar cúrcuma na sua comida.	She likes to use turmeric in her food.
O estatístico deseja normalizar os dados.	The statistician would like to normalize the data.
O museu tem uma porta que parece um círculo .	The museum has a door that looks like a circle.
A banda tocou um arranjo de uma peça de Beethoven.	The band played an arrangement of a piece by Beethoven.
O barco foi amarrado ao cais.	The boat was tied to the dock.
A baleia azul saltou da água.	The blue whale jumped out of the water.

A4. Consent forms and surveys

A4.1 Salvador consent form-Portuguese

FICHA DE INFORMAÇÕES DE PESQUISA PARA A UNIVERSIDADE DE INDIANA

[Os Róticos do Dialeto de Salvador do Português Brasileiro]

Você está sendo convidado a participar de um estudo de pesquisa. Os cientistas pesquisam para responder a perguntas importantes que podem ajudar a mudar ou melhorar a maneira como fazemos as coisas no futuro. Este documento fornecerá informações sobre o estudo para ajudá-lo(a) a decidir se deseja participar. Leia este formulário e tire suas dúvidas antes de concordar em participar do estudo.

Todas as pesquisas são voluntárias. Você pode optar por não participar deste estudo. Se decidir participar, você pode mudar de ideia mais tarde e sair do estudo a qualquer momento. Você não será penalizado ou perderá quaisquer benefícios se decidir não participar ou optar por deixar o estudo mais tarde.

O objetivo deste estudo é observar a pronúncia dos falantes de Salvador, Bahia.

Estamos perguntando se você deseja participar deste estudo porque é falante nativo do dialeto soteropolitano e mora em Salvador. O estudo está sendo conduzido por Kenneth de Jong, professor de linguística, e Trey Jagiella, doutorando, da Universidade de Indiana em Bloomington, Indiana, EUA.

Se concordar em participar do estudo, você fará o seguinte.

- Ler palavras individuais de slides em um dispositivo de gravação de áudio.
- Ler frases de slides em um dispositivo de gravação de áudio.
- Preencher uma breve pesquisa sobre suas informações demográficas.

Estima-se que a realização dessas tarefas levará cerca de 30 minutos para ser concluída.

Os riscos de participar desta pesquisa são o cansaço visual de olhar para a tela do computador e que alguém possa identificar sua voz se ouvir sua gravação.

Não achamos que você terá quaisquer benefícios pessoais ao participar deste estudo, mas esperamos aprender coisas que ajudarão os pesquisadores no futuro.

Você não será pago por participar deste estudo. Não há custo para participar do estudo.

Vamos proteger suas informações e faremos todos os esforços para manter a confidencialidade de suas informações pessoais, mas não podemos garantir confidencialidade absoluta. Nenhuma informação que possa identificá-lo será compartilhada em publicações sobre este estudo. Apenas os pesquisadores deste projeto terão acesso às gravações. Essas gravações serão destruídas após serem analisadas para determinar a pronúncia nelas.

Suas informações pessoais podem ser compartilhadas fora do estudo de pesquisa, se exigido por lei. Também podemos precisar compartilhar seus registros de pesquisa com outros grupos para garantia de qualidade ou análise de dados. Esses grupos incluem o Conselho de Revisão Institucional da Universidade de Indiana ou seus designados, e agências estaduais ou federais que podem precisar acessar os registros de pesquisa (conforme permitido por lei).

Se você tiver dúvidas sobre o estudo ou encontrar um problema com a pesquisa, entre em contato com o pesquisador, Trey Jagiella em + 1-219-669-5640 ou fjagiell@iu.edu.

Para perguntas sobre seus direitos como participante de pesquisa, para discutir problemas, reclamações ou preocupações sobre um estudo de pesquisa, ou para obter informações ou oferecer sugestões, entre em contato com o escritório do Programa de Proteção em Pesquisa Humana da Universidade de Indiana em + 1-800-696-2949 ou em irb@iu.edu.

CONSENTIMENTO DO PARTICIPANTE

Em consideração a todos os itens acima, concordo em participar deste estudo de pesquisa. Receberei uma cópia deste documento de consentimento informado para manter em meus registros.

Nome do Participante: _____

Assinatura do Participante: _____ **Data:** _____

Nome da Pessoa que Obtém Consentimento: _____

Assinatura da Pessoa que Obtém Consentimento: _____ **Data:** _____

A4.2 Salvador consent form-English

INDIANA UNIVERSITY STUDY INFORMATION SHEET FOR RESEARCH

[The Rhotics of the Salvador Dialect of Brazilian Portuguese]

You are being asked to participate in a research study. Scientists do research to answer important questions that might help change or improve the way we do things in the future. This document will give you information about the study to help you decide whether you want to participate. Please read this form, and ask any questions you have, before agreeing to be in the study.

All research is voluntary. You can choose not to take part in this study. If you decide to participate, you can change your mind later and leave the study at any time. You will not be penalized or lose any benefits if you decide not to participate or choose to leave the study later.

The purpose of this study is to observe the pronunciation of the speakers of Salvador, Bahia.

We are asking you if you want to be in this study because you are a native speaker of the Salvador dialect and you live in Salvador. The study is being conducted by Kenneth de Jong, Professor of Linguistics, and Trey Jagiella, doctoral student, at Indiana University in Bloomington, Indiana, USA.

If you agree to be in the study, you will do the following things.

- Read individual words from slides into an audio recording device.
- Read sentences from slides into an audio recording device.
- Complete a brief survey on your demographic information.

It is estimated that doing these tasks will take about 30 minutes to complete.

The risks of participating in this research are eye strain from looking at the computer screen and that someone may be able to identify your voice if they hear your recording.

We don't think you will have any personal benefits from taking part in this study, but we hope to learn things that will help researchers in the future.

You will not be paid for participating in this study. There is no cost to participate in the study.

We will protect your information and make every effort to keep your personal information confidential, but we cannot guarantee absolute confidentiality. No information which could identify you will be shared in publications about this study. Only the researchers on this project will have access to the recordings. These recordings will be destroyed after they have been analyzed to determine the pronunciation in the recordings.

Your personal information may be shared outside the research study if required by law. We also may need to share your research records with other groups for quality assurance or data analysis. These groups include the Indiana University Institutional Review Board or its designees, and state or federal agencies who may need to access the research records (as allowed by law).

If you have questions about the study or encounter a problem with the research, contact the researcher, Trey Jagiella at +1-219-669-5640 or fjagiell@iu.edu.

For questions about your rights as a research participant, to discuss problems, complaints, or concerns about a research study, or to obtain information or to offer input, please contact the IU Human Research Protection Program office at +1-800-696-2949 or at irb@iu.edu.

PARTICIPANT'S CONSENT

In consideration of all of the above, I agree to participate in this research study. I will be given a copy of this informed consent document to keep for my records.

Participant's Printed Name: _____

Participant's Signature: _____ **Date:** _____

Printed Name of Person Obtaining Consent: _____

Signature of Person Obtaining Consent: _____ **Date:** _____

A4.3 Salvador post-recording survey-Portuguese

PESQUISA PÓS-GRAVAÇÃO

Os Róticos do Dialeto de Salvador do Português Brasileiro

Quantos anos o senhor/a senhora tem? _____

Qual é o seu gênero? (Por favor escolha um.) _____ Masculino
_____ Feminino
_____ Outro (Por favor especifique)

O que melhor descreve sua raça?
_____ Amarelo
_____ Branco
_____ Indígena
_____ Pardo
_____ Preto

Qual das opções melhor descreve sua condição socioeconômica?
_____ Classe baixa
_____ Classe média
_____ Classe alta

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A4.4 Salvador post-recording survey-English

POST-RECORDING SURVEY

The Rhotics of the Salvador Dialect of Brazilian Portuguese

What is your age? _____

What is your gender? (Please select one) _____ Male
_____ Female
_____ Other (please specify)

Which best describes your race?
_____ Asian
_____ White
_____ Indigenous
_____ Mixed Race/Brown
_____ Black

Which best describes your socioeconomic status?
_____ Lower class
_____ Middle class
_____ Upper class

Were you born and raised in Salvador? _____ Yes

_____ No

Have you lived in Salvador for your entire life? _____ Yes

_____ No

A4.5 Salvador extra questions-Portuguese

O que você consideraria ser sua ascendência? (por exemplo, africano, brasileiro, português, japonês, etc.)

De onde é a sua mãe?

De onde é a seu pai?

De onde é sua avó materna?

De onde é seu avô materno?

De onde é sua avó paterna?

De onde é seu avô paterno?

A4.6 Salvador extra questions-English

What would you consider your heritage to be? (e.g., African, Brazilian, Portuguese, Japanese, etc.)

Where is your mother from?

Where is your father from?

Where is your maternal grandmother from?

Where is your maternal grandfather from?

Where is your paternal grandmother from?

Where is your paternal grandfather from?

A4.7 São Paulo consent form-Portuguese

FICHA DE INFORMAÇÕES DE PESQUISA PARA A UNIVERSIDADE DE INDIANA

[Os Róticos do Português Brasileiro]

Você está sendo convidado a participar de um estudo de pesquisa. Cientistas pesquisam para responder a perguntas importantes que podem ajudar a mudar ou melhorar a maneira como fazemos as coisas no futuro. Este documento fornecerá informações sobre o estudo para ajudá-lo(a) a decidir se deseja participar. Leia este formulário e tire suas dúvidas antes de concordar em participar do estudo.

Todas as pesquisas são voluntárias. Você pode optar por não participar deste estudo. Se decidir participar, você pode mudar de ideia mais tarde e sair do estudo a qualquer momento. Você não será penalizado ou perderá quaisquer benefícios se decidir não participar ou optar por deixar o estudo mais tarde.

O objetivo deste estudo é observar a pronúncia dos falantes **da cidade de São Paulo**.

Estamos perguntando se você deseja participar deste estudo porque é falante nativo do dialeto **da cidade de São Paulo**. O estudo está sendo conduzido por Kenneth de Jong, professor de linguística, e Trey Jagiella, doutorando, da Universidade de Indiana em Bloomington, Indiana, EUA.

Se concordar em participar do estudo, você fará o seguinte.

- Ler palavras individuais de slides em um dispositivo de gravação de áudio.
- Ler frases de slides em um dispositivo de gravação de áudio.
- Preencher uma breve pesquisa sobre suas informações demográficas.

Estima-se que a realização dessas tarefas levará cerca de 20 minutos para ser concluída.

Os riscos de participar desta pesquisa são o cansaço visual se você tiver problemas para olhar para a tela do computador e que alguém possa identificar sua voz se ouvir sua gravação.

Em termos de benefícios, esperamos que sua participação contribua para uma maior compreensão do português brasileiro.

Você não será pago por participar deste estudo. Não há custo para participar do estudo.

Vamos proteger suas informações e faremos todos os esforços para manter a confidencialidade de suas informações pessoais, mas não podemos garantir confidencialidade absoluta. Nenhuma informação que possa identificá-lo será compartilhada em publicações sobre este estudo. Apenas os pesquisadores deste projeto terão acesso às gravações. Essas gravações serão destruídas após serem analisadas para determinar a pronúncia nelas.

Suas informações pessoais podem ser compartilhadas fora do estudo de pesquisa, se exigido por lei. Também podemos precisar compartilhar seus registros de pesquisa com outros grupos para garantia de qualidade ou análise de dados. Esses grupos incluem o Conselho de Revisão Institucional da Universidade de Indiana ou seus designados, e agências estaduais ou federais que podem precisar acessar os registros de pesquisa (conforme permitido por lei).

Se você tiver dúvidas sobre o estudo ou encontrar um problema com a pesquisa, entre em contato com o pesquisador, Trey Jagiella em + 1-219-669-5640 ou fjagiell@iu.edu.

Para perguntas sobre seus direitos como participante de pesquisa, para discutir problemas, reclamações ou preocupações sobre um estudo de pesquisa, ou para obter informações ou oferecer sugestões, entre em contato com o escritório do Programa de Proteção em Pesquisa Humana da Universidade de Indiana em + 1-800-696-2949 ou em irb@iu.edu.

CONSENTIMENTO DO PARTICIPANTE

Considerando todos os itens acima, concordo em participar deste estudo de pesquisa.

Receberei uma cópia deste documento de consentimento para manter em meus registros.

Nome do Participante: _____

Assinatura do Participante: _____ **Data:** _____

Nome da Pessoa que Obtém Consentimento: _____

Assinatura da Pessoa que Obtém Consentimento: _____ **Data:** _____

A4.8 São Paulo consent form-English

INDIANA UNIVERSITY STUDY INFORMATION SHEET FOR RESEARCH

[The Rhotics of Brazilian Portuguese]

You are being asked to participate in a research study. Scientists do research to answer important questions that might help change or improve the way we do things in the future. This document will give you information about the study to help you decide whether you want to participate. Please read this form, and ask any questions you have, before agreeing to be in the study.

All research is voluntary. You can choose not to take part in this study. If you decide to participate, you can change your mind later and leave the study at any time. You will not be penalized or lose any benefits if you decide not to participate or choose to leave the study later.

The purpose of this study is to observe the pronunciation of the speakers of **the city of São Paulo**.

We are asking you if you want to be in this study because you are a native speaker of the dialect of **the city of São Paulo**. The study is being conducted by Kenneth de Jong, Professor of Linguistics, and Trey Jagiella, doctoral student, at Indiana University in Bloomington, Indiana, USA.

If you agree to be in the study, you will do the following things.

- Read individual words from slides into an audio recording device.
- Read sentences from slides into an audio recording device.
- Complete a brief survey on your demographic information.

It is estimated that doing these tasks will take about 20 minutes to complete.

The risks of participating in this research are eye strain if you have trouble looking at a computer screen and that someone may be able to identify your voice if they hear your recording.

In terms of benefits, we hope that your participation will help contribute to a greater understanding of Brazilian Portuguese.

You will not be paid for participating in this study. There is no cost to participate in the study.

We will protect your information and make every effort to keep your personal information confidential, but we cannot guarantee absolute confidentiality. No information which could identify you will be shared in publications about this study. Only the researchers on this project will have access to the recordings. These recordings will be destroyed after they have been analyzed to determine the pronunciation in the recordings.

Your personal information may be shared outside the research study if required by law. We also may need to share your research records with other groups for quality assurance or data analysis. These groups include the Indiana University Institutional Review Board or its designees, and state or federal agencies who may need to access the research records (as allowed by law).

If you have questions about the study or encounter a problem with the research, contact the researcher, Trey Jagiella at +1-219-669-5640 or fjagiell@iu.edu.

For questions about your rights as a research participant, to discuss problems, complaints, or concerns about a research study, or to obtain information or to offer input, please contact the IU Human Research Protection Program office at +1-800-696-2949 or at irb@iu.edu.

PARTICIPANT'S CONSENT

In consideration of all of the above, I agree to participate in this research study. I will be given a copy of this informed consent document to keep for my records.

Participant's Printed Name: _____

Participant's Signature: _____ **Date:** _____

Printed Name of Person Obtaining Consent: _____

Signature of Person Obtaining Consent: _____ **Date:** _____

A4.9 São Paulo post-recording survey-Portuguese

PESQUISA PÓS-GRAVAÇÃO

Os Róticos do Português Brasileiro

Quantos anos você tem? _____

Qual é o seu gênero? (Por favor escolha um.) _____ Masculino
_____ Feminino
_____ Outro (Por favor especifique)

Por favor, indique uma ou mais opções que descrevam sua raça:

_____ Amarelo
_____ Branco
_____ Indígena
_____ Pardo
_____ Preto

Qual das opções melhor descreve sua condição socioeconômica?

_____ Classe baixa
_____ Classe média
_____ Classe alta

Você nasceu na cidade de São Paulo? _____ Sim

_____ Não

Você cresceu na cidade de São Paulo? _____ Sim

_____ Não

Você morou na cidade de São Paulo a vida toda? _____ Sim

_____ Não

Se não, onde você morou ao longo de sua vida e por volta de que idade você morou lá?

O que você consideraria ser sua ascendência? (por exemplo, brasileiro, português, japonês, etc.)

De onde é a sua mãe?

De onde é a seu pai?

De onde é sua avó materna?

De onde é seu avô materno?

De onde é sua avó paterna?

De onde é seu avô paterno?

A4.10 São Paulo post-recording survey-English

POST-RECORDING SURVEY

The Rhotics of Brazilian Portuguese

What is your age? _____

What is your gender? (Please select one) _____

Male

Female

Other (please specify)

Please indicate any that describe your race:

Asian

White

Indigenous

Pardo

Black

Which best describes your socioeconomic status?

Lower class

Middle class

Upper class

Were you born in the city of São Paulo?

Yes

No

Were you raised in the city of São Paulo?

Yes

No

Have you lived in the city of São Paulo for your entire life?

Yes

No

If not, where have you lived throughout your life and at around what ages did you live there?

What would you consider your heritage to be? (e.g., Brazilian, Portuguese, Japanese, etc.)

Where is your mother from?

Where is your father from?

Where is your maternal grandmother from?

Where is your maternal grandfather from?

Where is your paternal grandmother from?

Where is your paternal grandfather from?

Francis Edward Jagiella III

EDUCATION

Indiana University, Bloomington, IN

2024: PhD in Linguistics with a minor in French Linguistics

- Primary Advisor: Kenneth de Jong
- Research committee members: Monica Nesbitt, J. Clancy Clements, Barbara Vance

2020: M.A. in Linguistics with a concentration in Computational Linguistics

Purdue University, West Lafayette, IN

2018: B.S. in Applied Statistics and Actuarial Science with minors in French and Linguistics

- Included in this time is a semester studying at Université Paul-Valéry-Montpellier-III in Montpellier, France (January-May 2016).

TEACHING EXPERIENCE

Associate Instructor, Interpersonal Communication (ANTH-A 122)

Fall 2023-present

- Introducing students to linguistic and cultural anthropology, supervising and grading course projects, leading class discussions based on readings.
- Supervisor: Jennifer Meta Robinson

Associate Instructor, Public Oral Communication (COLL-P 155)

Fall 2021-Spring 2023

- Teaching sessions for students to practice enhancing public speaking skills, grading speeches, providing feedback on various preparatory assignments and speeches.
- Supervisor: Cynthia Smith

Associate Instructor, Introductory Graduate Phonetics (LING-L 541)

Spring 2021

- Planning and teaching laboratory sessions (3 sections, once per week), creating and grading homework assignments, organizing and grading exams on phonetic transcription, reviewing progress on group projects and providing feedback.
- Supervisor: Kenneth de Jong.

OTHER EMPLOYMENT

Lead Math Editor/Quality Assurance

February 2019-January 2021

- *Indiana University Assistive Technology and Accessibility Center*: Formatting course documents, to be accessible to visually impaired students at Indiana University; supervising other e-Text editors, answering their questions, and reviewing their work.

X-Force Fellow

Summer 2020

- *National Security Innovation Network*: Internship working with the naval base in Crane, Indiana, to help improve computer code used to complete national security work on base, working along with two other interns remotely.

Graduate Live-In Advisor

August 2018-May 2020

- *Indiana Chapter of Acacia Fraternity*: Assist undergraduate members in operations of chapter, serve on chapter advisory board.

PUBLICATIONS

Peer-Reviewed Papers

Jagiella, T. and Clements, J. C. (accepted, under revision). "The rhotics of the Salvador, Bahia variety of Brazilian Portuguese." *Journal of Pidgin and Creole Languages*.

Zhou, H., Tian, Z., and Jagiella, T. "Tone 4 Sandhi in Heze Chinese." *IULC Working Papers*,

24(1). 1-27.

Jagiella, T. 2019 "Developing a Constraint Grammar for Lithuanian." *Proceedings of the NoDaLiDa 2019 Workshop on Constraint Grammar - Methods, Tools and Applications*.

Conference and Workshop Presentations

Jagiella, T. (upcoming) "Portuguese rhotic variation in the Brazilian cities of Salvador and São Paulo." 186th Meeting of the Acoustical Society of America/2024 Acoustics Week in Canada. Ottawa, Ontario, Canada. May 13-17, 2024.

Jagiella, T. "Rhotic deletion and demographics in the Brazilian cities of Salvador and São Paulo." 2024 Kaleidoscope 18th Annual Graduate Student Conference. Madison, WI. March 8-9, 2024.

Jagiella, T. "Brazilian Portuguese Rhotic Variation and Deletion in Salvador and São Paulo." 12th Central Kentucky Linguistics Conference. Lexington, KY. March 1-2, 2024.

Jagiella, T. "Rhotic Deletion and Demographics in the Salvador, Bahia, Variety of Brazilian Portuguese." 28th Mid-Continental Phonetics and Phonology Conference. West Lafayette, IN. October 20-21, 2023.

Jagiella, T. "Rhotics in the Salvador, Bahia, Dialect of Brazilian Portuguese." 184th Meeting of Acoustical Society of America. Chicago, IL. May 8-12, 2023.

Jagiella, T. "The Rhotics in the Salvador, Bahia Dialect of Brazilian Portuguese." 10th Central Kentucky Linguistics Conference. Lexington, KY. April 1-2, 2022.

Jagiella, T. "The Acoustics of Rhotics in the Salvador, Bahia Dialect of Brazilian Portuguese." 26th Mid-Continental Phonetics and Phonology Conference. Chicago, IL. September 18, 2021.

Zhou, H., Tian, Z., and Jagiella, T. "Tone 4 Sandhi in Heze Chinese." 24th Mid-Continental Phonetics and Phonology Conference. Milwaukee, WI. October 4-5, 2019.

Jagiella, T. "Developing a Constraint Grammar for Lithuanian." Constraint Grammar - Methods, Tools and Applications Workshop at the 22nd Nordic Conference on Computational Linguistics (NoDaLiDa'19). Turku, Finland. September 30, 2019.

HONORS, AWARDS AND GRANTS/FUNDING RECEIVED

Student Organization Grant Allocation Board (Purdue University)

- As president of the Purdue Chapter of Acacia Fraternity in 2017, secured a \$7500 grant for a professional development workshop for members.

College of Arts and Sciences Travel Award

- Received \$500 from the college to aid in travelling to the 53rd Linguistic Symposium on Romance Language (LSRL 53) in Paris, France.

Department of Linguistics Travel Award

- Received \$800 from the department to aid in travelling to LSRL 53.

ASA (Acoustical Society of America) School 2024

- Acoustical Society of America event for graduate students and early career acousticians in all areas of acoustics to learn about and discuss a wide variety of topics

Department of Linguistics at Indiana University: 2019 Householder Paper Honorable Mention

- Awarded based on exceptional quality of work for a paper on a project which explored the sandhi on Tone 4 of the Heze dialect of Chinese.

Acacia Fraternity Order of Pythagoras (2018)

- Established in 1962, The Order of Pythagoras recognizes contributions to Acacia Fraternity by those "serving beyond the ordinary call of duty."

Acacia Fraternity Senior Brother of the Year (2018)

- Awarded to graduating seniors of the fraternity who contributed consistently to the organization through their time as undergraduates.

PROFESSIONAL SERVICE

Scholarly Service

Indiana University Graduate and Professional Student Government Travel Awards March 2024

- Reviewed and rated applications for \$500 awards for academic travel

IULC Working Papers: Ad-hoc Reviewer April 2023

- Reviewed manuscript for publication on phonetics in Andalusian Spanish

Linguistics Club at Indiana University President 2021-22 school year

- Student representative to faculty meetings in the Department of Linguistics, organized colloquium with external speaker, served as Indiana University liaison for the Central Kentucky Linguistics conference, coordinated campus visits for prospective new students, oversaw election of new board members, coordinated club events.

Linguistics Club at Indiana University Treasurer 2020-21 school year

- Oversaw financial operations of the club alongside the president, made purchases for club events, handled sales of club merchandise.

Central Kentucky Linguistics Conference (CKLiC) 2021: Abstract Reviewer March 2021

- Reviewed abstract submissions

Other Service

Purdue Acacia Building Association, Inc.: Ex-Officio Board Member September 2019-present

- Serve as an advisor to the active fraternity members, provide input on facility management decisions.

Darold W. Larson Acacia Leadership Academy: Fellow July 31-August 3, 2019

- Served as a group leader for undergraduates attending the academy by facilitating several discussions relevant to life as a fraternity member.

Kelley School of Business Undergraduate Recognition Ceremony: Name Reader May 3, 2019

- : Announced names of graduating undergraduate students alongside Carolyn Calloway-Thomas.

60th Biennial Conclave of Acacia: Purdue Chapter Alumni Representative August 1-5, 2018

- Vote on changes in policies, elect international officers of Acacia Fraternity to two to four years terms.

CAMPUS INVOLVEMENT AT INDIANA UNIVERSITY

VIDA (Vision, Identity, Drama, and Art) Theatre Group Spring 2022-present

- Performed in *O Rei da Vela (The King of the Candle)* by Oswald de Andrade and *A Falecida (The Deceased)* by Nelson Rodrigues, plays written in the Portuguese language, as well as an adaptation of *A Cartomante (The Fortuneteller)* by Joaquim Maria Machado de Assis.

WIUX Student Radio Station Fall 2018-Spring 2020

- Host on air; record various liners, public service announcements, etc. for production committee; prepare content for and present the WIUX weekly newscast.

SKILLS AND ACHIEVEMENTS

- Passed Actuarial Exam Financial Mathematics/Exam 2 in August 2015.
- Experience working with Python, R, and Matlab, as well as working in a Linux environment.
- Proficient in French.
- Intermediate level in Portuguese.

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